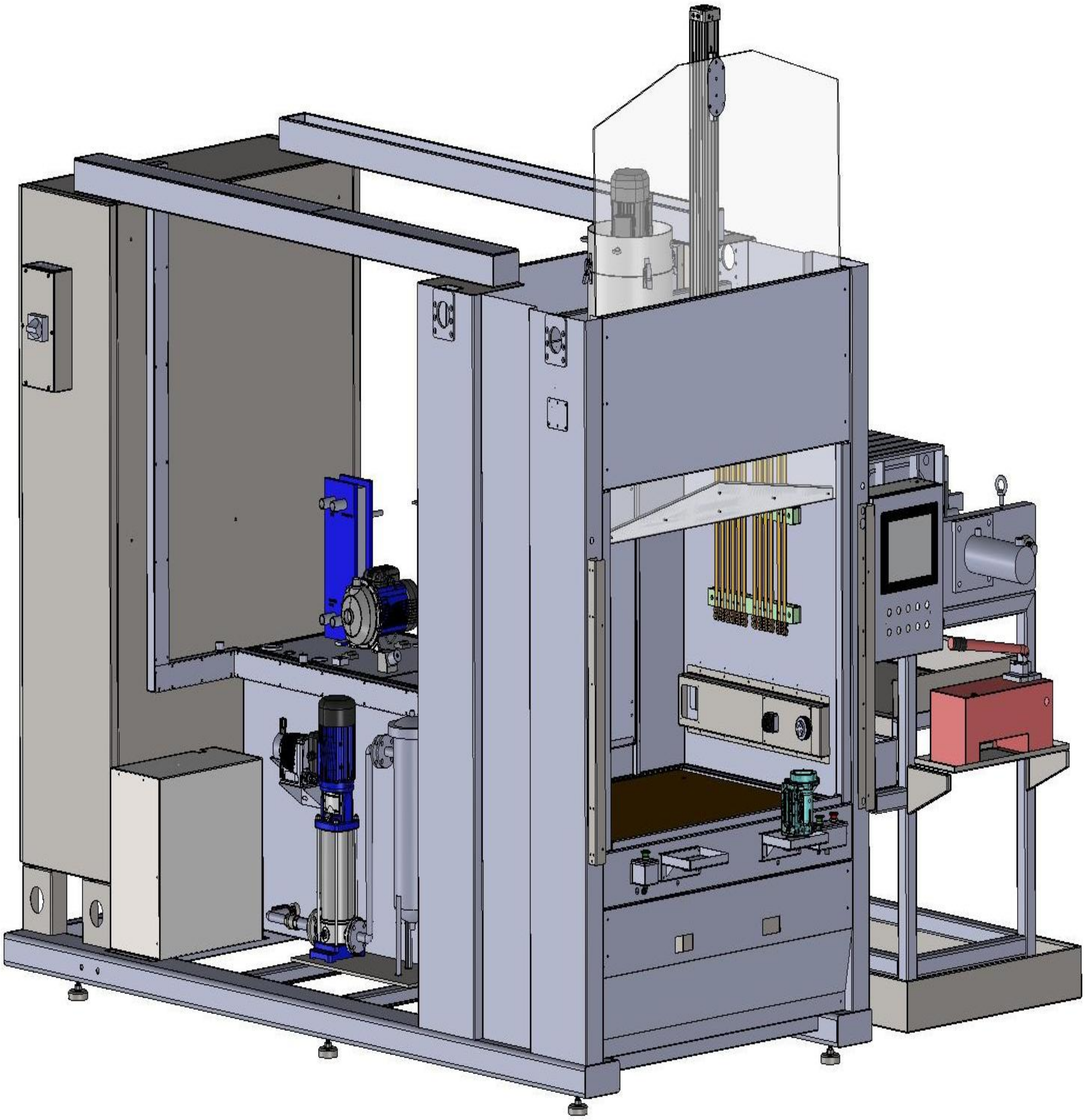
VAMTEC
MACHINES AND AUTOMATION PVT LTD

OPERATION AND MAINTENANCE MANUAL
MINDA ECM-1



August 16, 2024

MINDA CORPORATION LTD , PUNE

DOCUMENT INFORMATION

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TABLE OF CONTENT

1. GENERAL INFORMATION'S

- This manual is considered as a part of this machine (Minda corporation Ltd, Delphi line). Here you will inhale this machine.
- This manual is designed to provide the knowledge to use and maintain ECM at its full efficiency and ingredients are listed in this manual to get full efficiency from this machine.
- Before handling this machine make sure that your knowledge is well enough to operate/use or to carry over the maintenance activities. Well trained operators and engineers should be using this machine
- This SPM machine constructed based on Minda Corporation Ltd - Pune, in order to meet their requirements

2. GENERAL SAFETY WARNINGS

- The device may only be operated, serviced and maintained by people who are familiar with.
- If in doubt, always acquaint oneself with the entire documentation is absolutely necessary!
- In addition, these people must have a respective professional qualification for the task they are to carry out.
- The device with its mechanical, electrical components is equipped with all the necessary safety devices and corresponds to the current technical standards. Hazards to person and property can occur through electrical system parts, however, if they are not handled correctly.
- Changes to the protective devices are not permitted. Operating and servicing instructions must be followed exactly. Make sure only authorized and instructed people have access to the device!
- An operation of the machine is not permitted If the machine is defective Without protection cover by unauthorized modifications to the machine.
- Conversion work or changes to the machine are only permitted after written agreement with the manufacturer.
- In the event of a hazard, the machine must be brought to a standstill by using the emergency stop. Further measures must be specified by the owner/operator.
- The device must be brought to a standstill before any servicing, maintenance or repair work is carried out!
- The separate devices on which the machine consists of are only to be used by one operator. A use at the same time by one operator is inadmissible.

3. GENERAL OPERATION GUIDELINES

3.1. Transportation of the machine

- Since this machine considered as larger, the machine can be split as two portions by separating front and rear side for transportation of the machine. While separating front and rear side ensure that the cable connections and waterline connections are separated from control panel.
- For lifting the machine there are lifting holes available at the top right and left side for front portion. To lift rear end lifting holes available at the bottom of control panel and water tanks.

3.2. Switching the machine power on

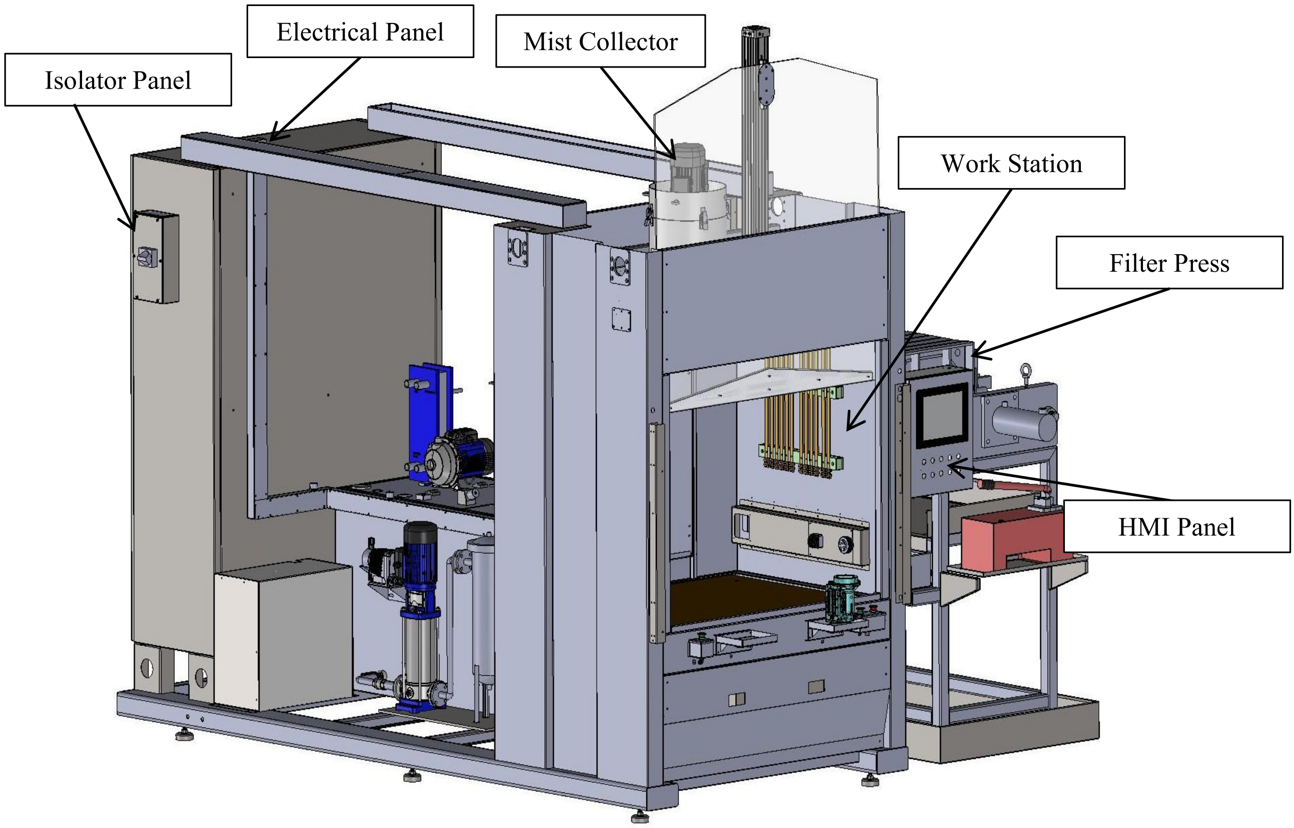
- After the installation of machine at your place make sure the proper electrical connection to the machine. Main power input to be connected to isolator breaker which is mounted on top of left side in control panel.
- Potential earth must be connected to where it requires. Potential earth point is indicated by below mentioned symbol. Before turn on the machine make sure that the voltage between Earth and Neutral is not beyond then 0.3v.
- Along with this machine there is another one power supply cabinet available. This cabinet body should be connected with potential earth at the top.

3.3. Deactivate the machine

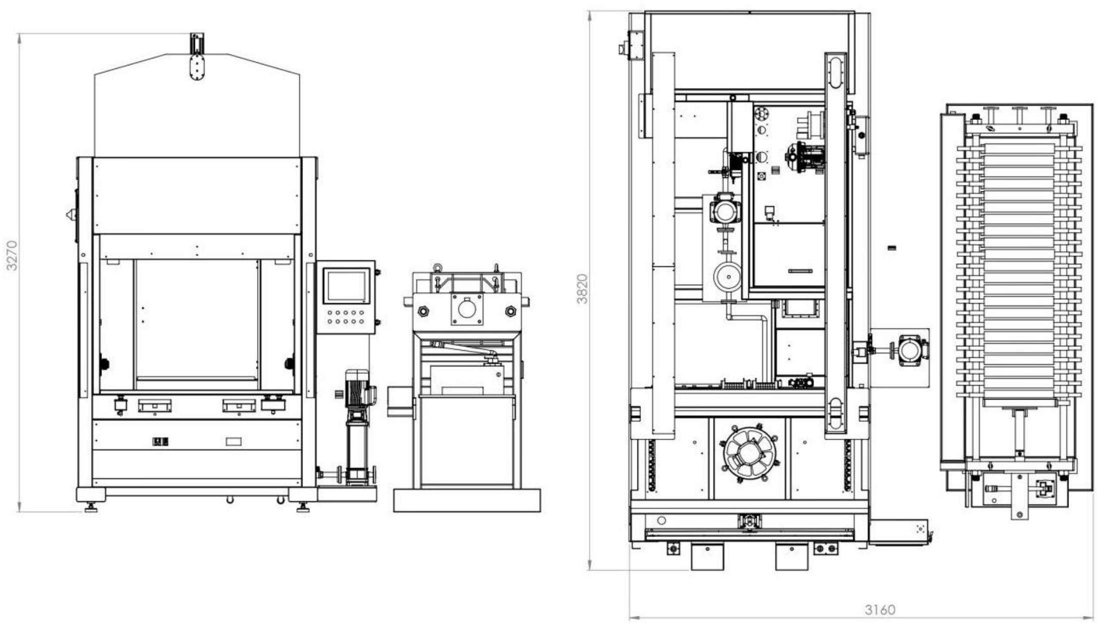
- Before beginning with the dis-assembly, the machine must be disconnected from anycompressed air and power supplies.
- The power connection must be removed!
- Drain the electrolyte from clean and dirty tank using drain valve.
- Parts that are removed must be disposed of according to environmental regulations.
- Materials must be separated (aluminum, steel, plastic, etc.) so that the materials can beprocessed in a recycling system for the protection of resources.

4. GENERAL FUNCTION DESCRIPTION

4.1. Construction of Machine



4.2. Machine Layout



4.3. ECM machining process procedure

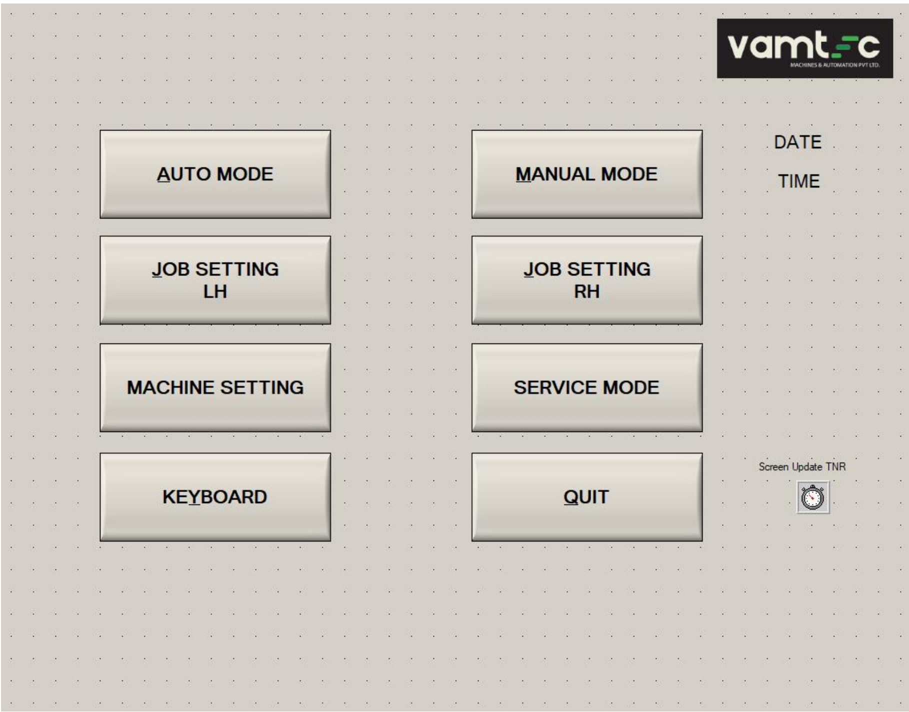
- Check the proper power supply and air supply to the machine.
- Select the Auto Mode in the HMI screen and follow the messages displayed.
- If machine is not in home condition press reset button to get homing process.
- After homing a message will be displayed in the HMI "**SCAN THE INPUT COMPONENT**".
- Pick one fresh component and scan by 2d barcode scanner.
- After scanning, a message will display in the HMI "**LOAD THE COMPONENT**".
- Place the component in the fixture.
- After placing the component, a message will display in the HMI "**WAITING FOR CYCLE START**".
- When the cycle start is pressed, initially the fixture operation will done.
- Then the door will close as it detects the down fixture cylinders reed switch.
- After detecting the door down reed switch, ECM process will start and run for the preset ECM process timing.
- After completion of the deburring process an electrolyte will be drain.
- After that electrolyte drain delay take place the fixture cylinders get reversed.
- The door will get open, a message will be displayed in the HMI "UNLOAD THE COMPONENT".
- The final step is take component and after that the machine will wait for the next cycle start.

5. SELECTION AND DESCRIPTION OF THE OPERATION MODES

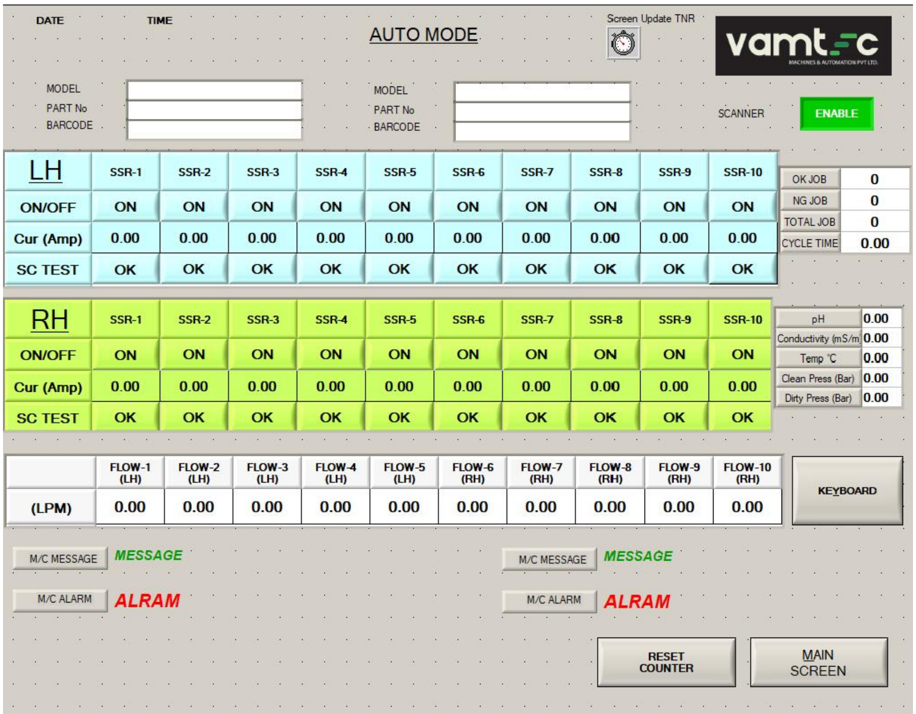
- ❖ This machine has two stations as station 1 and station 2. Both stations are suitable to run both Housing and Front plate components. But at a time either one of the stations can be runnable.
- ❖ While loading the fixtures they must be connected to machine as below mentioned to square connectors available RH and LH side of each station.
- ❖ After connecting fixture to machine select the connected model in corresponded JOB SETTINGS screen.
- ❖ Machine functions are split into six portions as follow shows in main menu

1. Main Menu
2. Auto Mode
3. Manual Mode
4. Job Setting LH
5. Job Setting RH
6. Machine Setting
7. Service Mode

5.1. Main Menu



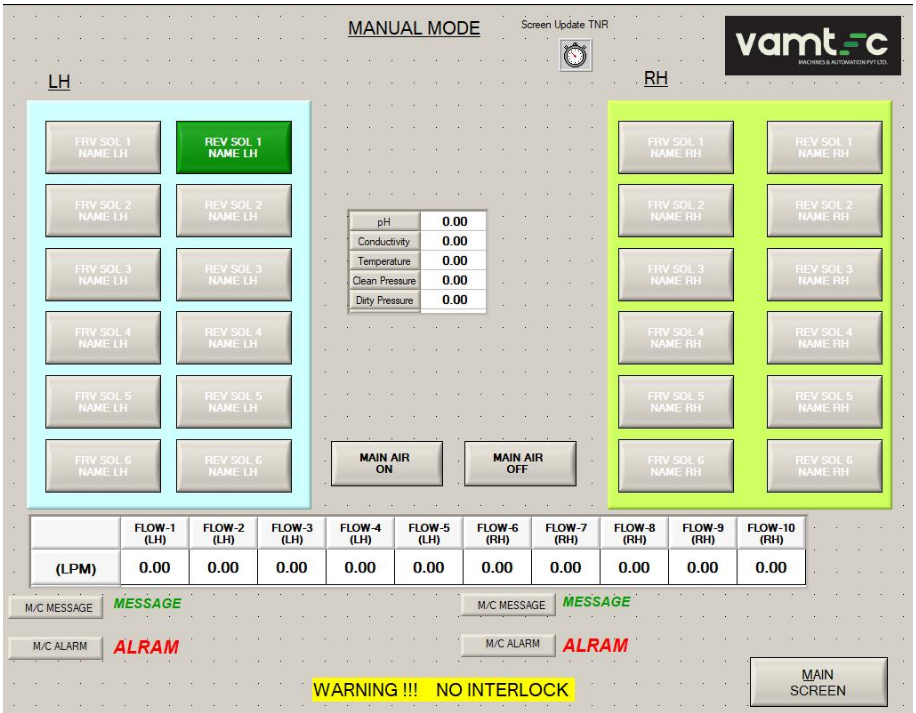
5.2. Auto Mode



❖ Machine will be ready for auto cycle. By fulfilling the necessary conditions auto cycle can be completed. For auto run instruction are displayed in HMI as MESSAGE. While selecting one machine function others can't be functional.

NOTE: Once fixtures were loaded to machine, they should be wash with available wash guns before auto run.

5.3. Manual Mode



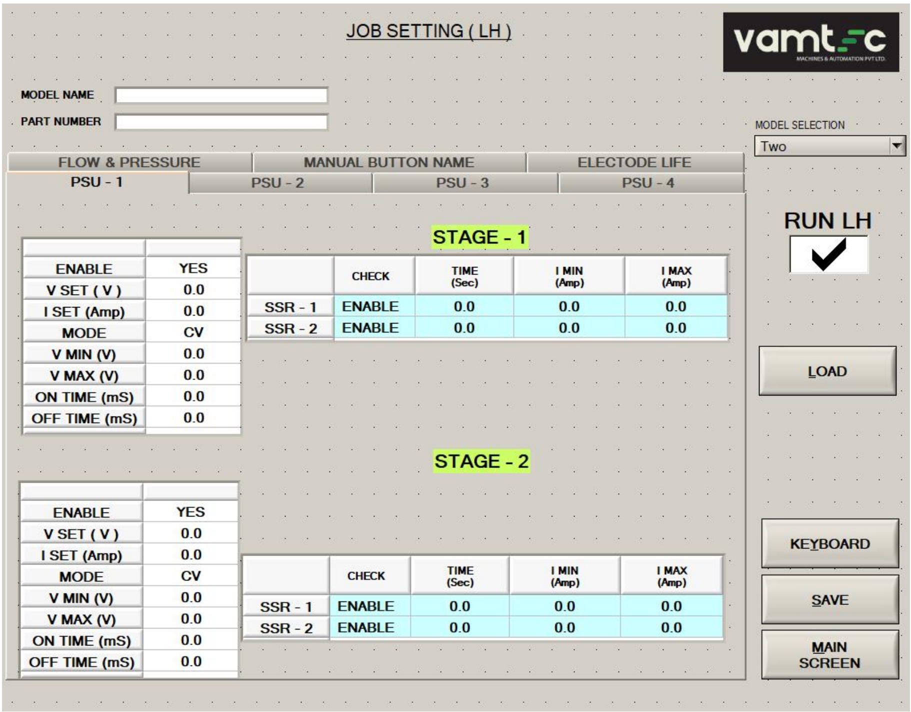
❖ Manual operations are can be operated by selecting manual mode. Once manual mode selected initially all pneumatic solenoids will be turned off. As per requirement we can operate

- fixture cylinders,
- Anode cylinders,
- Safety door open/close,
- Dirty, Clean motor on/off,
- Wash motor on/off,
- Acid motors on/Off,
- Mist collector on/off,
- Chiller on/off.
- Main Air on/off.

5.4. Job Setting LH

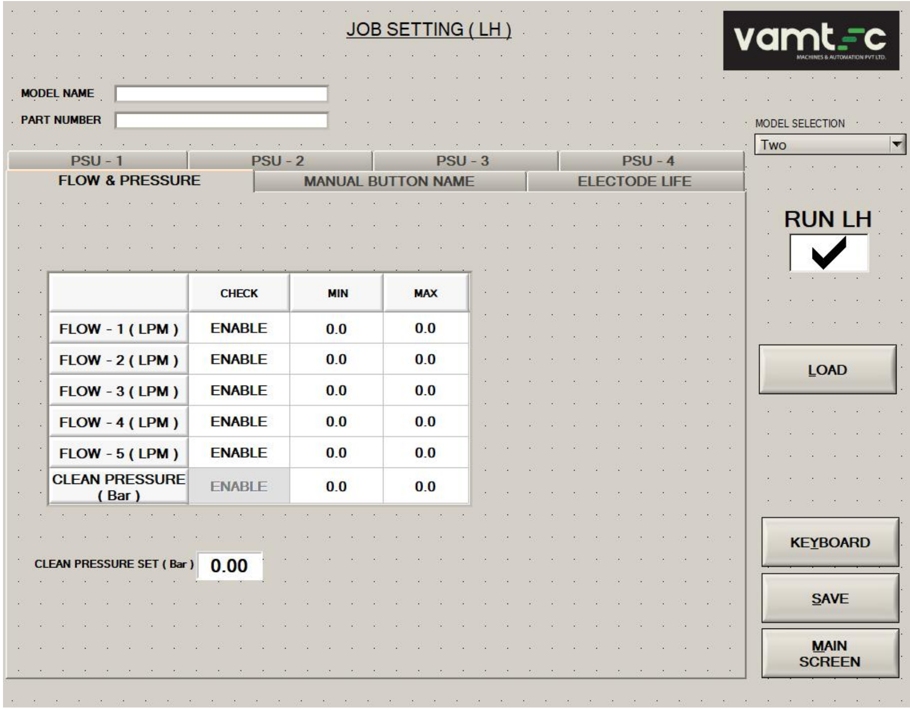
❖ Here job-related settings are available. You can modify those settings. When fixtures are loaded related settings should be selected for auto cycle. In those settings if any changes required you can modify. In Job setting LH screen we can control the following parameters.

5.4.1. PSU Settings.



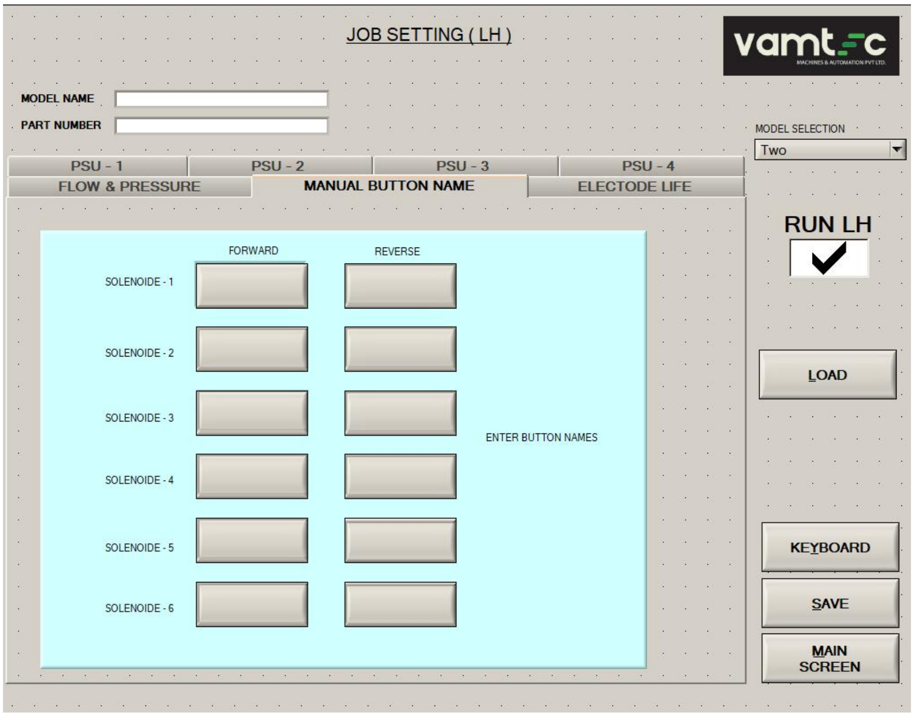
In PSU settings, we can change the mode of PSU, enable or disable the SSR stages, ECM process time, voltage, current, and min/max limits for voltage and current.

5.4.2. Flow and Pressure.



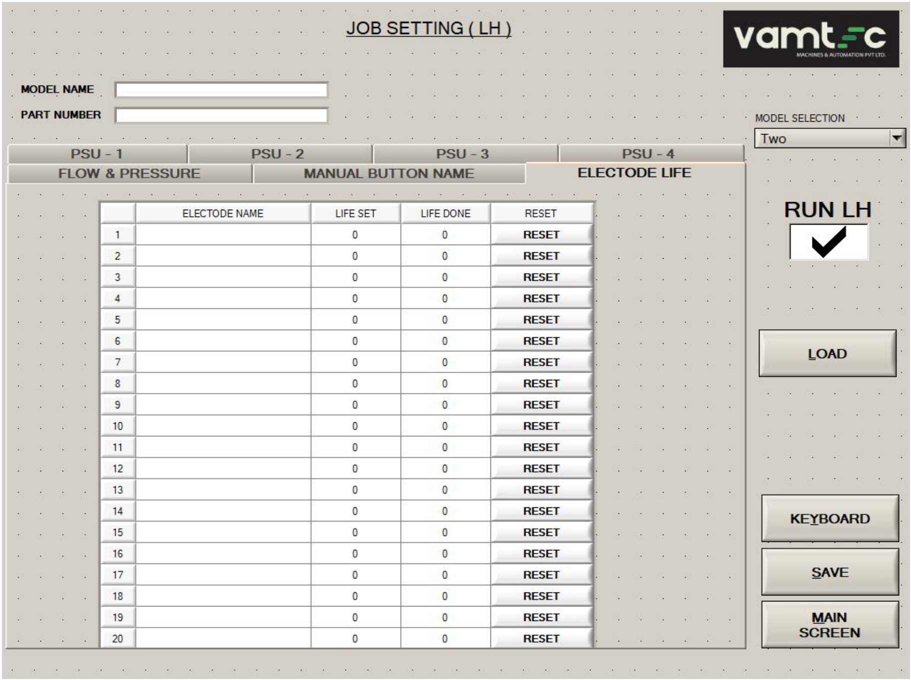
In Flow & Pressure settings we can set the Min & Max values for Flow and set pressure for clean pressure.

5.4.3. Manual Button Name.



In the manual button name, we can set the button name for the manual screen for different models.

5.4.4. Electrode Life Monitoring.

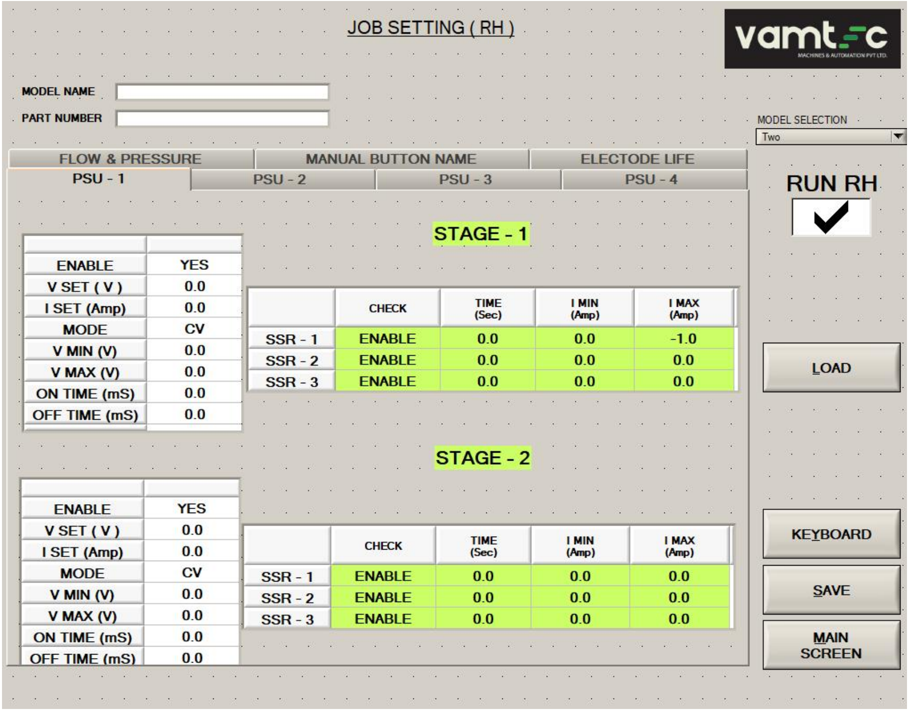


In the Electrode Life Screen, we can monitor the electrode life in the Life done area, set the life in the Life set, and reset the life count.

5.5. Job Setting RH

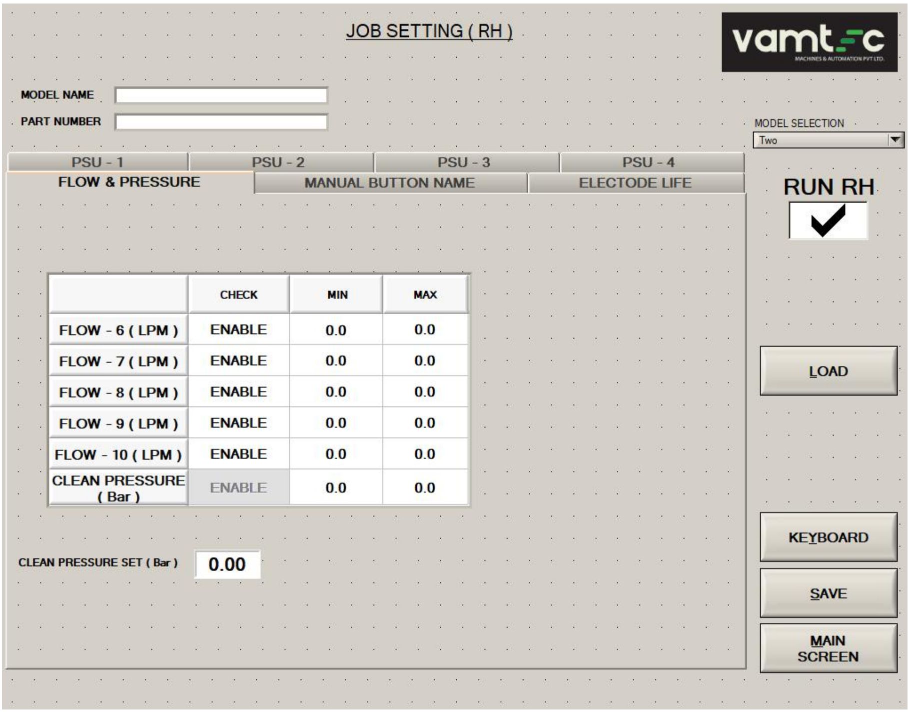
❖ Here job-related settings are available. You can modify those settings. When fixtures are loaded related settings should be selected for auto cycle. In those settings if any changes required you can modify. In Job setting RH screen we can control the following parameters.

5.5.1. PSU Settings.



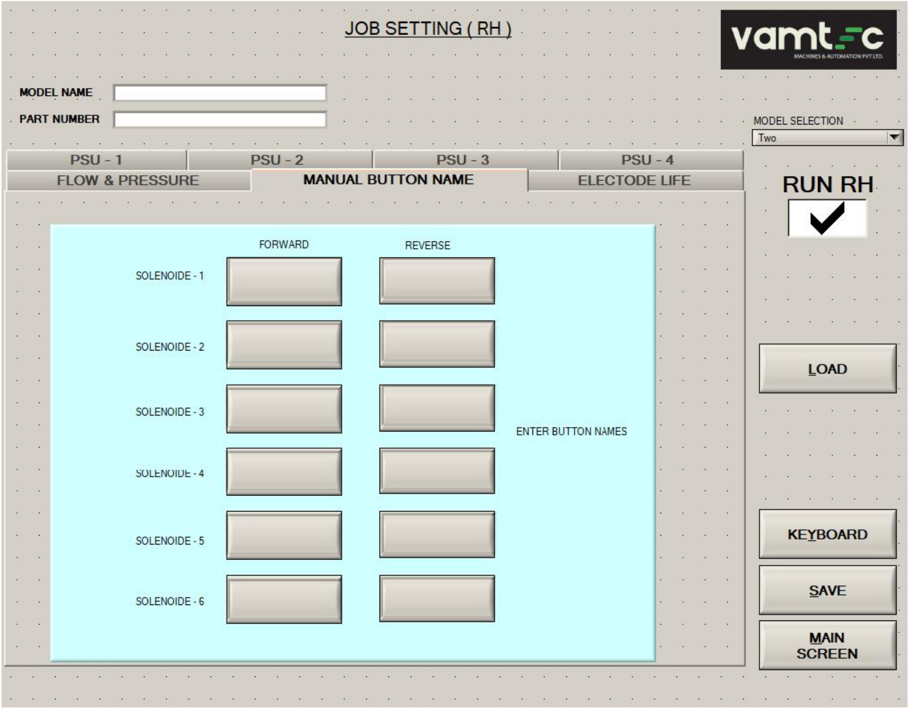
In PSU settings, we can change the mode of PSU, enable or disable the SSR stages, ECM process time, voltage, current, and min/max limits for voltage and current.

5.5.2. Flow and Pressure.



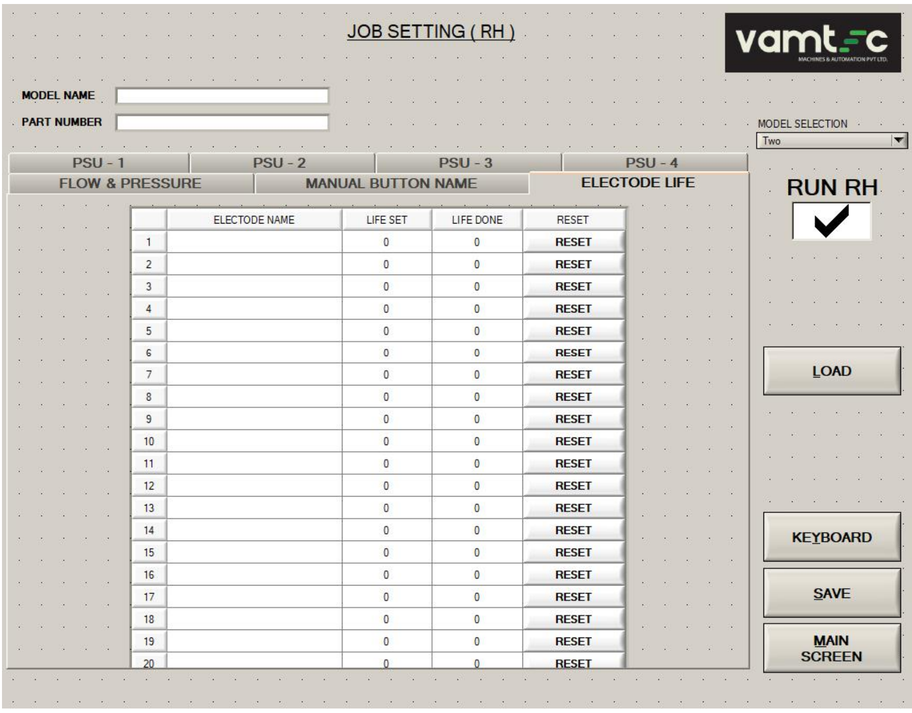
In Flow & Pressure settings we can set the Min & Max values for Flow and set pressure for clean pressure.

5.5.3. Manual Button Name.



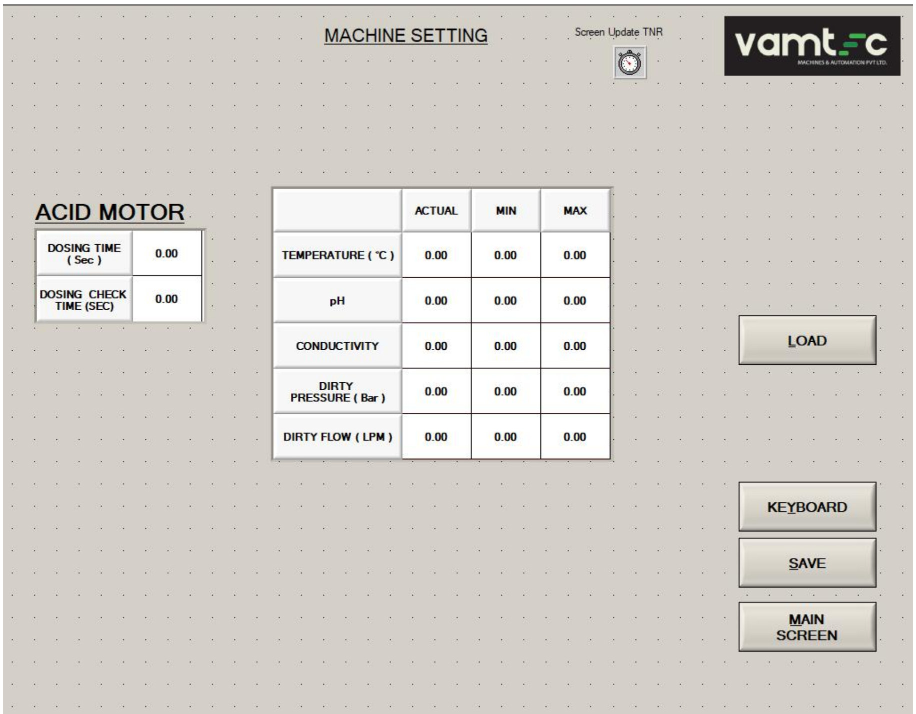
In the manual button name, we can set the button name for the manual screen for different models.

5.5.4. Electrode Life Monitoring.



In the Electrode Life Screen, we can monitor the electrode life in the Life done area, set the life in the Life set, and reset the life count.

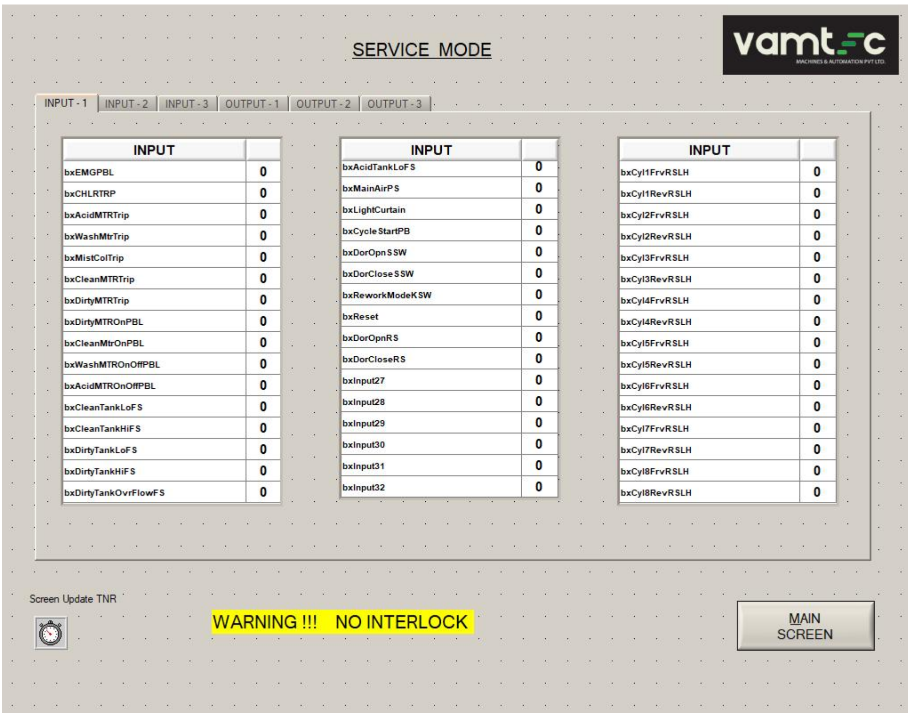
5.6. Machine Setting



Here machine related settings are listed.

- ❖ Dirty Motor.
- ❖ Settings for min, max, and actual values for pressure and flow.
- ❖ Acid Motor
- ❖ Settings for min, max, and actual values for pressure and flow.
- ❖ Electrolyte Temperature.
- ❖ Settings for min, max, and actual values for pressure and flow.
- ❖ Electrolyte pH.
- ❖ Settings for min, max, and actual values for pressure and flow.
- ❖ Electrolyte Conductivity.
- ❖ Settings for min, max, and actual values for pressure and flow.

5.7. Service Mode



In service mode we can monitor the status of each digital inputs and outputs available in the machine

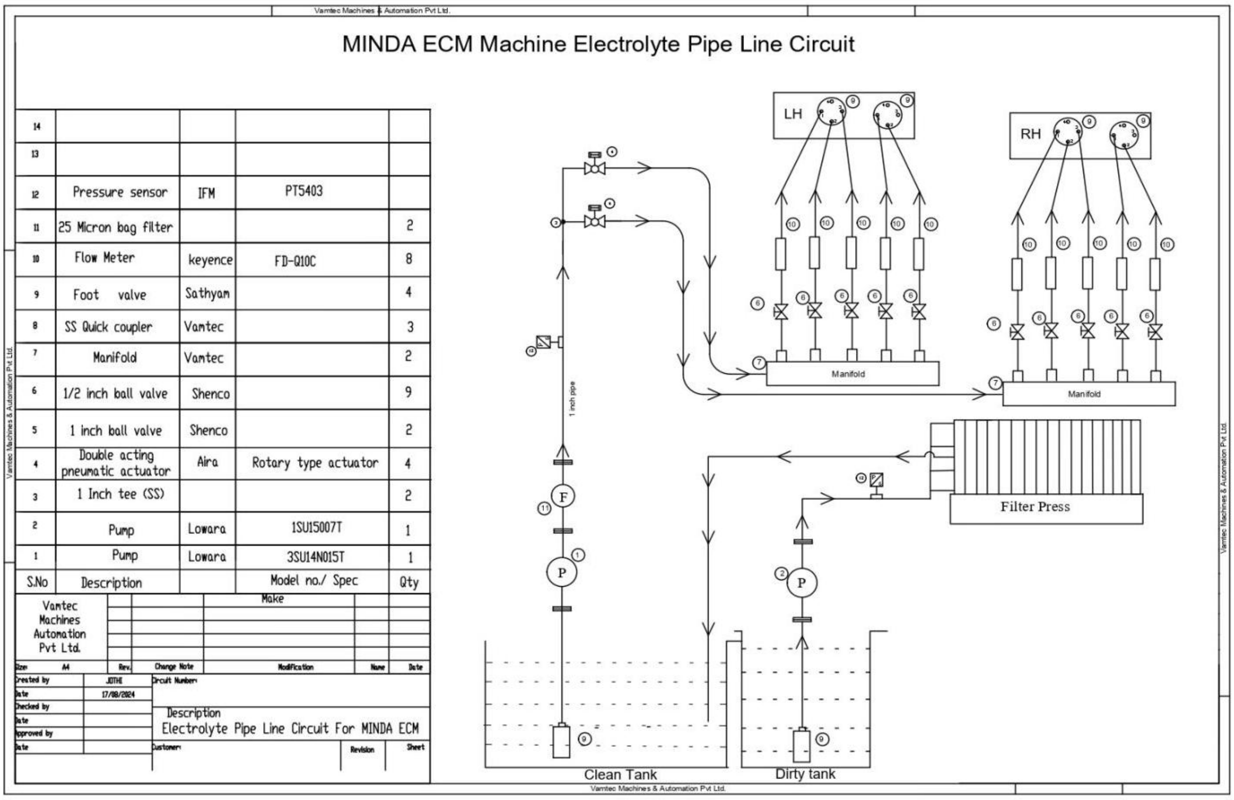
5.8. Measures after an emergency-stop situation

- ❖ When emergency-stop push button has been pressed, the operator must have a clarity on the reason behind the emergency push button being pressed and need to eliminate the causebehind the malfunction of the machine. Until finding out the cause of action, it is prohibited to release emergency-stop and to put the machine back into operation!

- ❖ After the machine has been stopped by emergency-stop, the emergency-stop switch mustbe put back into operating position and press the reset button for operation.

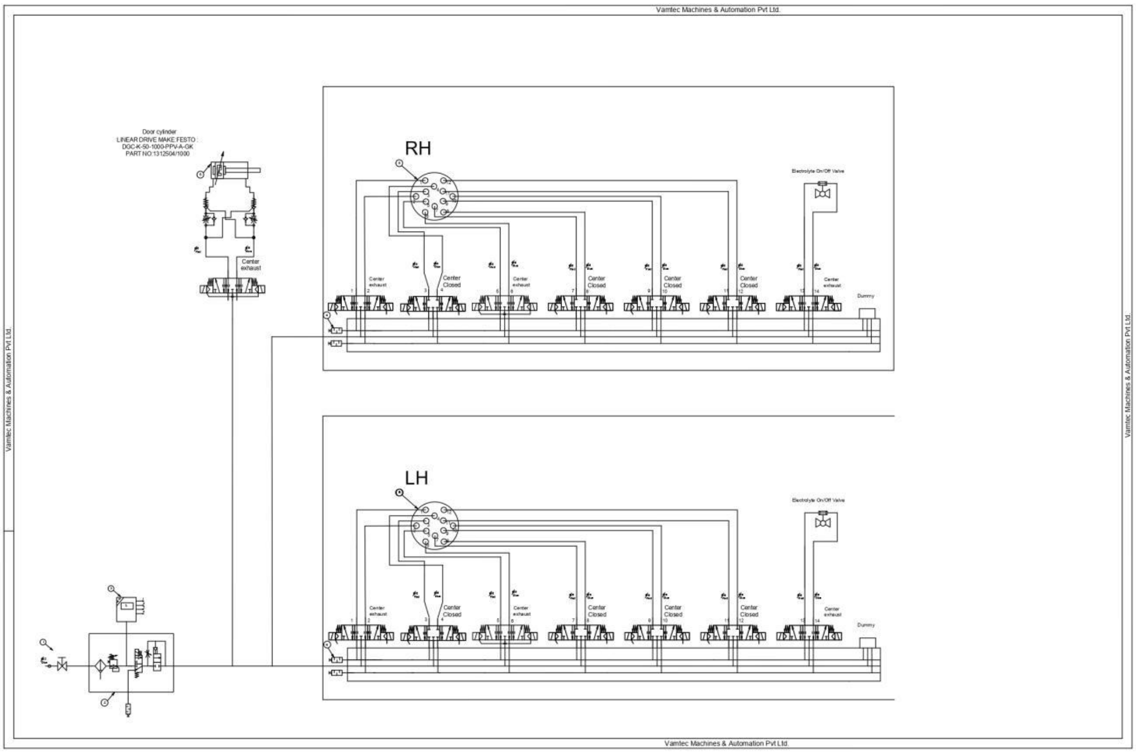
6. ELECTROLYTE CONNECTIONS FOR MACHINE

- ❖ In this ECM machine Electrolyte connections should be connected as follow since parameter were interlocked based on the following order.
- ❖ Each station constructed with five electrolyte flow line diagram as follow,



- ❖ Any variant components are run in any of the station. Electrolyte connection to be done as follow.

7. PNEUMATIC CONNECTIONS FOR MACHINE



- In this ECM machine Pneumatic connections should be connected as follow since parameter were interlocked based on the following order.
- Each station constructed with 7 Solenoid values and one for future expansions.
- The 6 Solenoid values are used for fixture operation and one is used for Electrolyte direction selection.

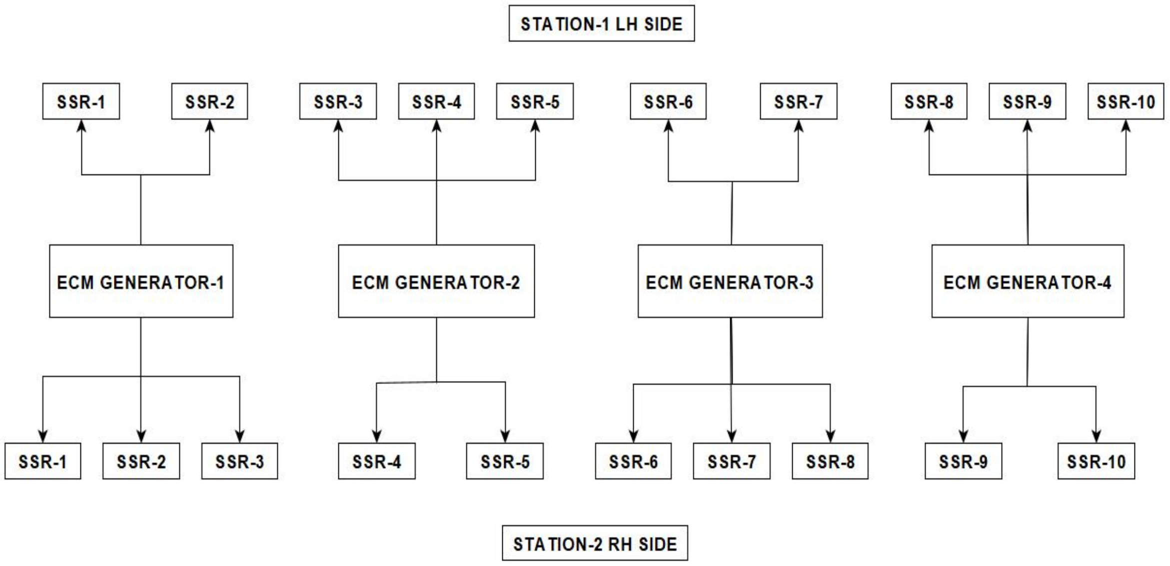
8. ECM GENERATIOR INTERFACING DETAILS

- DC Power supply

  Make: VAMTEC

  Model: 60V 60A

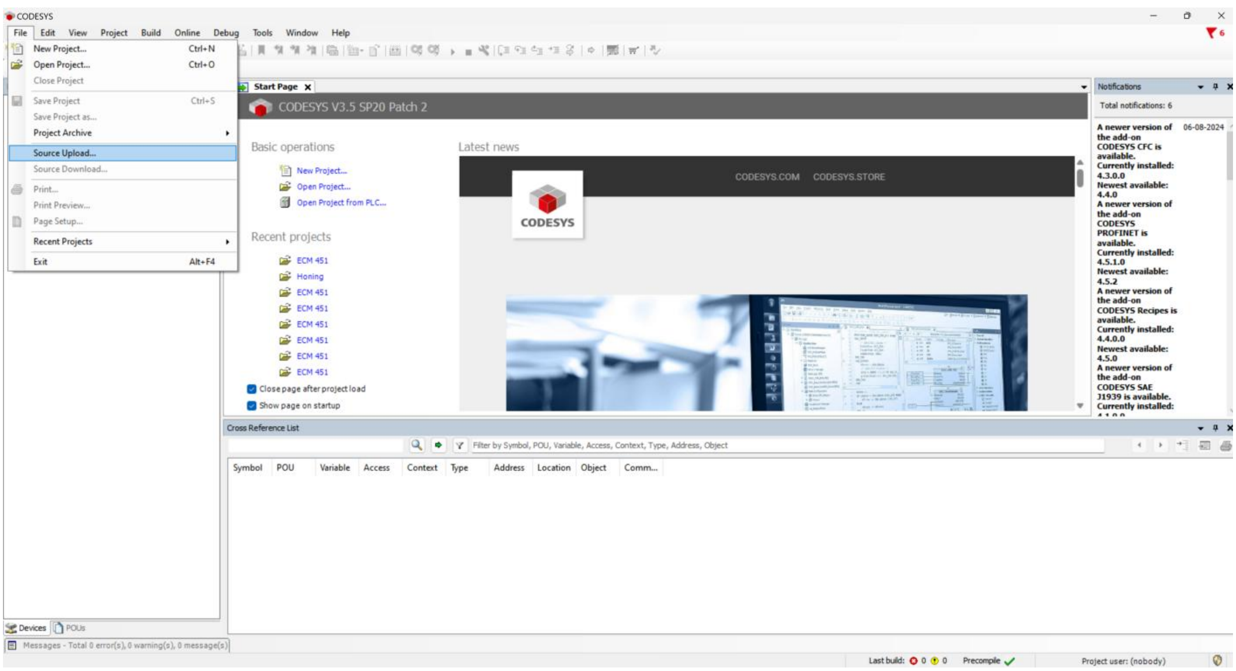
- In this machine four Vamtec ECM Generators are used. Those power sources are connected as follow



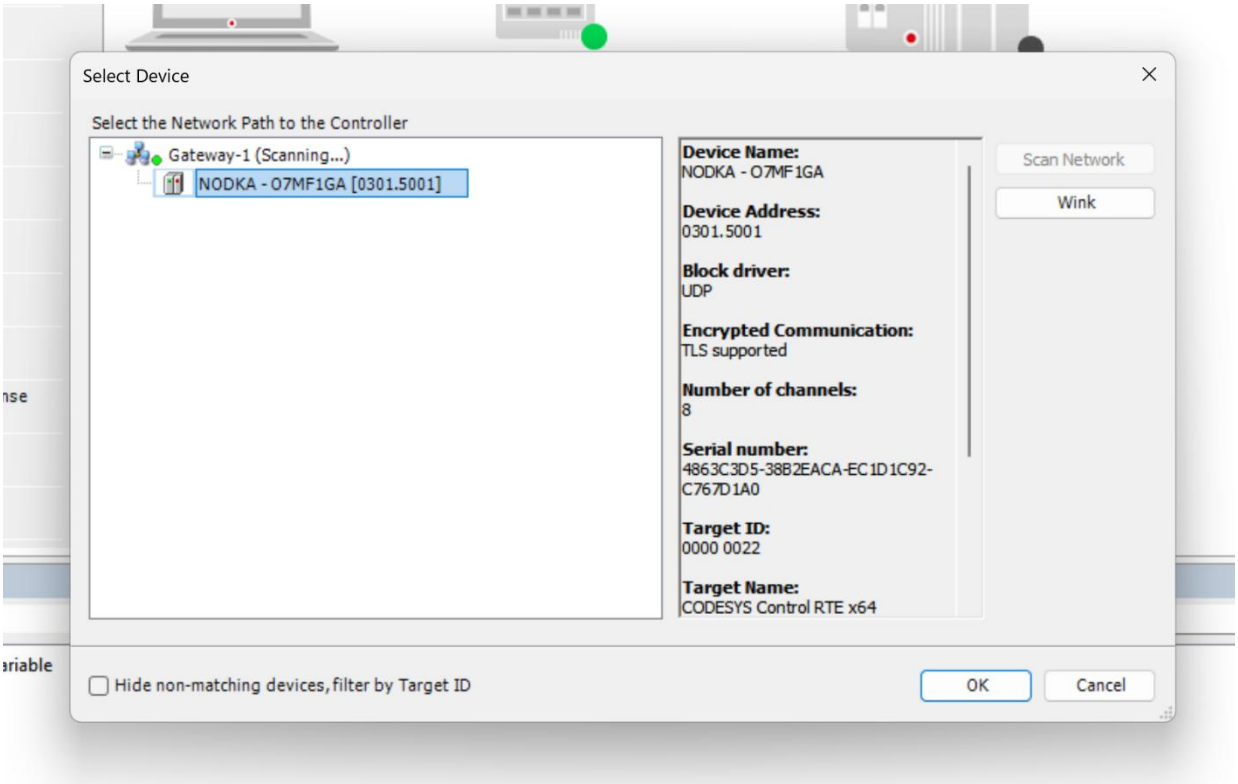
- In this machine four ECM Generators were used. Each can deliver max. of 60 Amps Current with variable Voltage of 0 - 60 VDC. Voltage and Current can be controlled by Modbus RS485 Communication.

9. PROCEDURE FOR TAKING PLC ONLINE

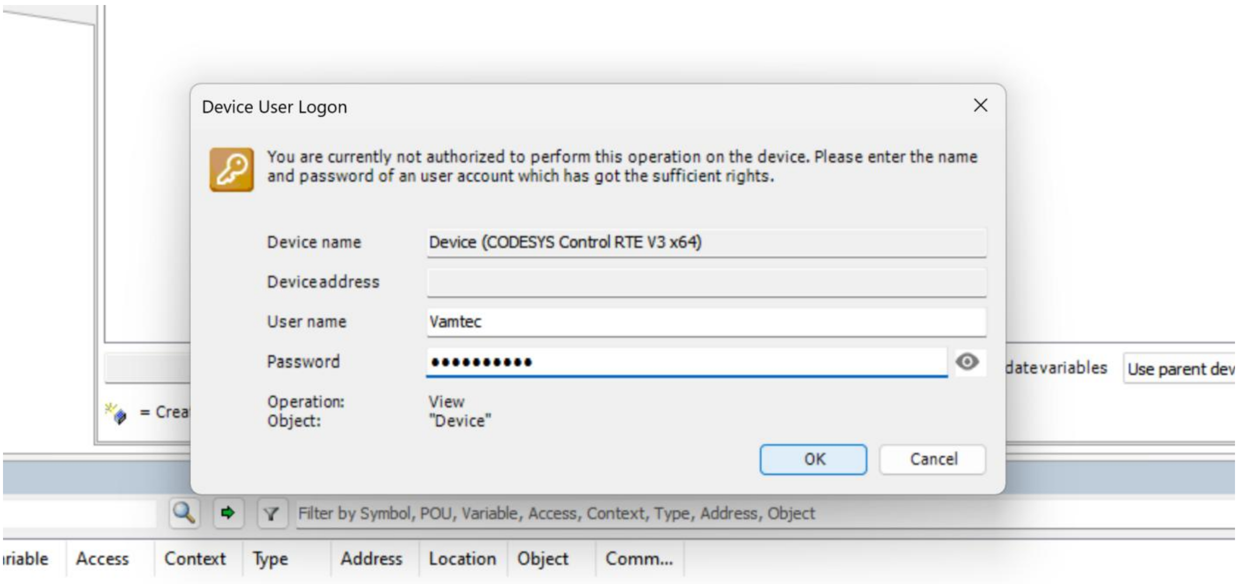
- Connect the Laptop to the Ethernet Switch via Ethernet cable which is located inside the Electric panel.
- Set the Static IP for **PC** as **192.168.0.10** .
- The IP address for **PLC** is **192.168.0.1** .
- Open **CODESYS V3.5 SP20** and navigate to File>Source Upload.



Select the Device Showed in the Dialog box.



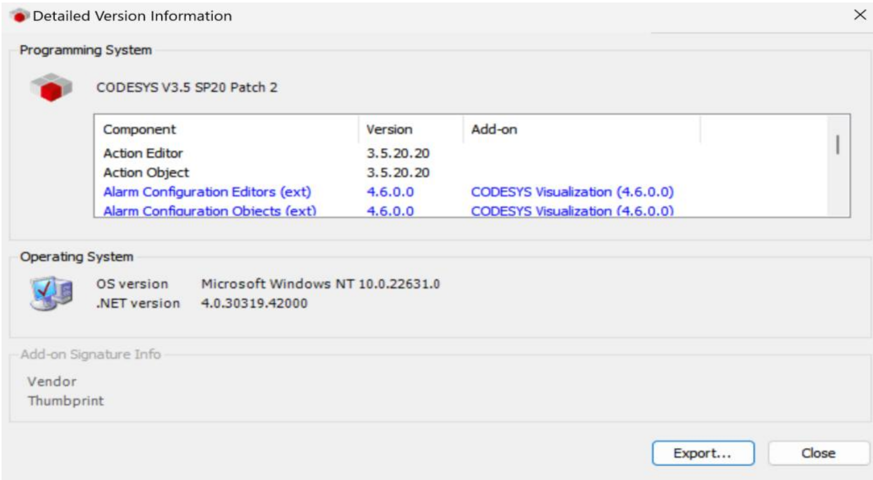
- ❖ Login the device by the following Username and Password.
- ❖ Username : Vamtec.
- ❖ Password : Vamtec@123.



Now the PLC is ready to take online.

10. VERSION INFORMATION OF SOFTWARES

10.1. Version Information of PLC Software



10.2. Version Information of HMI Software

11. I/O LIST

11.1 Digital Inputs

S.NO	LABEL NAME	MODULE NO	TAG
1	bxEMGPBL		6/95
2	bxCHLRTRP		1/95
3	bxAcidMTRTrip		4/95
4	bxWashMtrTrip		7/95
5	bxMistColTrip		8/95
6	bxCleanMTRTrip		2/95
7	bxDirtyMTRTrip		3/95
8	bxDirtyMTROnPBL		9/95
9	bxCleanMtrOnPBL		2/97
10	bxWashMTROnOffPBL		3/97
11	bxAcidMTROnOffPBL		4/97
12	bxCleanTankLoFS		6/97
13	bxCleanTankHiFS		7/97
14	bxDirtyTankLoFS		8/97
15	bxDirtyTankHiFS		9/97
16	bxDirtyTankOvrFlowFS	1	10/97
17	bxAcidTankLoFS		1/99
18	bxMainAirPS		2/99
19	bxLightCurtain		3/99
20	bxCycleStartPB		4/99
21	bxDorOpnSSW		5/99
22	bxDorCloseSSW	2	6/99
23	bxReworkModeKSW		7/99
24	bxReset		8/99
25	bxDorOpnRS		1/101
26	bxDorCloseRS		2/101

27	bxInput27		3/101
28	bxInput28		4/101
29	bxInput29		5/101
30	bxInput30		6/101
31	bxInput31		7/101
32	bxInput32		8/101
33	bxCyl1FrvRSLH	3	1/103
34	bxCyl1RevRSLH		2/103
35	bxCyl2FrvRSLH		3/103
36	bxCyl2RevRSLH		4/103
37	bxCyl3FrvRSLH		5/103
38	bxCyl3RevRSLH		6/103
39	bxCyl4FrvRSLH		7/103
40	bxCyl4RevRSLH		8/103
41	bxCyl5FrvRSLH		1/105
42	bxCyl5RevRSLH		2/105
43	bxCyl6FrvRSLH		3/105
44	bxCyl6RevRSLH		4/105
45	bxCyl7FrvRSLH		5/105
46	bxCyl7RevRSLH		6/105
47	bxCyl8FrvRSLH		7/105
48	bxCyl8RevRSLH		8/105
49	bxJobProxiLH	4	1/107
50	bxSpare1LH		2/107
51	bxSpare2LH		3/107
52	bxSpare3LH		4/107
53	bxSpare4LH		5/107
54	bxSpare5LH		6/107
55	bxFixIden1LH		7/107

56	bxFixIden2LH		8/107
57	bxFixIden3LH		1/109
58	bxFixIden4LH		2/109
59	bxFixIden5LH		3/109
60	bxCyl1FrvRSRH		4/109
61	bxCyl1RevRSRH		5/109
62	bxCyl2FrvRSRH		6/109
63	bxCyl2RevRSRH		7/109
64	bxCyl3FrvRSRH		8/109
65	bxCyl3RevRSRH		1/111
66	bxCyl4FrvRSRH		2/111
67	bxCyl4RevRSRH		3/111
68	bxCyl5FrvRSRH		4/111
69	bxCyl5RevRSRH		5/111
70	bxCyl6FrvRSRH		6/111
71	bxCyl6RevRSRH		7/111
72	bxCyl7FrvRSRH	5	8/111
73	bxCyl7RevRSRH		1/113
74	bxCyl8FrvRSRH		2/113
75	bxCyl8RevRSRH		3/113
76	bxJobProxiRH		4/113
77	bxSpare1RH		5/113
78	bxSpare2RH		6/113
79	bxSpare3RH		7/113
80	bxSpare4RH		8/113
81	bxSpare5RH		1/115
82	bxFixIden1RH	6	2/115
83	bxFixIden2RH		3/115
84	bxFixIden3RH		4/115

85	bxFixIden4RH		5/115
86	bxFixIden5RH		6/115
87	bxSmpsACF		7/115
88	bxSmpsDCF		8/115
89	bxSmpsBatLw		1/117
90	bxSmpsCHRF		2/117
91	bxPsu1On		3/117
92	bxPsu1Fault		4/117
93	bxPsu2On		5/117
94	bxPsu2Fault		6/117
95	bxPsu3On		7/117
96	bxPsu3Fault		8/117
97	bxPsu4On	7	1/118
98	bxPsu4Fault		2/118
99	bxInput99		3/118
100	bxInput100		4/118
101	bxInput101		5/118
102	bxInput102		6/118
103	bxInput103		7/118
104	bxInput104		8/118
105	bxInput105		1/118A
106	bxInput106		2/118A
107	bxInput107		3/118A
108	bxInput108		4/118A
109	bxInput109		5/118A
110	bxInput110		6/118A
111	bxInput111		7/118A
112	bxInput112		8/118A

11.2 Digital Outputs

S.NO	LABEL NAME	MODULE NO	TAG
1	bySSR1On	1	1/119
2	bySSR2On		2/119
3	bySSR3On		3/119
4	bySSR4On		4/119
5	bySSR5On		5/119
6	bySSR6On		6/119
7	bySSR7On		7/119
8	bySSR8On		8/119
9	bySSR9On		1/121
10	bySSR10On		2/121
11	bySSR11On		3/121
12	bySSR12On		4/121
13	bySSR13On		5/121
14	bySSR14On		6/121
15	bySSR15On		7/121
16	bySSR16On		8/121
17	bySSR17On	2	1/123
18	bySSR18On		2/123
19	bySSR19On		3/123
20	bySSR20On		4/123
21	bySCTestRly1		5/123
22	bySCTestRly2		6/123
23	bySCTestRly3		7/123
24	bySCTestRly4		8/123
25	bySCTestRly5		1/125
26	bySCTestRly6		2/125

27	bySCTestRly7		3/125
28	bySCTestRly8		4/125
29	bySCTestRly9		5/125
30	bySCTestRly10		6/125
31	bySCTestRly11		7/125
32	bySCTestRly12		8/125
33	bySCTestRly13		1/127
34	bySCTestRly14		2/127
35	bySCTestRly15		3/127
36	bySCTestRly16		4/127
37	bySCTestRly17		5/127
38	bySCTestRly18		6/127
39	bySCTestRly19		7/127
40	bySCTestRly20		8/127
41	byBtryTmrOn	3	1/129
42	byOutput42		2/129
43	byOutput43		3/129
44	byOutput44		4/129
45	byOutput45		5/129
46	byOutput46		6/129
47	byOutput47		7/129
48	byOutput48		8/129
49	byMistCollectorOn		1/133
50	byDoorOpenSol		2/133
51	byDoorCloseSol		3/133
52	byTwrLmpRed	4	4/133
53	byTwrLmpYellow		5/133
54	byTwrLmpGreen		6/133
55	byMainAirSolOn		7/133

56	byCleanMtrOn		8/133
57	byDirtyMtrOn		1/137
58	byWashMtrOn		2/137
59	byAcidMtrOn		3/137
60	byChillerOn		4/137
61	byScTestSlecRlyLH		5/137
62	byScTestSlecRlyRH		6/137
63	byOutput63		7/137
64	byOutput64		8/137
65	bySol1FrvLH	5	1/141
66	bySol1RevLH		2/141
67	bySol2FrvLH		3/141
68	bySol2RevLH		4/141
69	bySol3FrvLH		5/141
70	bySol3RevLH		6/141
71	bySol4FrvLH		7/141
72	bySol4RevLH		8/141
73	bySol5FrvLH		1/145
74	bySol5RevLH		2/145
75	bySol6FrvLH		3/145
76	bySol6RevLH		4/145
77	bySol7FrvLH		5/145
78	bySol7RevLH		6/145
79	bySol8FrvLH		7/145
80	bySol8RevLH		8/145
81	bySol1FrvRH	6	1/149
82	bySol1RevRH		2/149
83	bySol2FrvRH		3/149
84	bySol2RevRH		4/149

85	bySol3FrvRH		5/149
86	bySol3RevRH		6/149
87	bySol4FrvRH		7/149
88	bySol4RevRH		8/149
89	bySol5FrvRH		1/153
90	bySol5RevRH		2/153
91	bySol6FrvRH		3/153
92	bySol6RevRH		4/153
93	bySol7FrvRH		5/153
94	bySol7RevRH		6/153
95	bySol8FrvRH		7/153
96	bySol8RevRH		8/153
97	byPsu1Rst	7	1/157
98	byPsu1Run		2/157
99	byPsu2Rst		3/157
100	byPsu2Run		4/157
101	byPsu3Rst		5/157
102	byPsu3Run		6/157
103	byPsu4Rst		7/157
104	byPsu4Run		8/157
105	byOutput105		1/161
106	byOutput106		2/161
107	byOutput107		3/161
108	byOutput108		4/161
109	byOutput109		5/161
110	byOutput110		6/161
111	byOutput111		7/161
112	byOutput112		8/161

11.3 Analog Inputs

S.NO	LABEL NAME	MODEL	MODULE NO	TAG
1	nfAdCount1	EL3104	1	5/85
2	nfAdCount2			6/85
3	nfAdCount3			1/85
4	nfAdCount4			3/85
5	nfAdCount5	EL3104	2	1/87
6	nfAdCount6			2/87
7	nfAdCount7			3/87
8	nfAdCount8			4/87
9	nfAdCount9	EL3104	3	1/89
10	nfAdCount10			2/89
11	nfAdCount11			3/89
12	nfAdCount12			4/89
13	nfAdCount13	EL3104	4	4/91
14	nfAdCount14			5/91
15	nfAdCount15			3/91
16	nfAdCount16			1/91
17	nfAdCount17	EL3104	5	2/45
18	nfAdCount18			4/45
19	nfAdCount19			6/45
20	nfAdCount20			8/45
21	nfAdCount21	EL3124	6	10/45
22	nfAdCount22			12/45
23	nfAdCount23			14/45
24	nfAdCount24			16/45
25	nfAdCount25	EL3124	7	1/47
26	nfAdCount26			3/47
27	nfAdCount27			5/47

28	nfAdCount28			7/47
29	nfAdCount29	EL3124	8	9/47
30	nfAdCount30			11/47
31	nfAdCount31			13/47
32	nfAdCount32			15/47
33	nfAdCount33	EL3124	9	1/49
34	nfAdCount34			3/49
35	nfAdCount35			5/49
36	nfAdCount36			7/49

11.4 Analog Outputs

S.NO	LABEL NAME	MODEL	TAG
1	nfDaCount1	EL4022	1/93
2	nfDaCount2		3/93

12. WEAR PARTS LIST

S.No	Part Name	Make	Qty used in Machine	Recommended Qty for stock
1	400V/200v (500VA)Transformer	Jayanthi Transformer	1	1
2	DRS-480-24 SMPS	MeanWell	2	1
3	TDR-240-24 SMPS	MeanWell	1	1
4	400V/80V (16KVA) Transformer	Jayanthi Transformer	1	1
5	1.0HP Pump	Lowara	1	1
6	1.5HP Pump	Lowara	1	1
7	24V Fixture Lamp	Craftech	2	1
8	Programable DC Power Supply	Vamtec	4	1
9	ElectroMechanic Slim Relay	Phoenix contact	28	5
10	Short Circuit PCB	Vamtec	1	1
11	Solid State Relay	ERI	20	1
12	8 Ch Current sensor PCB	Vamtec	2	1
13	4 Ch Current sensor PCB	Vamtec	1	1
14	VLT Micro Drive FC-51	Danfoss	2	1
15	Light Curtain GL-R47F-T	Keyence	1	1
16	USB to RS422 Converter	ICP CON	1	1
17	Panel PC	Advantech	1	1
18	EK1100 EtherCAT Coupler	Beckhoff	2	1
19	pH-600-1 Meter & probe	Sansel	1	1
20	Conductivity meter CC603-1	Sansel	1	1
21	EL3124 Analog Input module	Beckhoff	5	1
22	EL3104 Analog Input module	Beckhoff	5	1
23	EL4012 Analog Output module	Beckhoff	1	1
24	EL1809 Digital Input Module	Beckhoff	7	1
25	EL2809 Digital Outout module	Beckhoff	7	1
26	EL9550 Surge Filter	Beckhoff	2	1

27	FD-Q10C Flow sensor	Keyence	10	1
28	FD-Q32C Flow sensor	Keyence	1	1
29	PT5403 Pressure sensor	IFM Electronic	2	1
30	8 Ch Driver Card	Vamtec	7	3
31	2 Pole MCB 4A	Schneider	1	1
32	2 Pole MCB 6A	Schneider	3	1
33	3 Pole MCB 63A	Schneider	1	1
34	4 Pole MCB 16A	Schneider	1	1
35	3 Pole MCB 6A	Schneider	1	1
36	3 Pole MCB 20A	Schneider	4	1
37	3 Pole MCB 10A	Schneider	2	1
38	LC1D09BD Contactor	Schneider	3	1
39	ISE20-P-01-LD Pressure Switch	SMC	1	1
40	Acid Dosing Pump	PRC	1	1
41	ZBE-101 NO Terminal for Push Button	Schneider	10	2
42	ZBE-101 NC Terminal for Push Button	Schneider	1	1
43	16 Sq mm cable for electrode	Vamtec	12	2
44	25 Sq mm cable for electrode	Vamtec	2	1
45	16 Sq mm M6 SS lug	Vamtec	24	10
46	25 Sm mm M6 SS lug	Vamtec	4	2
47	Panel Cooler (TURBO-1000-M23)	Advance Cooling	1	1
48	Chiller (WC-9000-407-T-CITSS-AC-CSS-ET3-85774)	Advance Cooling	1	1
49	Wash Motor (CEA80/5/0)	Lowara	1	1
50	Mist Collector	FilterMist	1	1
51	Mechanical Relay (MY4N-GS)	Omron	1	1
52	RTD	Vamtec	1	1
53	Reed Switch	SMC	12	2
54	Peak Proximity Sensor	Vamtec	2	1

55	Tower Lamp (T63-HGV0T2-RAG)	Green	1	1
56	Lead Acid Battery 12V, 7Ah	Chloride Safeguard	2	1

13. ALARMS AND ACTIONS

S.NO	ALARMS	ACTIONS
1	OVERALL SHORT CIRCUIT FAIL	Overall short circuit fail
2	SSR SHORT CIRCUIT FAIL	Individual short circuit fail
3	EMERGENCY PRESSED	Emergency pressed release the emergency button and press reset button
4	BATTERY CHARGE LOW	Battery charge low or Battery not connected, Connect the battery
5	CHILLER TRIPPED	Chiller tripped, Reset the MPCB
6	DC FAIL	DC fail / Battery not connected
7	MIST COLLECTOR TRIPPED	Reset the Mist collector MPCB
8	ACID MOTOR TRIPPED	Reset the Acid Motor MCB
9	WASH MOTOR TRIPPED	Reset the Wash motor MPCB
10	CLEAN TANK LEVEL LOW	Add electrolyte in the clean tank
11	CLEAN TANK LEVEL HIGH	Drain electrolyte in the clean tank
12	EMERGENCY PRESSED	Release the Emergency and press Reset
13	AC FAIL	AC input power fail
14	DIRTY TANK OVERFLOW	Wait for dirty tank low level, Check the Dirty motor flow
15	AIR PRESSURE FAULT	Air Pressure not in limit
16	LIGHT CURTAIN INTERRUPT	Light curtain interrupt during door close
17	SSR CURRENT HIGH	SSR current reaches max value, Check the Burr in the component
18	SSR CURRENT LOW	SSR current reaches min value, Check the SSR cable
19	PSU FAULT	PSU Fault, Check the Burr in the component

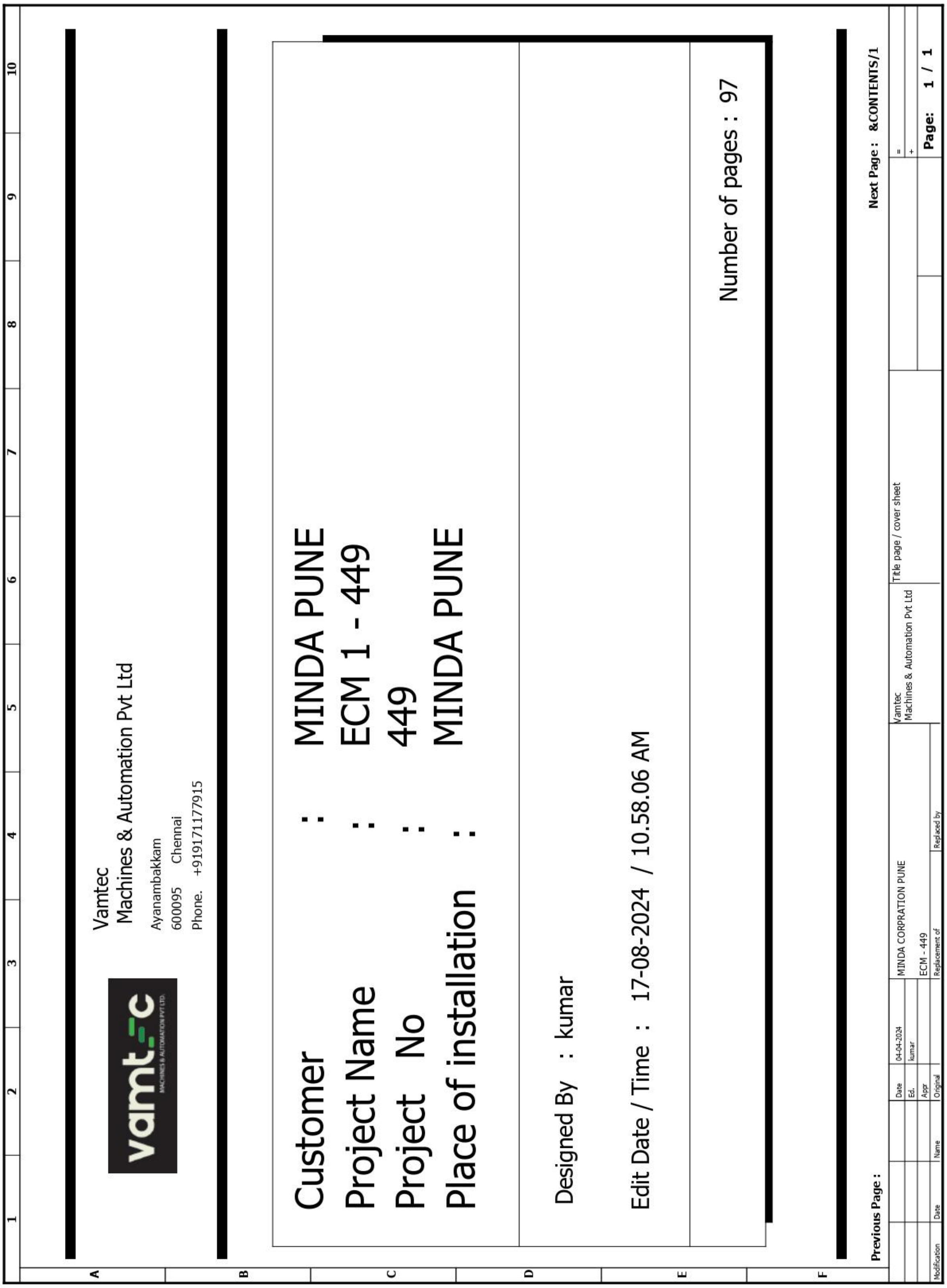
20	FLOW LOW	Electrolyte flow reaches min limit Check the Electrolyte hose
21	FLOW HIGH	Electrolyte flow reaches max limit, Adjust the flow valve
22	ELECTRODE LIFE OVER	Reset the electrode life count in the Job setting screen
23	CLEAN PRESSURE LOW	Clean pressure reaches min value adjust the Clean motor frequency
24	CLEAN PRESSURE HIGH	Clean pressure reaches max value adjust the Clean motor frequency
25	SSR SHORT CIRCUIT FAIL	Check the Burr in the component

14. MESSAGES AND ACTIONS

SL.NO	MESSAGES	ACTIONS
1	OVERALL SHORT CIRCUIT CHECKING	Overall Short Circuit process is going on
2	OVERALL SHORT CIRCUIT PASS	Overall Short Circuit Process pass
3	OVERALL SHORT CIRCUIT FAIL	Overall Short Circuit Process fail
4	ELECTROLYTE DRAINING	Electrolyte draining process starts
5	WRONG PART	Wrong part Scanned, Scan the correct part
6	WAIT FOR DIRTY TANK DRAIN	Dirty tank full wait for Dirty tank drain
7	MODEL NOT SELECTED	Select the model in Job Setting screen
8	FILLING ELECTROLYTE	Electrolyte filling in the component
9	CONFIRMATION NOT GET	Reed switch confirmation not get check the cylinder/reed switch position
10	DOOR CLOSING KEEP AWAY	Door closing keep away the hands from the door
11	ECM PROCESSING	ECM Processing starts
12	RESET THE ELECTRODE LIFE	Electrode life over reset the Electrode Life Count

13	SSR INDIVIDUAL SHORT CIRCUIT CHECK	Individual short circuit check starts
14	`CYLINDERS NOT WORK	Cylinders not work when Light Curtain get interrupt
15	SCAN THE COMPONENT	Scan the input component by 2d barcode scanner
16	UNLOAD THE COMPONENT	Unload the component from the fixture
17	LOAD THE JOB	Load the Job in the fixture
18	PRESS THE CYCLE START	Press the cycle start after loading the fixture

15. ELECTRICAL DIAGRAM



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			Appr		ECM - 449		
Modification	Date	Name	Original		Replacement of / Replaced by		

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Ed. kumar
Appr
Original

MINDA CORPRATION PUNE

ECM - 449

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Modification | Date | Name

Vamtec
Machines & Automation Pvt Ltd

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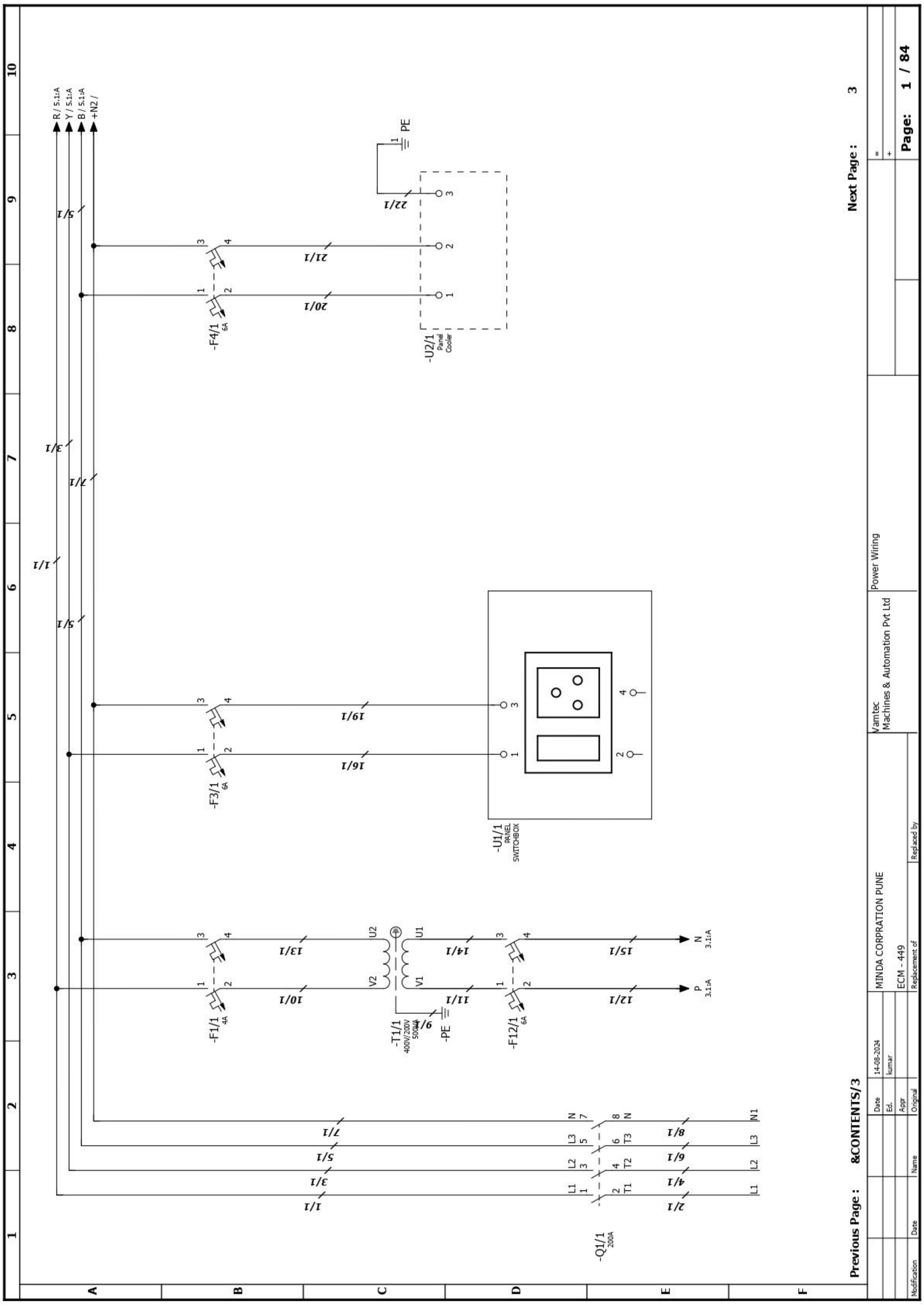

Power Wiring
Vamtec
Machines & Automation Pvt Ltd
MINDA CORPRATION PUNE
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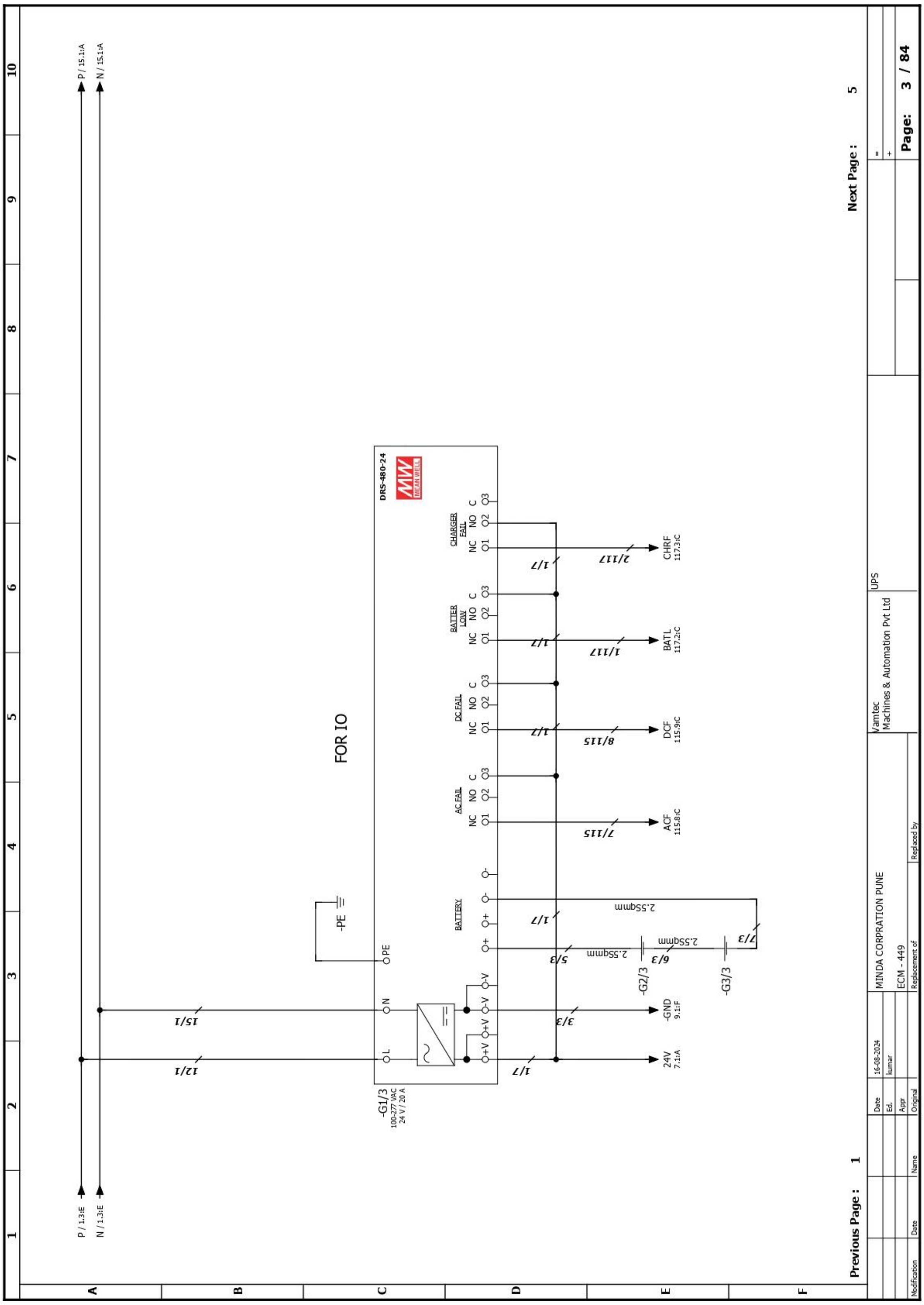
Previous Page : &CONTENTS/3
Date 14-08-2024
Ed. kumar
-Q1/1
200A
-F1/1
4A
-T1/1
400V/200V
-F12/1
6A
-F3/1
6A
-U1/1
PANEL
SWITCHBOX
-F4/1
6A
-U2/1
Panel
Cooler
PE
R / 5.1:A
Y / 5.1:A
B / 5.1:A
+N2 /


P / 15.1:A
N / 15.1:A
P / 1.3:E
N / 1.3:E
FOR IO
DRS-480-24
MEAN WELL
-G1/3
100-277 VAC
24 V / 20 A
L
N
PE
-PE
+V
+V
-V
-V
BATTERY
AC FAIL
DC FAIL
BATTER LOW
CHARGER FAIL
NC NO C
ACF
115.8:C
DCF
115.9:C
BATL
117.2:C
CHRF
117.3:C
7/115
8/115
1/117
2/117
1/7
12/1
15/1
3/3
5/3
6/3
7/3
2.5Sqmm
-G2/3
-G3/3
24V
7.1:A
-GND
9.1:F
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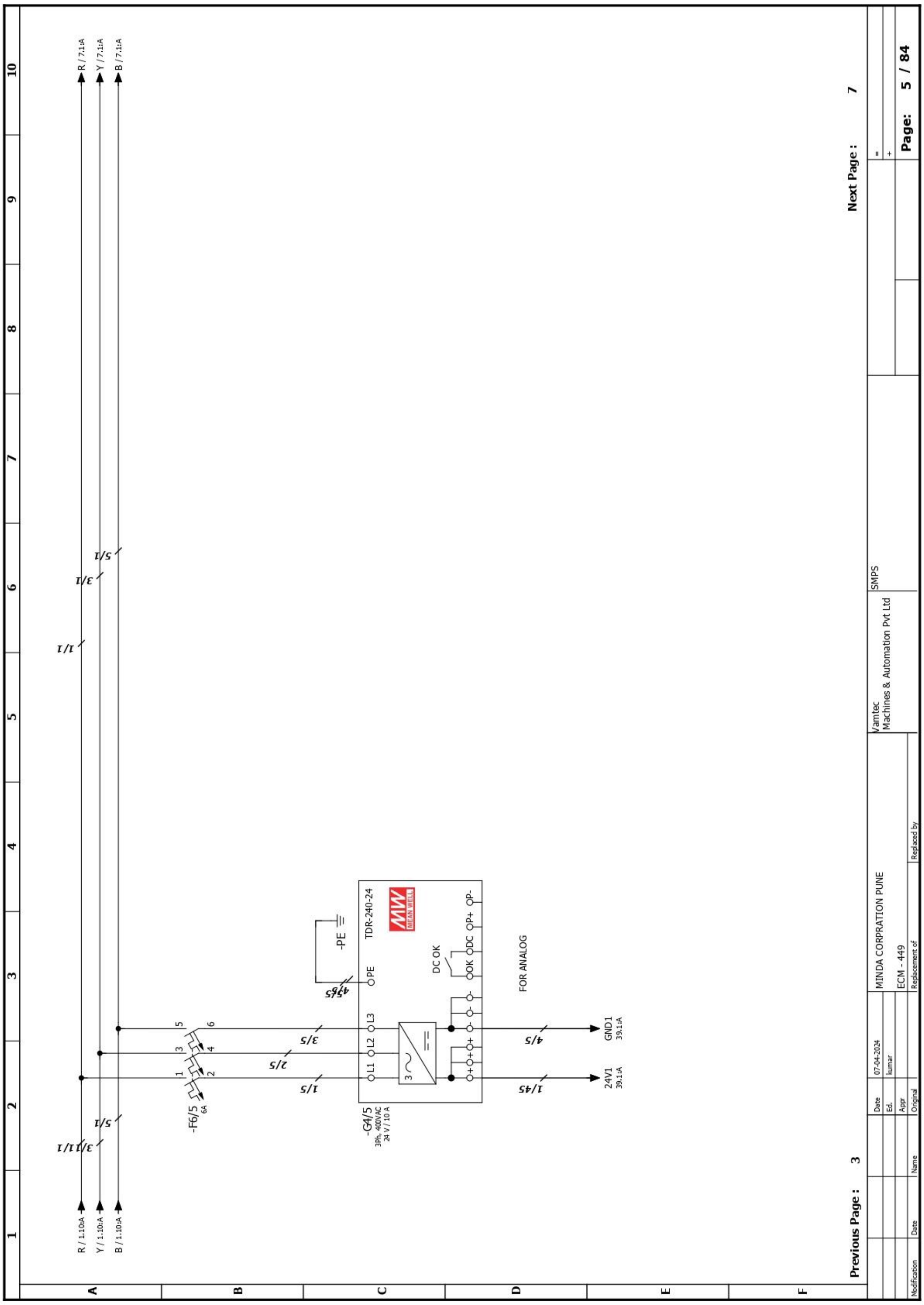
Machines & Automation Pvt Ltd
UPS



R / 1.10:A
Y / 1.10:A
B / 1.10:A
R / 7.1:A
Y / 7.1:A
B / 7.1:A
1/1
3/1
5/1
3/1 1/1
-F6/5
6A
-G4/5
3Ph, 400VAC
24 V / 10 A
TDR-240-24
MEAN WELL
-PE
PE
L1
L2
L3
DC OK
OK
DC
P+
P-
FOR ANALOG
1/5
2/5
3/5
4/5
1/45
24V1
39.1:A
GND1
39.1:A
Previous Page : 3
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Ed. kumar
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Replacement of
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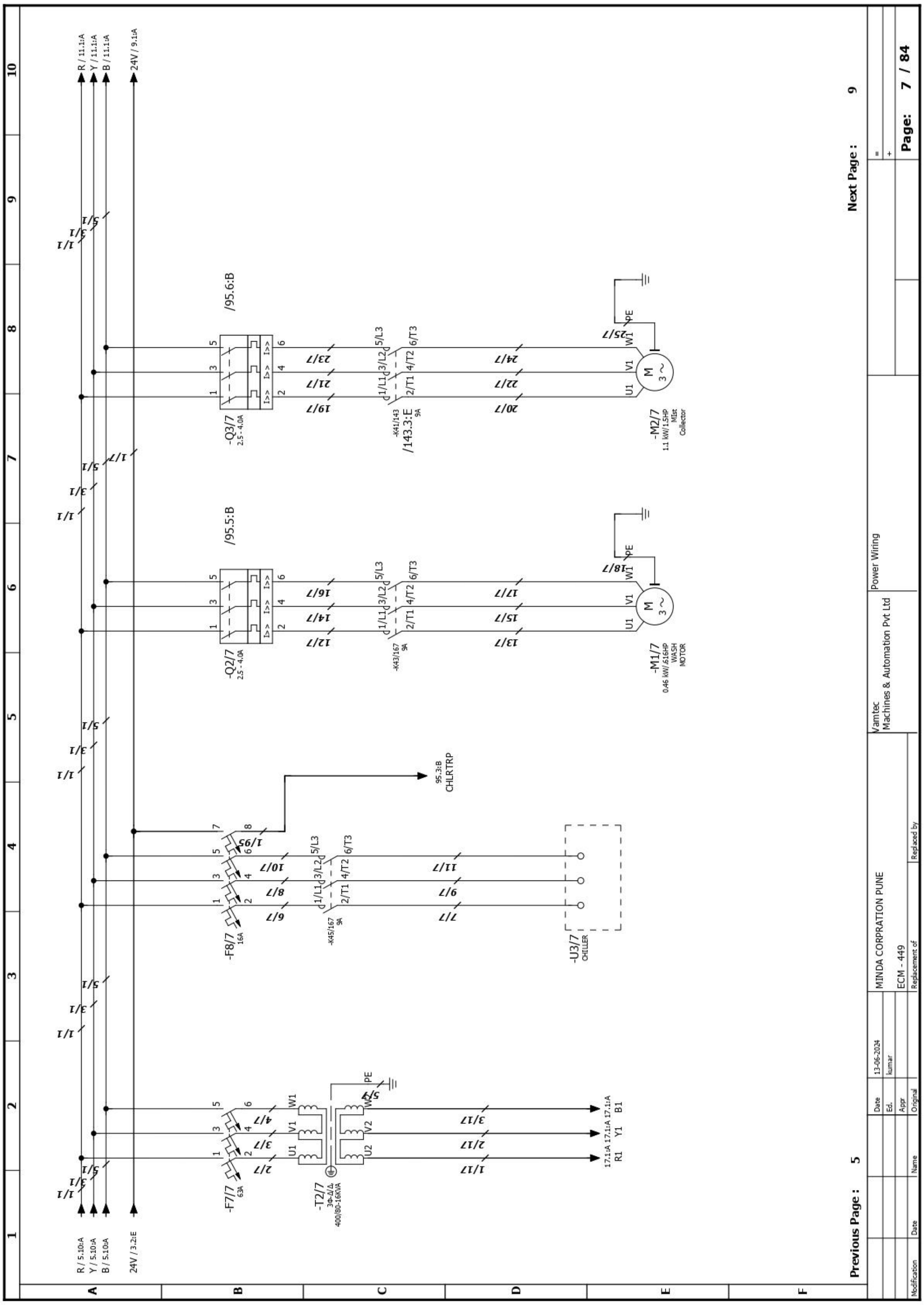
Machines & Automation Pvt Ltd
SMPS
Modification
Date
Name


Power Wiring
Vamtec
Machines & Automation Pvt Ltd
MINDA CORPRATION PUNE
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13-06-2024
kumar
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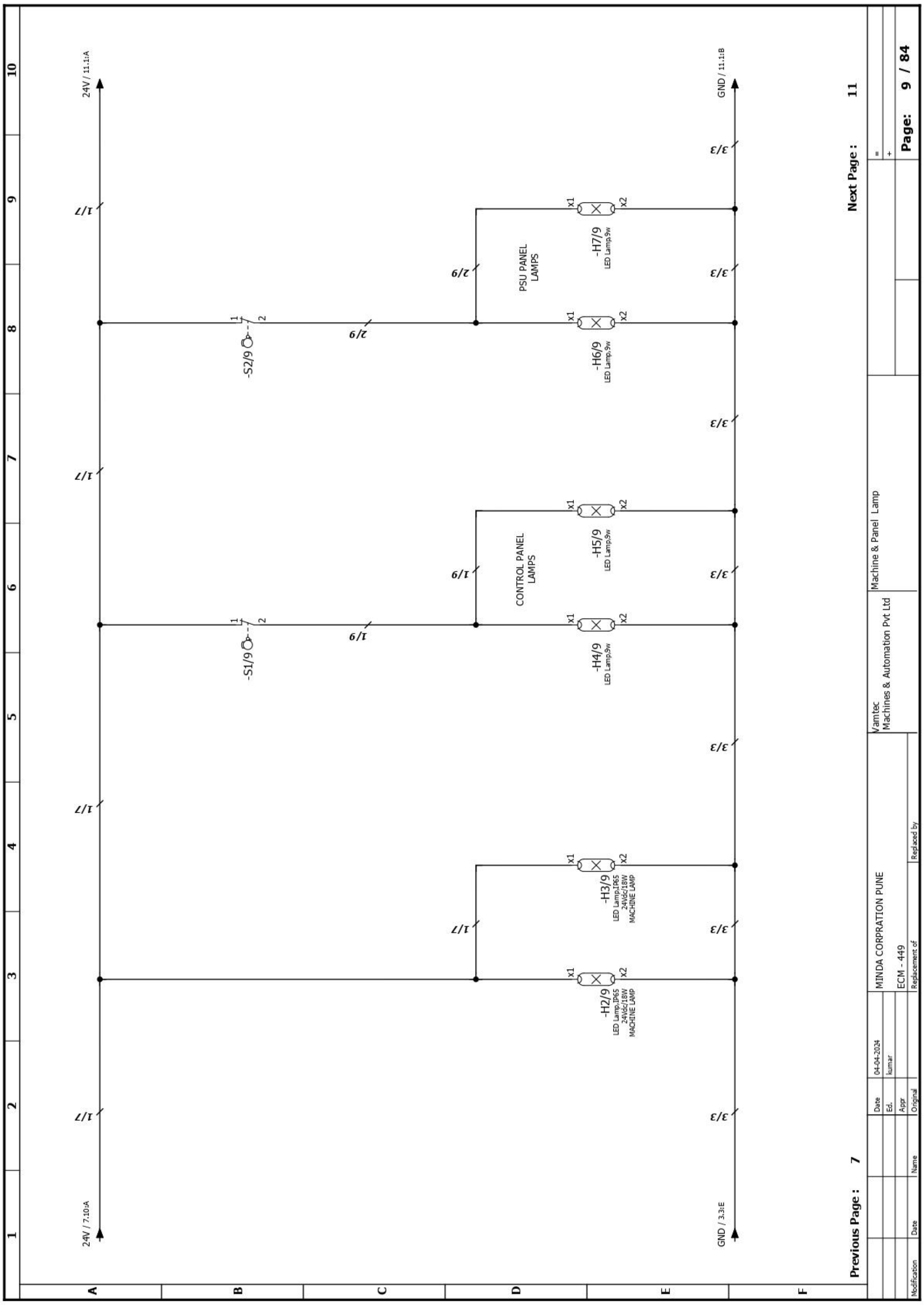
R / 5.10:A
Y / 5.10:A
B / 5.10:A
24V / 3.2:E
R / 11.1:A
Y / 11.1:A
B / 11.1:A
24V / 9.1:A
-F7/7
63A
-T2/7
-F8/7
16A
-U3/7
CHILLER
95.3:B
CHLRTRP
-Q2/7
2.5 - 4.0A
/95.5:B
-M1/7
0.46 kW/.616HP
WASH
MOTOR
-Q3/7
2.5 - 4.0A
/95.6:B
/143.3:E
-M2/7
1.1 kW/1.5HP
Mist
Collector



24V / 7.10:A
24V / 11.1:A
GND / 3.3:E
GND / 11.1:B
-S1/9
-S2/9
-H2/9
LED Lamp,IP65
24Vdc/18W
MACHINE LAMP
-H3/9
LED Lamp,IP65
24Vdc/18W
MACHINE LAMP
-H4/9
LED Lamp,9w
-H5/9
LED Lamp,9w
CONTROL PANEL
LAMPS
-H6/9
LED Lamp,9w
-H7/9
LED Lamp,9w
PSU PANEL
LAMPS
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Machines & Automation Pvt Ltd
Machine & Panel Lamp
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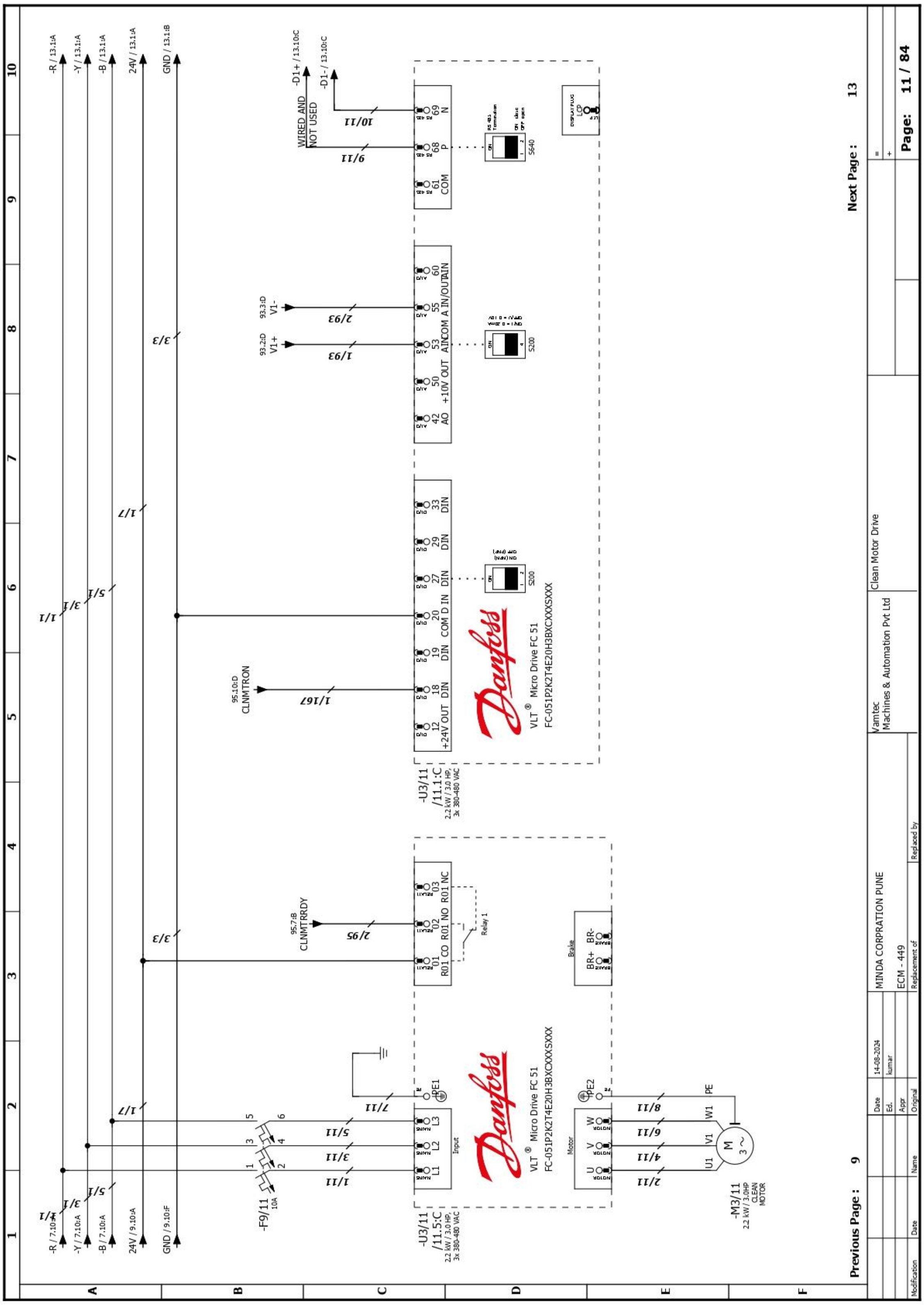

VLT® Micro Drive FC 51
FC-051P2K2T4E20H3BXCXXXSXXX
-U3/11 /11.5:C
2.2 kW / 3.0 HP, 3x 380-480 VAC
-U3/11 /11.1:C
-F9/11 10A
-M3/11 2.2 kW / 3.0HP CLEAN MOTOR
CLNMTRRDY
CLNMTRON
WIRED AND NOT USED
Relay 1
Brake
Input
Motor
Vamtec Machines & Automation Pvt Ltd
Clean Motor Drive
MINDA CORPRATION PUNE
ECM - 449
14-08-2024
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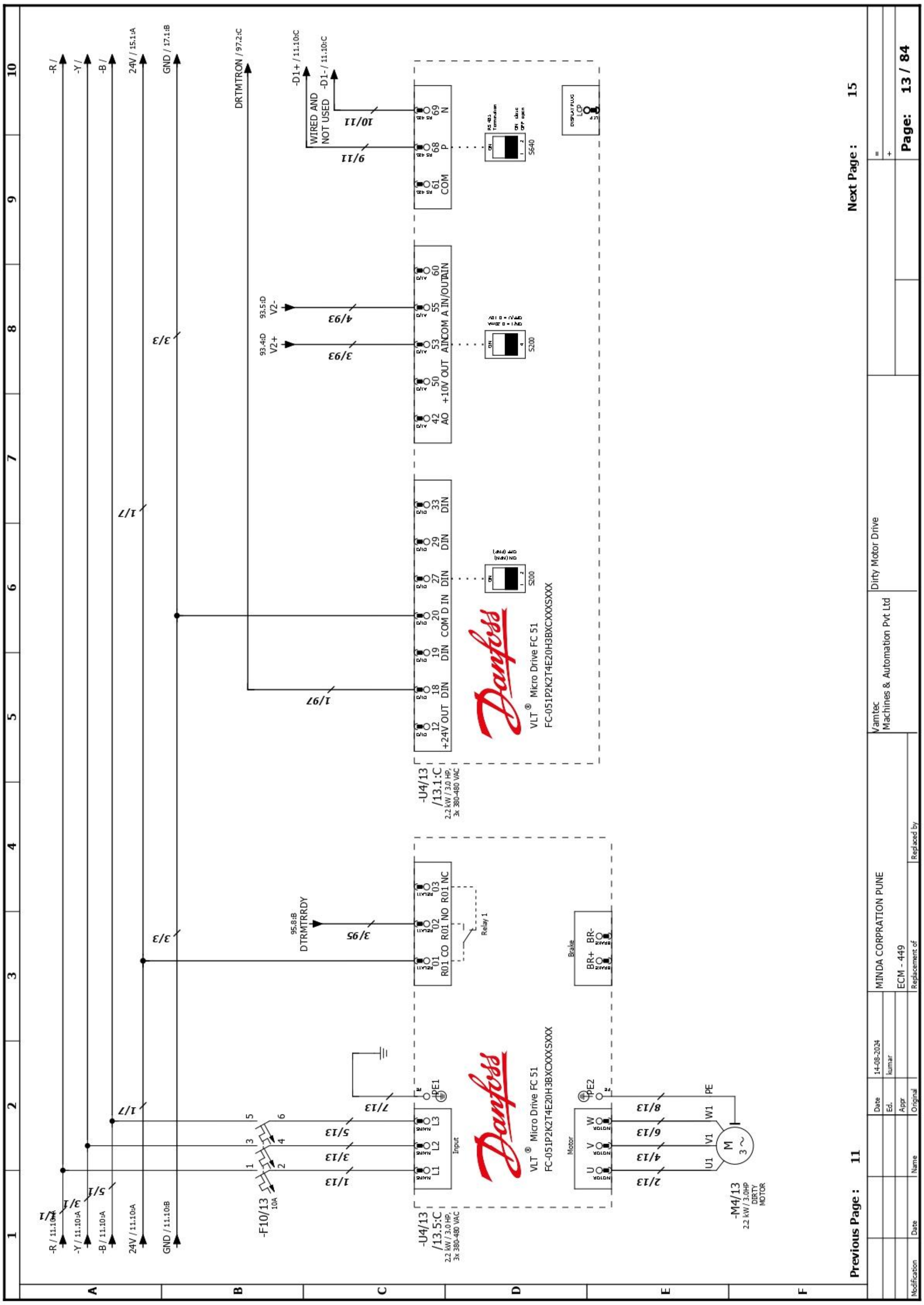
-U4/13
VLT® Micro Drive FC 51
FC-051P2K2T4E20H3BXCXXXSXXX
-F10/13
-M4/13
2.2 kW / 3.0HP
DIRTY
MOTOR
DTRMTRON / 97.2:C
DTRMTRRDY
WIRED AND
NOT USED
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Dirty Motor Drive
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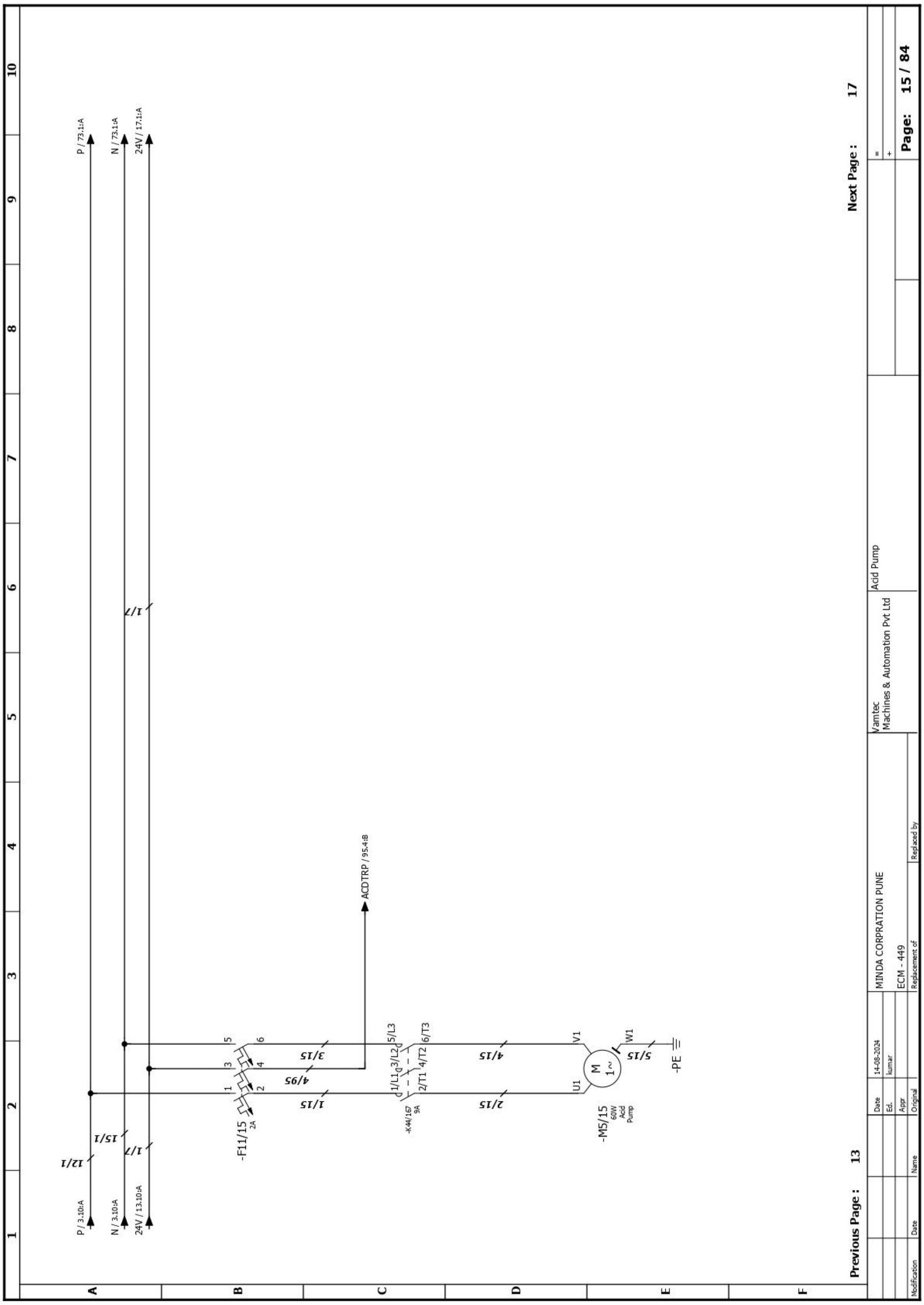

P / 3.10:A
N / 3.10:A
24V / 13.10:A
P / 73.1:A
N / 73.1:A
24V / 17.1:A
12/1
15/1
1/7
-F11/15
2A
3/15
4/95
1/15
ACDTRP / 95.4:B
1/L1
3/L2
5/L3
2/T1
4/T2
6/T3
-K44/167
9A
4/15
2/15
5/15
V1
U1
W1
M
1~
-M5/15
60W
Acid
Pump
-PE
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Acid Pump
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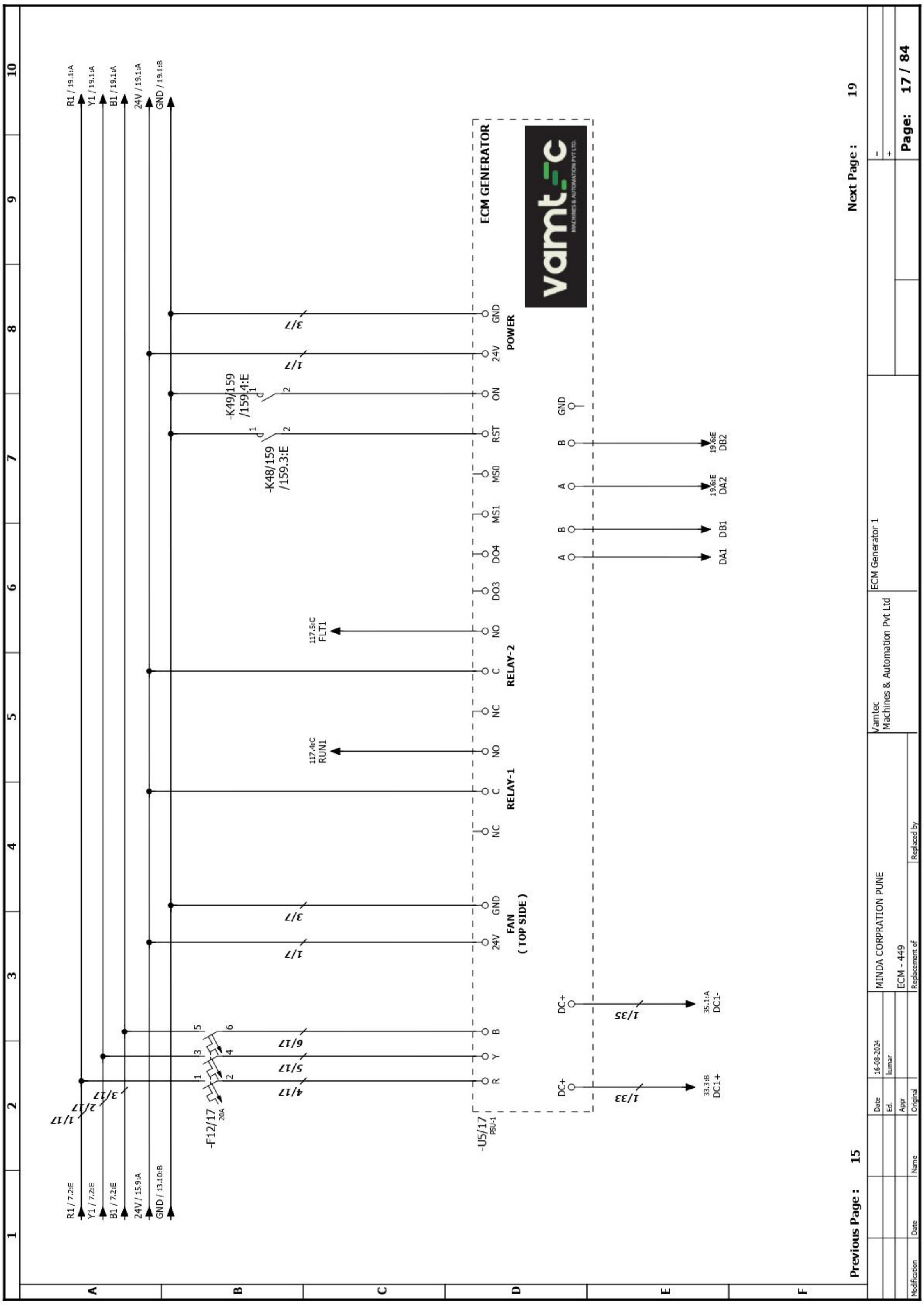

ECM GENERATOR
POWER
24V
GND
ON
RST
MS0
MS1
DO4
DO3
RELAY-2
RELAY-1
NC
C
NO
FAN
(TOP SIDE)
-K49/159
/159.4:E
-K48/159
/159.3:E
FLT1
RUN1
DA1
DB1
DA2
DB2
DC1+
DC1-
-F12/17
-U5/17
R1 / 19.1:A
Y1 / 19.1:A
B1 / 19.1:A
24V / 19.1:A
GND / 19.1:B
R1 / 7.2:E
Y1 / 7.2:E
B1 / 7.2:E
24V / 15.9:A
GND / 13.10:B
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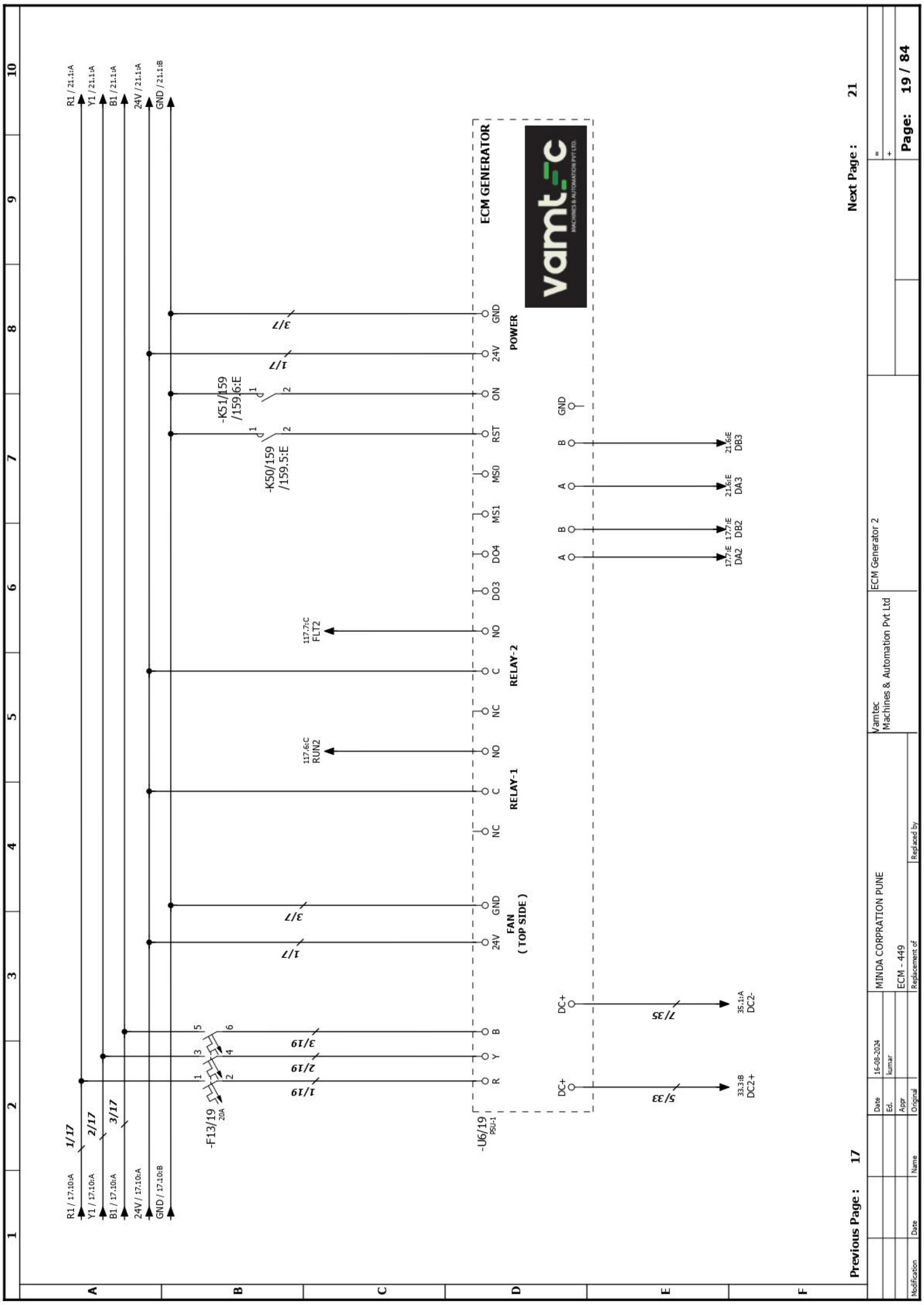
MINDA CORPRATION PUNE
ECM - 449
Vamtec
Machines & Automation Pvt Ltd
ECM Generator 1
Date 16-08-2024
Ed. kumar


ECM GENERATOR
POWER
GND
24V
ON
RST
MS0
MS1
DO4
DO3
NO
C
NC
RELAY-2
RELAY-1
FAN
(TOP SIDE)
DC+
DC2-
DC2+
DA2
DB2
DA3
DB3
-U6/19
PSU-1
-F13/19
20A
-K51/159
/159.6:E
-K50/159
/159.5:E
R1 / 17.10:A
Y1 / 17.10:A
B1 / 17.10:A
24V / 17.10:A
GND / 17.10:B
R1 / 21.1:A
Y1 / 21.1:A
B1 / 21.1:A
24V / 21.1:A
GND / 21.1:B
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ECM Generator 2
Vamtec
Machines & Automation Pvt Ltd
MINDA CORPRATION PUNE
ECM - 449
Date 16-08-2024
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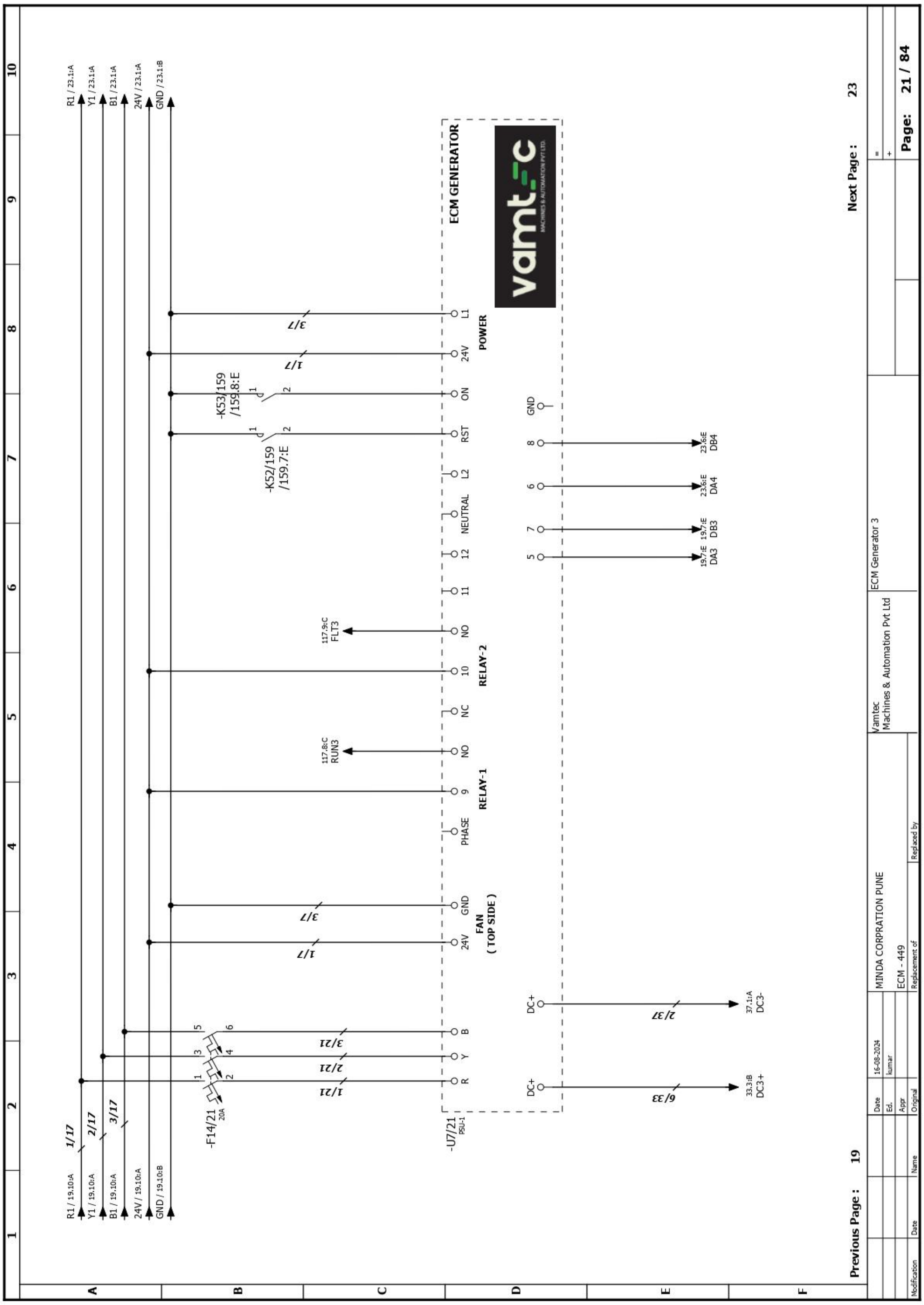
ECM GENERATOR
POWER
RELAY-2
RELAY-1
FAN
(TOP SIDE)
-U7/21
-F14/21
-K53/159
/159.8:E
-K52/159
/159.7:E
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MINDA CORPRATION PUNE
ECM - 449
Vamtec
Machines & Automation Pvt Ltd
ECM Generator 3
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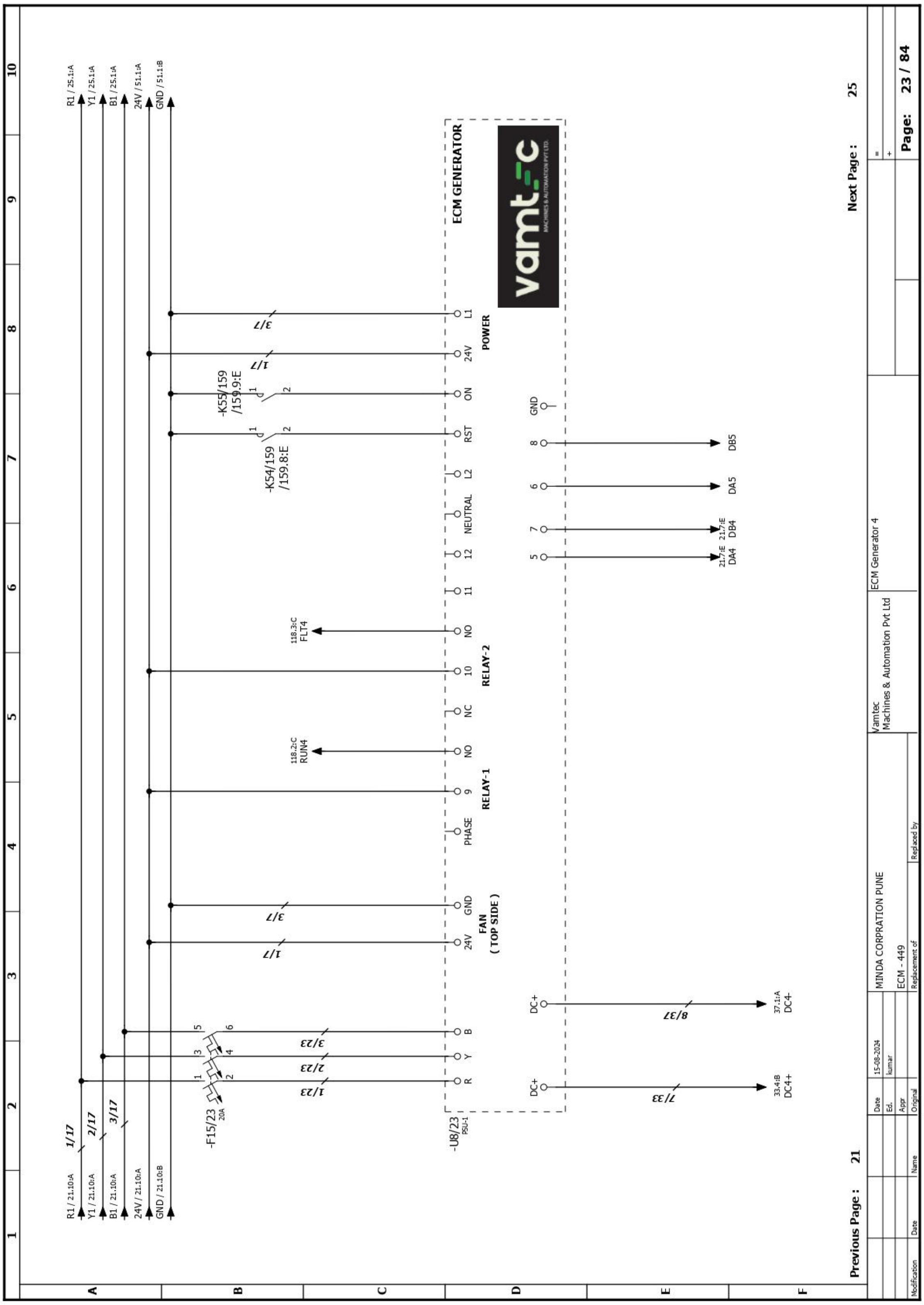
ECM GENERATOR
POWER
RELAY-2
RELAY-1
FAN (TOP SIDE)
-U8/23
-F15/23
-K55/159 /159.9:E
-K54/159 /159.8:E
DC4+
DC4-
DA4
DB4
DA5
DB5
FLT4
RUN4
Previous Page : 21
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MINDA CORPRATION PUNE
ECM - 449
Vamtec Machines & Automation Pvt Ltd
ECM Generator 4
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kumar
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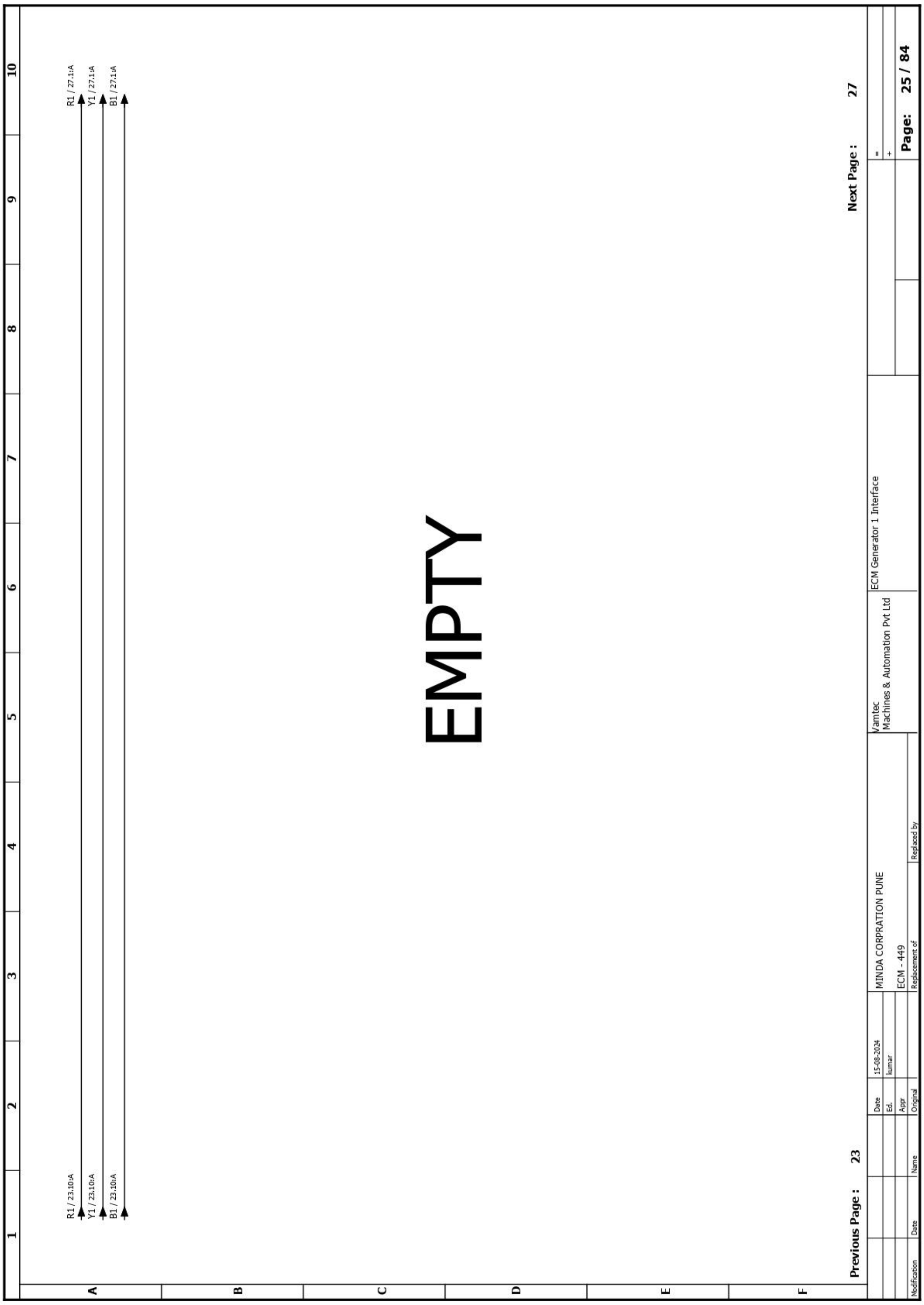
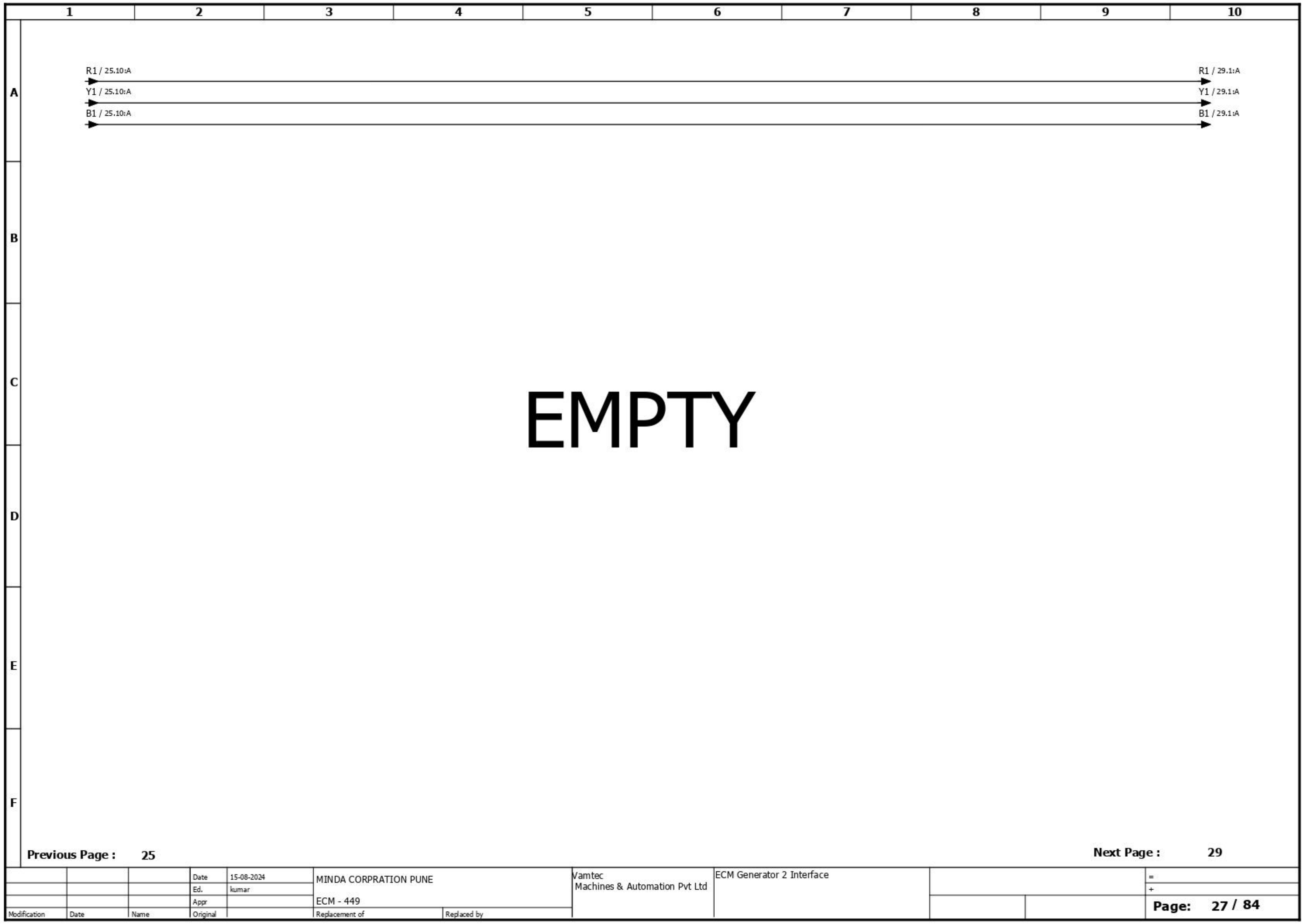
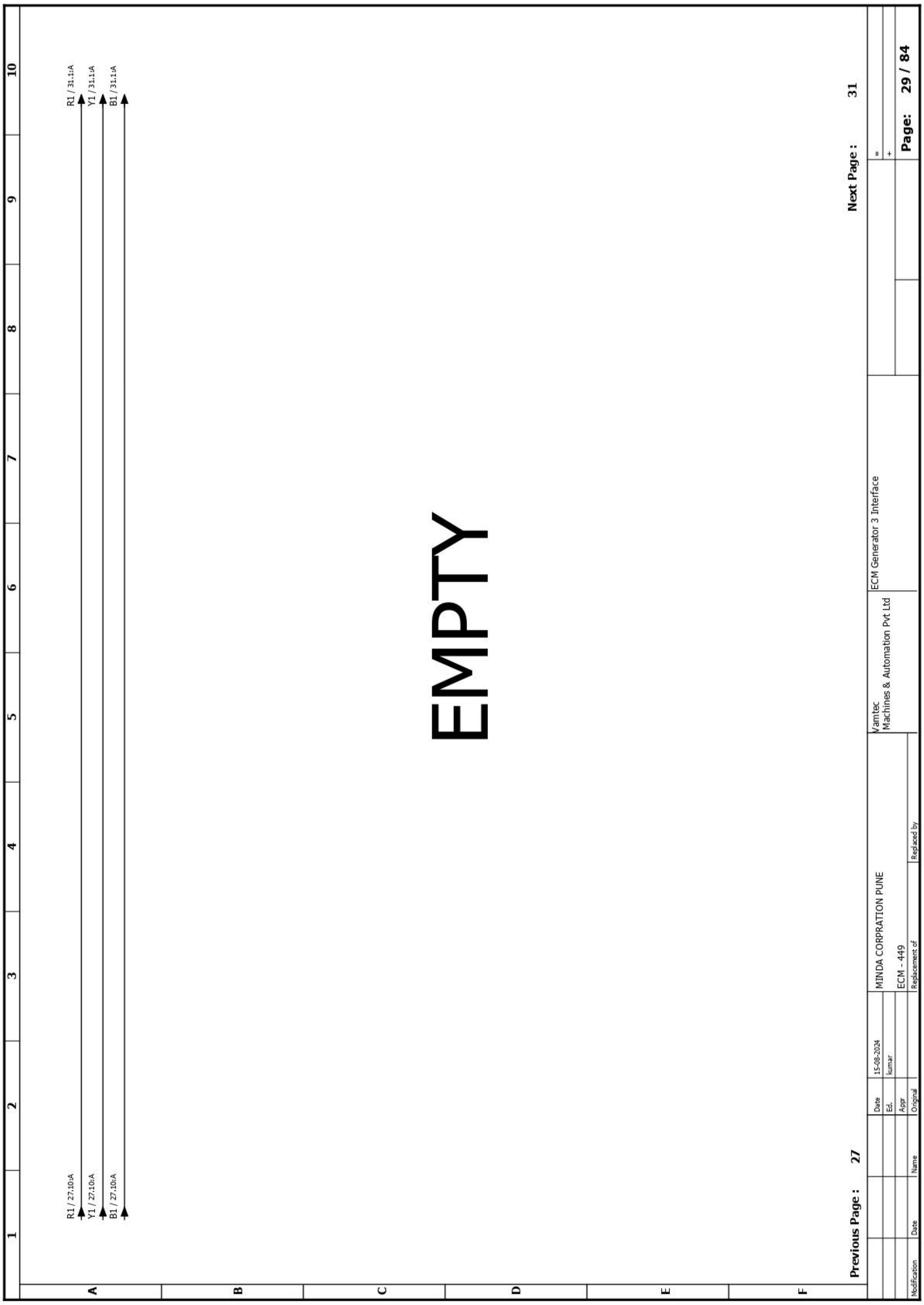
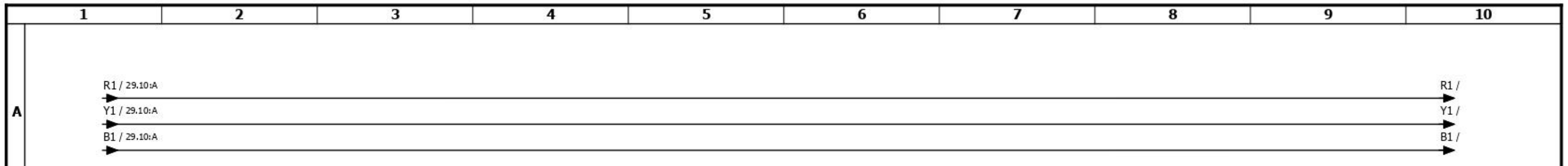

R1 / 23.10:A
Y1 / 23.10:A
B1 / 23.10:A
R1 / 27.1:A
Y1 / 27.1:A
B1 / 27.1:A
EMPTY
Previous Page : 23
Next Page : 27
Date 15-08-2024
Ed. kumar
Appr
MINDA CORPRATION PUNE
ECM - 449
Vamtec
Machines & Automation Pvt Ltd
ECM Generator 1 Interface
Modification
Date
Name
Original
Replacement of
Replaced by
Page: 25 / 84


1
2
3
4
5
6
7
8
9
10
A
B
C
D
E
F
R1 / 25.10:A
Y1 / 25.10:A
B1 / 25.10:A
R1 / 29.1:A
Y1 / 29.1:A
B1 / 29.1:A
EMPTY
Previous Page : 25
Next Page : 29
Date 15-08-2024
Ed. kumar
Appr
Original
Modification
Date
Name
MINDA CORPRATION PUNE
ECM - 449
Replacement of
Replaced by
Vamtec
Machines & Automation Pvt Ltd
ECM Generator 2 Interface
=
+
Page: 27 / 84


EMPTY
R1 / 27.10:A
Y1 / 27.10:A
B1 / 27.10:A
R1 / 31.1:A
Y1 / 31.1:A
B1 / 31.1:A
Previous Page : 27
Next Page : 31
Date 15-08-2024
Ed. kumar
Appr
MINDA CORPRATION PUNE
ECM - 449
Vamtec
Machines & Automation Pvt Ltd
ECM Generator 3 Interface
Page: 29 / 84
Modification
Date
Name
Original
Replacement of
Replaced by

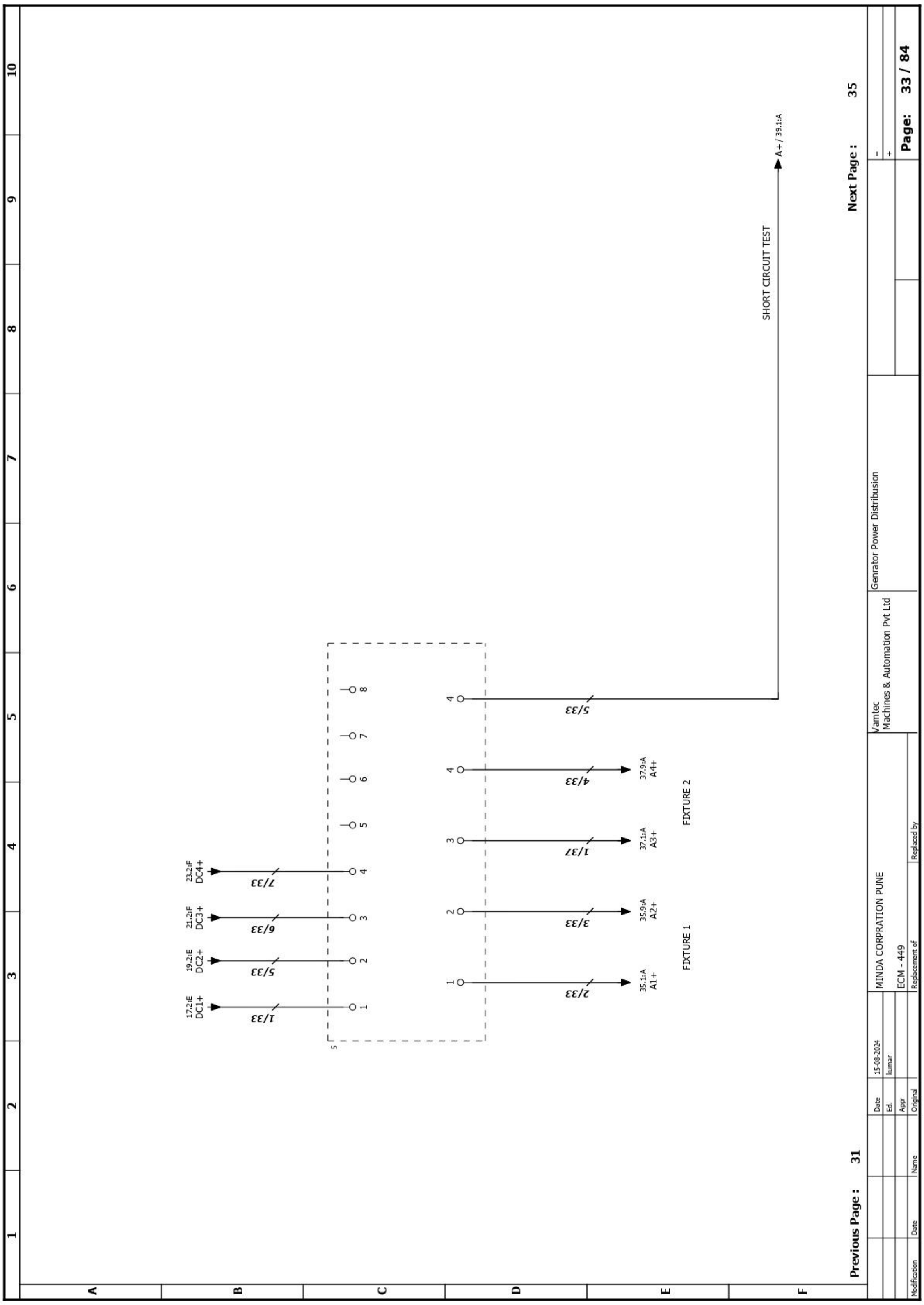


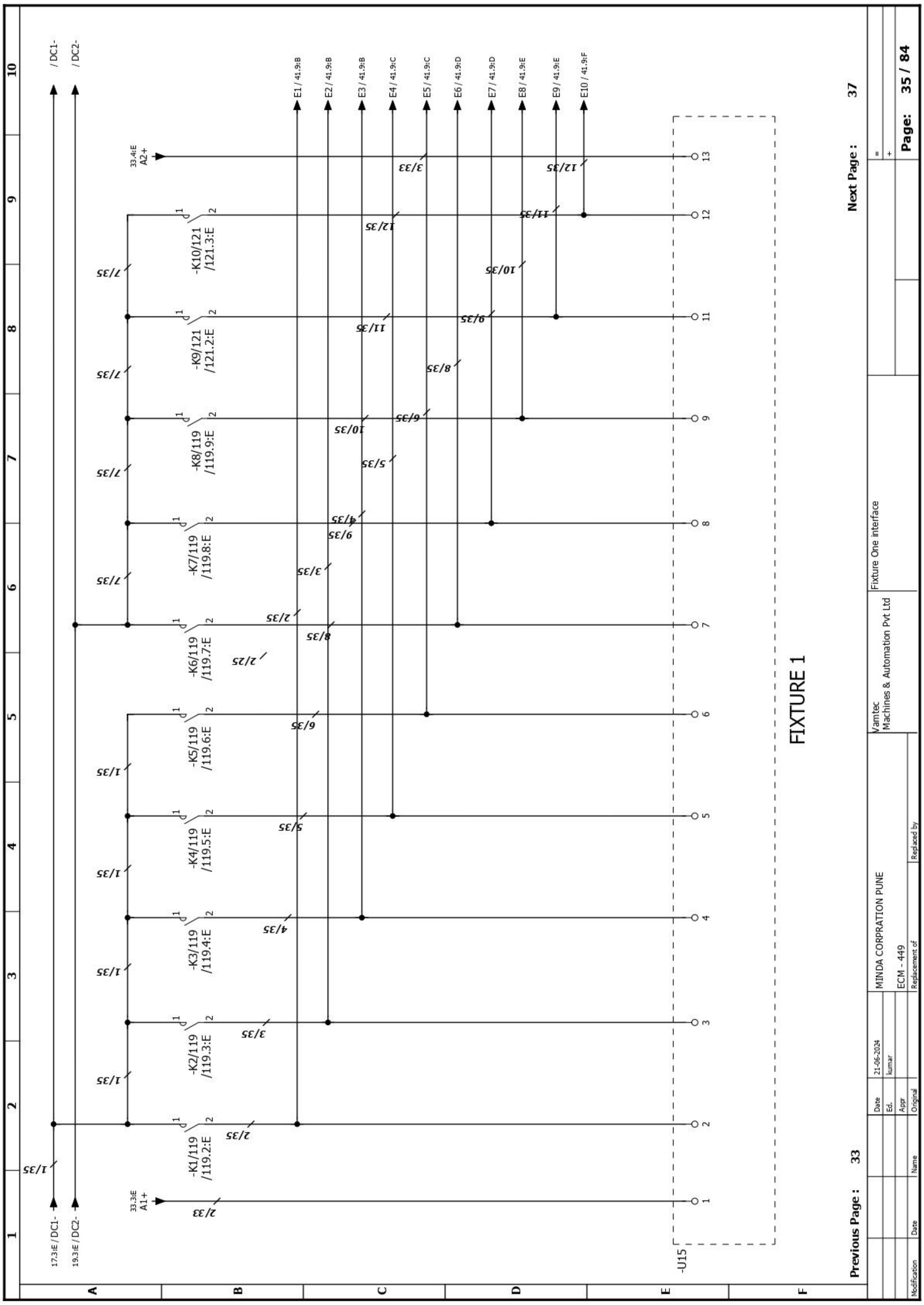
EMPTY

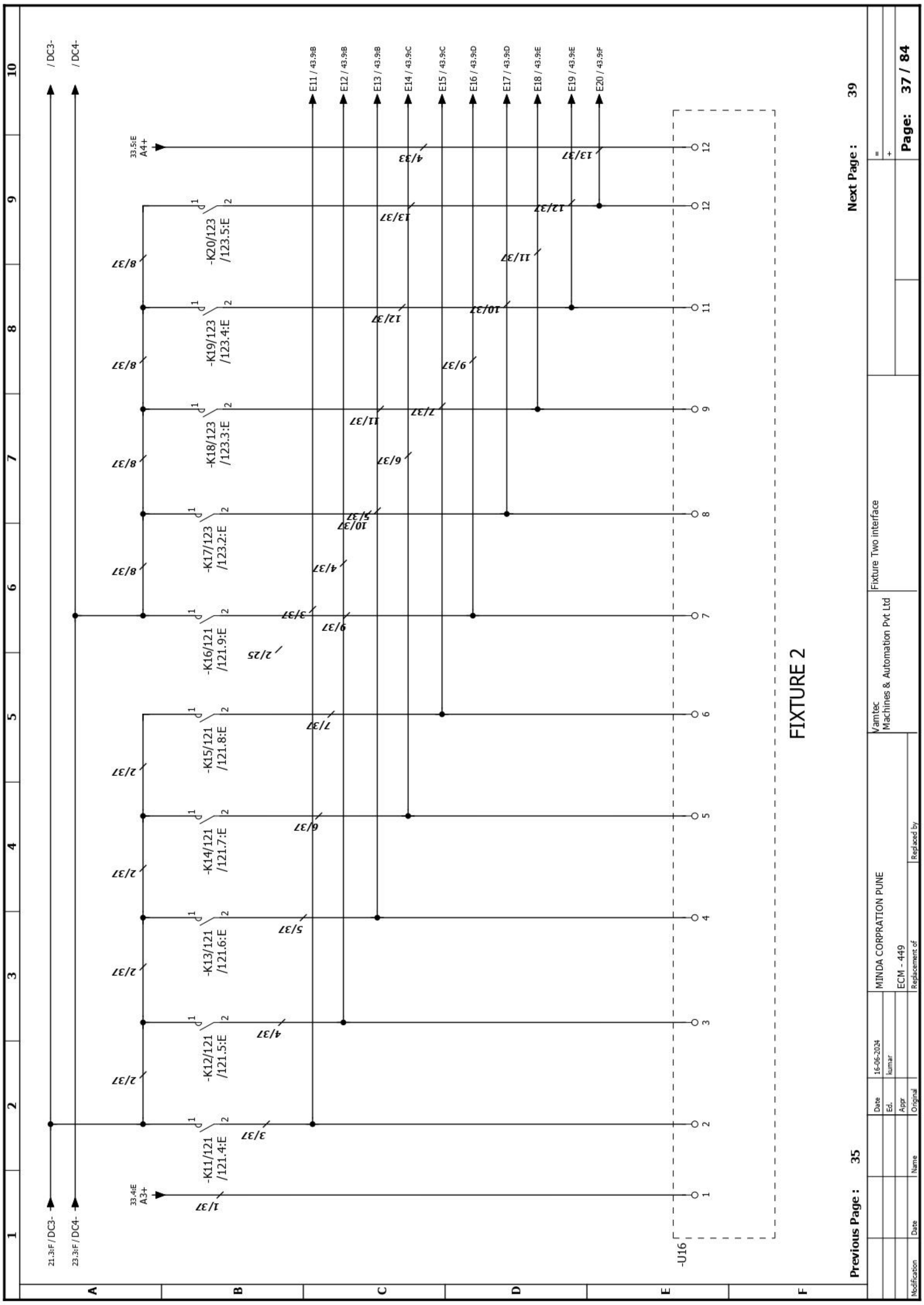
Previous Page : 29

Next Page : 33

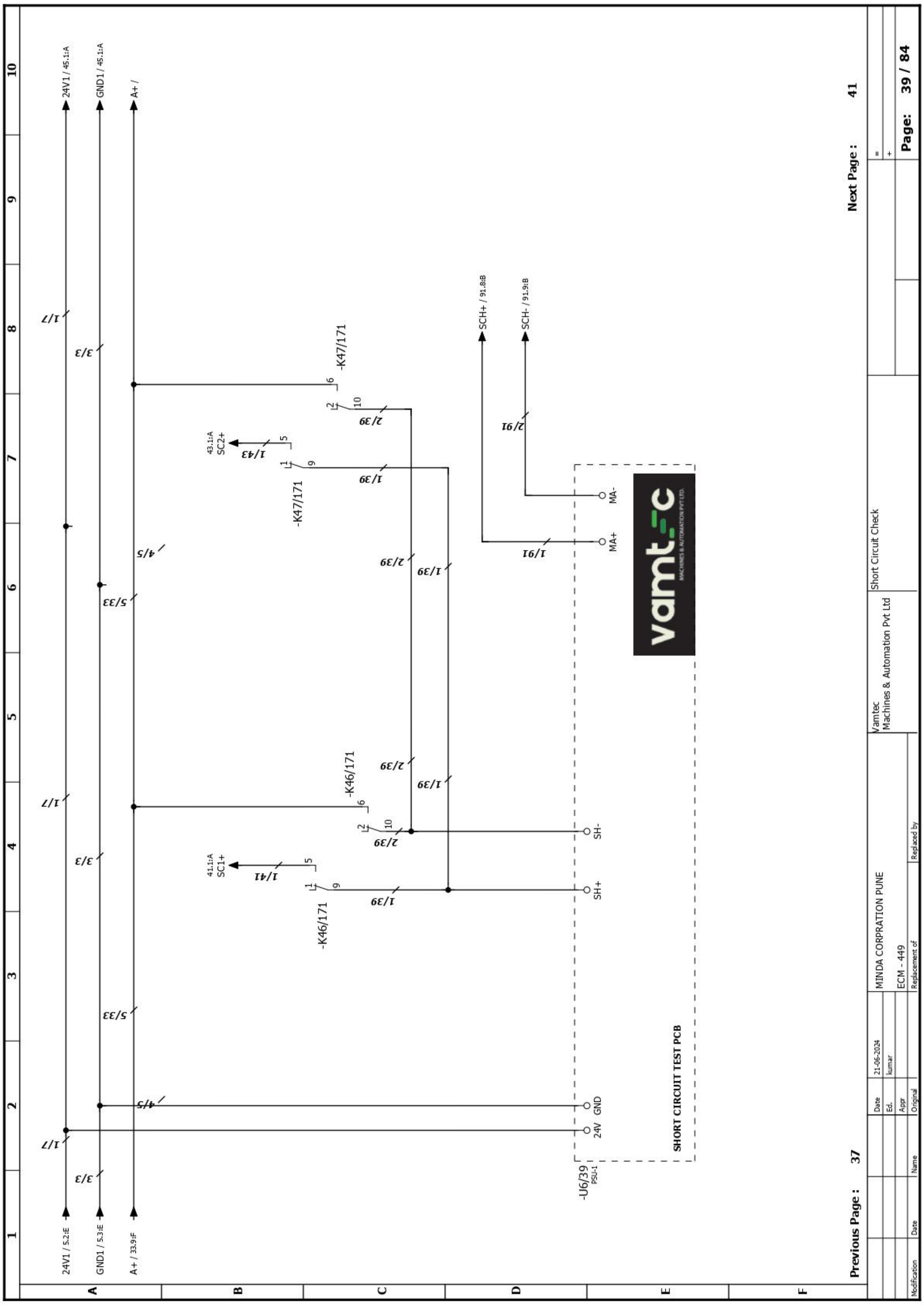
			Date	15-08-2024	MINDA CORPRATION PUNE	Vamtec Machines & Automation Pvt Ltd	ECM Generator 4 Interface	=
			Ed.	kumar				+
			Appr		ECM - 449			Page: 31 / 84
Modification	Date	Name	Original		Replacement of / Replaced by			

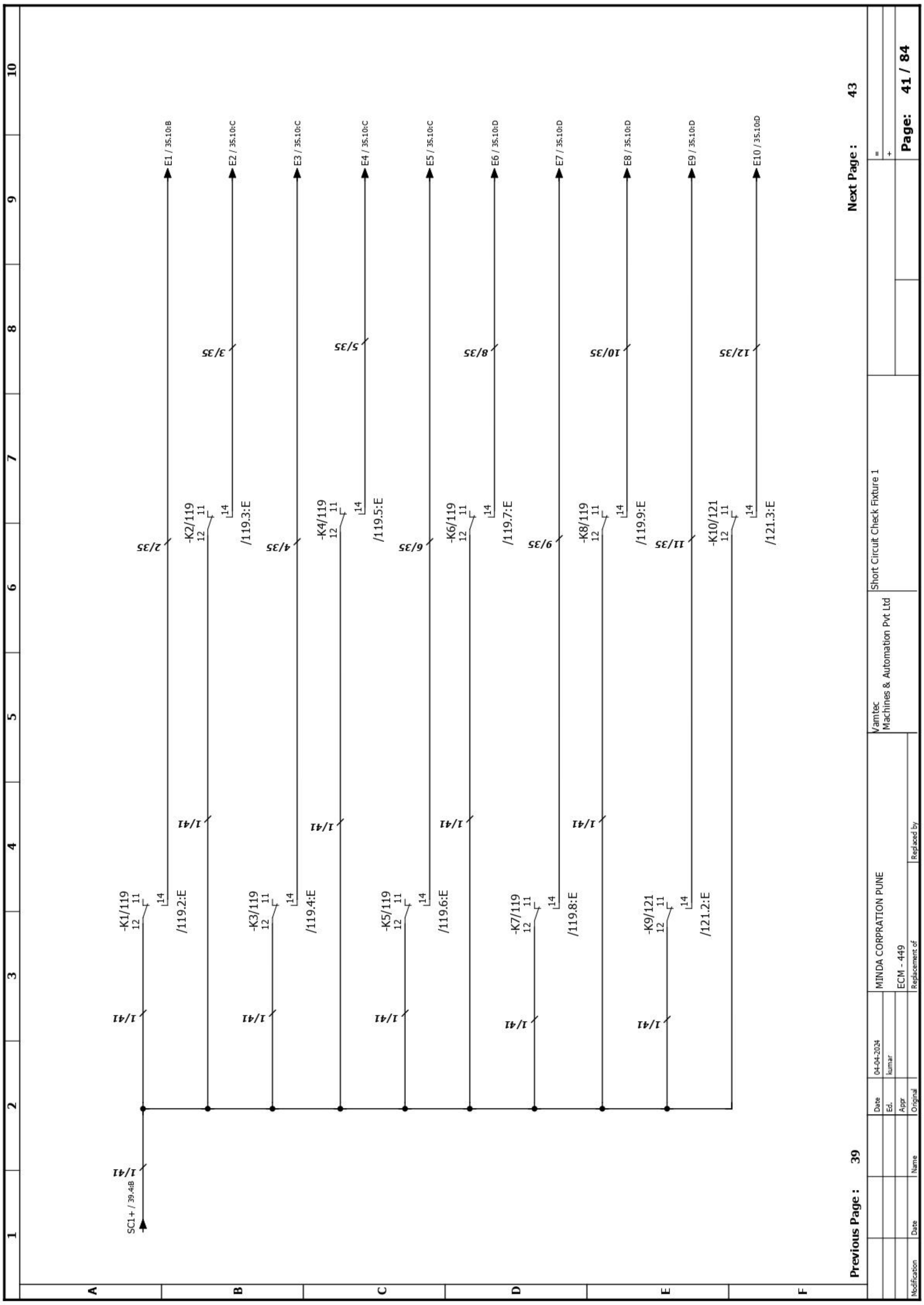

SHORT CIRCUIT TEST
A+ / 39.1:A
FIXTURE 1
FIXTURE 2
DC1+
DC2+
DC3+
DC4+
A1+
A2+
A3+
A4+
Previous Page : 31
Next Page : 35
Page: 33 / 84
MINDA CORPRATION PUNE
ECM - 449
Vamtec
Machines & Automation Pvt Ltd
Genrator Power Distribution
15-08-2024
kumar


FIXTURE 1
-U15
-K1/119
/119.2:E
-K2/119
/119.3:E
-K3/119
/119.4:E
-K4/119
/119.5:E
-K5/119
/119.6:E
-K6/119
/119.7:E
-K7/119
/119.8:E
-K8/119
/119.9:E
-K9/121
/121.2:E
-K10/121
/121.3:E
17.3:E / DC1-
19.3:E / DC2-
/ DC1-
/ DC2-
33.3:E A1+
33.4:E A2+
E1 / 41.9:B
E2 / 41.9:B
E3 / 41.9:B
E4 / 41.9:C
E5 / 41.9:C
E6 / 41.9:D
E7 / 41.9:D
E8 / 41.9:E
E9 / 41.9:E
E10 / 41.9:F
Previous Page : 33
Next Page : 37
Page: 35 / 84
Fixture One interface
Vamtec
Machines & Automation Pvt Ltd
MINDA CORPRATION PUNE
ECM - 449
Date 21-06-2024
Ed. kumar
Appr
Original
Replacement of
Replaced by
Modification
Date
Name

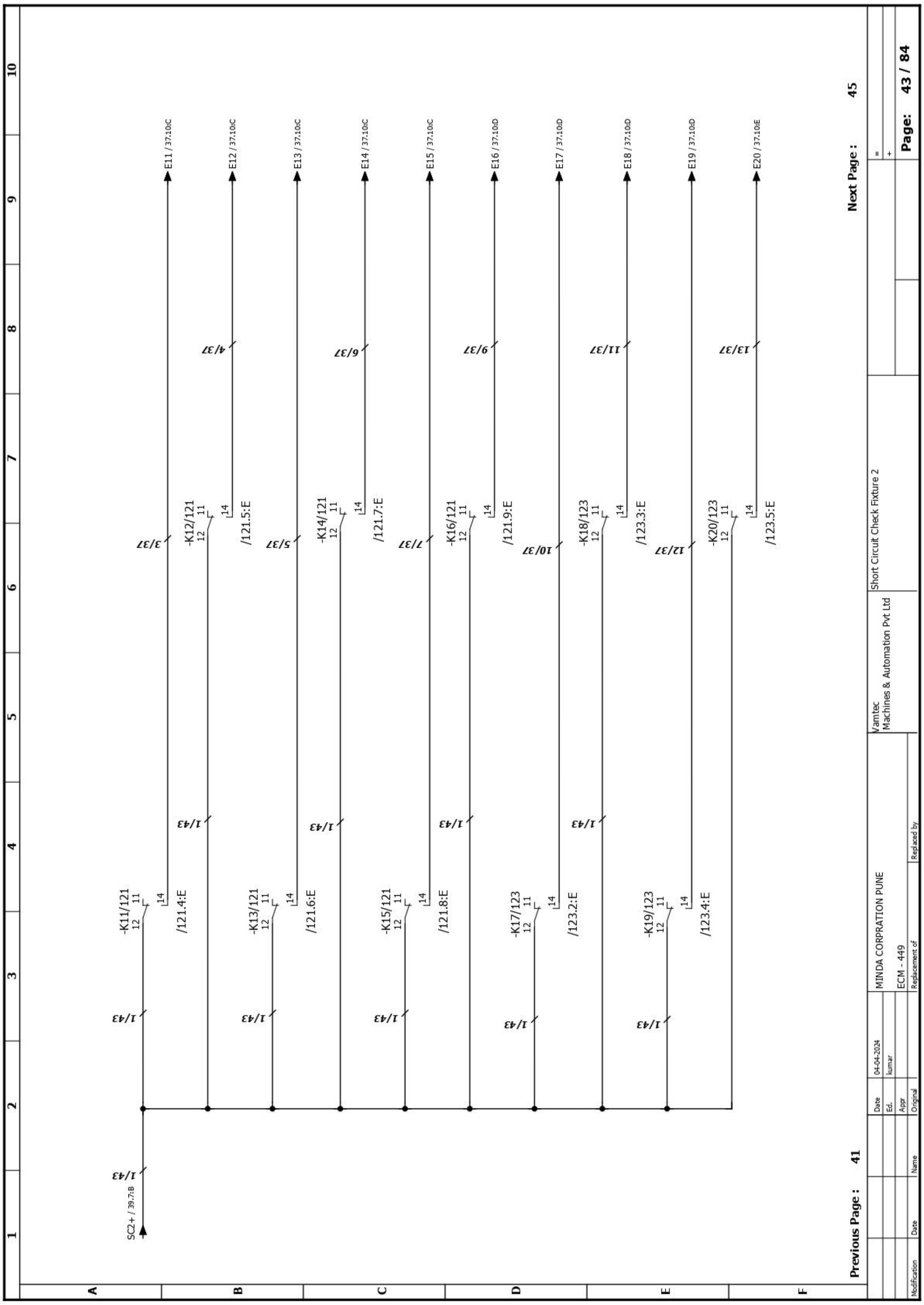


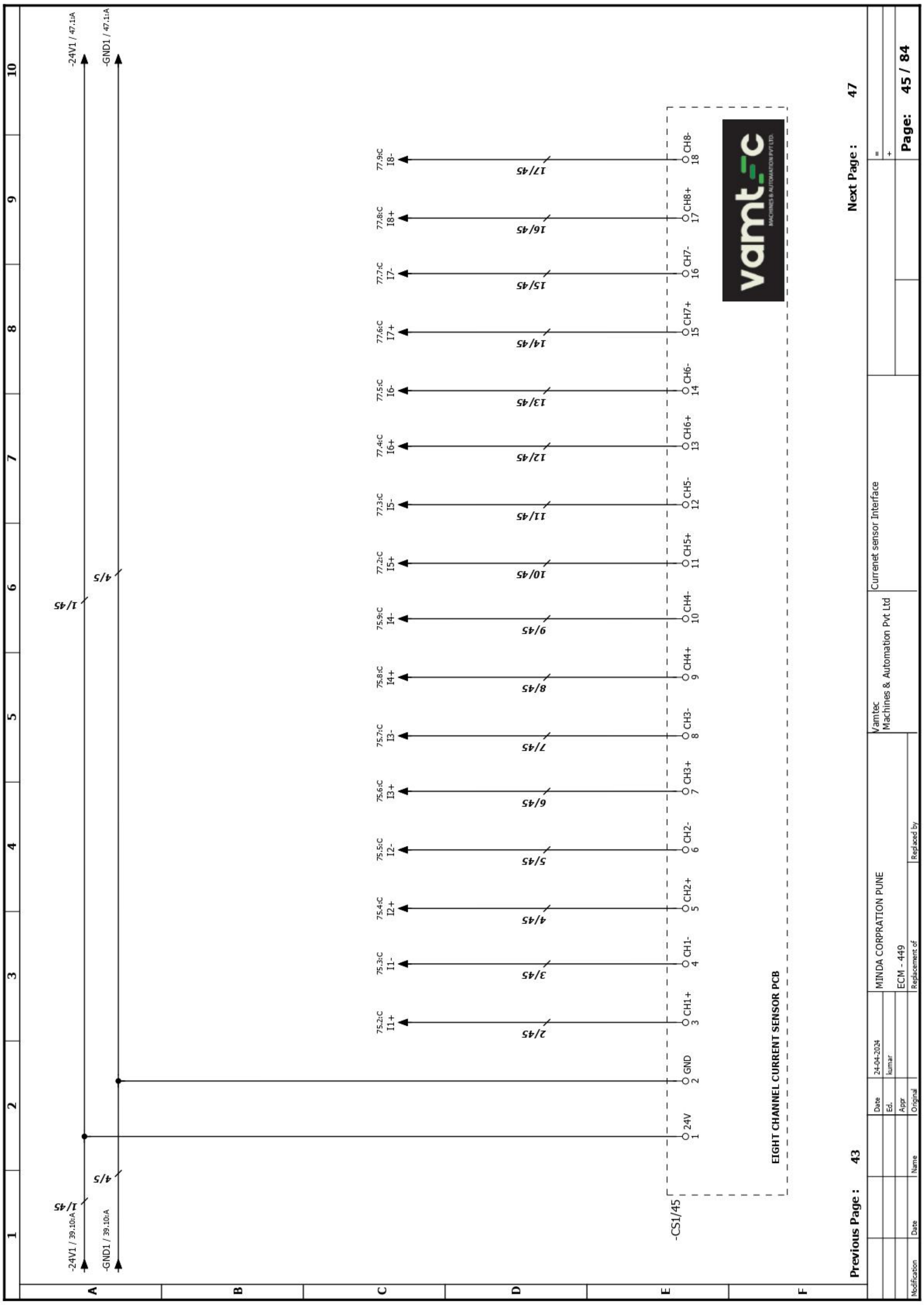
FIXTURE 2
-U16
21.3:F / DC3-
23.3:F / DC4-
/ DC3-
/ DC4-
33.4:E A3+
33.5:E A4+
-K11/121 /121.4:E
-K12/121 /121.5:E
-K13/121 /121.6:E
-K14/121 /121.7:E
-K15/121 /121.8:E
-K16/121 /121.9:E
-K17/123 /123.2:E
-K18/123 /123.3:E
-K19/123 /123.4:E
-K20/123 /123.5:E
E11 / 43.9:B
E12 / 43.9:B
E13 / 43.9:B
E14 / 43.9:C
E15 / 43.9:C
E16 / 43.9:D
E17 / 43.9:D
E18 / 43.9:E
E19 / 43.9:E
E20 / 43.9:F
Previous Page : 35
Next Page : 39
Page: 37 / 84
MINDA CORPRATION PUNE
ECM - 449
Vamtec Machines & Automation Pvt Ltd
Fixture Two interface
Date 16-06-2024
Ed. kumar
Appr
Original
Replacement of
Replaced by
Modification
Date
Name

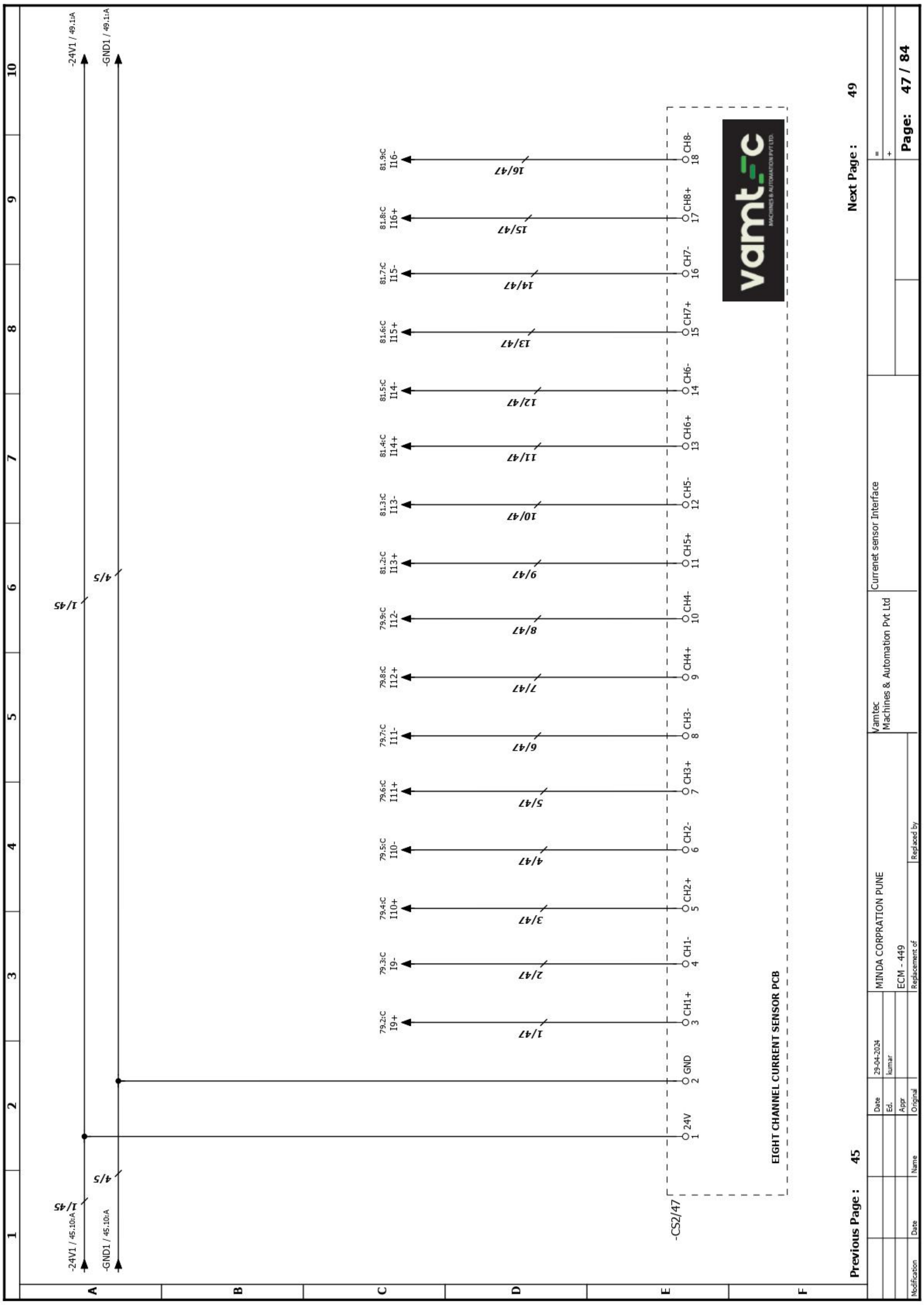

24V1 / 5.2:E
GND1 / 5.3:E
A+ / 33.9:F
24V1 / 45.1:A
GND1 / 45.1:A
A+ /
-K46/171
-K47/171
SC1+
SC2+
SCH+ / 91.8:B
SCH- / 91.9:B
24V
GND
SH+
SH-
MA+
MA-
-U6/39
PSU-1
SHORT CIRCUIT TEST PCB
Previous Page : 37
Next Page : 41
Page: 39 / 84
Short Circuit Check
Vamtec
Machines & Automation Pvt Ltd
MINDA CORPRATION PUNE
ECM - 449
21-06-2024
kumar

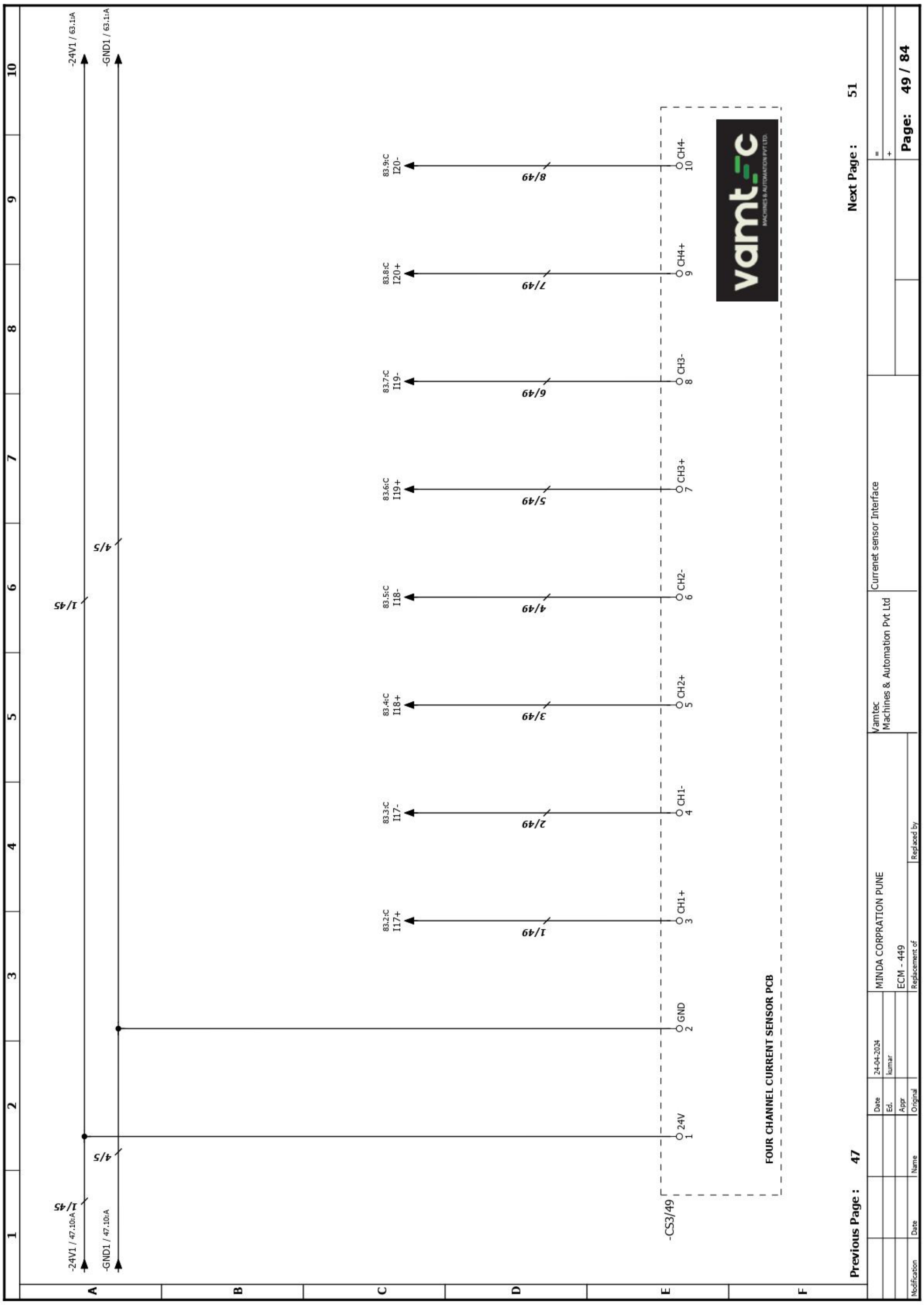


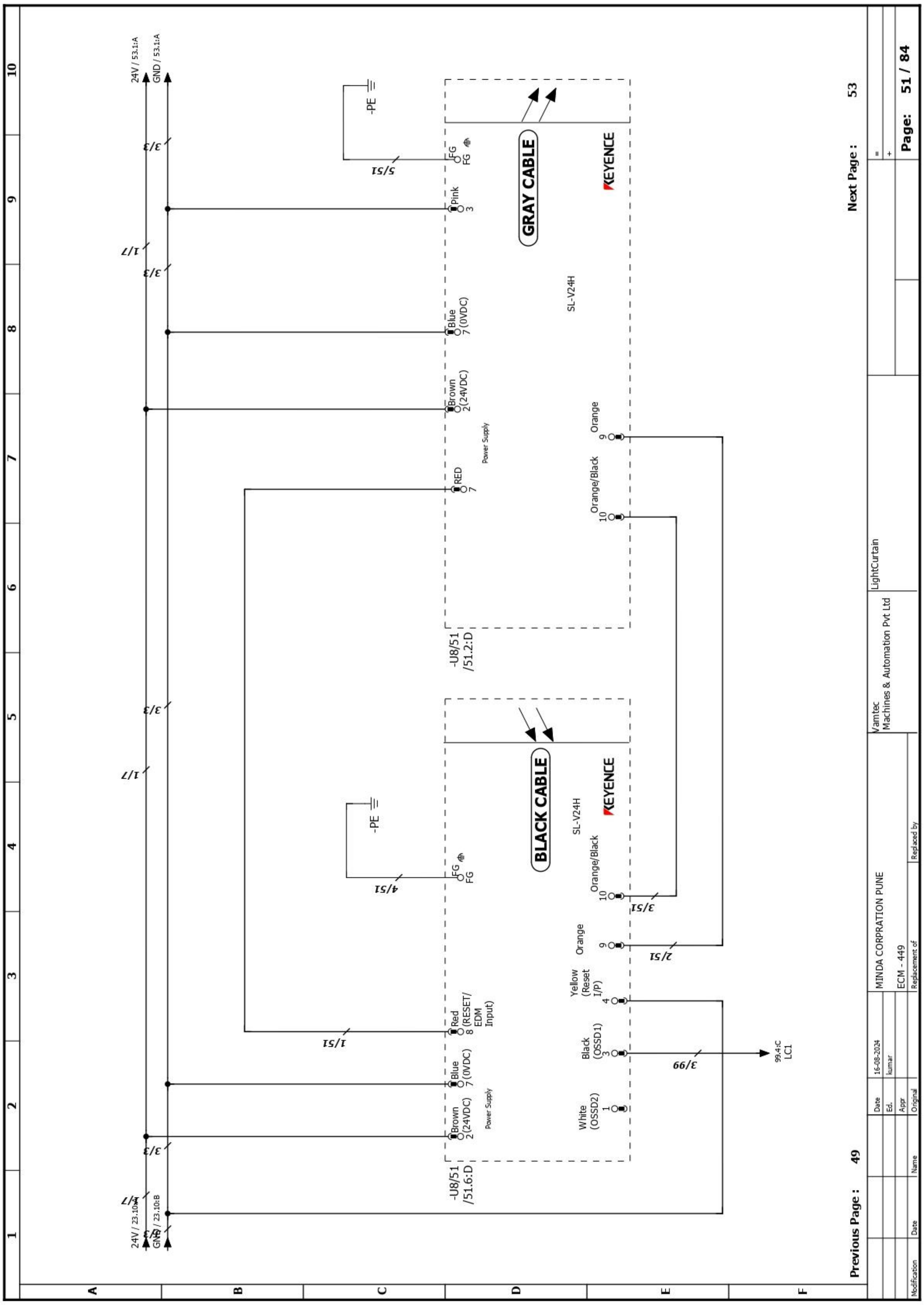
-K1/119
-K2/119
-K3/119
-K4/119
-K5/119
-K6/119
-K7/119
-K8/119
-K9/121
-K10/121
Previous Page : 39
Next Page : 43
MINDA CORPRATION PUNE
ECM - 449
Vamtec
Machines & Automation Pvt Ltd
Short Circuit Check Fixture 1
Page: 41 / 84

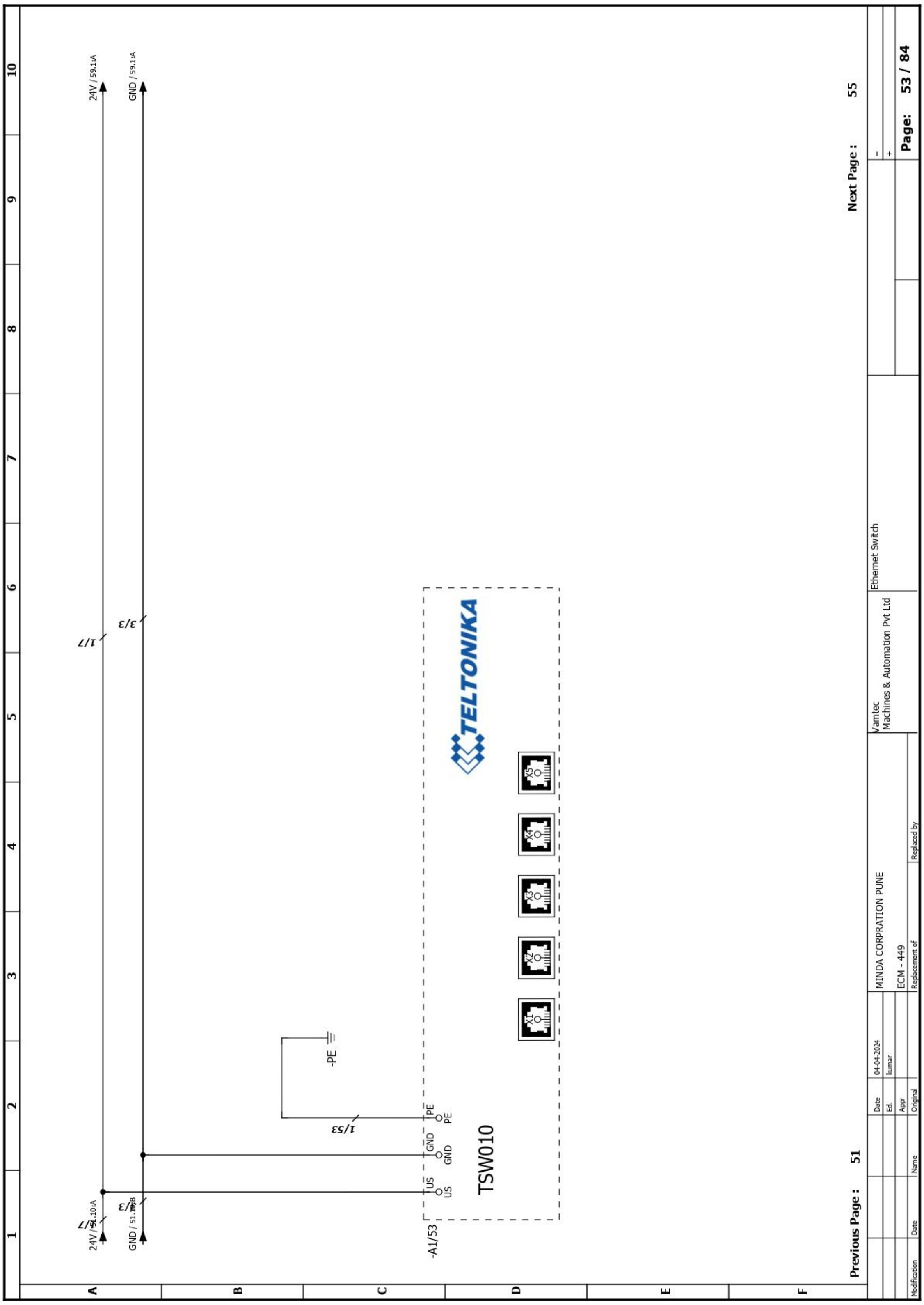

SC2+ / 39.7:B
1/43
-K11/121
/121.4:E
-K12/121
/121.5:E
-K13/121
/121.6:E
-K14/121
/121.7:E
-K15/121
/121.8:E
-K16/121
/121.9:E
-K17/123
/123.2:E
-K18/123
/123.3:E
-K19/123
/123.4:E
-K20/123
/123.5:E
3/37
4/37
5/37
6/37
7/37
9/37
10/37
11/37
12/37
13/37
E11 / 37.10:C
E12 / 37.10:C
E13 / 37.10:C
E14 / 37.10:C
E15 / 37.10:C
E16 / 37.10:D
E17 / 37.10:D
E18 / 37.10:D
E19 / 37.10:D
E20 / 37.10:E
Previous Page : 41
Next Page : 45
Date 04-04-2024
Ed. kumar
Appr
Original
Modification
Date
Name
MINDA CORPRATION PUNE
ECM - 449
Replacement of
Replaced by
Vamtec
Machines & Automation Pvt Ltd
Short Circuit Check Fixture 2
Page: 43 / 84

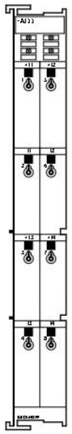

-24V1 / 47.1:A
-GND1 / 47.1:A
-24V1 / 39.10:A
-GND1 / 39.10:A
1/45
4/5
-CS1/45
EIGHT CHANNEL CURRENT SENSOR PCB
24V 1
GND 2
CH1+ 3
CH1- 4
CH2+ 5
CH2- 6
CH3+ 7
CH3- 8
CH4+ 9
CH4- 10
CH5+ 11
CH5- 12
CH6+ 13
CH6- 14
CH7+ 15
CH7- 16
CH8+ 17
CH8- 18
2/45
3/45
4/45
5/45
6/45
7/45
8/45
9/45
10/45
11/45
12/45
13/45
14/45
15/45
16/45
17/45
I1+ 75.2:C
I1- 75.3:C
I2+ 75.4:C
I2- 75.5:C
I3+ 75.6:C
I3- 75.7:C
I4+ 75.8:C
I4- 75.9:C
I5+ 77.2:C
I5- 77.3:C
I6+ 77.4:C
I6- 77.5:C
I7+ 77.6:C
I7- 77.7:C
I8+ 77.8:C
I8- 77.9:C
vamtec
Previous Page : 43
Next Page : 47
Page: 45 / 84
Date 24-04-2024
Ed. kumar
Appr
Original
Modification
Date
Name
Replacement of
Replaced by
MINDA CORPRATION PUNE
ECM - 449
Vamtec
Machines & Automation Pvt Ltd
Currenet sensor Interface


-24V1 / 45.10:A
-GND1 / 45.10:A
-24V1 / 49.1:A
-GND1 / 49.1:A
1/45
4/5
24V 1
GND 2
CH1+ 3
CH1- 4
CH2+ 5
CH2- 6
CH3+ 7
CH3- 8
CH4+ 9
CH4- 10
CH5+ 11
CH5- 12
CH6+ 13
CH6- 14
CH7+ 15
CH7- 16
CH8+ 17
CH8- 18
I9+ 79.2:C
I9- 79.3:C
I10+ 79.4:C
I10- 79.5:C
I11+ 79.6:C
I11- 79.7:C
I12+ 79.8:C
I12- 79.9:C
I13+ 81.2:C
I13- 81.3:C
I14+ 81.4:C
I14- 81.5:C
I15+ 81.6:C
I15- 81.7:C
I16+ 81.8:C
I16- 81.9:C
1/47
2/47
3/47
4/47
5/47
6/47
7/47
8/47
9/47
10/47
11/47
12/47
13/47
14/47
15/47
16/47
-CS2/47
EIGHT CHANNEL CURRENT SENSOR PCB
vamt c
Previous Page : 45
Next Page : 49
Date 29-04-2024
Ed. kumar
Appr
Original
Modification
Date
Name
MINDA CORPRATION PUNE
ECM - 449
Replacement of
Replaced by
Vamtec
Machines & Automation Pvt Ltd
Currenet sensor Interface
Page: 47 / 84


-24V1 / 63.1:A
-GND1 / 63.1:A
-24V1 / 47.10:A
-GND1 / 47.10:A
1/45
4/5
FOUR CHANNEL CURRENT SENSOR PCB
-CS3/49
24V 1
GND 2
CH1+ 3
CH1- 4
CH2+ 5
CH2- 6
CH3+ 7
CH3- 8
CH4+ 9
CH4- 10
1/49
2/49
3/49
4/49
5/49
6/49
7/49
8/49
I17+ 83.2:C
I17- 83.3:C
I18+ 83.4:C
I18- 83.5:C
I19+ 83.6:C
I19- 83.7:C
I20+ 83.8:C
I20- 83.9:C
Previous Page : 47
Next Page : 51
Date 24-04-2024
Ed. kumar
Appr
Original
Modification
Date
Name
MINDA CORPRATION PUNE
ECM - 449
Replacement of
Replaced by
Vamtec
Machines & Automation Pvt Ltd
Currenet sensor Interface
Page: 49 / 84


GRAY CABLE
BLACK CABLE
KEYENCE
KEYENCE
SL-V24H
SL-V24H
-U8/51 /51.2:D
-U8/51 /51.6:D
Vamtec Machines & Automation Pvt Ltd
MINDA CORPRATION PUNE
ECM - 449
LightCurtain
16-08-2024
kumar
Next Page : 53
Previous Page : 49
Page: 51 / 84


24V / 59.1:A
GND / 59.1:A
1/7
3/3
TELTONIKA
X1
X2
X3
X4
X5
-PE
1/53
PE
GND
US
TSW010
-A1/53
Previous Page : 51
Next Page : 55
Date 04-04-2024
Ed. kumar
Appr
Original
MINDA CORPRATION PUNE
ECM - 449
Replacement of
Replaced by
Vamtec
Machines & Automation Pvt Ltd
Ethernet Switch
Page: 53 / 84



1. PANEL PC TO BE ADDED.
2. USB to RS485 Isolator.

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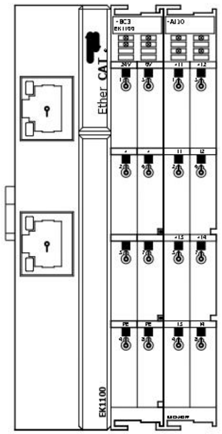
Next Page : 57

			Date	15-08-2024	MINDA CORPRATION PUNE		Vamtec Machines & Automation Pvt Ltd	DIGITAL PLC RACK	=
			Ed.	kumar					+
			Appr		ECM - 449				**Page:** 55 / 84
Modification	Date	Name	Original		Replacement of	Replaced by			



1. PANEL PC TO BE ADDED.
2. USB to RS485 Isolator.

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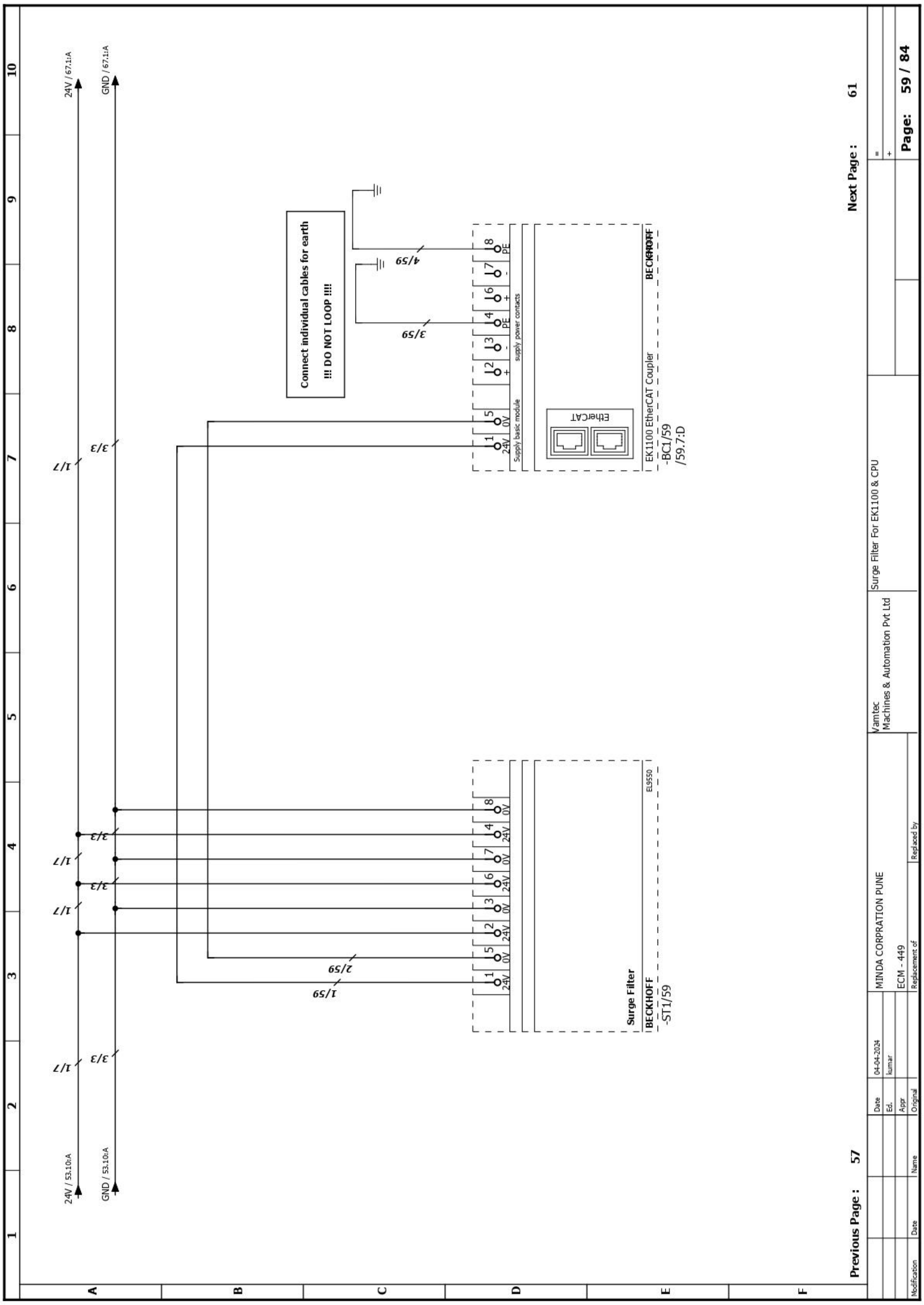
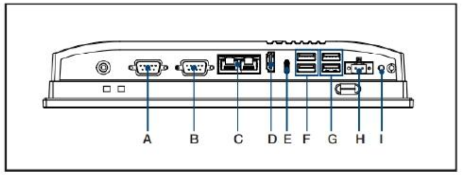
Next Page : 59

			Date	15-08-2024	MINDA CORPRATION PUNE	Vamtec Machines & Automation Pvt Ltd	ANALOG PLC RACK	=
			Ed.	kumar				+
			Appr		ECM - 449			Page: 57 / 84
Modification	Date	Name	Original		Replacement of / Replaced by			


Connect individual cables for earth
!!! DO NOT LOOP !!!!
EK1100 EtherCAT Coupler
-BC1/59
/59.7:D
BECKHOFF
EtherCAT
Supply basic module
supply power contacts
Surge Filter
BECKHOFF
-ST1/59
EL9550
24V / 53.10:A
GND / 53.10:A
24V / 67.1:A
GND / 67.1:A
1/59
2/59
3/59
4/59
Previous Page : 57
Next Page : 61
Page: 59 / 84
MINDA CORPRATION PUNE
ECM - 449
Vamtec
Machines & Automation Pvt Ltd
Surge Filter For EK1100 & CPU
04-04-2024
kumar



A. 1 x RS-232
B. 1 x RS-232/422/485
C. 2 x LAN
D. 1 x HDMI
E. 1 x Type C (USB only)
F. 2 x USB 3.1
G. 2 x USB 2.0
H. DC-In
I. Power Switch

-PPC

PANEL PC

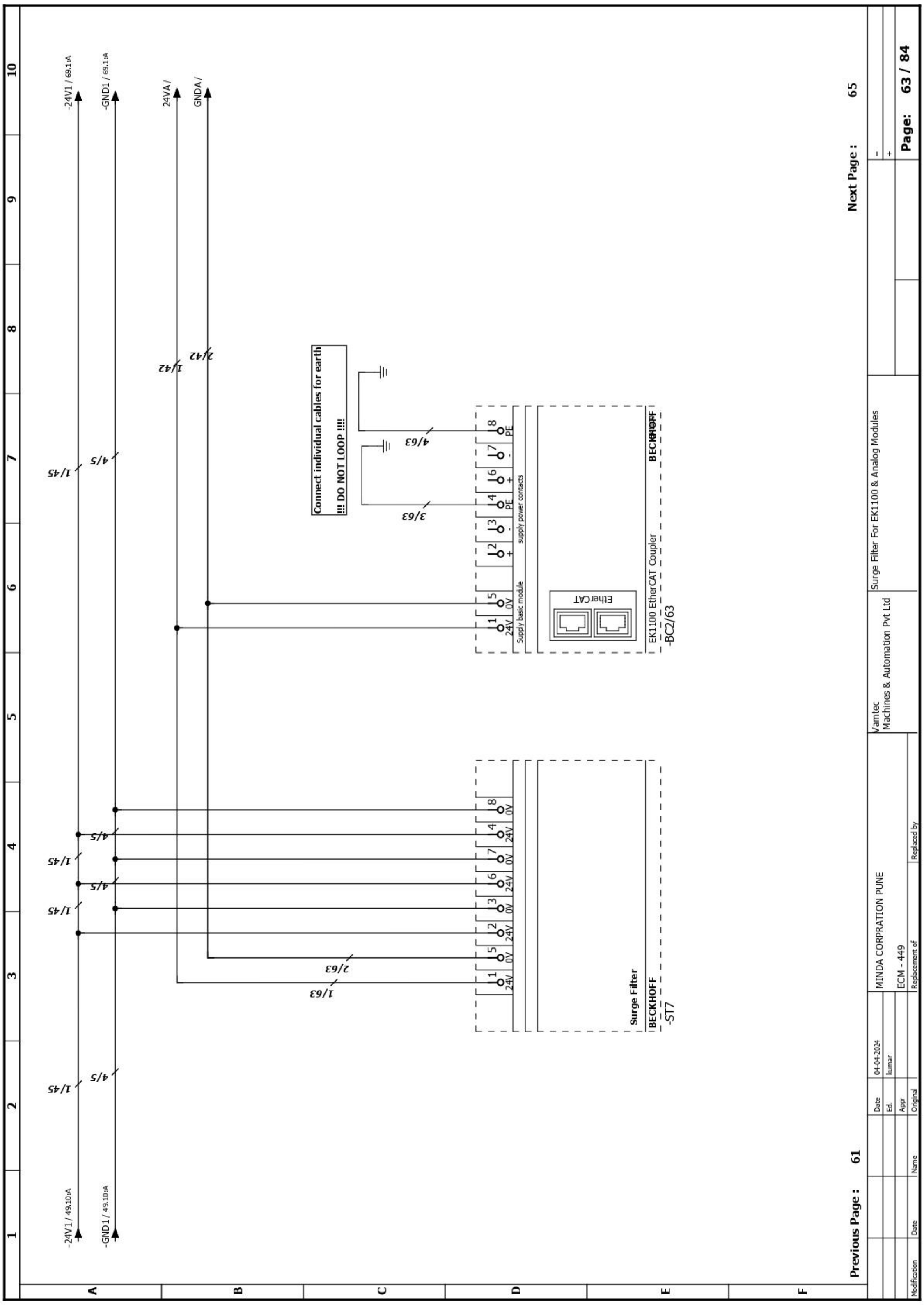
PPC-312-PJ60A

Previous Page : 59

Next Page : 63

			Date	15-08-2024	MINDA CORPRATION PUNE	Vamtec Machines & Automation Pvt Ltd	CPU Wiring
			Ed.	kumar	ECM - 449		
			Appr				
Modification	Date	Name	Original		Replacement of / Replaced by		

=
+
Page: 61 / 84



-24V1 / 49.10:A
-GND1 / 49.10:A
-24V1 / 69.1:A
-GND1 / 69.1:A
24VA /
GNDA /
1/45
4/5
1/42
2/42
1/63
2/63
3/63
4/63
Connect individual cables for earth
!!! DO NOT LOOP !!!
Surge Filter
BECKHOFF
-ST7
EK1100 EtherCAT Coupler
BECKHOFF
-BC2/63
EtherCAT
Supply basic module
supply power contacts
24V
0V
PE
Previous Page : 61
Next Page : 65
Date 04-04-2024
Ed. kumar
Appr
Original
Modification
Date
Name
MINDA CORPRATION PUNE
ECM - 449
Replacement of
Replaced by
Vamtec
Machines & Automation Pvt Ltd
Surge Filter For EK1100 & Analog Modules
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EMPTY

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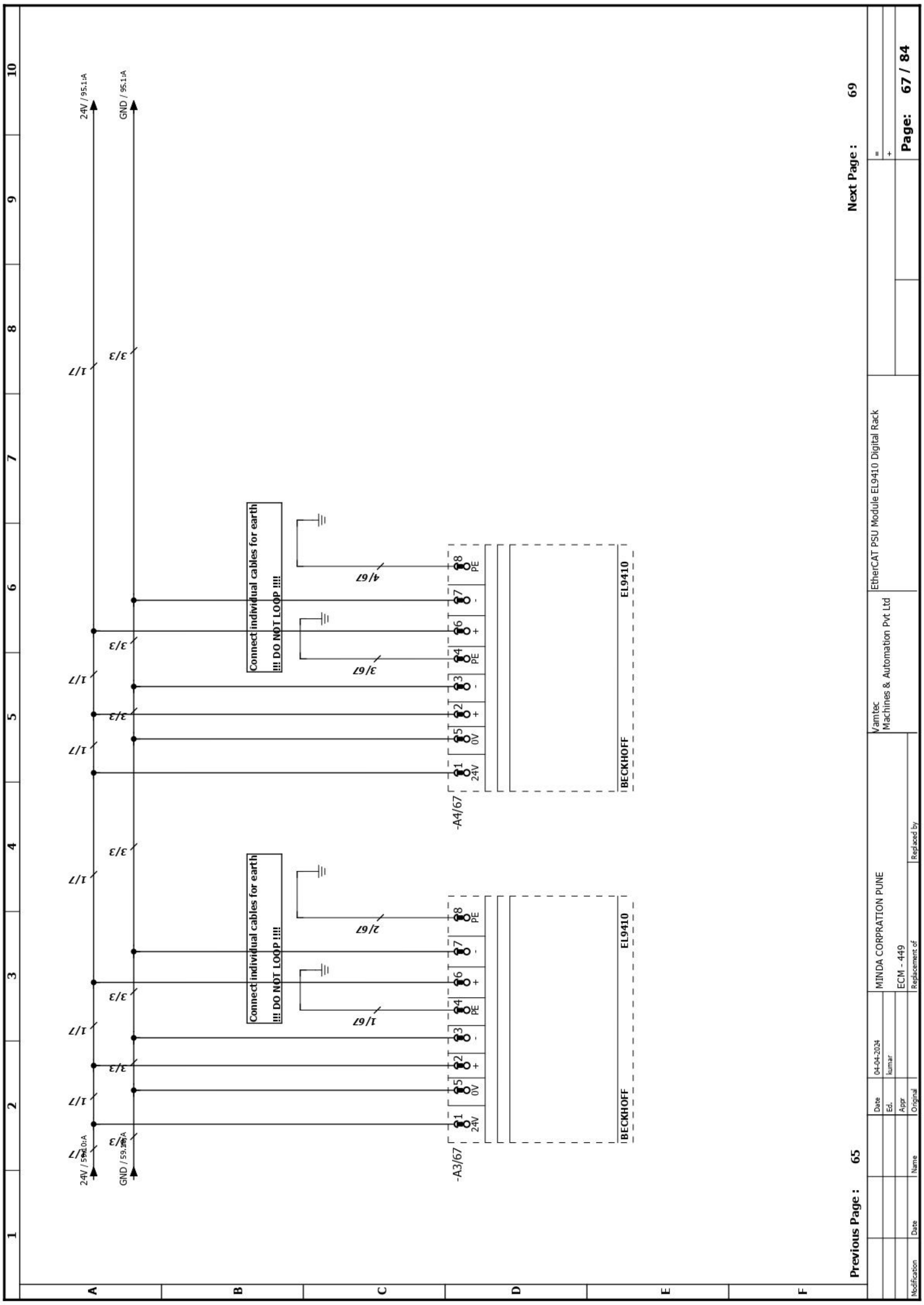
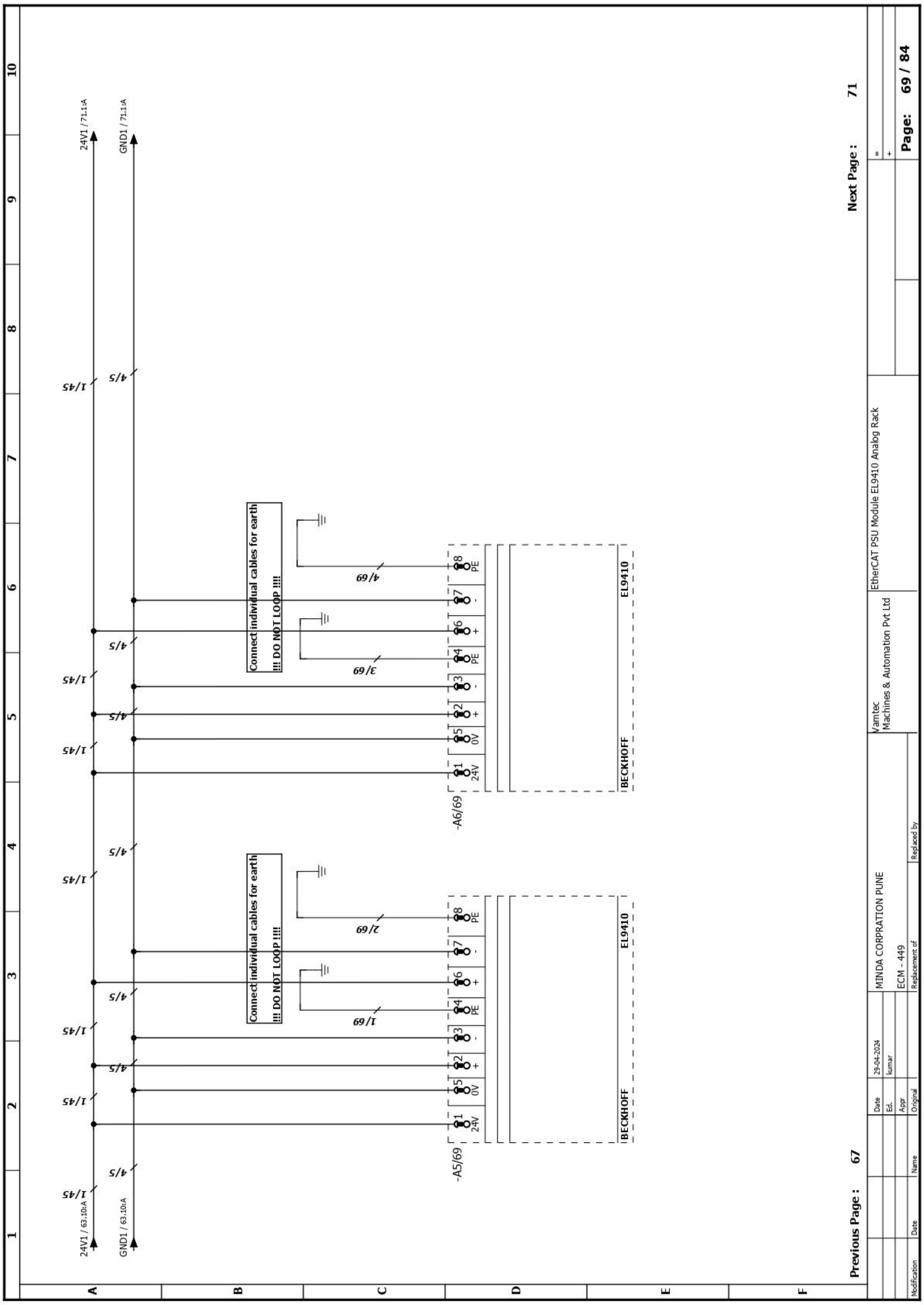
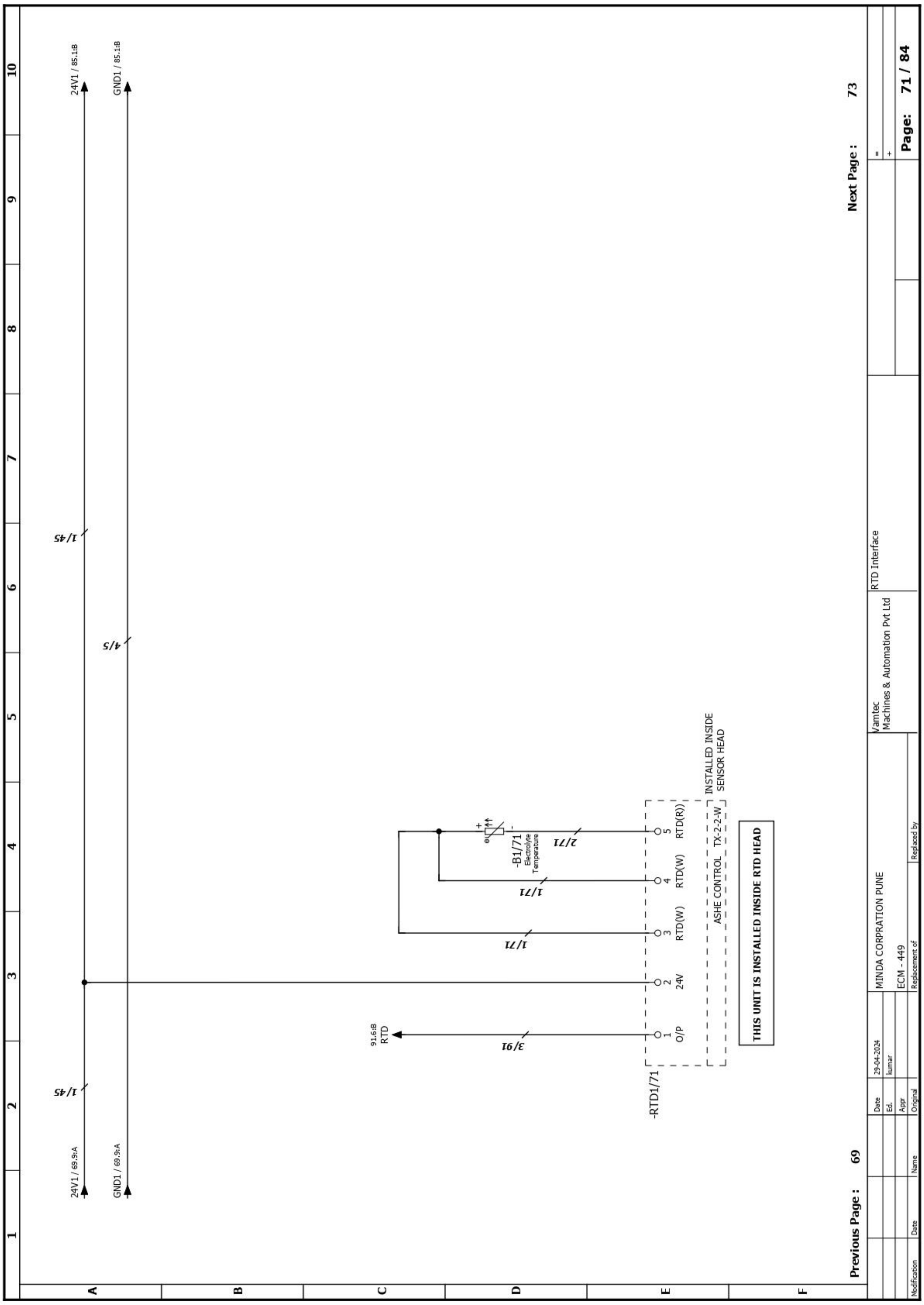
			Date	15-08-2024	MINDA CORPRATION PUNE	Vamtec Machines & Automation Pvt Ltd	RS422 /RS485 Interface
			Ed.	kumar	ECM - 449		
			Appr				
Modification	Date	Name	Original		Replacement of / Replaced by		

= +


24V / 95.1:A
GND / 95.1:A
Connect individual cables for earth
!!! DO NOT LOOP !!!!
-A3/67
-A4/67
BECKHOFF
EL9410
24V
0V
PE
Previous Page : 65
Next Page : 69
Date 04-04-2024
Ed. kumar
MINDA CORPRATION PUNE
ECM - 449
Vamtec
Machines & Automation Pvt Ltd
EtherCAT PSU Module EL9410 Digital Rack
Page: 67 / 84



24V1 / 63.10:A
GND1 / 63.10:A
24V1 / 71.1:A
GND1 / 71.1:A
1/45
4/5
Connect individual cables for earth
!!! DO NOT LOOP !!!!
-A5/69
-A6/69
1 24V
5 0V
2 +
3 -
4 PE
6 +
7 -
8 PE
1/69
2/69
3/69
4/69
BECKHOFF
EL9410
Previous Page : 67
Next Page : 71
Date 29-04-2024
Ed. kumar
Appr
Original
Modification
Date
Name
MINDA CORPRATION PUNE
ECM - 449
Replacement of
Replaced by
Vamtec
Machines & Automation Pvt Ltd
EtherCAT PSU Module EL9410 Analog Rack
=
+
Page: 69 / 84



24V1 / 85.1:B
GND1 / 85.1:B
24V1 / 69.9:A
GND1 / 69.9:A
1/45
4/5
-RTD1/71
O/P
24V
RTD(W)
RTD(W)
RTD(R))
3/91
1/71
1/71
2/71
91.6:B
RTD
-B1/71
Electrolyte
Temperature
ASHE CONTROL TX-2-2-W
INSTALLED INSIDE
SENSOR HEAD
THIS UNIT IS INSTALLED INSIDE RTD HEAD
Previous Page : 69
Next Page : 73
Date 29-04-2024
Ed. kumar
Appr
Original
MINDA CORPRATION PUNE
ECM - 449
Replacement of
Replaced by
Vamtec
Machines & Automation Pvt Ltd
RTD Interface
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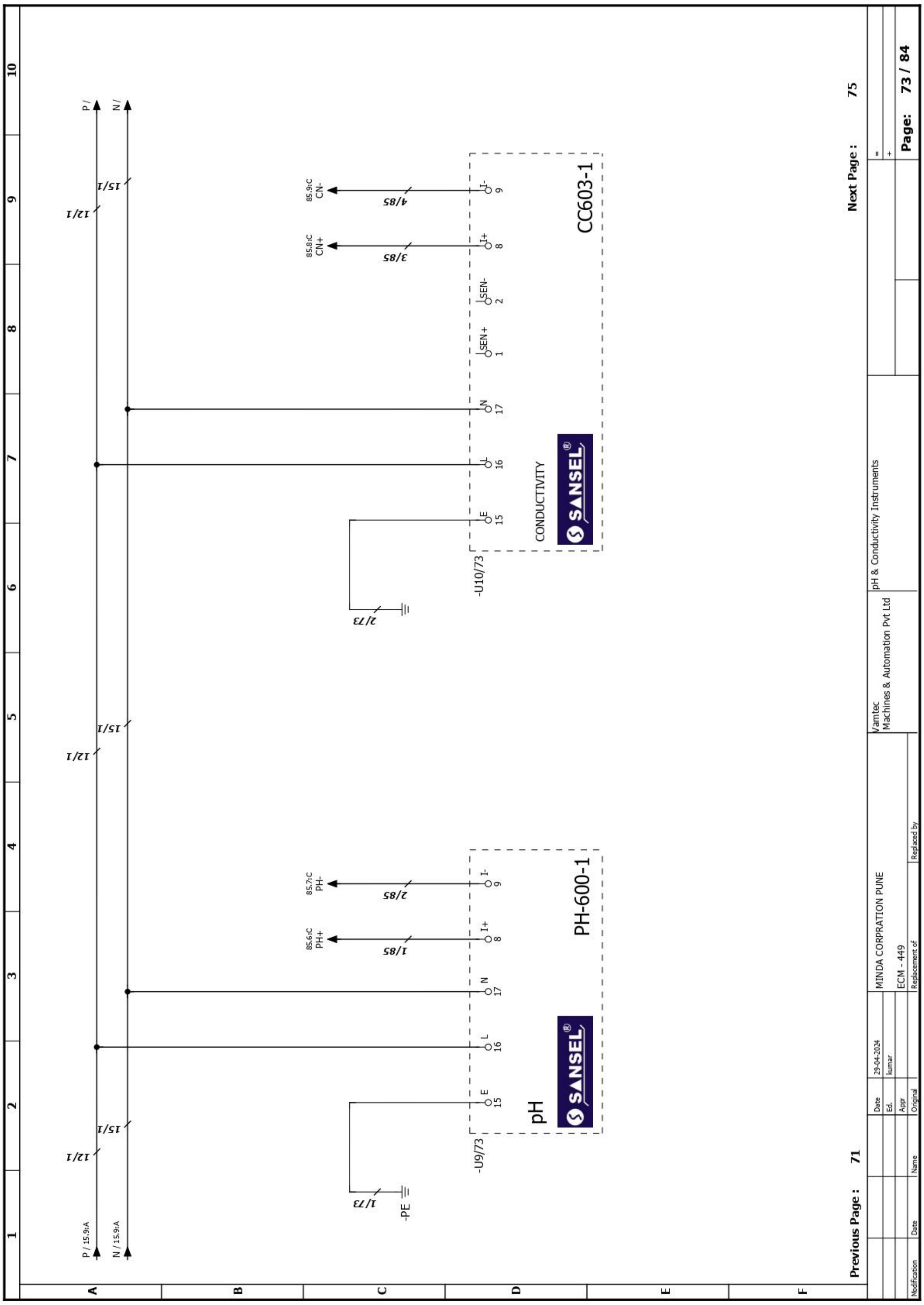
Name


CC603-1
CONDUCTIVITY
-U10/73
PH-600-1
pH
-U9/73
-PE
SEN+
SEN-
I+
I-
CN+
CN-
PH+
PH-
Previous Page : 71
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MINDA CORPRATION PUNE
ECM - 449
Vamtec
Machines & Automation Pvt Ltd
pH & Conductivity Instruments
Date 29-04-2024
Ed. kumar
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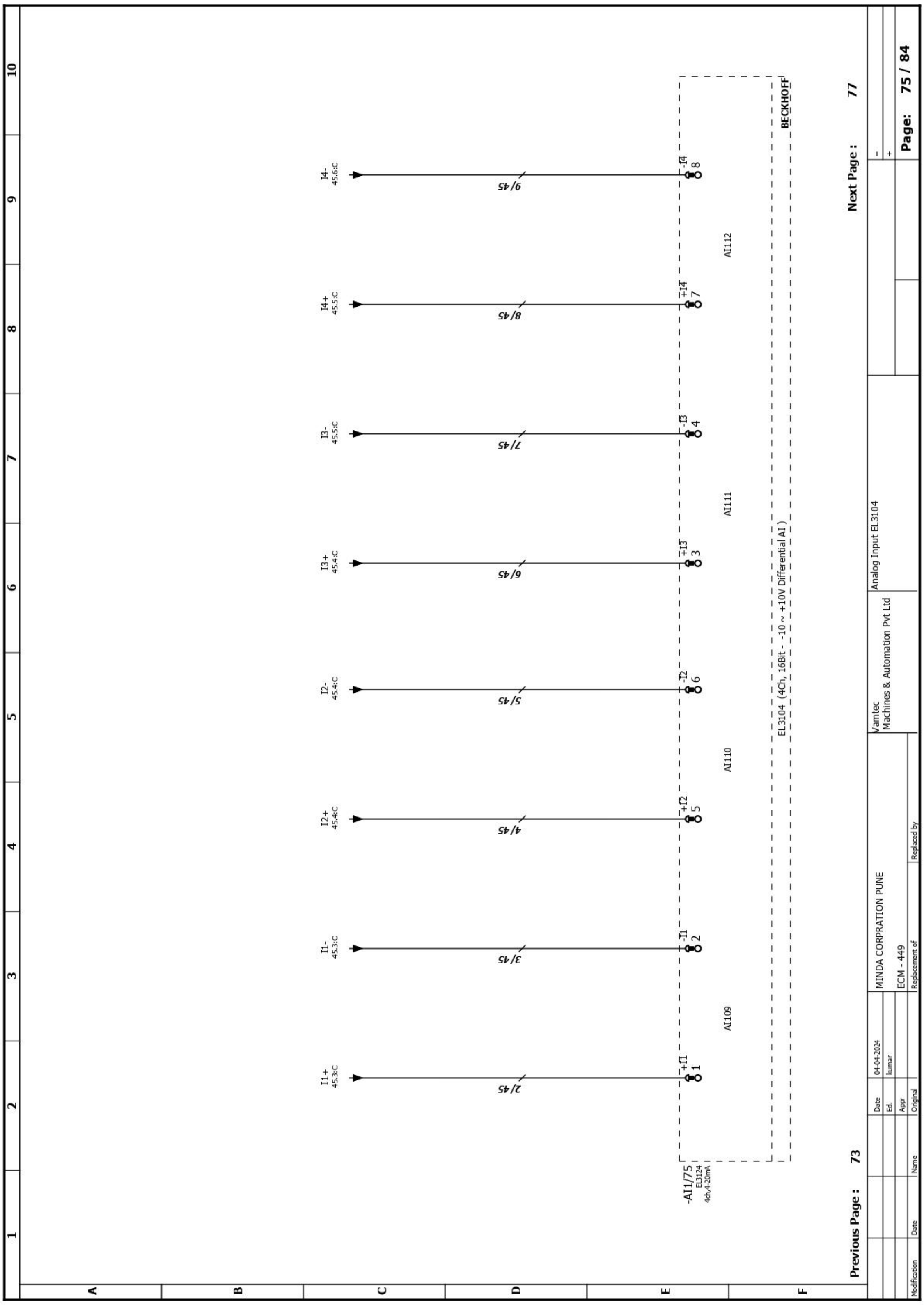
Original
Replacement of
Replaced by
Modification
Date
Name


I1+ 45.3:C
I1- 45.3:C
I2+ 45.4:C
I2- 45.4:C
I3+ 45.4:C
I3- 45.5:C
I4+ 45.5:C
I4- 45.6:C
2/45
3/45
4/45
5/45
6/45
7/45
8/45
9/45
+I1 1
-I1 2
+I2 5
-I2 6
+I3 3
-I3 4
+I4 7
-I4 8
AI109
AI110
AI111
AI112
EL3104 (4Ch, 16Bit - -10 ~ +10V Differential AI)
BECKHOFF
-AI1/75
EL3124
4ch,4-20mA
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Analog Input EL3104
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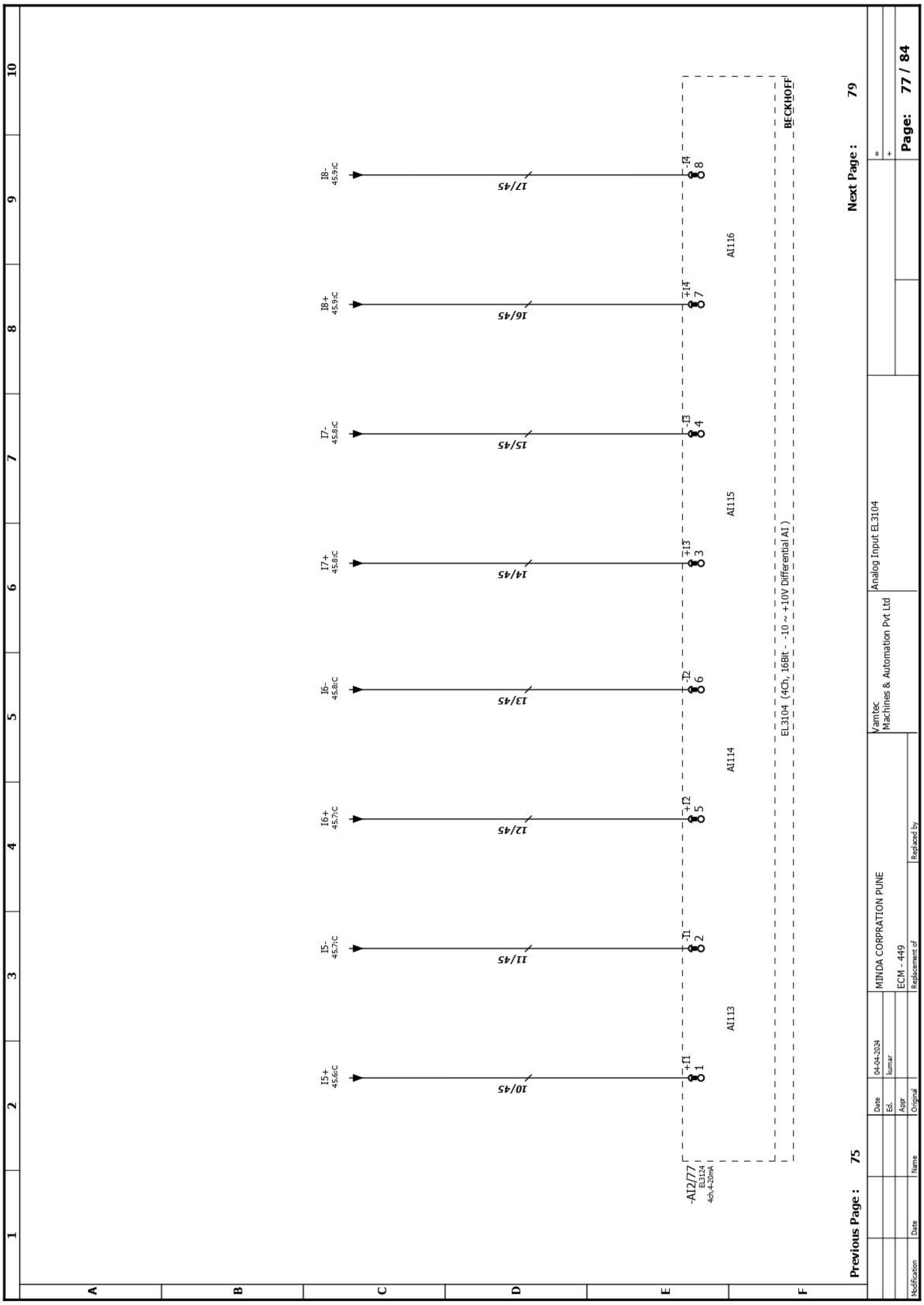
1
2
3
4
5
6
7
8
9
10
A
B
C
D
E
F



I5+ 45.6:C
I5- 45.7:C
I6+ 45.7:C
I6- 45.8:C
I7+ 45.8:C
I7- 45.8:C
I8+ 45.9:C
I8- 45.9:C
10/45
11/45
12/45
13/45
14/45
15/45
16/45
17/45
+I1 1
-I1 2
+I2 5
-I2 6
+I3 3
-I3 4
+I4 7
-I4 8
AI113
AI114
AI115
AI116
-AI2/77
EL3124
4ch,4-20mA
EL3104 (4Ch, 16Bit - -10 ~ +10V Differential AI)
BECKHOFF
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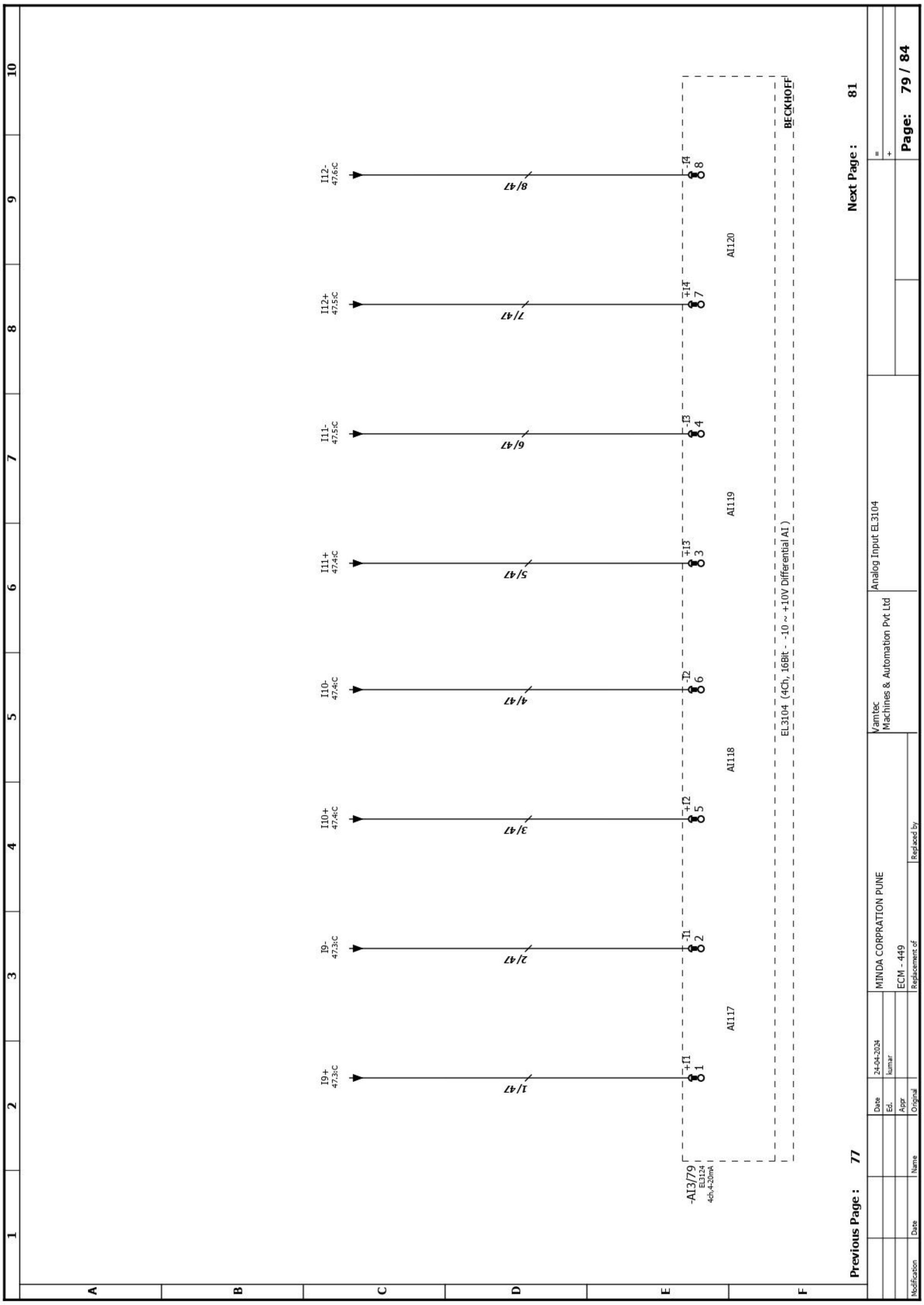
I9+ 47.3:C
I9- 47.3:C
I10+ 47.4:C
I10- 47.4:C
I11+ 47.4:C
I11- 47.5:C
I12+ 47.5:C
I12- 47.6:C
1/47
2/47
3/47
4/47
5/47
6/47
7/47
8/47
+I1 1
-I1 2
+I2 5
-I2 6
+I3 3
-I3 4
+I4 7
-I4 8
AI117
AI118
AI119
AI120
-AI3/79
EL3124
4ch,4-20mA
EL3104 (4Ch, 16Bit - -10 ~ +10V Differential AI)
BECKHOFF
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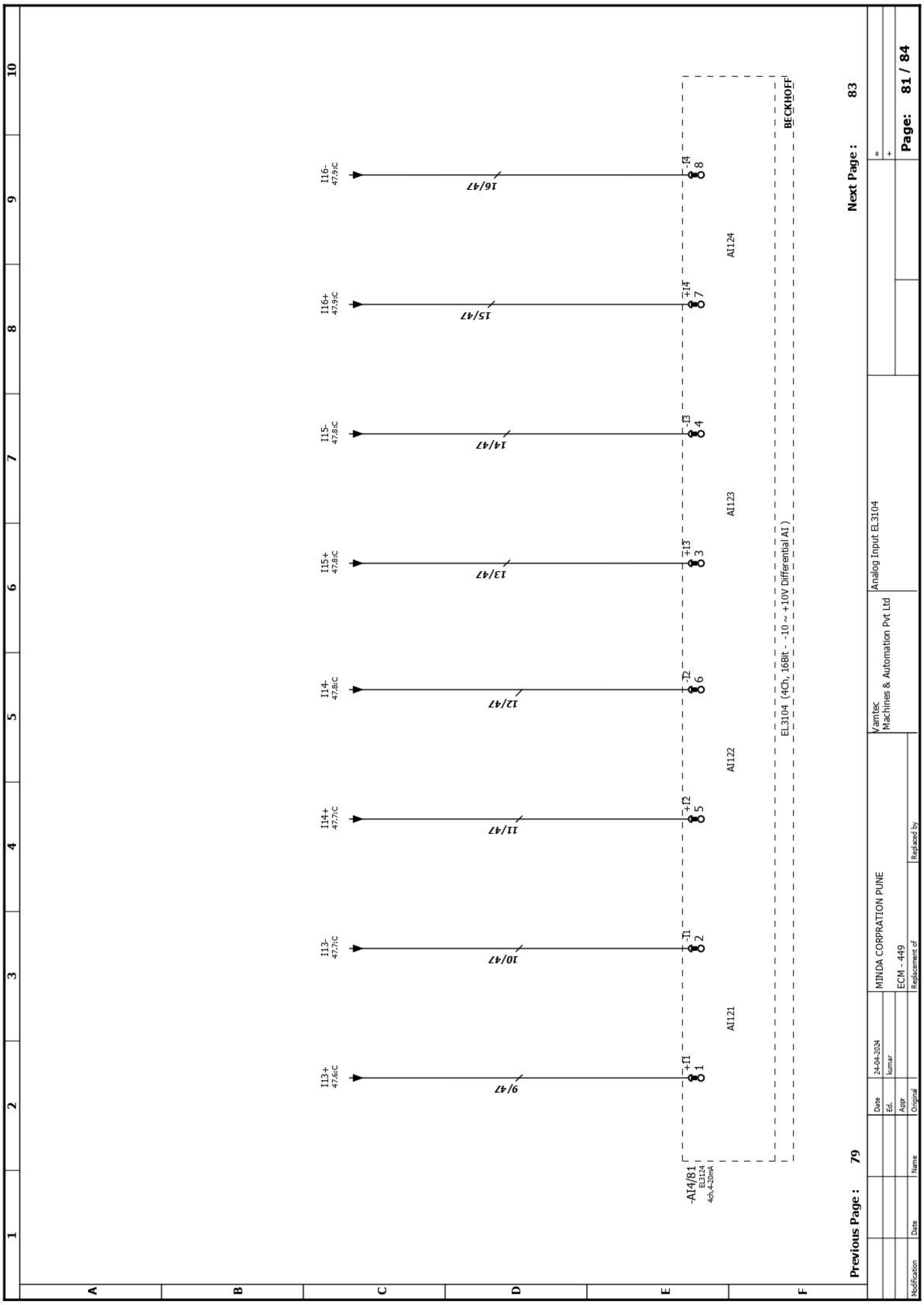

I13+ 47.6:C
I13- 47.7:C
I14+ 47.7:C
I14- 47.8:C
I15+ 47.8:C
I15- 47.8:C
I16+ 47.9:C
I16- 47.9:C
9/47
10/47
11/47
12/47
13/47
14/47
15/47
16/47
+I1 1
-I1 2
+I2 5
-I2 6
+I3 3
-I3 4
+I4 7
-I4 8
AI121
AI122
AI123
AI124
-AI4/81
EL3124
4ch,4-20mA
EL3104 (4Ch, 16Bit - -10 ~ +10V Differential AI)
BECKHOFF
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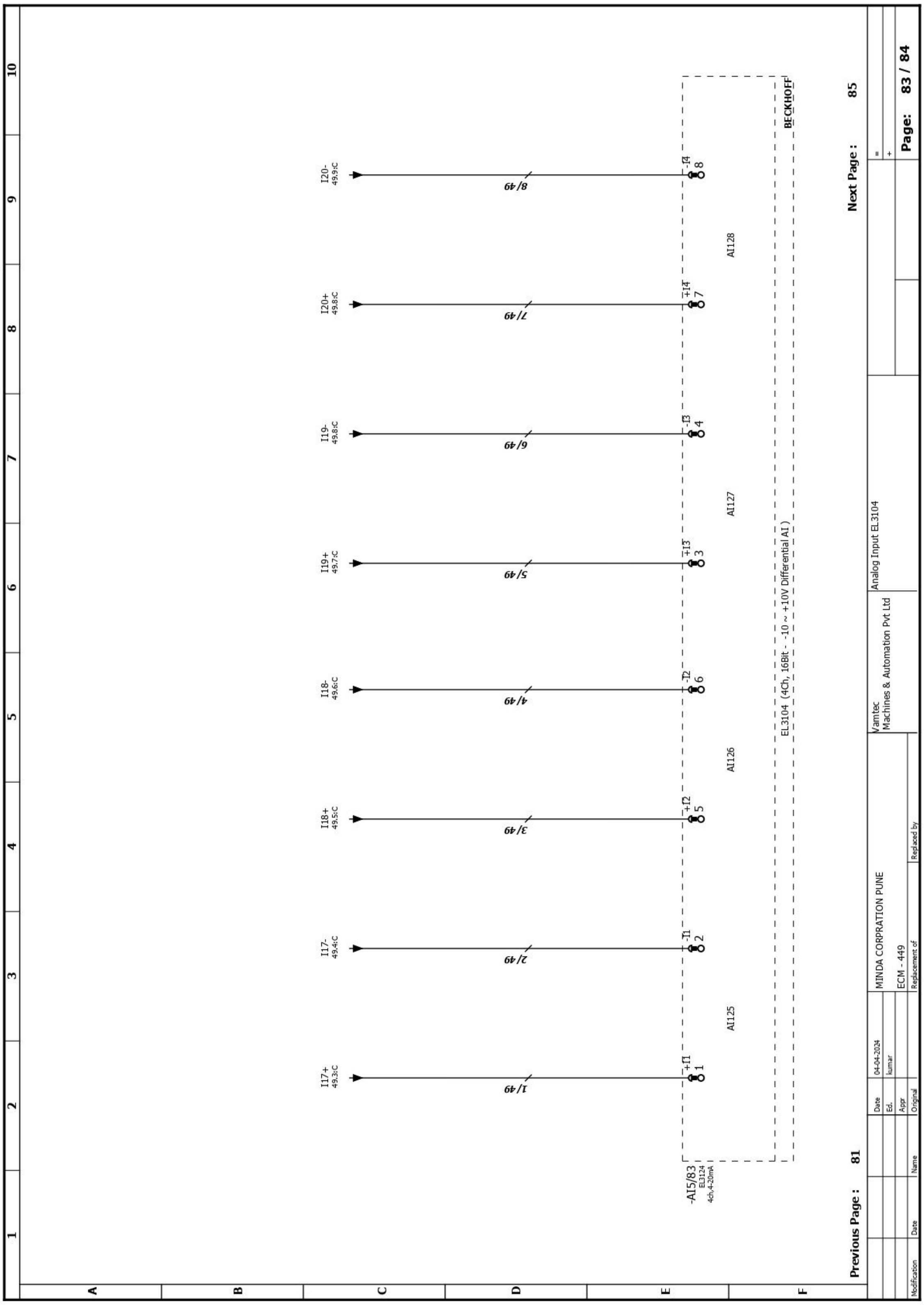
I17+ 49.3:C
I17- 49.4:C
I18+ 49.5:C
I18- 49.6:C
I19+ 49.7:C
I19- 49.8:C
I20+ 49.8:C
I20- 49.9:C
1/49
2/49
3/49
4/49
5/49
6/49
7/49
8/49
+I1 1
-I1 2
+I2 5
-I2 6
+I3 3
-I3 4
+I4 7
-I4 8
AI125
AI126
AI127
AI128
-AI5/83
EL3124
4ch,4-20mA
EL3104 (4Ch, 16Bit - -10 ~ +10V Differential AI)
BECKHOFF
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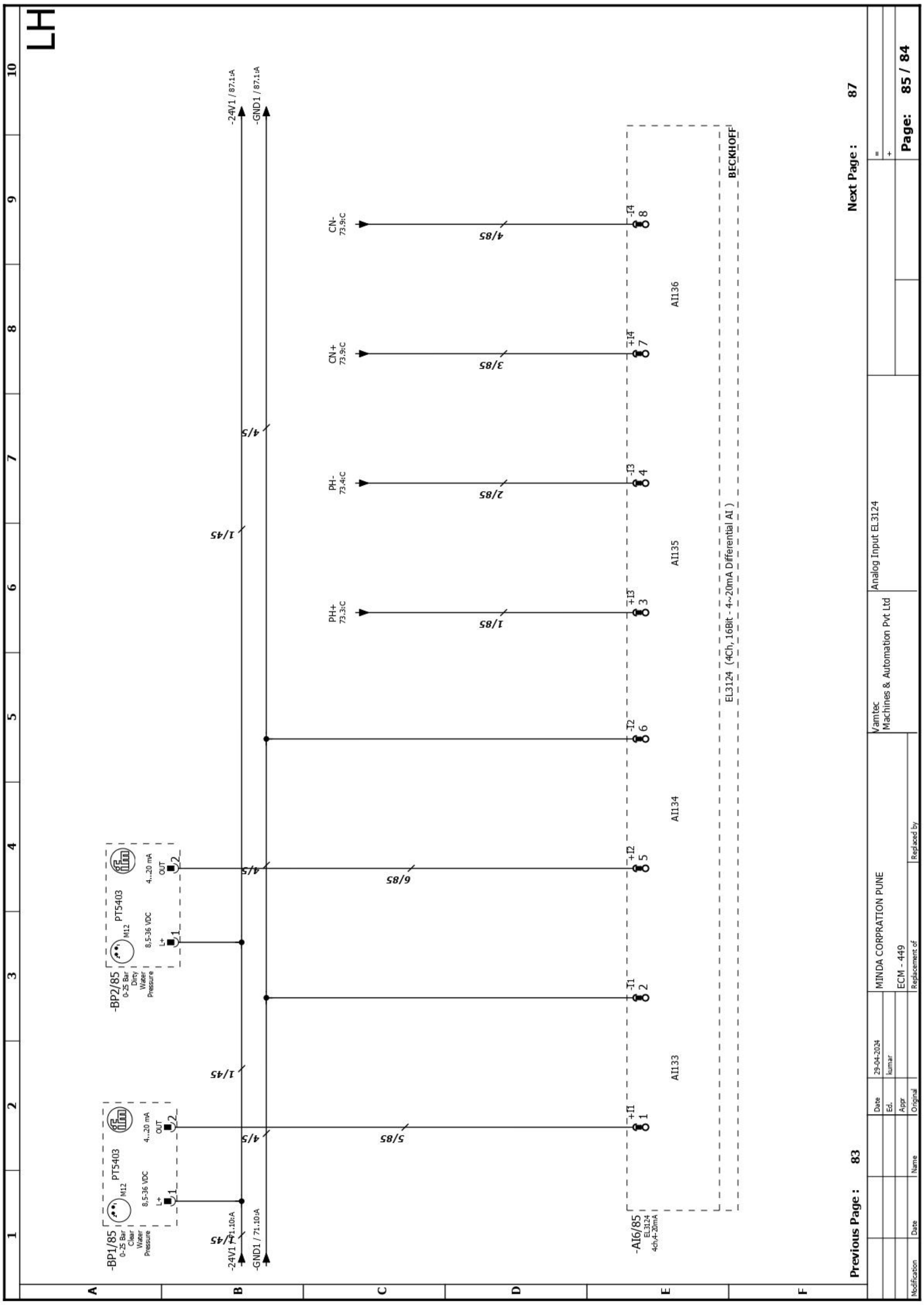

LH
-24V1 / 87.1:A
-GND1 / 87.1:A
-BP1/85
0-25 Bar
Clear
Water
Pressure
PT5403
M12
8.5-36 VDC
4...20 mA
-BP2/85
0-25 Bar
Dirty
Water
Pressure
CN-
73.9:C
CN+
73.9:C
PH-
73.4:C
PH+
73.3:C
1/85
2/85
3/85
4/85
5/85
6/85
1/45
4/5
-AI6/85
EL3124
4ch,4-20mA
AI133
AI134
AI135
AI136
EL3124 (4Ch, 16Bit - 4~20mA Differential AI)
BECKHOFF
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Next Page : 87
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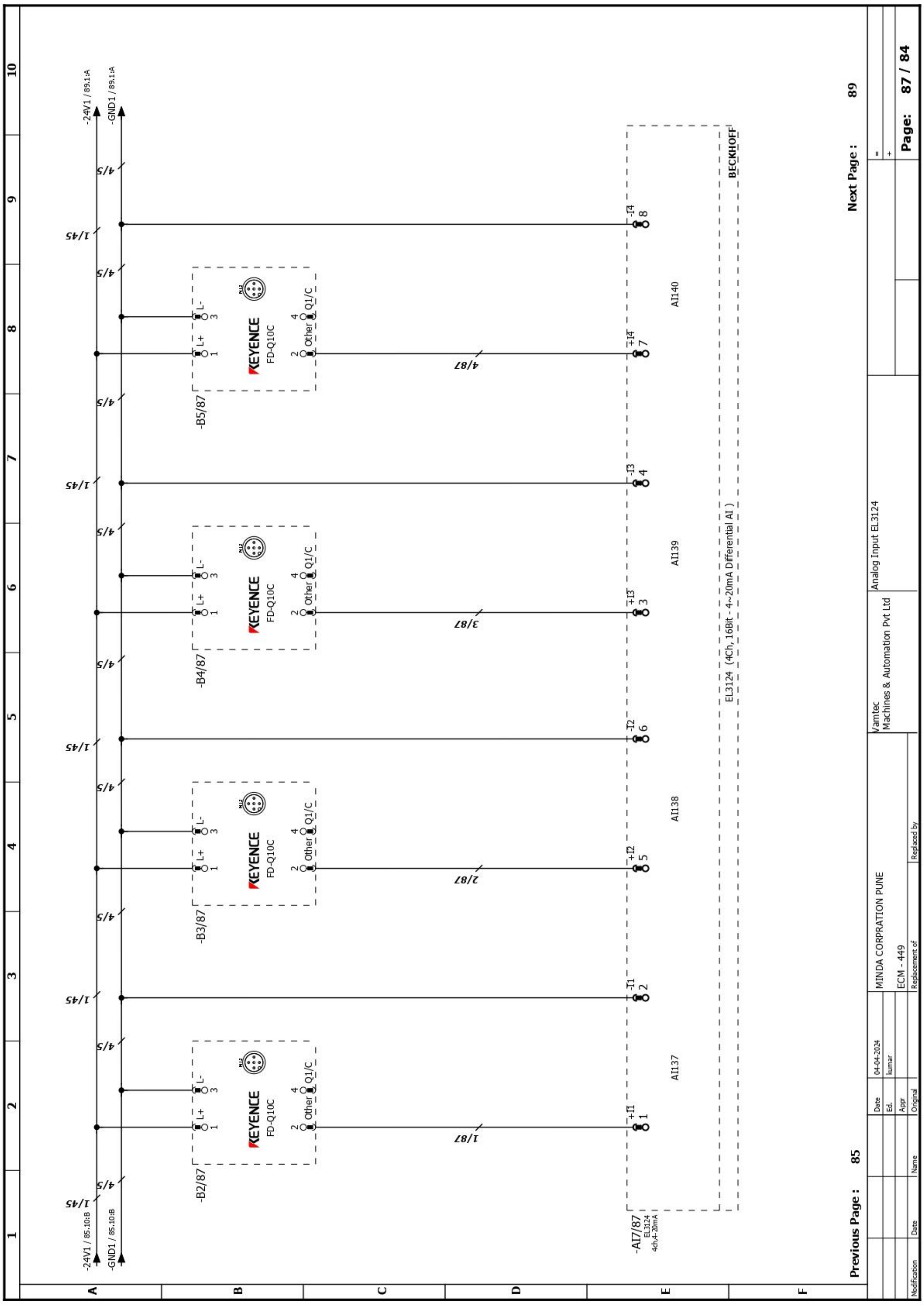
Replaced by



-24V1 / 89.1:A
-GND1 / 89.1:A
-24V1 / 85.10:B
-GND1 / 85.10:B
-B2/87
-B3/87
-B4/87
-B5/87
KEYENCE
FD-Q10C
L+
L-
Other
Q1/C
1/87
2/87
3/87
4/87
-AI7/87
EL3124
4ch,4-20mA
AI137
AI138
AI139
AI140
+I1
-I1
+I2
-I2
+I3
-I3
+I4
-I4
EL3124 (4Ch, 16Bit - 4~20mA Differential AI)
BECKHOFF
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Next Page : 89
Date 04-04-2024
Ed. kumar
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Original
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Analog Input EL3124
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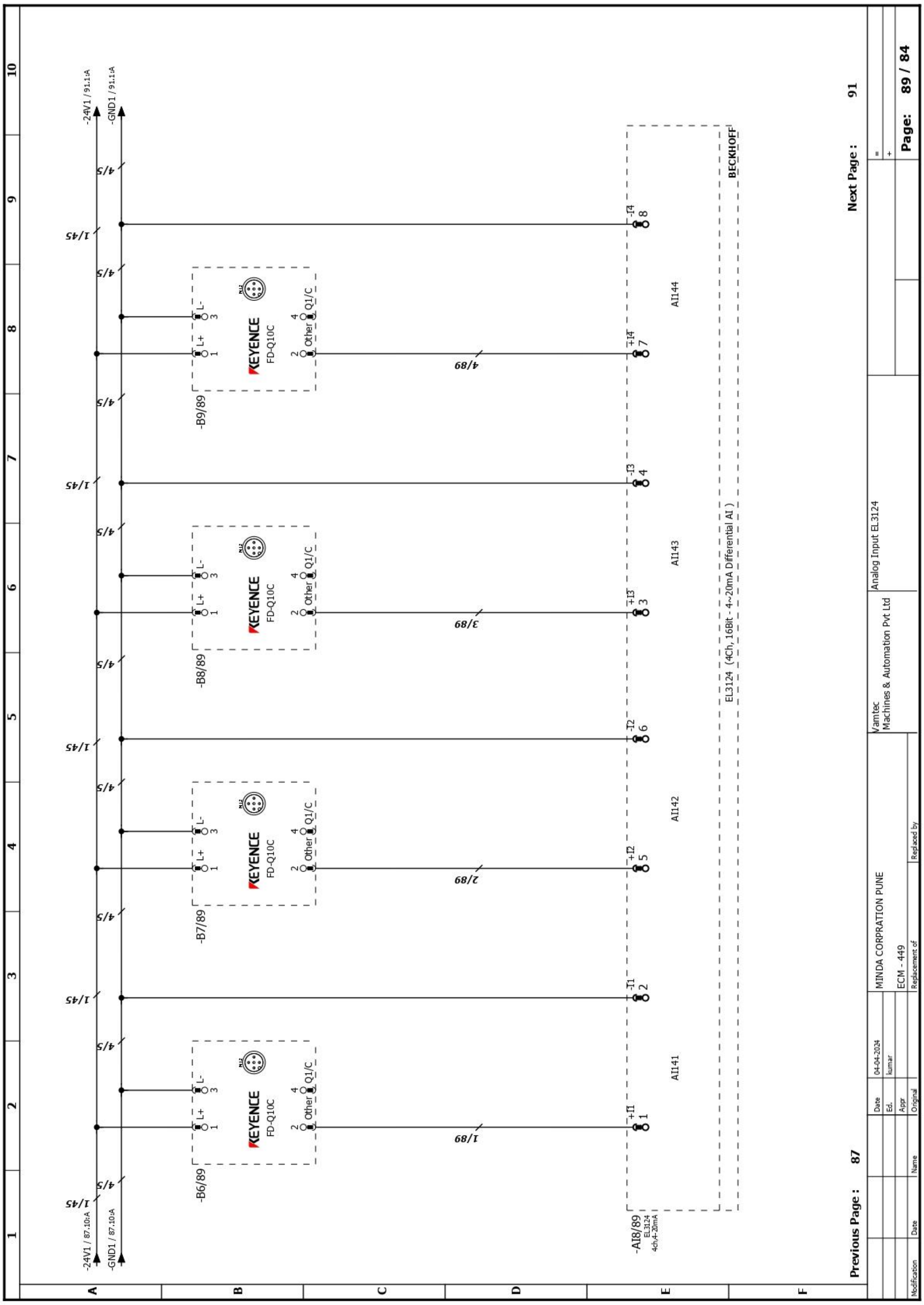
-24V1 / 87.10:A
-GND1 / 87.10:A
-24V1 / 91.1:A
-GND1 / 91.1:A
-B6/89
-B7/89
-B8/89
-B9/89
KEYENCE
FD-Q10C
L+
L-
Other
Q1/C
1/89
2/89
3/89
4/89
1/45
4/5
-AI8/89
EL3124
4ch,4-20mA
AI141
AI142
AI143
AI144
EL3124 (4Ch, 16Bit - 4~20mA Differential AI)
BECKHOFF
Previous Page : 87
Next Page : 91
Date 04-04-2024
Ed. kumar
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Replaced by
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Machines & Automation Pvt Ltd
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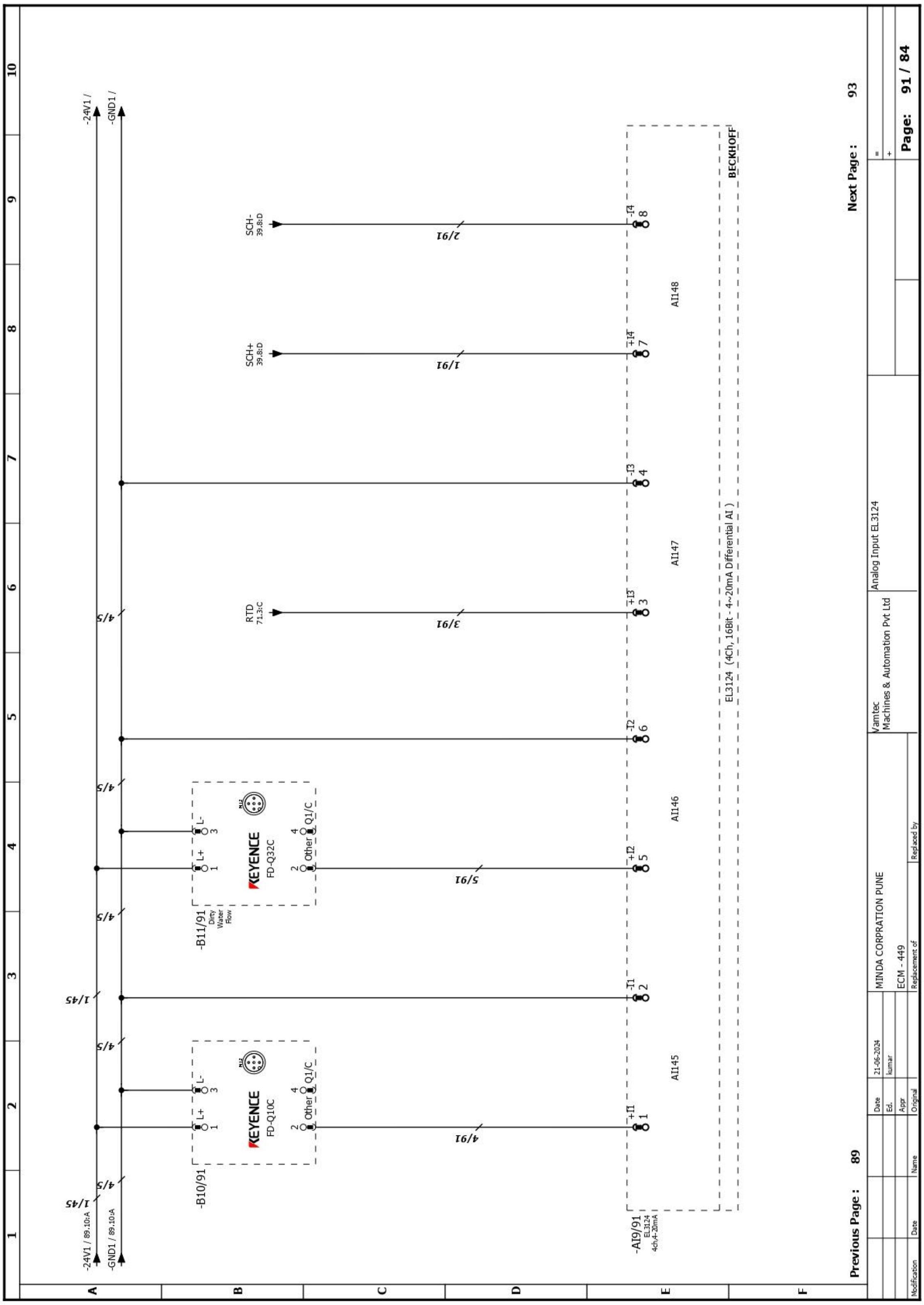
Name


-24V1 / 89.10:A
-GND1 / 89.10:A
-24V1 /
-GND1 /
-B10/91
KEYENCE
FD-Q10C
L+
L-
Other
Q1/C
-B11/91
Dirty Water Flow
KEYENCE
FD-Q32C
RTD 71.3:C
SCH+ 39.8:D
SCH- 39.8:D
-AI9/91
EL3124 4ch,4-20mA
AI145
AI146
AI147
AI148
EL3124 (4Ch, 16Bit - 4~20mA Differential AI)
BECKHOFF
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Next Page : 93
Date 21-06-2024
Ed. kumar
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Original
Modification
Date
Name
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Replaced by
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Analog Input EL3124
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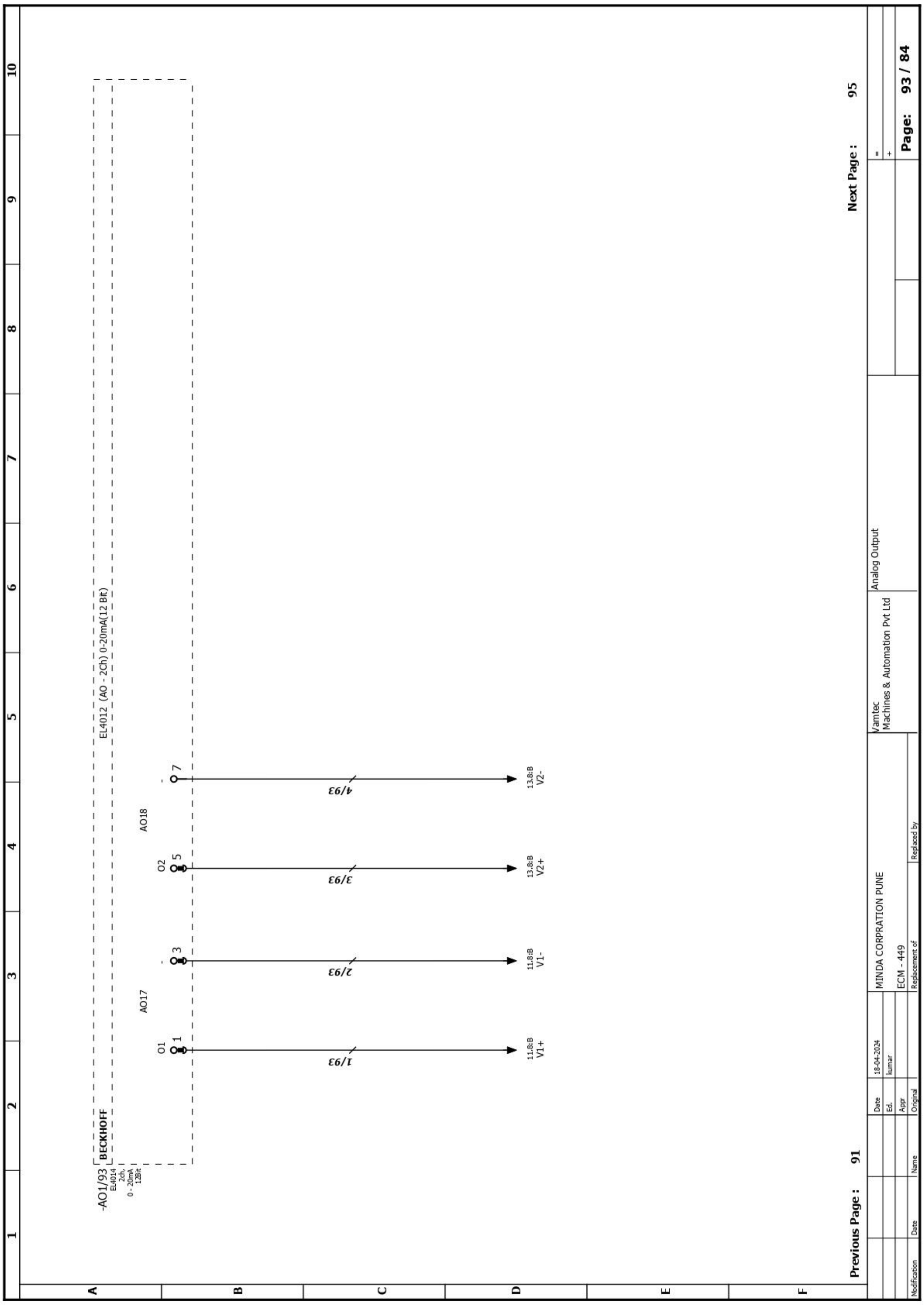
-AO1/93
EL4014
2ch.
0 - 20mA
12Bit
BECKHOFF
EL4012 (AO - 2Ch) 0-20mA(12 Bit)
AO17
AO18
O1 1
- 3
O2 5
- 7
1/93
2/93
3/93
4/93
11.8:B
V1+
11.8:B
V1-
13.8:B
V2+
13.8:B
V2-
Previous Page : 91
Next Page : 95
Date 18-04-2024
Ed. kumar
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Name
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Analog Output
=
+
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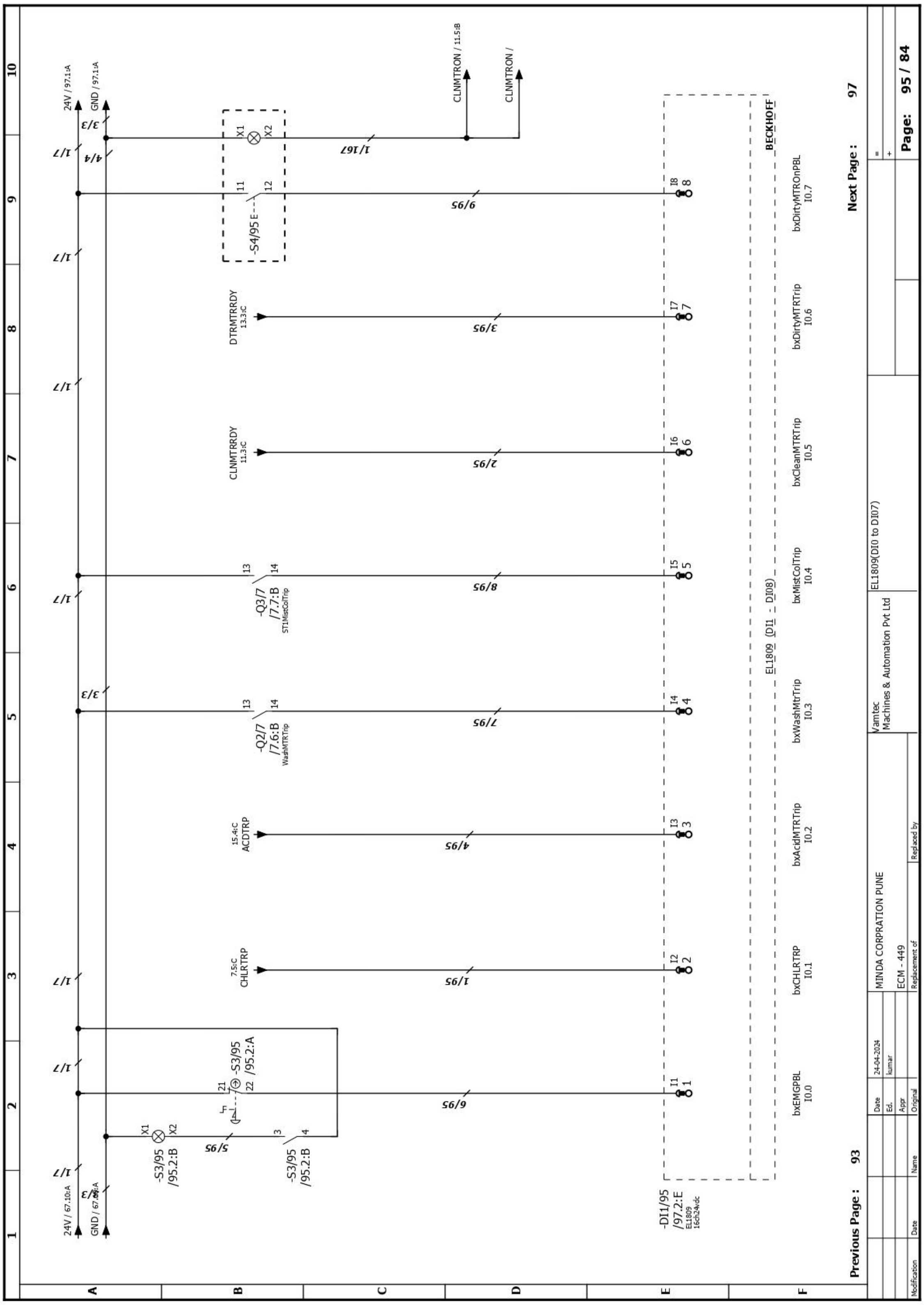
24V / 67.10:A
GND / 67.10:A
24V / 97.1:A
GND / 97.1:A
-S3/95
/95.2:B
-S3/95
/95.2:A
CHLRTRP
ACDTRP
-Q2/7
/7.6:B
WashMTRTrip
-Q3/7
/7.7:B
ST1MistColTrip
CLNMTRRDY
DTRMTRRDY
-S4/95 E
CLNMTRON / 11.5:B
CLNMTRON /
-DI1/95
/97.2:E
EL1809
16ch24vdc
EL1809 (DI1 - DI08)
BECKHOFF
bxEMGPBL 10.0
bxCHLRTRP 10.1
bxAcidMTRTrip 10.2
bxWashMtrTrip 10.3
bxMistColTrip 10.4
bxCleanMTRTrip 10.5
bxDirtyMTRTrip 10.6
bxDirtyMTROnPBL 10.7
Previous Page : 93
Next Page : 97
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Machines & Automation Pvt Ltd
EL1809(DI0 to DI07)
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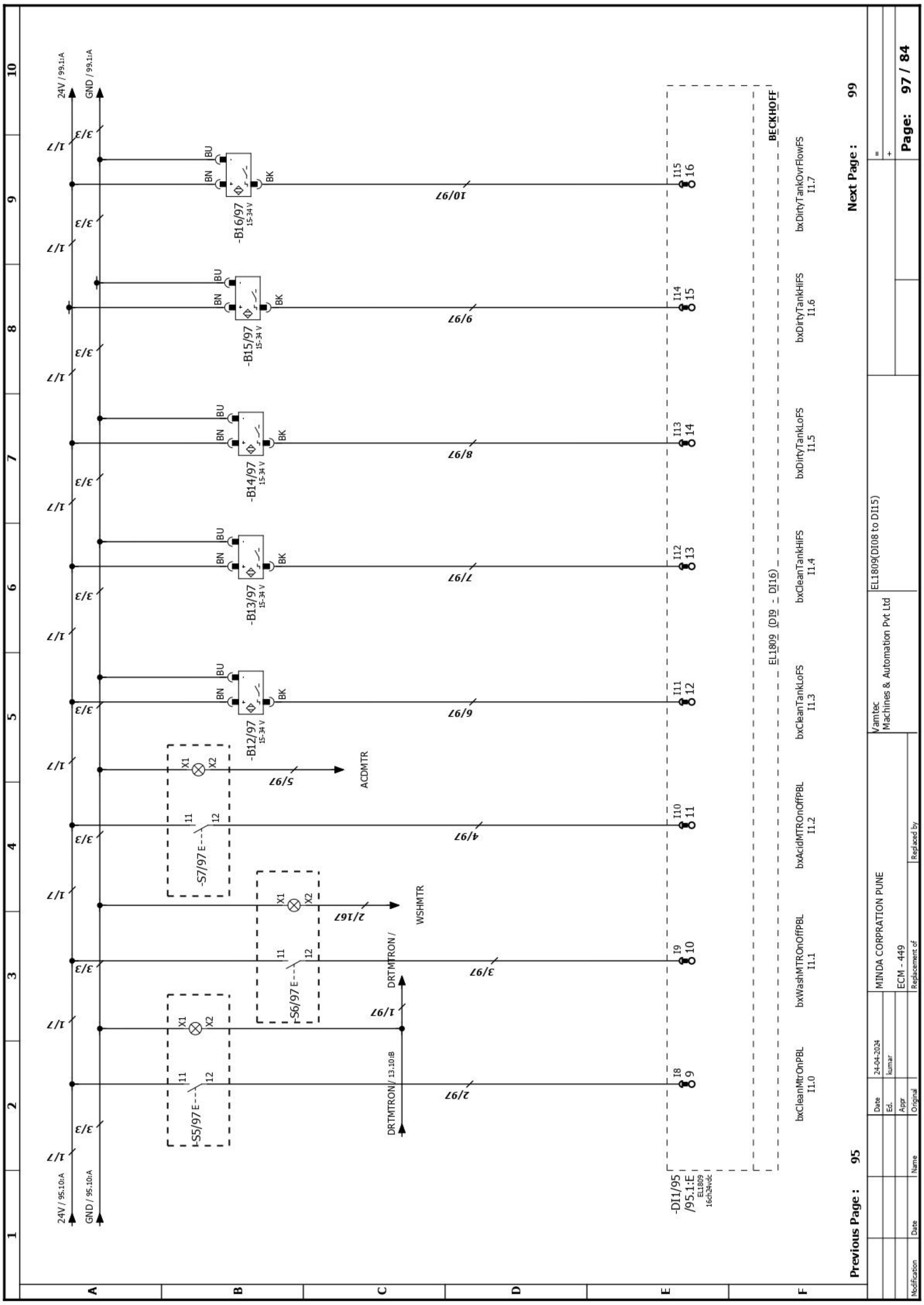
24V / 95.10:A
GND / 95.10:A
24V / 99.1:A
GND / 99.1:A
-S5/97
-S6/97
-S7/97
DRTMTRON / 13.10:B
DRTMTRON /
WSHMTR
ACDMTR
-B12/97
-B13/97
-B14/97
-B15/97
-B16/97
15-34 V
BN
BU
BK
-DI1/95
/95.1:E
EL1809
16ch24vdc
EL1809 (DI9 - DI16)
BECKHOFF
bxCleanMtrOnPBL
I1.0
bxWashMTROnOffPBL
I1.1
bxAcidMTROnOffPBL
I1.2
bxCleanTankLoFS
I1.3
bxCleanTankHiFS
I1.4
bxDirtyTankLoFS
I1.5
bxDirtyTankHiFS
I1.6
bxDirtyTankOvrFlowFS
I1.7
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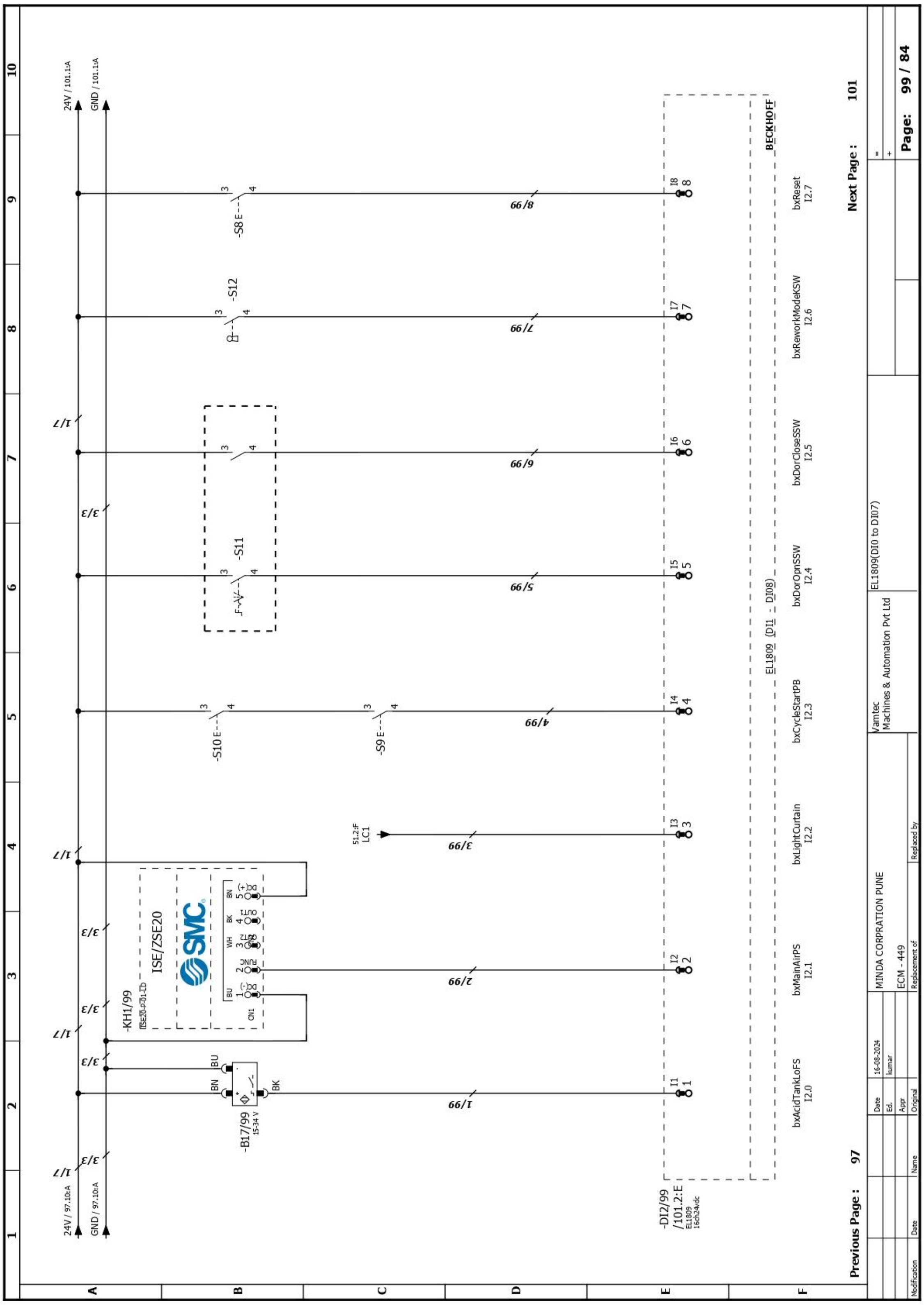
24V / 97.10:A
GND / 97.10:A
24V / 101.1:A
GND / 101.1:A
-B17/99
15-34 V
-KH1/99
ISE/ZSE20
SMC
-S10
-S9
-S11
-S12
-S8
LC1
-DI2/99
/101.2:E
EL1809
16ch24vdc
EL1809 (DI1 - DI08)
BECKHOFF
bxAcidTankLoFS
I2.0
bxMainAirPS
I2.1
bxLightCurtain
I2.2
bxCycleStartPB
I2.3
bxDorOpnSSW
I2.4
bxDorCloseSSW
I2.5
bxReworkModeSSW
I2.6
bxReset
I2.7
Previous Page : 97
Next Page : 101
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EL1809(DI0 to DI07)
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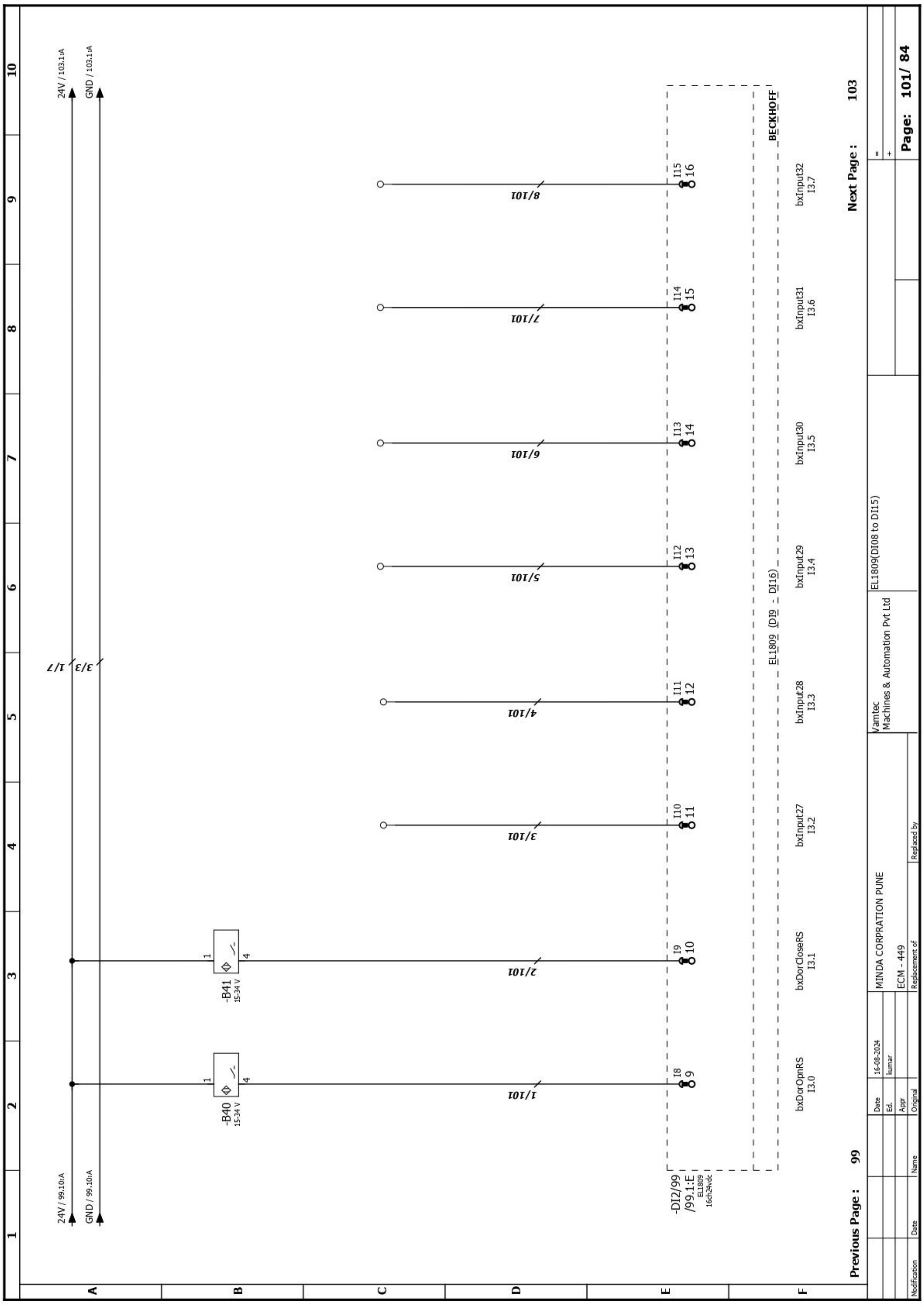
24V / 99.10:A
GND / 99.10:A
24V / 103.1:A
GND / 103.1:A
1/7
3/3
-B40
15-34 V
-B41
15-34 V
1/101
2/101
3/101
4/101
5/101
6/101
7/101
8/101
I8 9
I9 10
I10 11
I11 12
I12 13
I13 14
I14 15
I15 16
-DI2/99
/99.1:E
EL1809
16ch24vdc
EL1809 (DI9 - DI16)
BECKHOFF
bxDorOpnRS I3.0
bxDorCloseRS I3.1
bxInput27 I3.2
bxInput28 I3.3
bxInput29 I3.4
bxInput30 I3.5
bxInput31 I3.6
bxInput32 I3.7
Previous Page : 99
Next Page : 103
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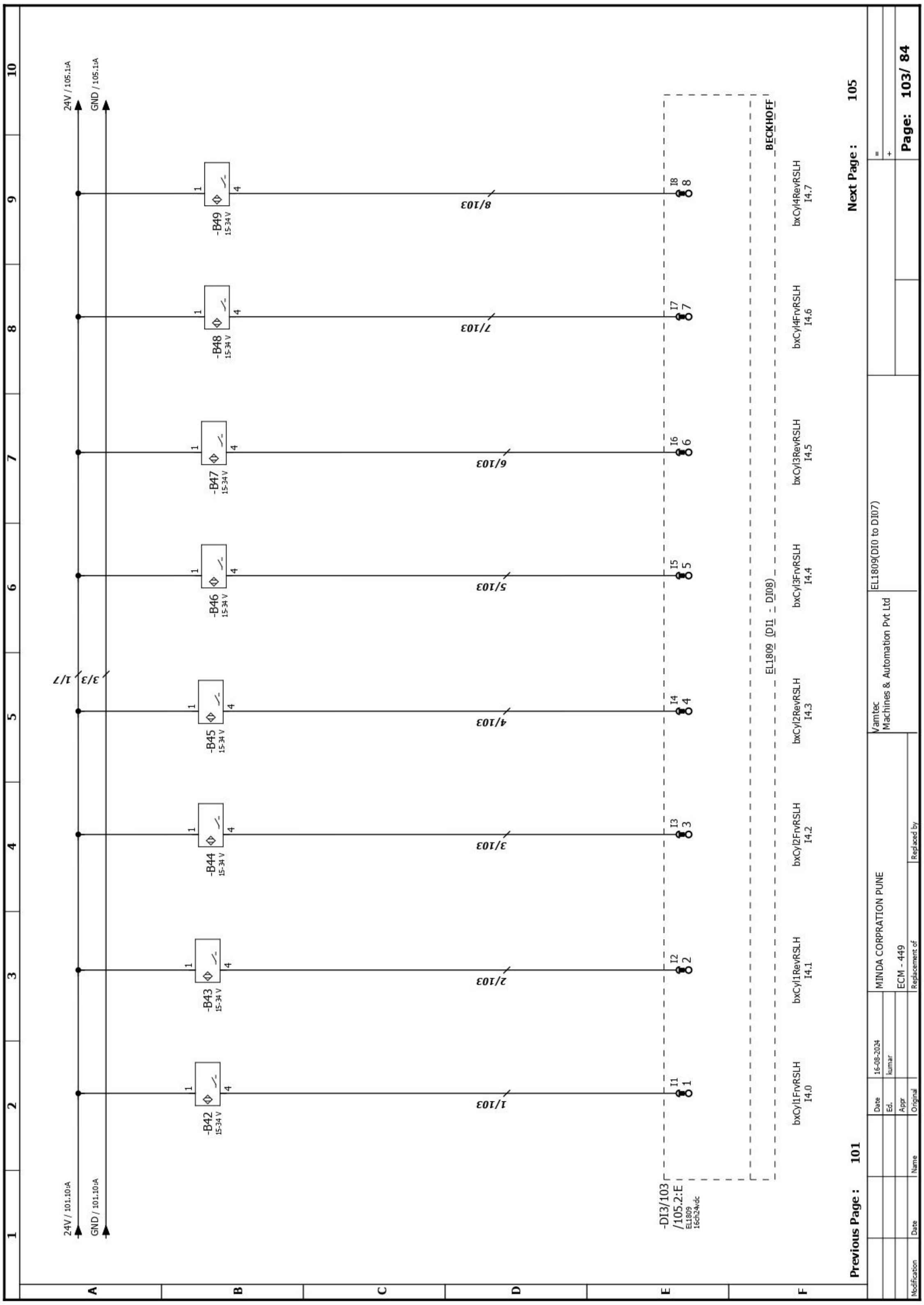
24V / 101.10:A
GND / 101.10:A
24V / 105.1:A
GND / 105.1:A
1/7
3/3
-B42
15-34 V
-B43
15-34 V
-B44
15-34 V
-B45
15-34 V
-B46
15-34 V
-B47
15-34 V
-B48
15-34 V
-B49
15-34 V
1/103
2/103
3/103
4/103
5/103
6/103
7/103
8/103
I1 1
I2 2
I3 3
I4 4
I5 5
I6 6
I7 7
I8 8
-DI3/103
/105.2:E
EL1809
16ch24vdc
EL1809 (DI1 - DI08)
BECKHOFF
bxCyl1FrvRSLH
I4.0
bxCyl1RevRSLH
I4.1
bxCyl2FrvRSLH
I4.2
bxCyl2RevRSLH
I4.3
bxCyl3FrvRSLH
I4.4
bxCyl3RevRSLH
I4.5
bxCyl4FrvRSLH
I4.6
bxCyl4RevRSLH
I4.7
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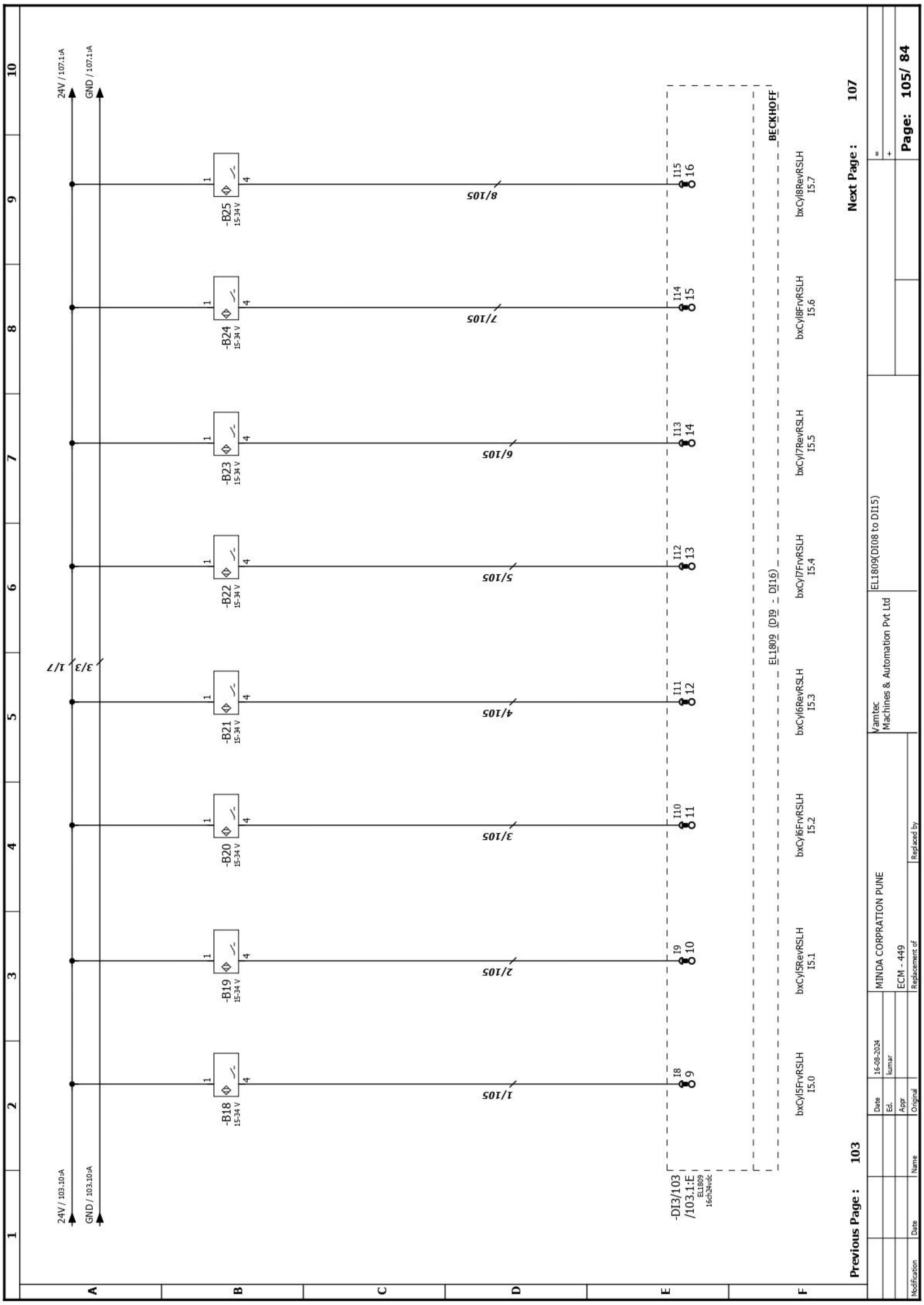

24V / 103.10:A
GND / 103.10:A
24V / 107.1:A
GND / 107.1:A
-B18
-B19
-B20
-B21
-B22
-B23
-B24
-B25
15-34 V
1/105
2/105
3/105
4/105
5/105
6/105
7/105
8/105
3/3
1/7
-DI3/103
/103.1:E
EL1809
16ch24vdc
EL1809 (DI9 - DI16)
BECKHOFF
bxCyl5FrvRSLH
I5.0
bxCyl5RevRSLH
I5.1
bxCyl6FrvRSLH
I5.2
bxCyl6RevRSLH
I5.3
bxCyl7FrvRSLH
I5.4
bxCyl7RevRSLH
I5.5
bxCyl8FrvRSLH
I5.6
bxCyl8RevRSLH
I5.7
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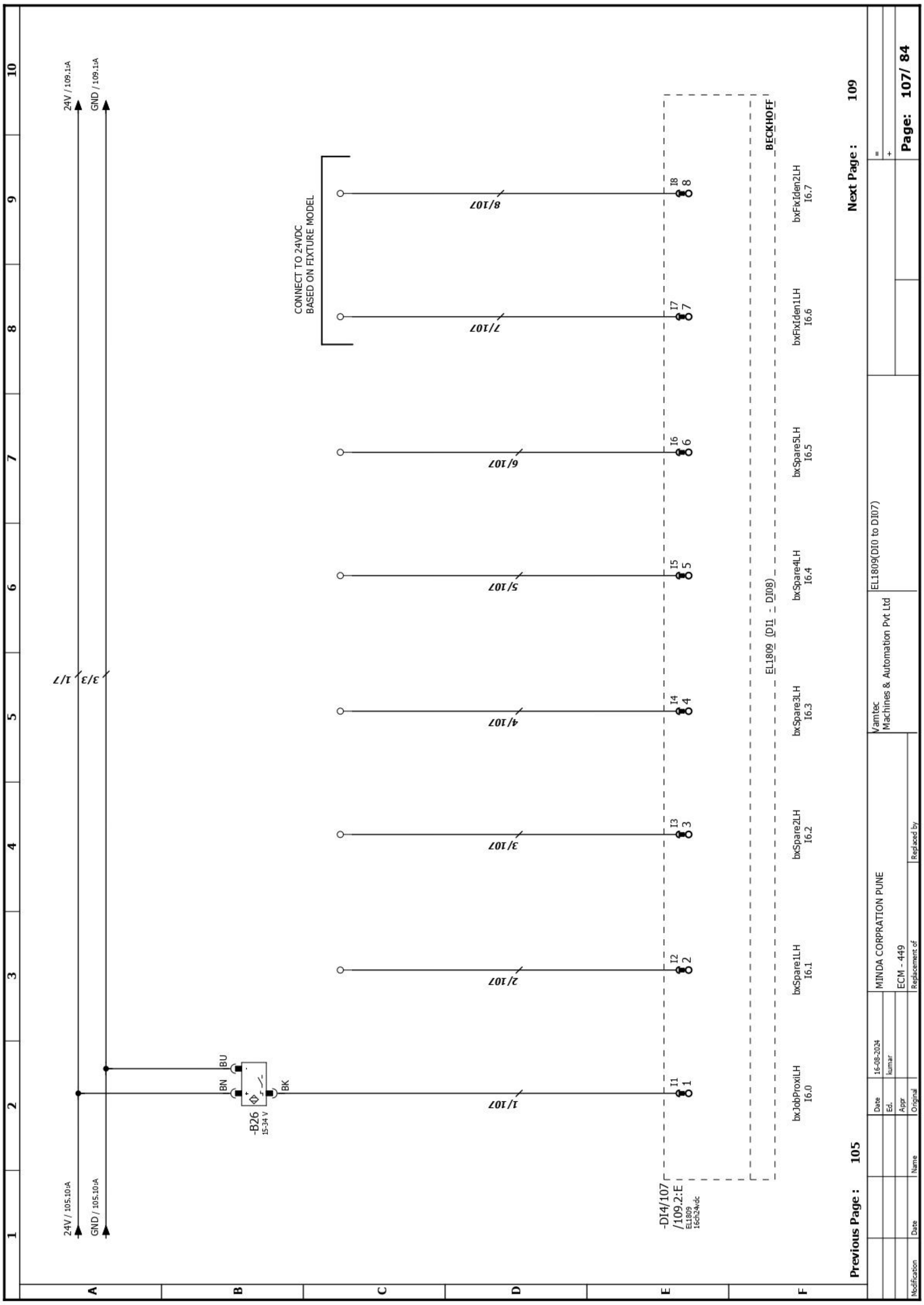

24V / 109.1:A
GND / 109.1:A
24V / 105.10:A
GND / 105.10:A
CONNECT TO 24VDC
BASED ON FIXTURE MODEL
BECKHOFF
EL1809 (DI1 - DI08)
-DI4/107
/109.2:E
EL1809
16ch24vdc
-B26
15-34 V
BN
BU
BK
1/107
2/107
3/107
4/107
5/107
6/107
7/107
8/107
bxJobProxiLH
I6.0
bxSpare1LH
I6.1
bxSpare2LH
I6.2
bxSpare3LH
I6.3
bxSpare4LH
I6.4
bxSpare5LH
I6.5
bxFixIden1LH
I6.6
bxFixIden2LH
I6.7
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MINDA CORPRATION PUNE
ECM - 449
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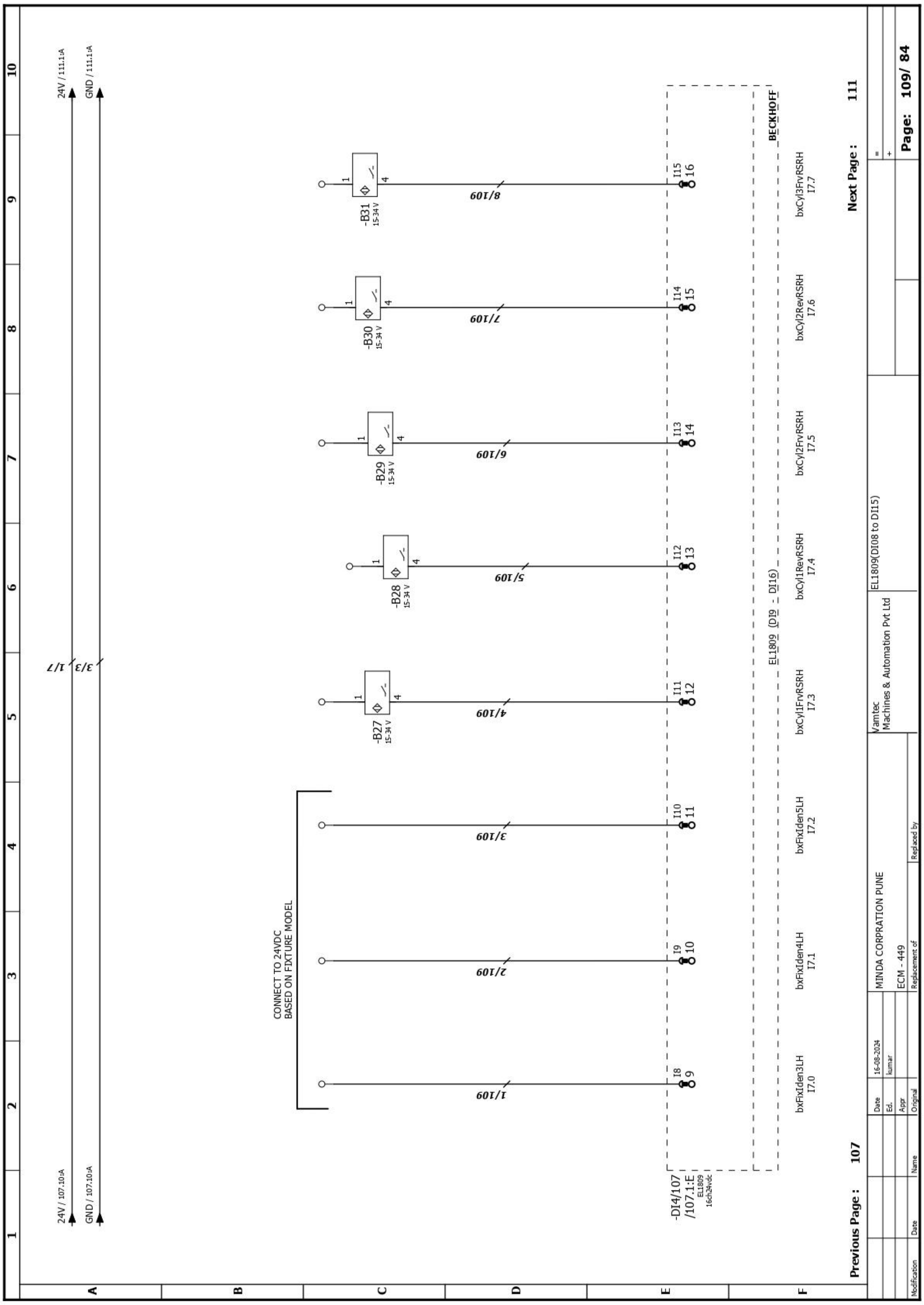
24V / 107.10:A
GND / 107.10:A
24V / 111.1:A
GND / 111.1:A
1/7
3/3
CONNECT TO 24VDC
BASED ON FIXTURE MODEL
-B27
-B28
-B29
-B30
-B31
15-34 V
1/109
2/109
3/109
4/109
5/109
6/109
7/109
8/109
I8 9
I9 10
I10 11
I11 12
I12 13
I13 14
I14 15
I15 16
-DI4/107
/107.1:E
EL1809
16ch24vdc
EL1809 (DI9 - DI16)
BECKHOFF
bxFixIden3LH I7.0
bxFixIden4LH I7.1
bxFixIden5LH I7.2
bxCyl1FrvRSRH I7.3
bxCyl1RevRSRH I7.4
bxCyl2FrvRSRH I7.5
bxCyl2RevRSRH I7.6
bxCyl3FrvRSRH I7.7
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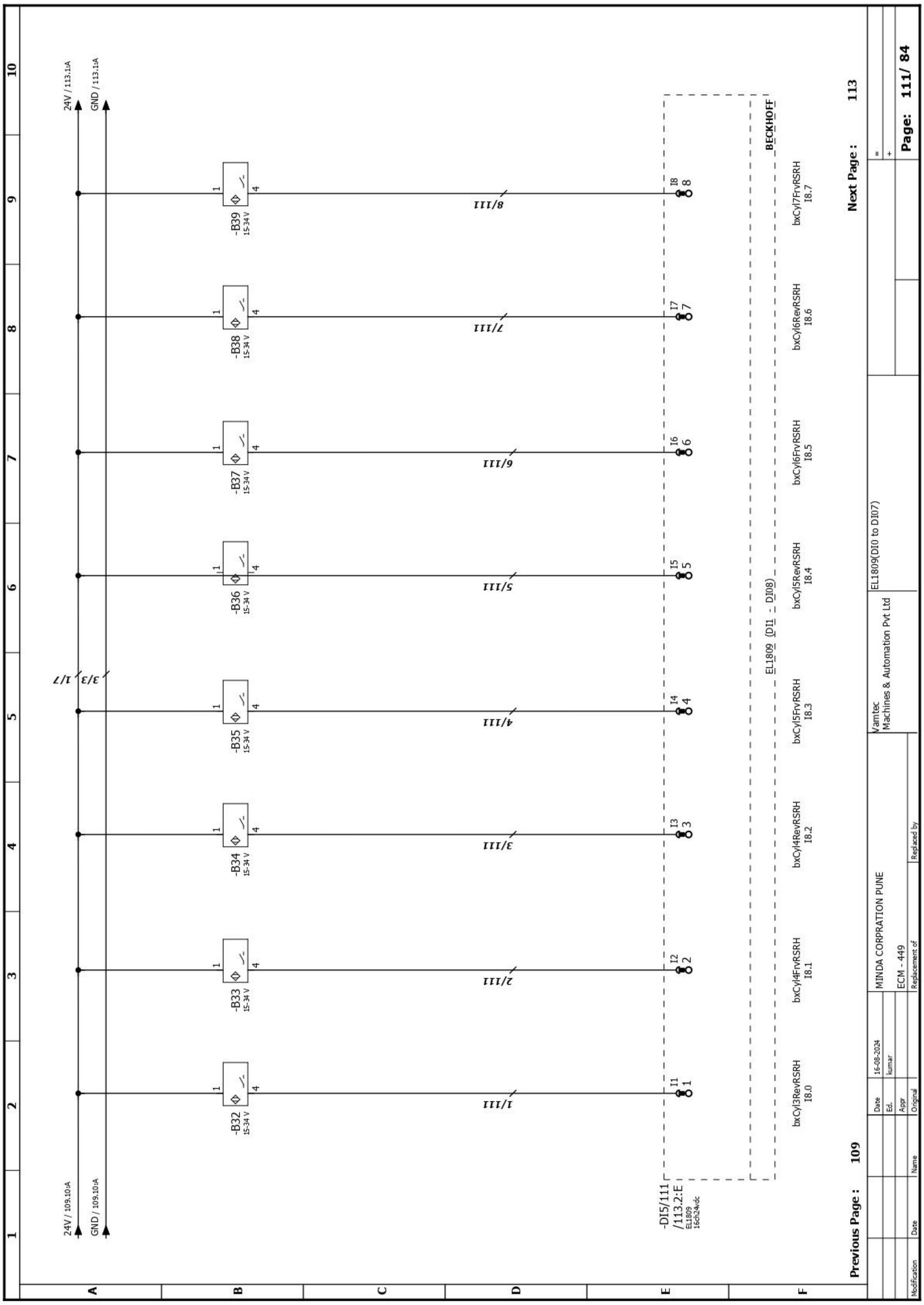

24V / 109.10:A
GND / 109.10:A
24V / 113.1:A
GND / 113.1:A
1/7
3/3
-B32 15-34 V
-B33 15-34 V
-B34 15-34 V
-B35 15-34 V
-B36 15-34 V
-B37 15-34 V
-B38 15-34 V
-B39 15-34 V
1/111
2/111
3/111
4/111
5/111
6/111
7/111
8/111
-DI5/111
/113.2:E
EL1809
16ch24vdc
EL1809 (DI1 - DI08)
BECKHOFF
bxCyl3RevRSRH I8.0
bxCyl4FrvRSRH I8.1
bxCyl4RevRSRH I8.2
bxCyl5FrvRSRH I8.3
bxCyl5RevRSRH I8.4
bxCyl6FrvRSRH I8.5
bxCyl6RevRSRH I8.6
bxCyl7FrvRSRH I8.7
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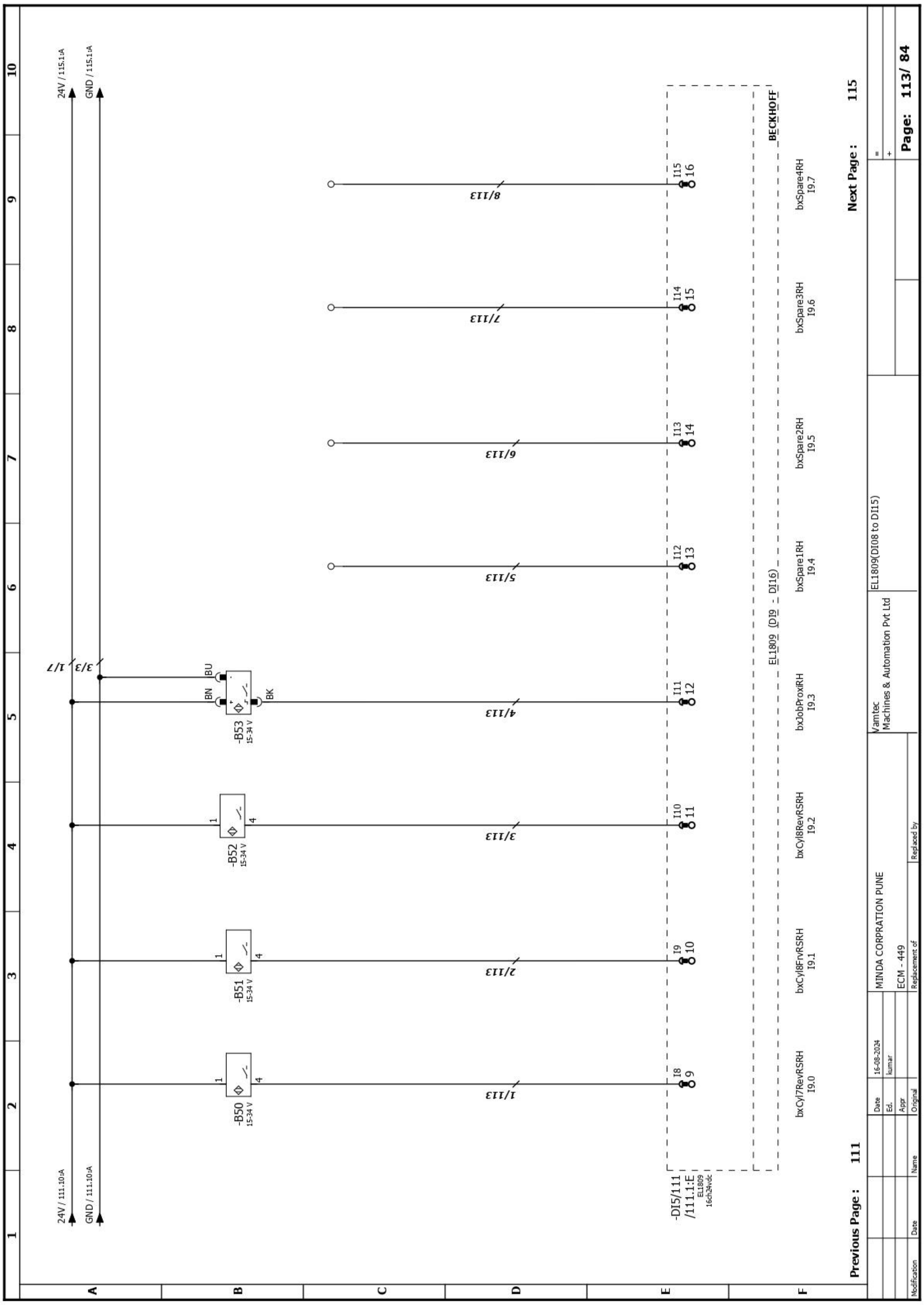
24V / 111.10:A
GND / 111.10:A
24V / 115.1:A
GND / 115.1:A
-B50
15-34 V
-B51
15-34 V
-B52
15-34 V
-B53
15-34 V
BN
BU
BK
1/113
2/113
3/113
4/113
5/113
6/113
7/113
8/113
-DI5/111
/111.1:E
EL1809
16ch24vdc
EL1809 (DI9 - DI16)
BECKHOFF
bxCyl7RevRSRH
I9.0
bxCyl8FrRSRH
I9.1
bxCyl8RevRSRH
I9.2
bxJobProxiRH
I9.3
bxSpare1RH
I9.4
bxSpare2RH
I9.5
bxSpare3RH
I9.6
bxSpare4RH
I9.7
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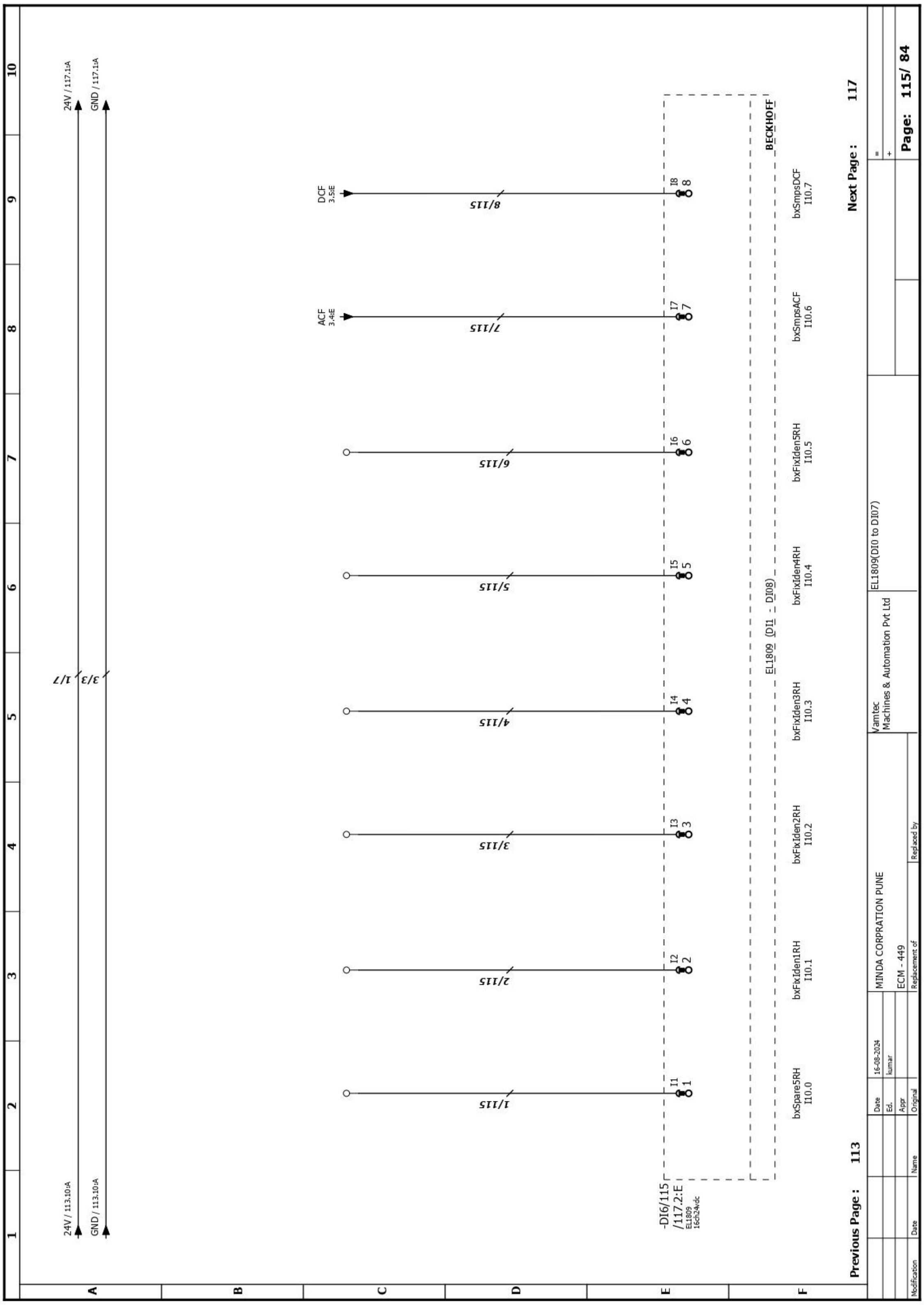

24V / 113.10:A
GND / 113.10:A
24V / 117.1:A
GND / 117.1:A
1/7
3/3
DCF
3.5:E
ACF
3.4:E
1/115
2/115
3/115
4/115
5/115
6/115
7/115
8/115
I1 1
I2 2
I3 3
I4 4
I5 5
I6 6
I7 7
I8 8
-DI6/115
/117.2:E
EL1809
16ch24vdc
EL1809 (DI1 - DI08)
BECKHOFF
bxSpareSRH
I10.0
bxFixIden1RH
I10.1
bxFixIden2RH
I10.2
bxFixIden3RH
I10.3
bxFixIden4RH
I10.4
bxFixIden5RH
I10.5
bxSmpsACF
I10.6
bxSmpsDCF
I10.7
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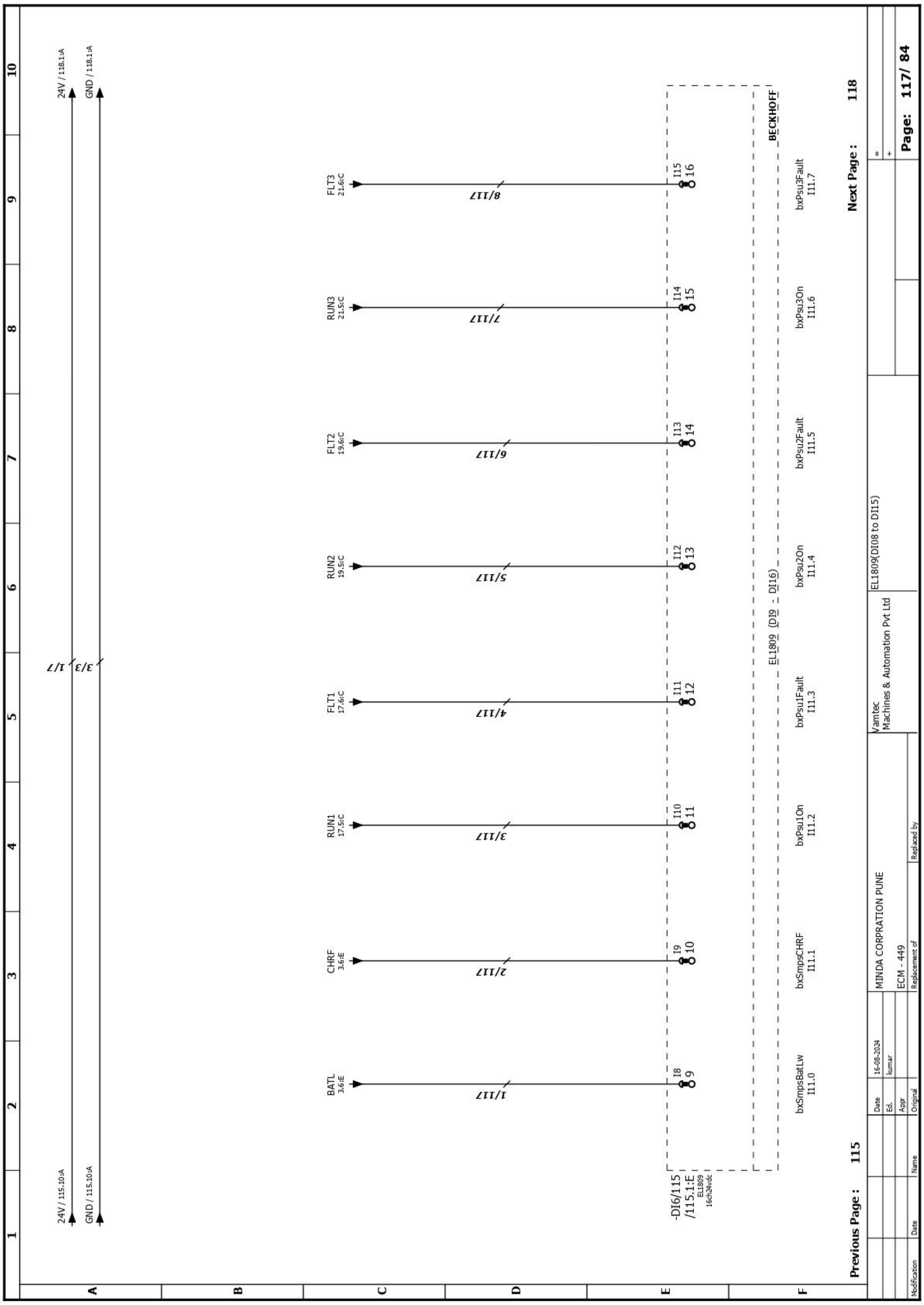
24V / 118.1:A
GND / 118.1:A
24V / 115.10:A
GND / 115.10:A
1/7
3/3
BATL 3.6:E
1/117
I8 9
bxSmpsBatLw I11.0
CHRF 3.6:E
2/117
I9 10
bxSmpsCHRF I11.1
RUN1 17.5:C
3/117
I10 11
bxPsu1On I11.2
FLT1 17.6:C
4/117
I11 12
bxPsu1Fault I11.3
RUN2 19.5:C
5/117
I12 13
bxPsu2On I11.4
FLT2 19.6:C
6/117
I13 14
bxPsu2Fault I11.5
RUN3 21.5:C
7/117
I14 15
bxPsu3On I11.6
FLT3 21.6:C
8/117
I15 16
bxPsu3Fault I11.7
EL1809 (DI9 - DI16)
BECKHOFF
-DI6/115
/115.1:E
EL1809
16ch24vdc
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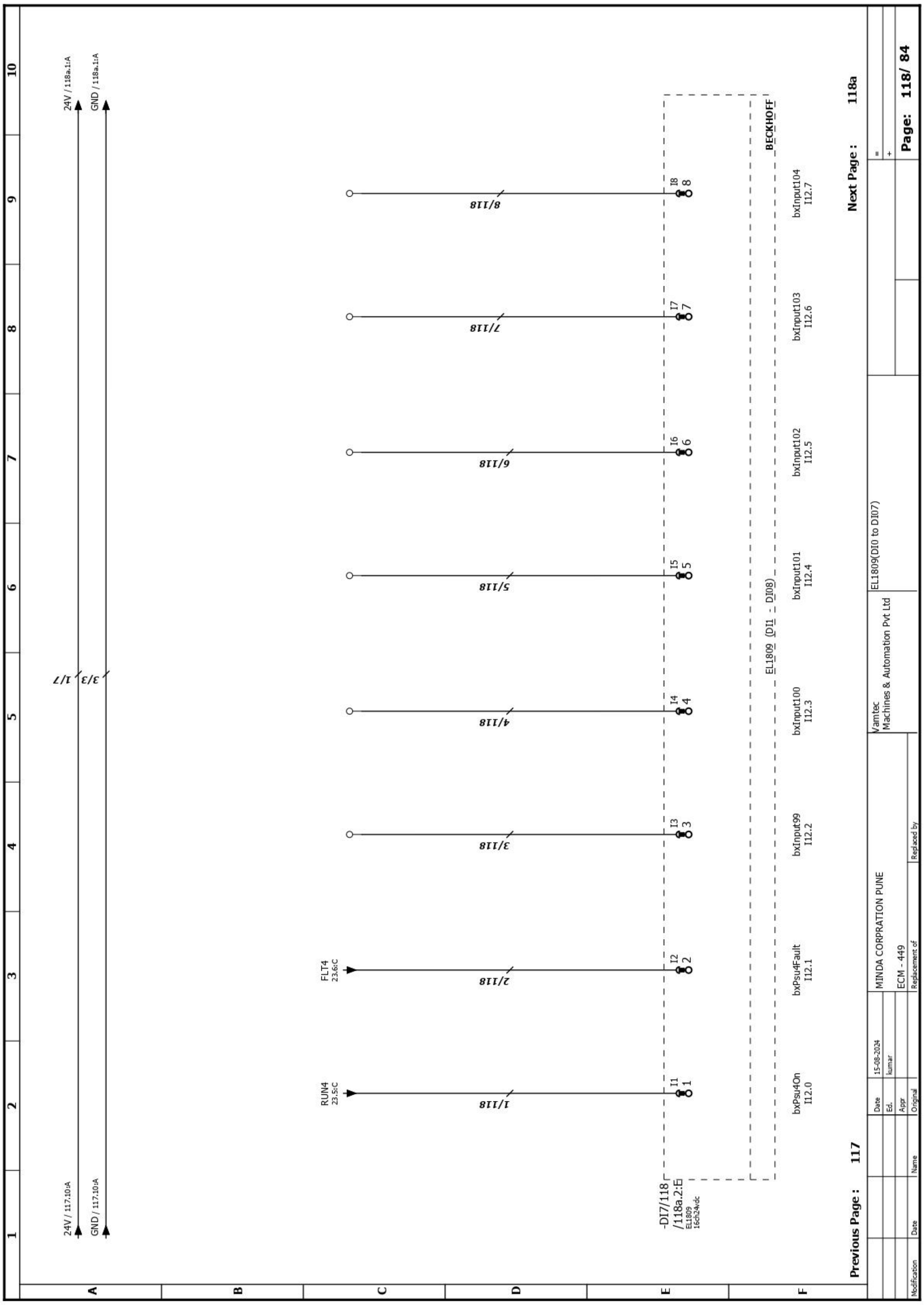

24V / 118a.1:A
GND / 118a.1:A
24V / 117.10:A
GND / 117.10:A
1/7
3/3
RUN4
23.5:C
FLT4
23.6:C
1/118
2/118
3/118
4/118
5/118
6/118
7/118
8/118
I1 1
I2 2
I3 3
I4 4
I5 5
I6 6
I7 7
I8 8
-DI7/118
/118a.2:E
EL1809
16ch24vdc
EL1809 (DI1 - DI08)
BECKHOFF
bxPsu4On I12.0
bxPsu4Fault I12.1
bxInput99 I12.2
bxInput100 I12.3
bxInput101 I12.4
bxInput102 I12.5
bxInput103 I12.6
bxInput104 I12.7
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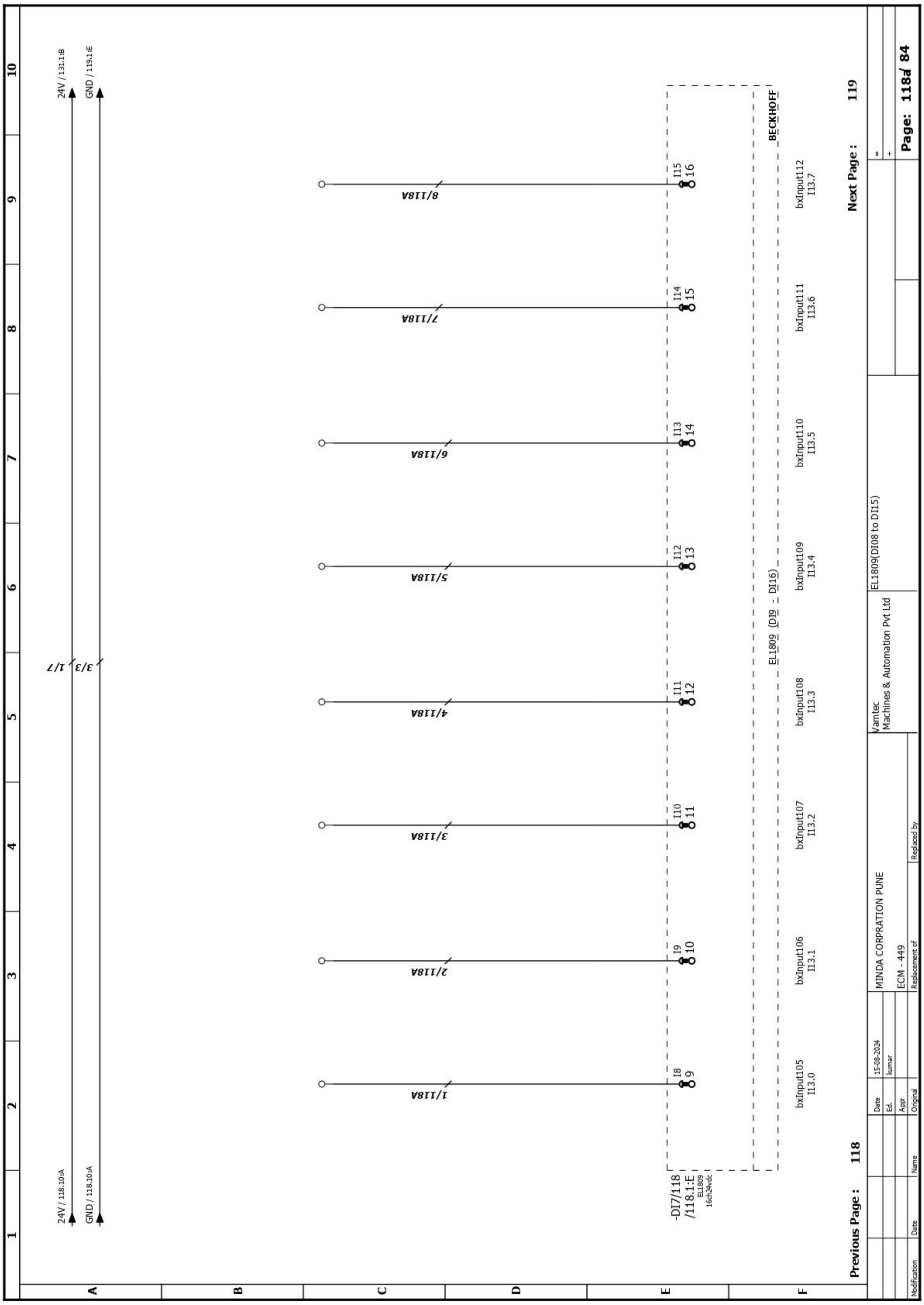
24V / 131.1:B
GND / 119.1:E
24V / 118.10:A
GND / 118.10:A
-DI7/118
/118.1:E
EL1809
16ch24vdc
EL1809 (DI9 - DI16)
BECKHOFF
1/118A
2/118A
3/118A
4/118A
5/118A
6/118A
7/118A
8/118A
I8 9
I9 10
I10 11
I11 12
I12 13
I13 14
I14 15
I15 16
bxInput105 I13.0
bxInput106 I13.1
bxInput107 I13.2
bxInput108 I13.3
bxInput109 I13.4
bxInput110 I13.5
bxInput111 I13.6
bxInput112 I13.7
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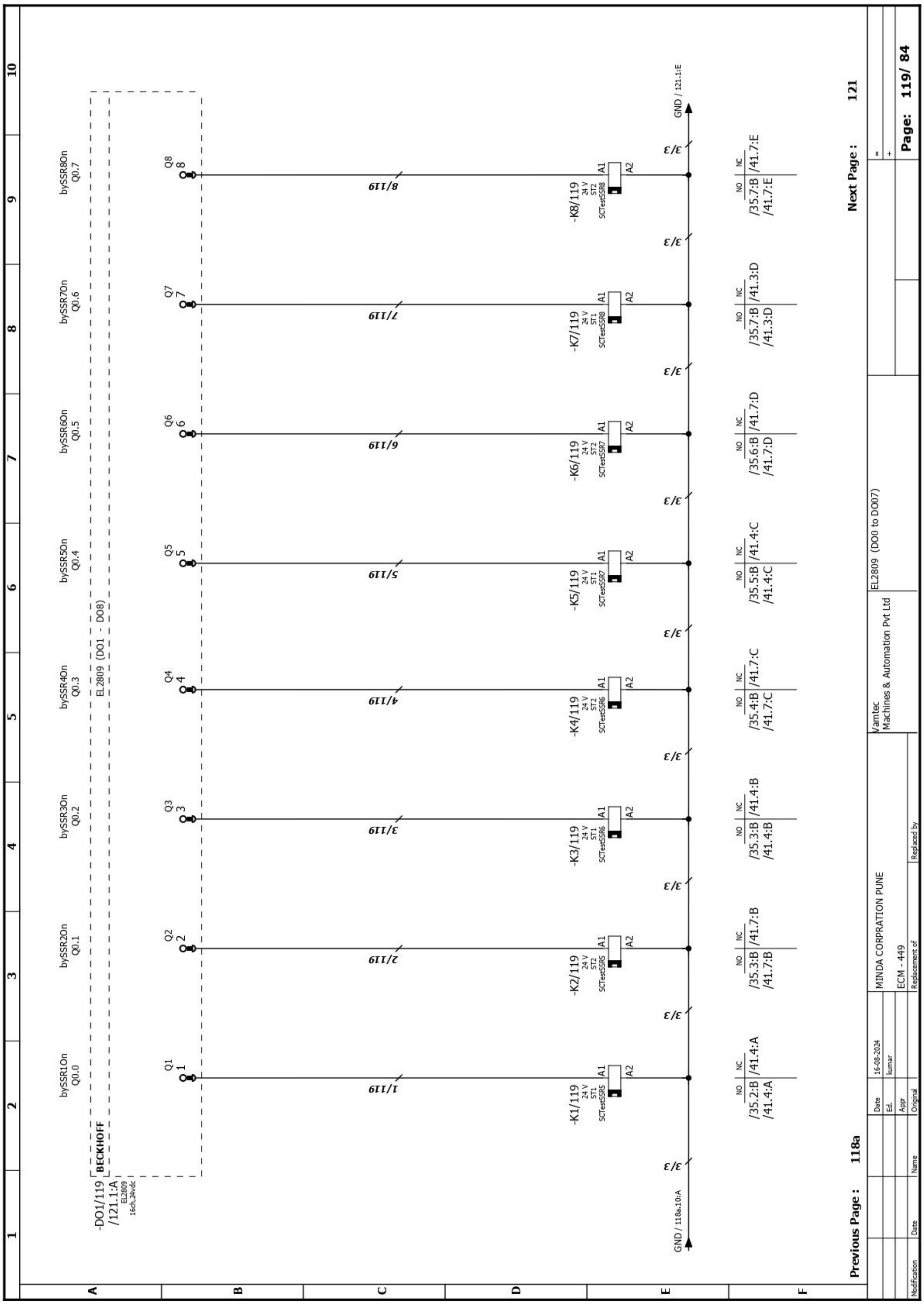
-DO1/119
/121.1:A
BECKHOFF
EL2809 (DO1 - DO8)
bySSR1On Q0.0
bySSR2On Q0.1
bySSR3On Q0.2
bySSR4On Q0.3
bySSR5On Q0.4
bySSR6On Q0.5
bySSR7On Q0.6
bySSR8On Q0.7
-K1/119
-K2/119
-K3/119
-K4/119
-K5/119
-K6/119
-K7/119
-K8/119
GND / 118a.10:A
GND / 121.1:E
/35.2:B /41.4:A
/35.3:B /41.7:B
/35.3:B /41.4:B
/35.4:B /41.7:C
/35.5:B /41.4:C
/35.6:B /41.7:D
/35.7:B /41.3:D
/35.7:B /41.7:E
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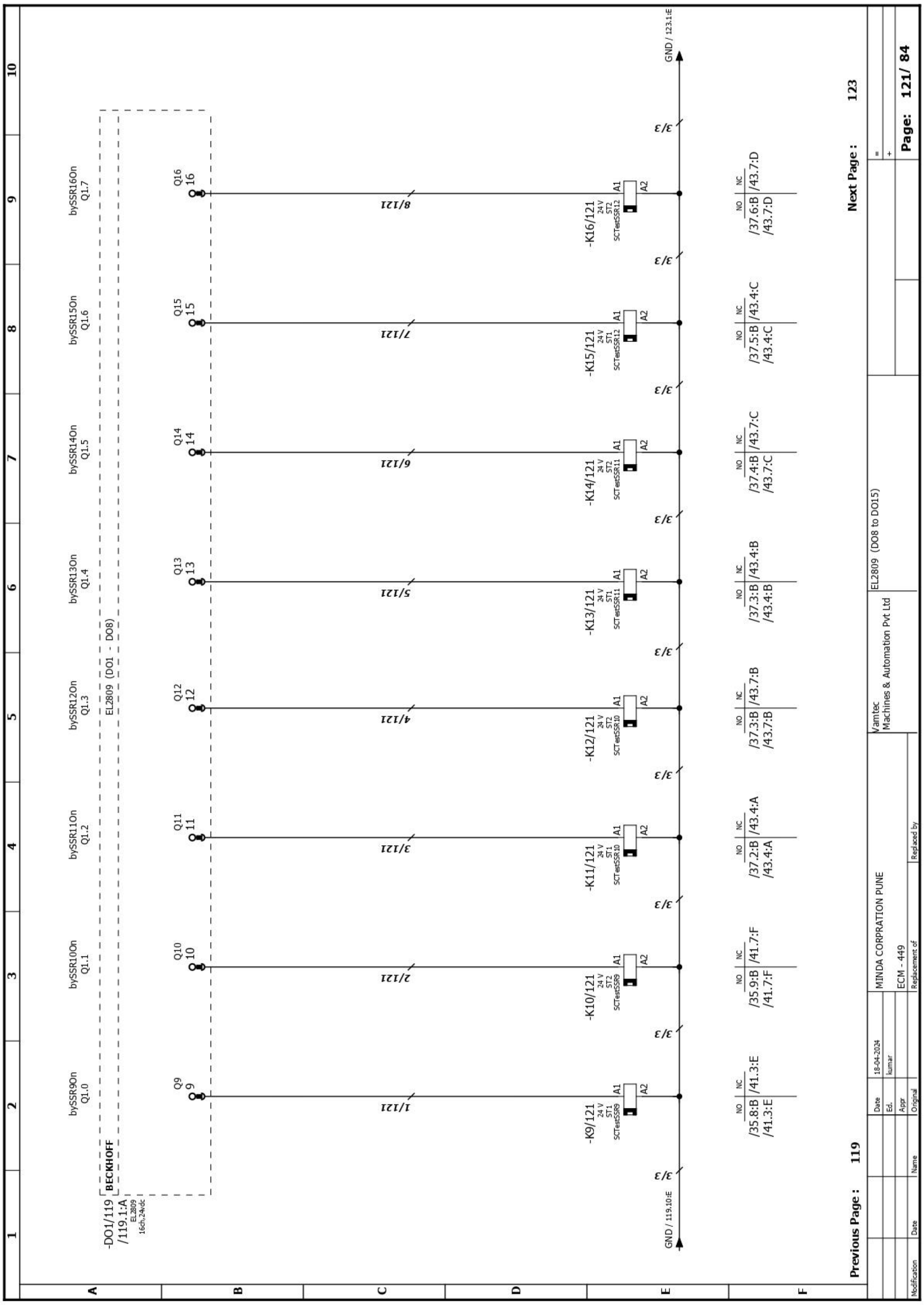
-DO1/119
/119.1:A
BECKHOFF
EL2809 (DO1 - DO8)
bySSR9On Q1.0
bySSR10On Q1.1
bySSR11On Q1.2
bySSR12On Q1.3
bySSR13On Q1.4
bySSR14On Q1.5
bySSR15On Q1.6
bySSR16On Q1.7
-K9/121
-K10/121
-K11/121
-K12/121
-K13/121
-K14/121
-K15/121
-K16/121
GND / 119.10:E
GND / 123.1:E
/35.8:B /41.3:E
/35.9:B /41.7:F
/37.2:B /43.4:A
/37.3:B /43.7:B
/37.3:B /43.4:B
/37.4:B /43.7:C
/37.5:B /43.4:C
/37.6:B /43.7:D
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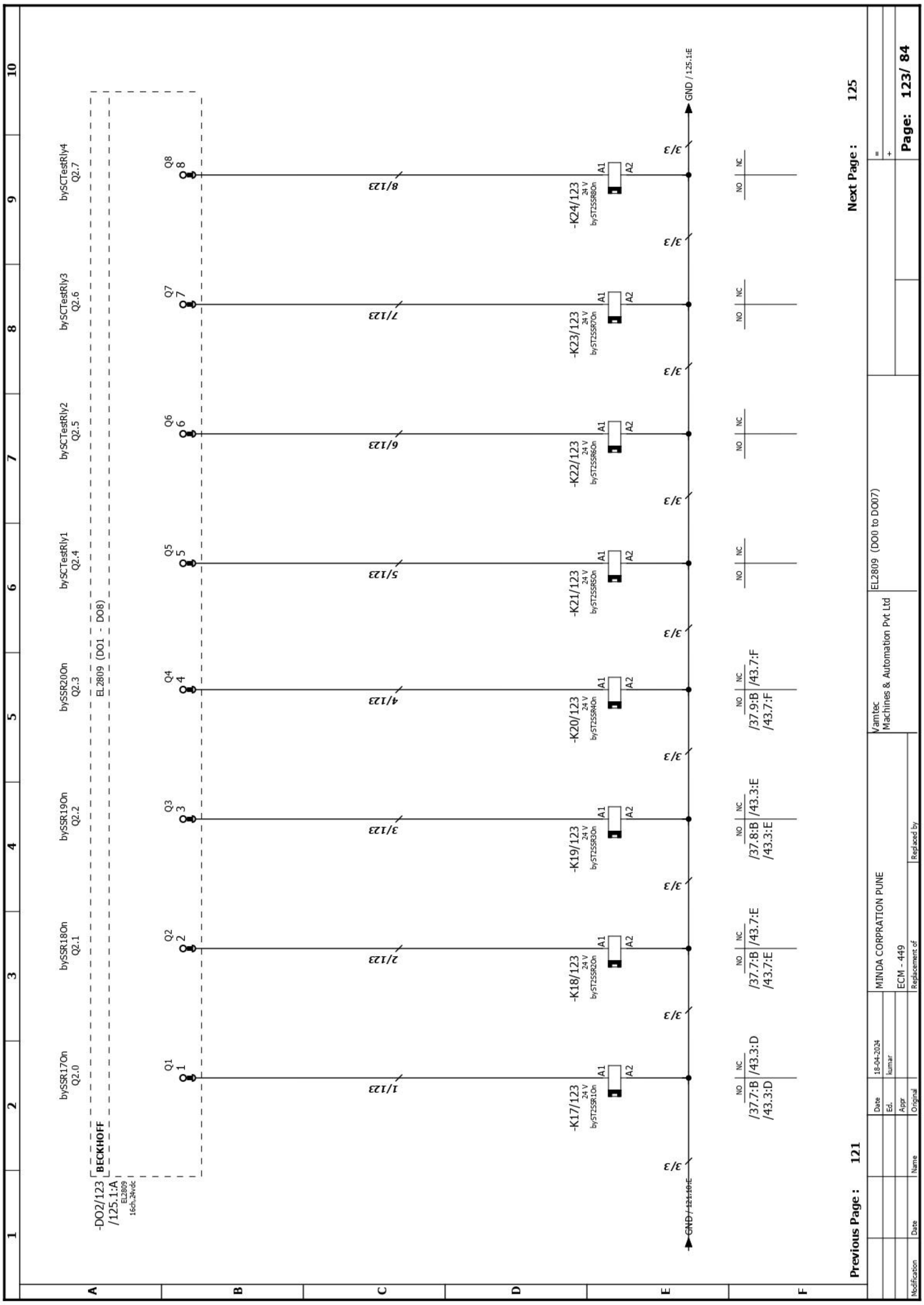
BECKHOFF
-DO2/123
/125.1:A
EL2809 16ch,24vdc
EL2809 (DO1 - DO8)
bySSR17On Q2.0
bySSR18On Q2.1
bySSR19On Q2.2
bySSR20On Q2.3
bySCTestRly1 Q2.4
bySCTestRly2 Q2.5
bySCTestRly3 Q2.6
bySCTestRly4 Q2.7
-K17/123
-K18/123
-K19/123
-K20/123
-K21/123
-K22/123
-K23/123
-K24/123
24 V
A1
A2
NO NC
/37.7:B /43.3:D
/43.3:D
/37.7:B /43.7:E
/43.7:E
/37.8:B /43.3:E
/43.3:E
/37.9:B /43.7:F
/43.7:F
GND / 121.10:E
GND / 125.1:E
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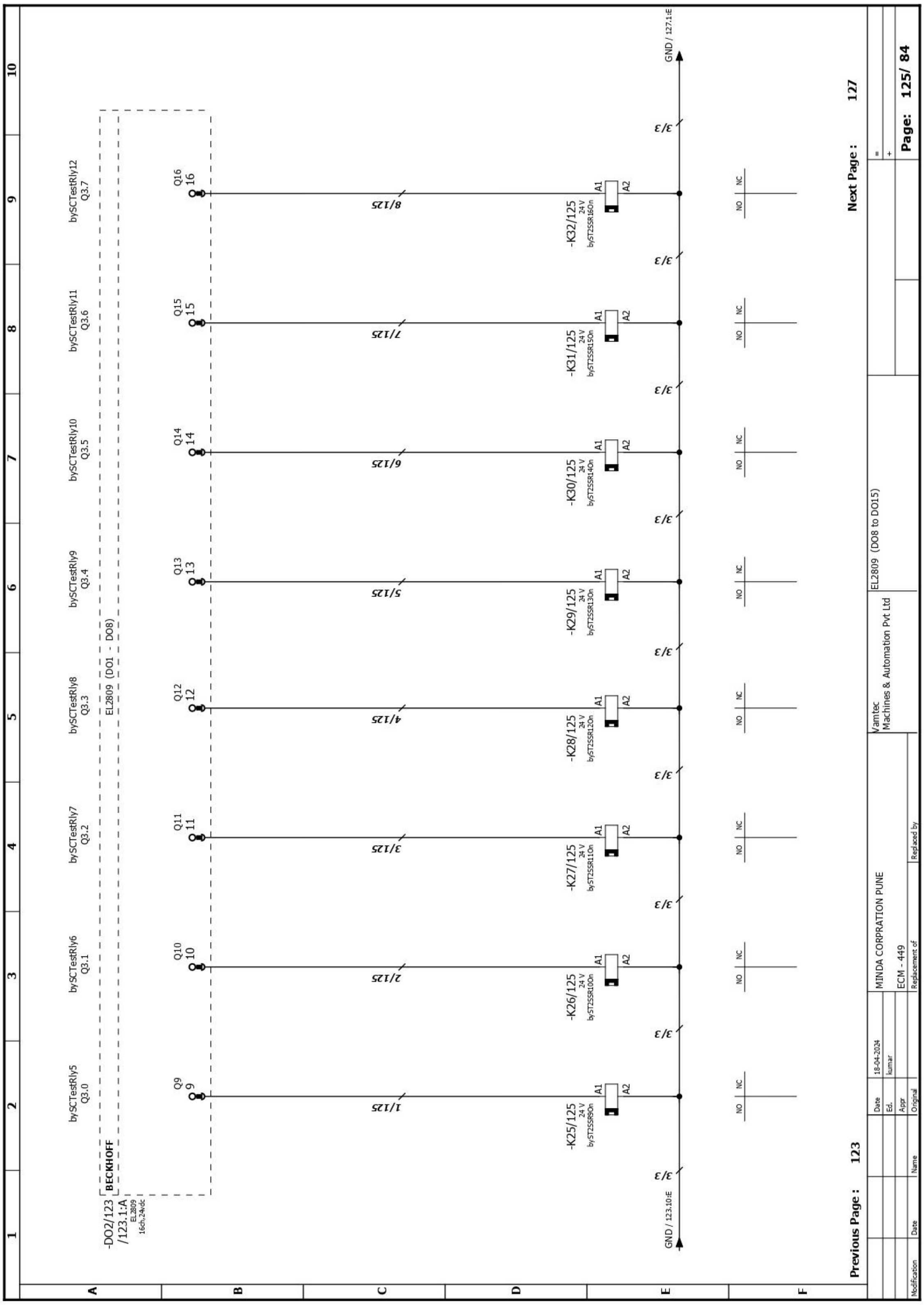
-DO2/123
/123.1:A
BECKHOFF
EL2809 (DO1 - DO8)
bySCTestRly5 Q3.0
bySCTestRly6 Q3.1
bySCTestRly7 Q3.2
bySCTestRly8 Q3.3
bySCTestRly9 Q3.4
bySCTestRly10 Q3.5
bySCTestRly11 Q3.6
bySCTestRly12 Q3.7
-K25/125
-K26/125
-K27/125
-K28/125
-K29/125
-K30/125
-K31/125
-K32/125
GND / 123.10:E
GND / 127.1:E
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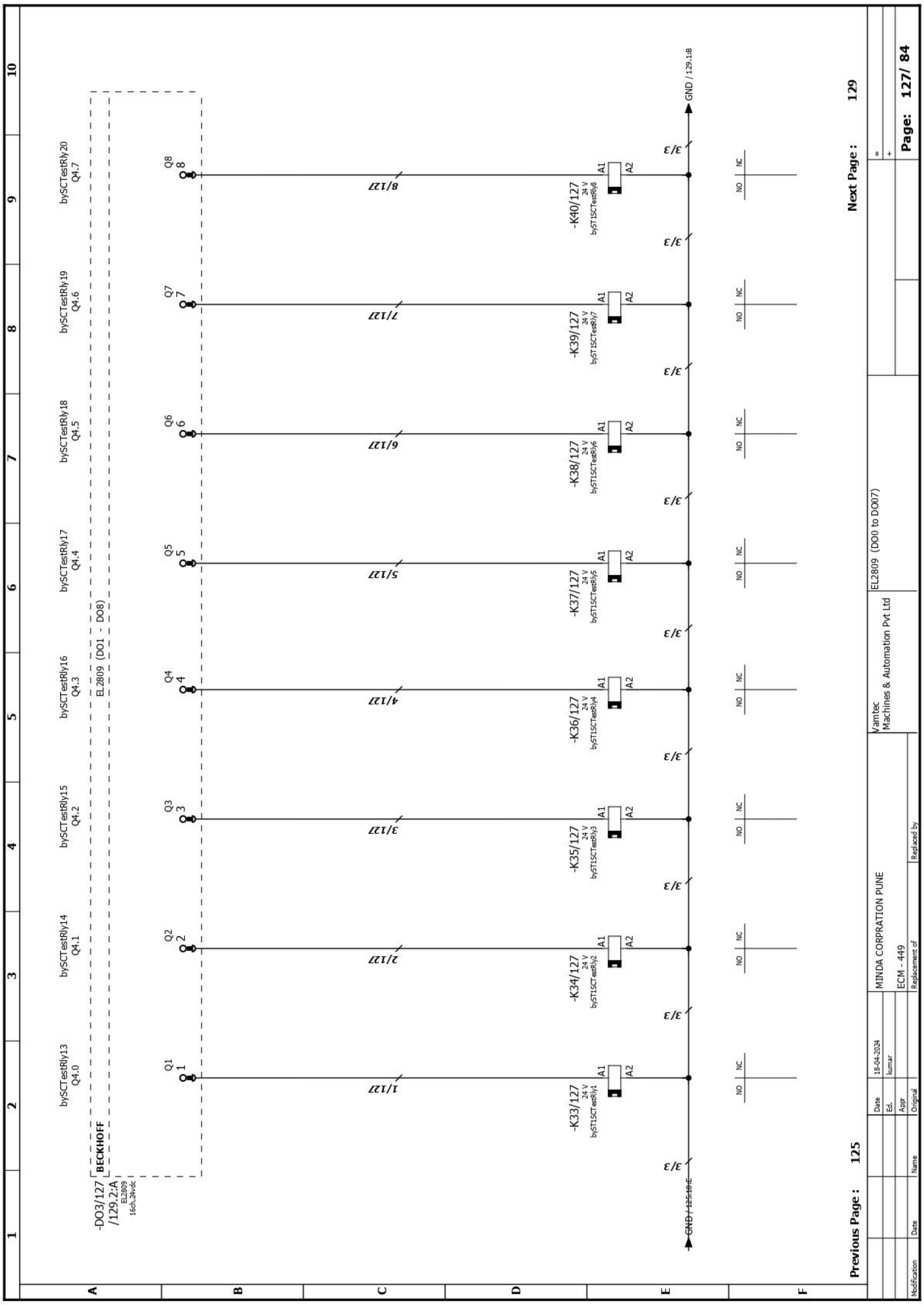

-DO3/127
/129.2:A
EL2809
16ch.24vdc
BECKHOFF
EL2809 (DO1 - DO8)
bySCTestRly13
Q4.0
bySCTestRly14
Q4.1
bySCTestRly15
Q4.2
bySCTestRly16
Q4.3
bySCTestRly17
Q4.4
bySCTestRly18
Q4.5
bySCTestRly19
Q4.6
bySCTestRly20
Q4.7
-K33/127
-K34/127
-K35/127
-K36/127
-K37/127
-K38/127
-K39/127
-K40/127
24 V
A1
A2
NO
NC
GND / 125.10:E
GND / 129.1:B
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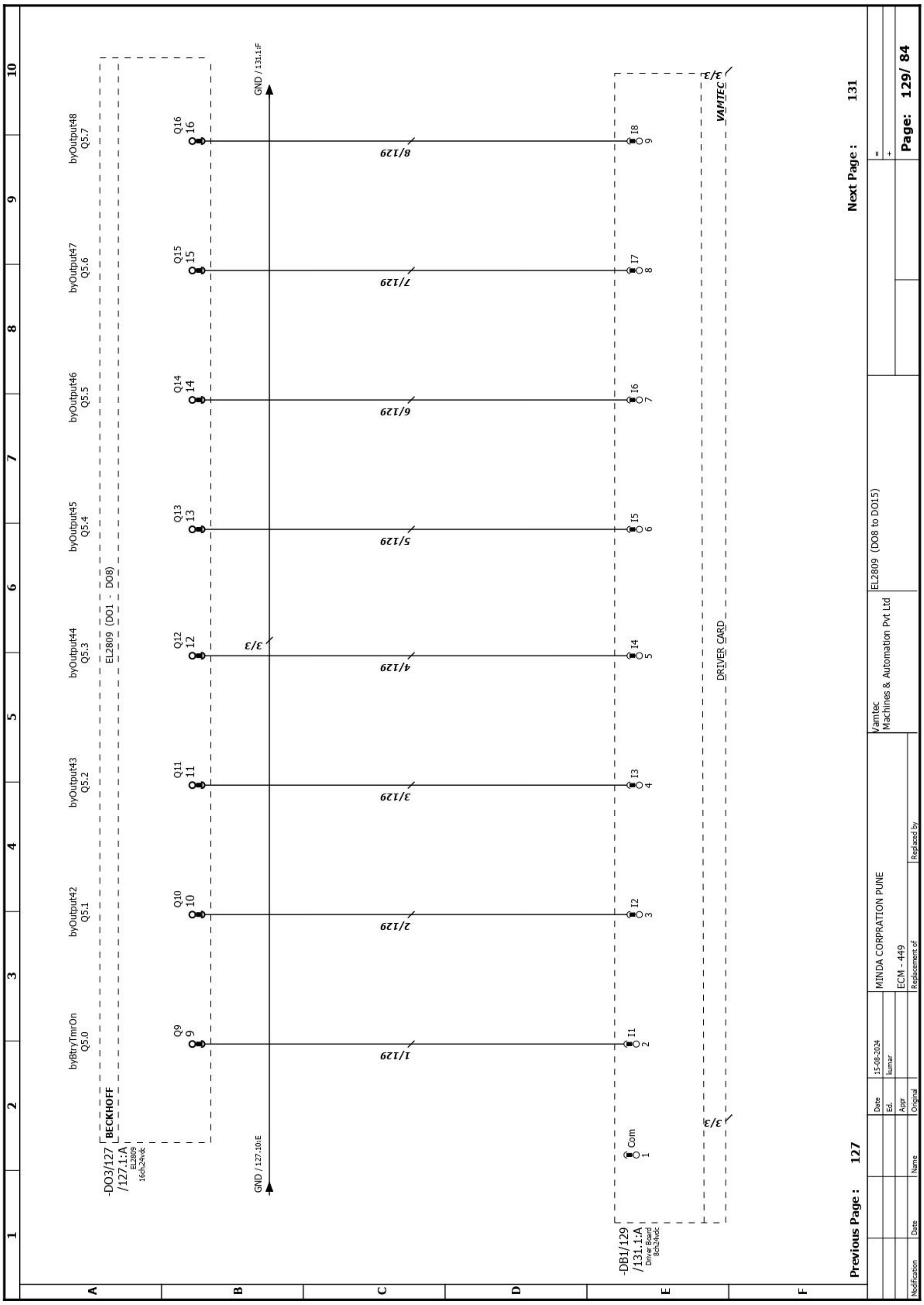

-DO3/127
/127.1:A
EL2809
16ch24vdc
BECKHOFF
EL2809 (DO1 - DO8)
byBtryTmrOn
Q5.0
byOutput42
Q5.1
byOutput43
Q5.2
byOutput44
Q5.3
byOutput45
Q5.4
byOutput46
Q5.5
byOutput47
Q5.6
byOutput48
Q5.7
Q9 9
Q10 10
Q11 11
Q12 12
Q13 13
Q14 14
Q15 15
Q16 16
GND / 127.10:E
GND / 131.1:F
3/3
1/129
2/129
3/129
4/129
5/129
6/129
7/129
8/129
-DB1/129
/131.1:A
Driver Board
8ch24vdc
Com 1
I1 2
I2 3
I3 4
I4 5
I5 6
I6 7
I7 8
I8 9
DRIVER CARD
VAMTEC
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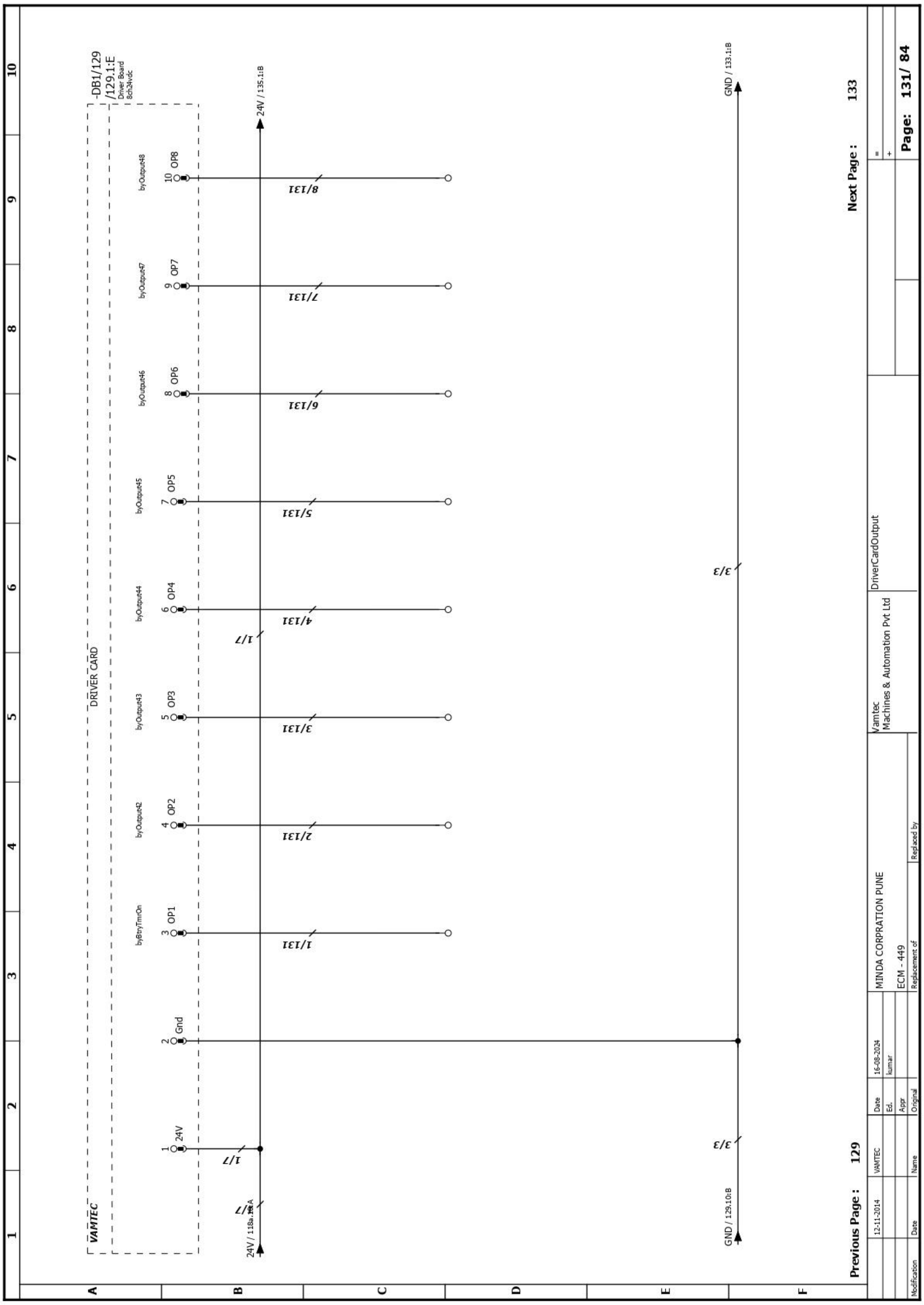

-DB1/129
/129.1:E
DRIVER CARD
VAMTEC
24V
Gnd
OP1
OP2
OP3
OP4
OP5
OP6
OP7
OP8
byOutput42
byOutput43
byOutput44
byOutput45
byOutput46
byOutput47
byOutput48
24V / 135.1:B
GND / 133.1:B
GND / 129.10:B
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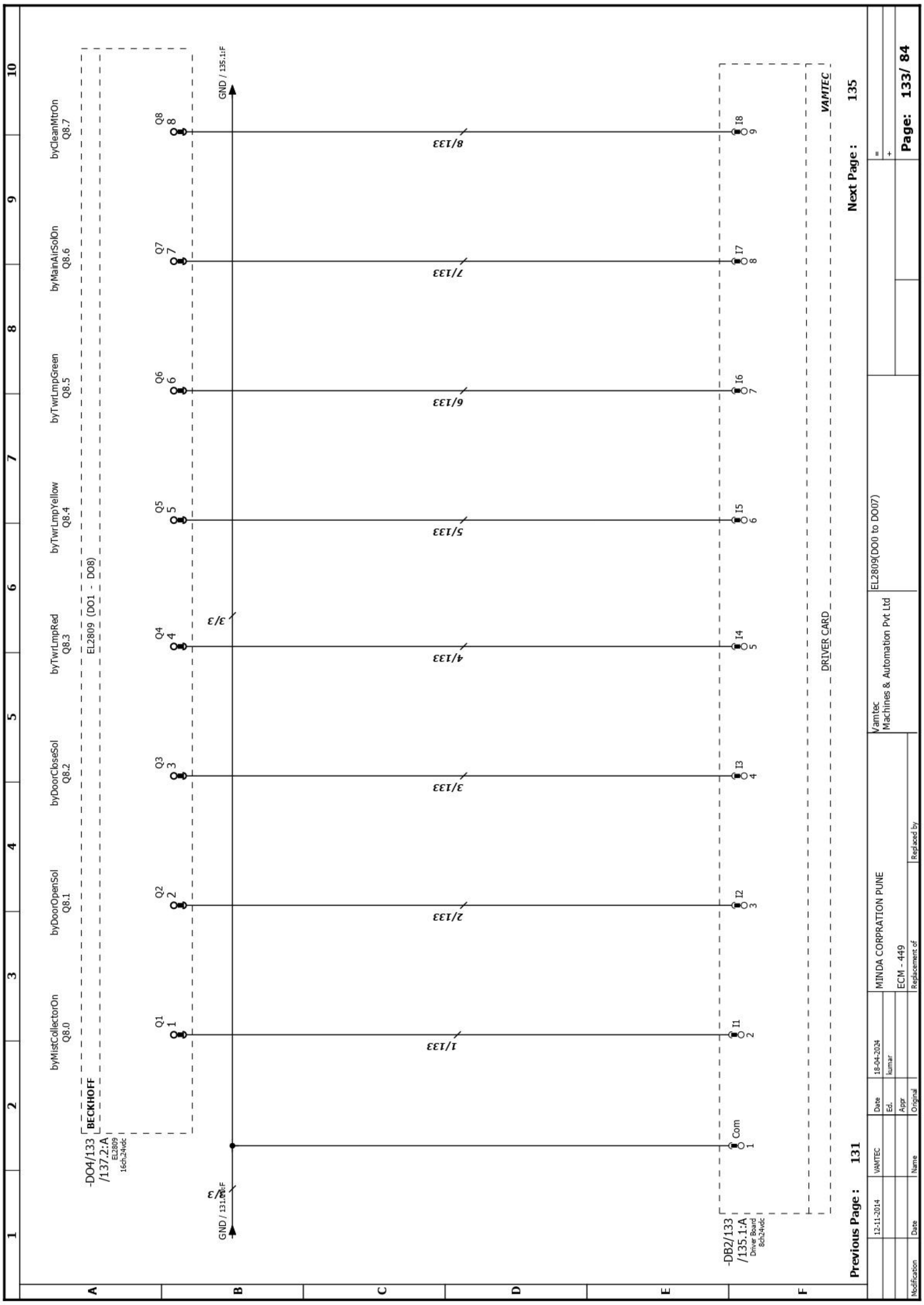

byMistCollectorOn
Q8.0
byDoorOpenSol
Q8.1
byDoorCloseSol
Q8.2
byTwrLmpRed
Q8.3
byTwrLmpYellow
Q8.4
byTwrLmpGreen
Q8.5
byMainAirSolOn
Q8.6
byCleanMtrOn
Q8.7
BECKHOFF
EL2809 (DO1 - DO8)
-DO4/133
/137.2:A
EL2809
16ch,24vdc
GND / 131.8:F
GND / 135.1:F
1/133
2/133
3/133
4/133
5/133
6/133
7/133
8/133
3/3
Com
DRIVER CARD
VAMTEC
-DB2/133
/135.1:A
Driver Board
8ch,24vdc
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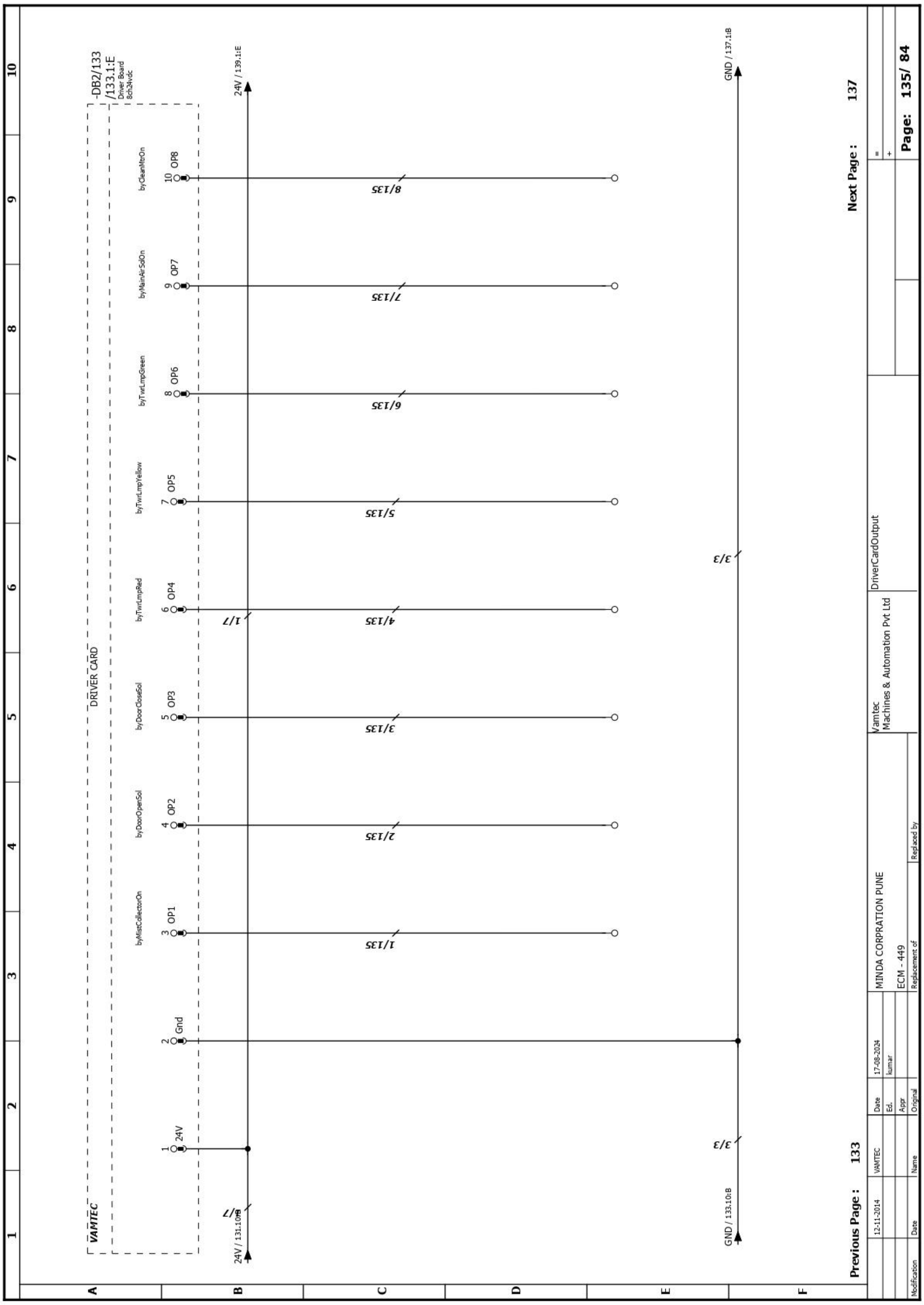

DRIVER CARD
-DB2/133
/133.1:E
24V / 139.1:E
GND / 137.1:B
24V / 131.10:B
GND / 133.10:B
24V
Gnd
OP1
OP2
OP3
OP4
OP5
OP6
OP7
OP8
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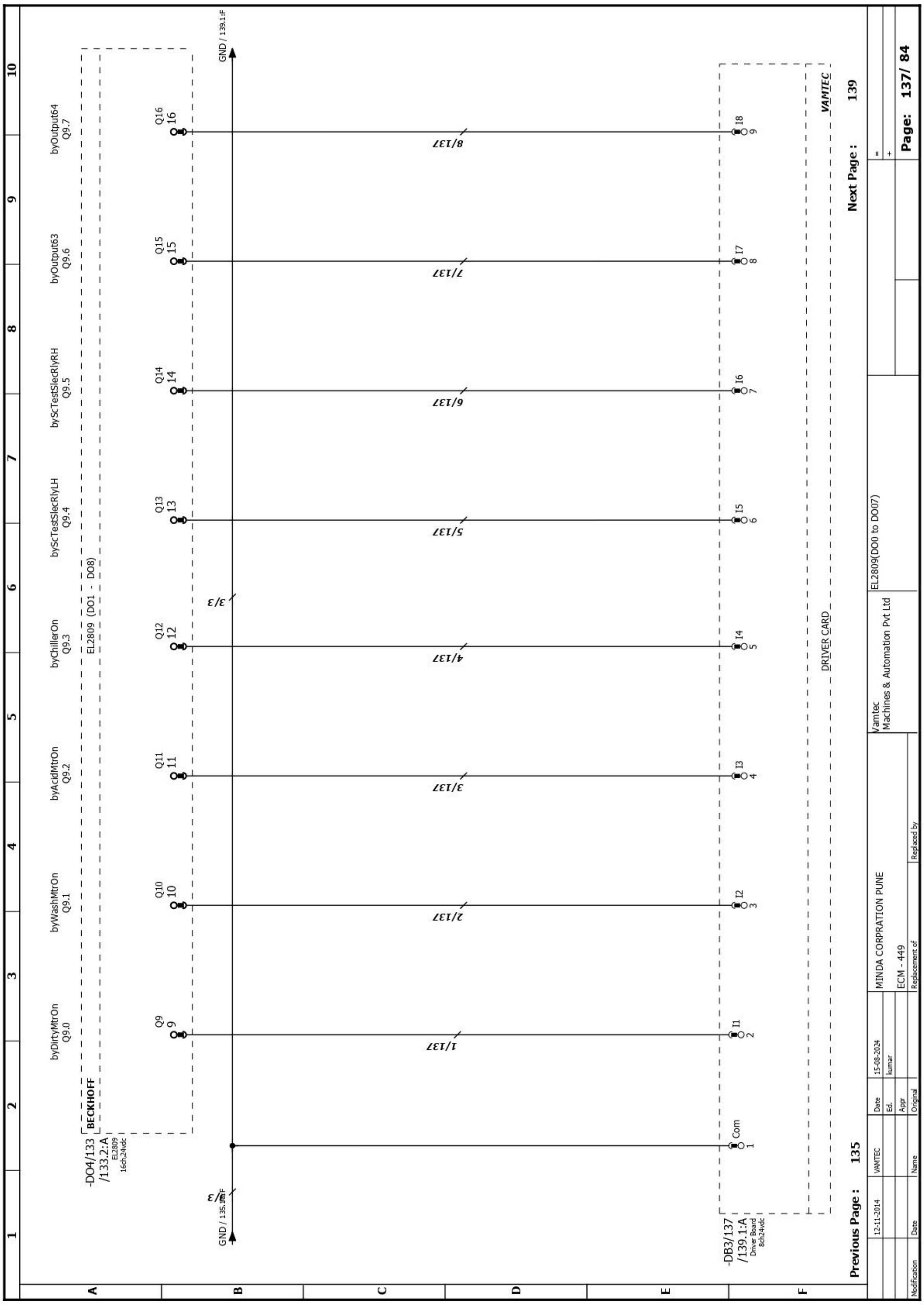
-DO4/133
/133.2:A
EL2809
16ch,24vdc
BECKHOFF
EL2809 (DO1 - DO8)
byDirtyMtrOn Q9.0
byWashMtrOn Q9.1
byAcidMtrOn Q9.2
byChillerOn Q9.3
byScTestSlecRlyLH Q9.4
byScTestSlecRlyRH Q9.5
byOutput63 Q9.6
byOutput64 Q9.7
Q9 9
Q10 10
Q11 11
Q12 12
Q13 13
Q14 14
Q15 15
Q16 16
GND / 135.2:F
GND / 139.1:F
3/3
1/137
2/137
3/137
4/137
5/137
6/137
7/137
8/137
Com 1
I1 2
I2 3
I3 4
I4 5
I5 6
I6 7
I7 8
I8 9
-DB3/137
/139.1:A
Driver Board
8ch,24vdc
DRIVER CARD
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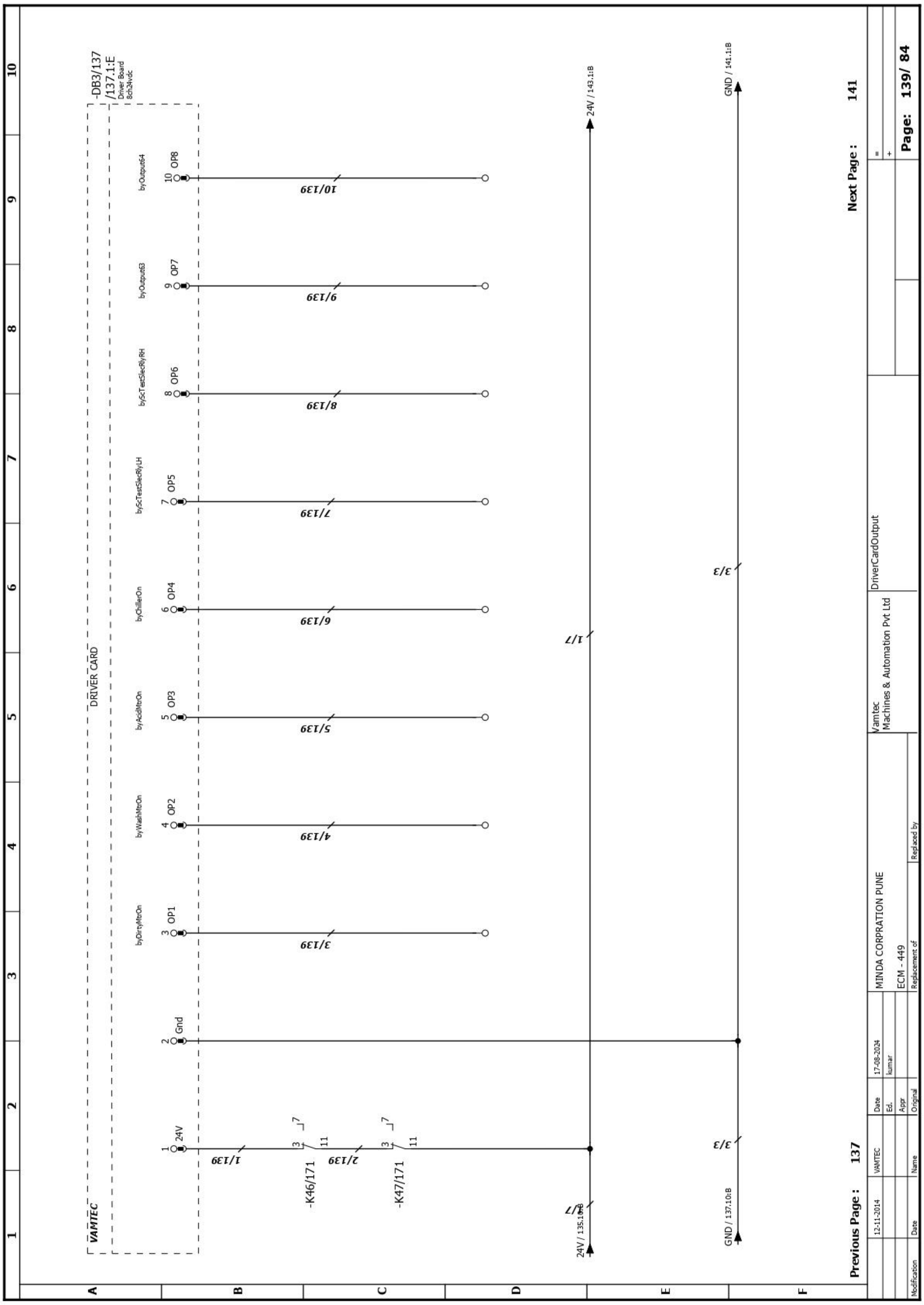
-DB3/137
/137.1:E
DRIVER CARD
VAMTEC
24V
Gnd
OP1
OP2
OP3
OP4
OP5
OP6
OP7
OP8
byDirtyMtrOn
byWashMtrOn
byAcidMtrOn
byChillerOn
byScTestSlecRlyLH
byScTestSlecRlyRH
byOutput63
byOutput64
-K46/171
-K47/171
1/139
2/139
3/139
4/139
5/139
6/139
7/139
8/139
9/139
10/139
1/7
3/3
24V / 143.1:B
GND / 141.1:B
GND / 137.10:B
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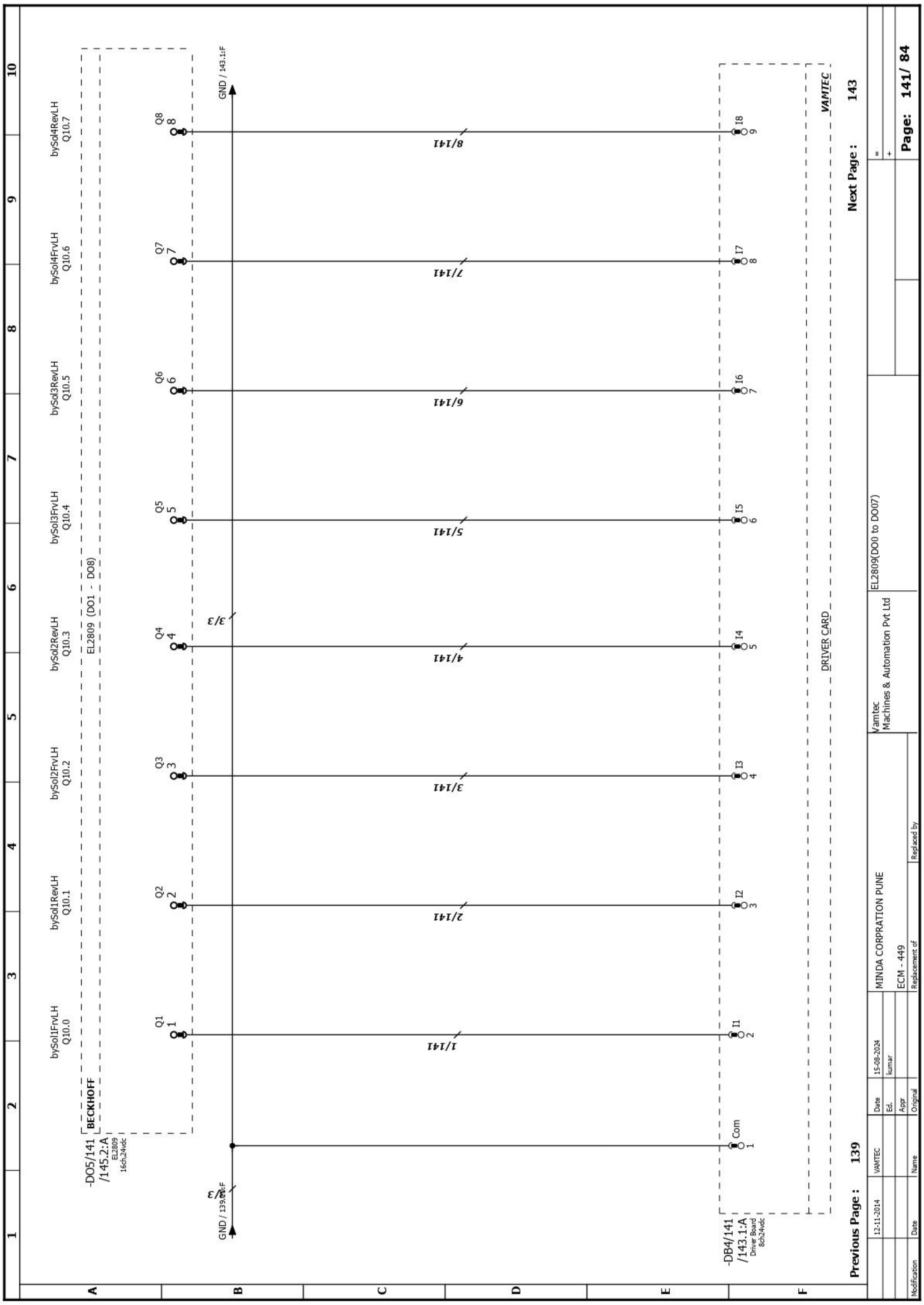
bySol1FrvLH Q10.0
bySol1RevLH Q10.1
bySol2FrvLH Q10.2
bySol2RevLH Q10.3
bySol3FrvLH Q10.4
bySol3RevLH Q10.5
bySol4FrvLH Q10.6
bySol4RevLH Q10.7
-DO5/141 /145.2:A
BECKHOFF
EL2809 (DO1 - DO8)
GND / 143.1:F
-DB4/141 /143.1:A
DRIVER CARD
VAMTEC
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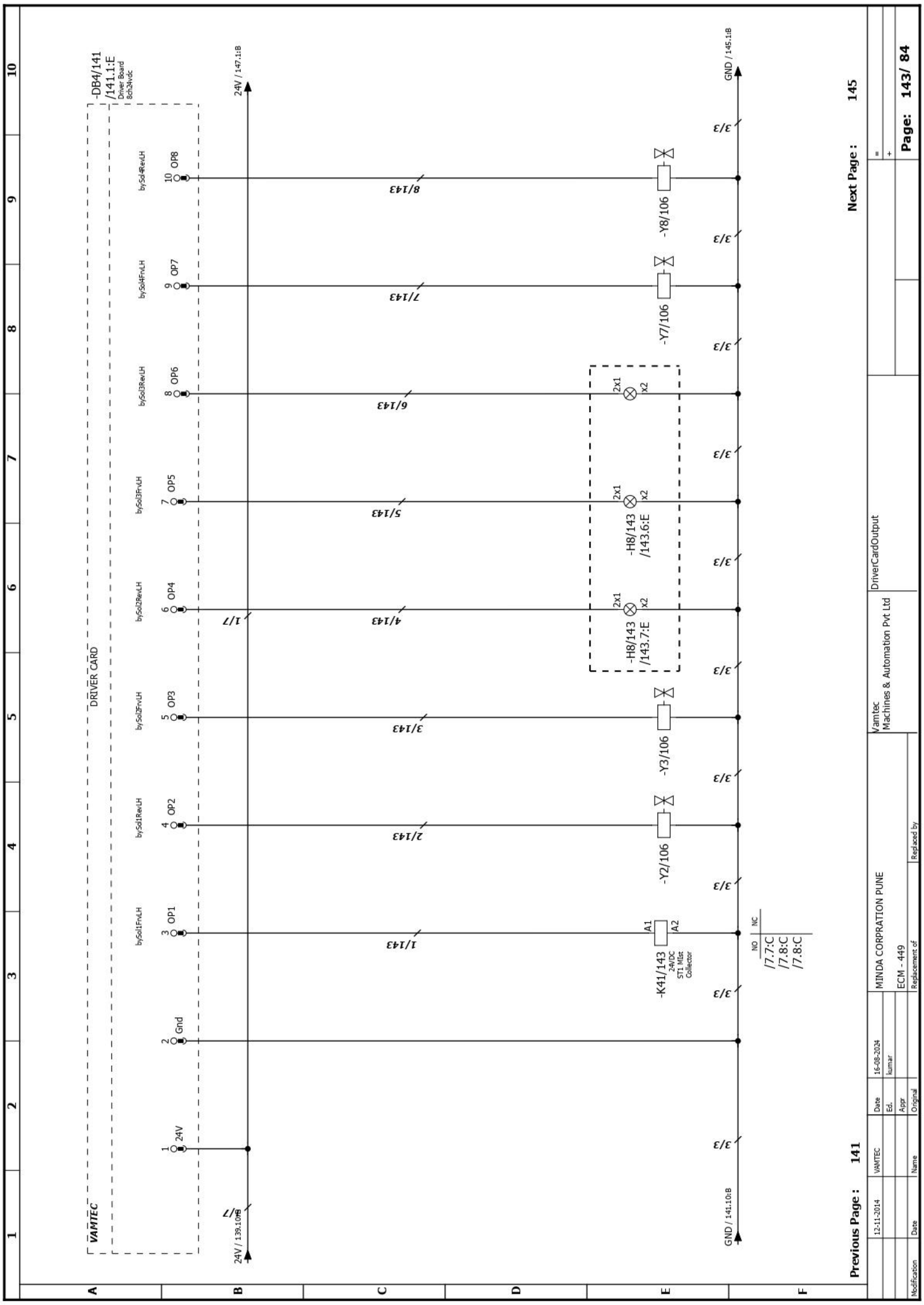
-DB4/141
/141.1:E
Driver Board
8ch24vdc
DRIVER CARD
VAMTEC
24V / 147.1:B
GND / 145.1:B
24V / 139.10:B
GND / 141.10:B
1 24V
2 Gnd
3 OP1
4 OP2
5 OP3
6 OP4
7 OP5
8 OP6
9 OP7
10 OP8
bySol1FrvLH
bySol1RevLH
bySol2FrvLH
bySol2RevLH
bySol3FrvLH
bySol3RevLH
bySol4FrvLH
bySol4RevLH
1/143
2/143
3/143
4/143
5/143
6/143
7/143
8/143
-K41/143
24VDC
ST1 Mtor
Collector
A1
A2
NO NC
/7.7:C
/7.8:C
/7.8:C
-Y2/106
-Y3/106
-H8/143
/143.7:E
-H8/143
/143.6:E
2x1
x2
-Y7/106
-Y8/106
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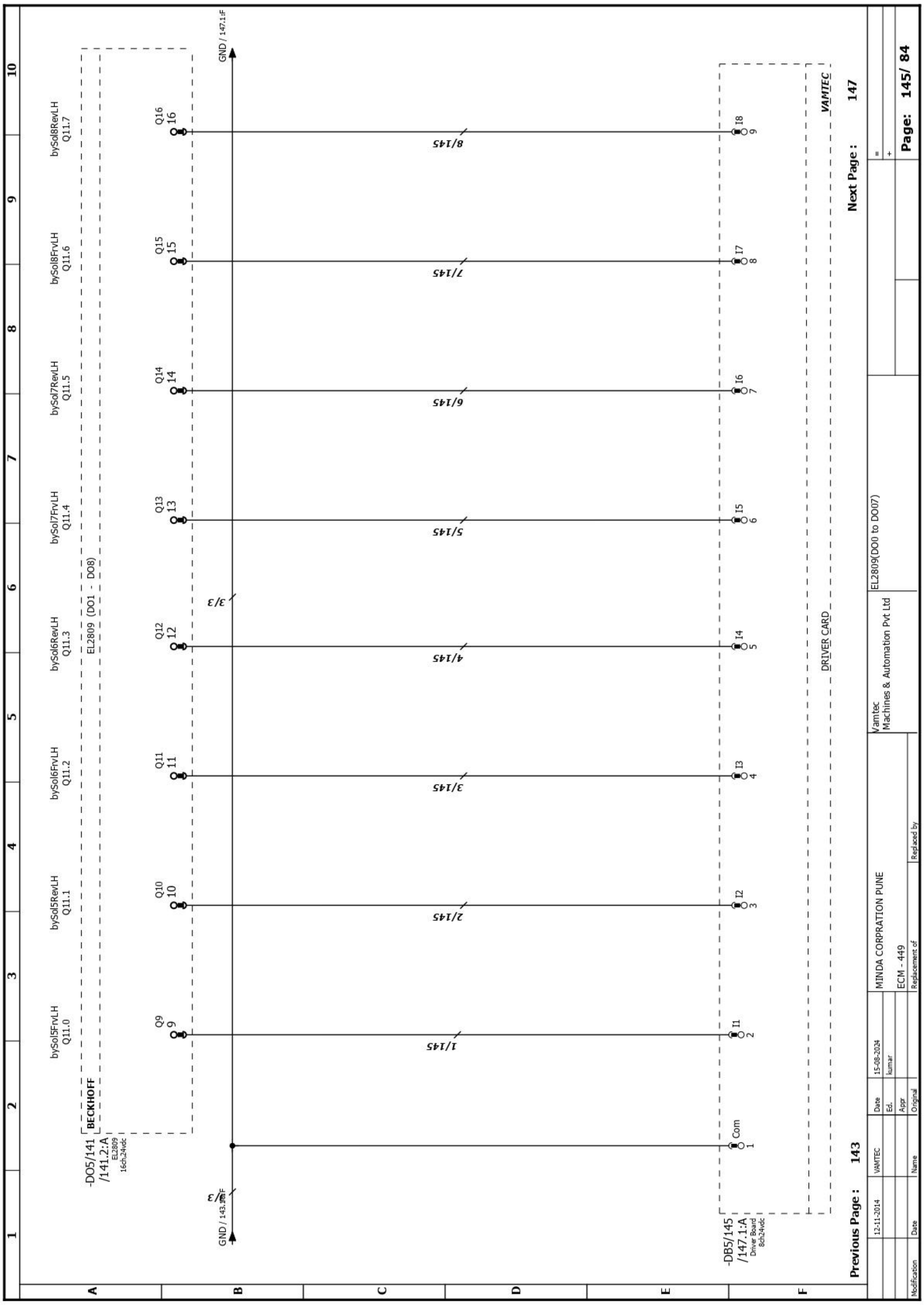

-DO5/141
/141.2:A
BECKHOFF
EL2809 (DO1 - DO8)
bySol5FrvLH
Q11.0
bySol5RevLH
Q11.1
bySol6FrvLH
Q11.2
bySol6RevLH
Q11.3
bySol7FrvLH
Q11.4
bySol7RevLH
Q11.5
bySol8FrvLH
Q11.6
bySol8RevLH
Q11.7
-DB5/145
/147.1:A
DRIVER CARD
VAMTEC
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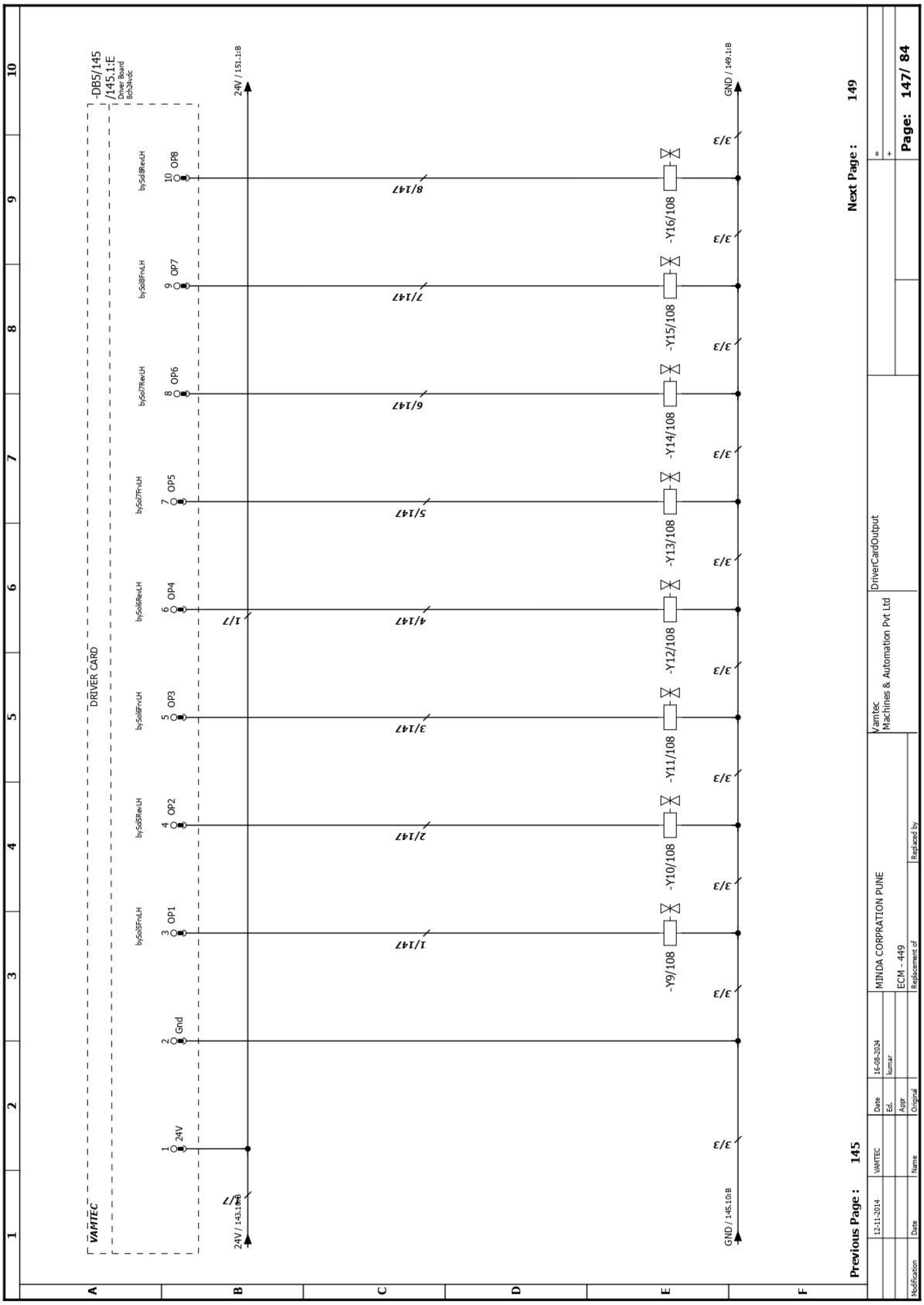

-DB5/145
/145.1:E
Driver Board
8ch24vdc
DRIVER CARD
VAMTEC
24V
Gnd
OP1
OP2
OP3
OP4
OP5
OP6
OP7
OP8
bySol5FrvLH
bySol5RevLH
bySol6FrvLH
bySol6RevLH
bySol7FrvLH
bySol7RevLH
bySol8FrvLH
bySol8RevLH
-Y9/108
-Y10/108
-Y11/108
-Y12/108
-Y13/108
-Y14/108
-Y15/108
-Y16/108
24V / 151.1:B
24V / 143.10:B
GND / 149.1:B
GND / 145.10:B
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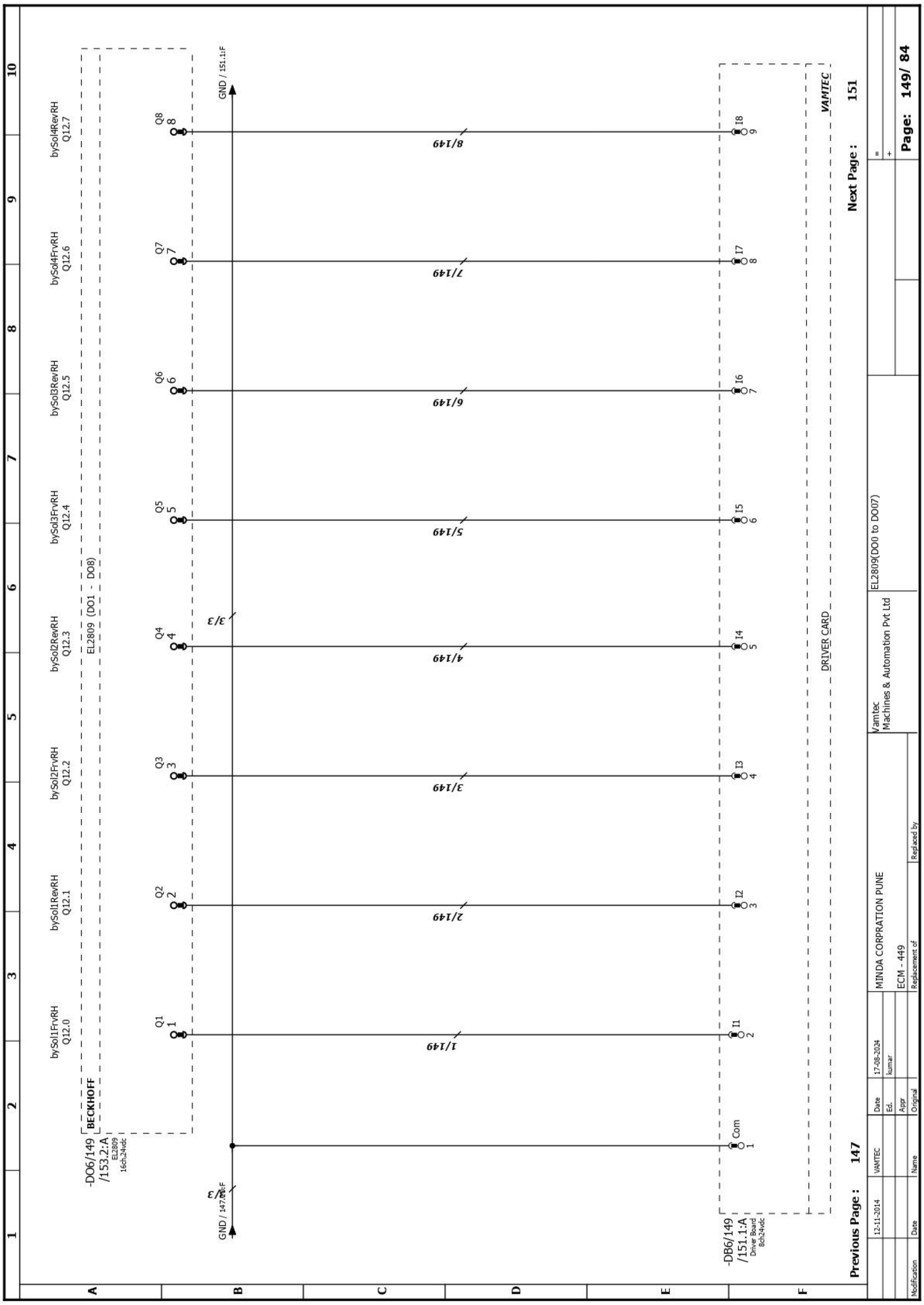

bySol1FrvRH Q12.0
bySol1RevRH Q12.1
bySol2FrvRH Q12.2
bySol2RevRH Q12.3
bySol3FrvRH Q12.4
bySol3RevRH Q12.5
bySol4FrvRH Q12.6
bySol4RevRH Q12.7
-DO6/149 /153.2:A
EL2809 16ch,24vdc
BECKHOFF
EL2809 (DO1 - DO8)
GND / 147.1:F
GND / 151.1:F
1/149
2/149
3/149
4/149
5/149
6/149
7/149
8/149
3/3
-DB6/149 /151.1:A
Driver Board 8ch,24vdc
DRIVER CARD
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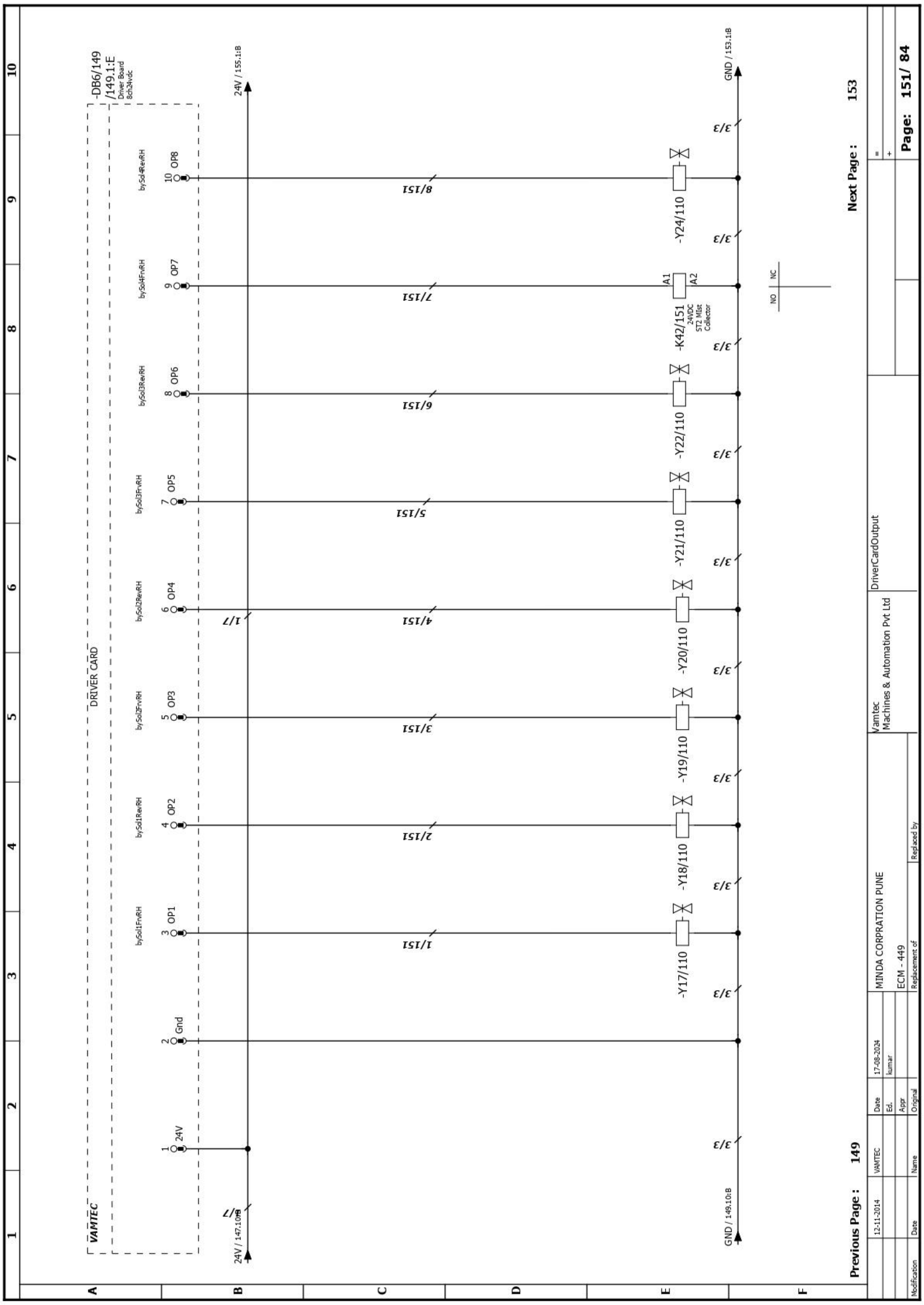

-DB6/149
/149.1:E
DRIVER CARD
VAMTEC
24V
Gnd
OP1
OP2
OP3
OP4
OP5
OP6
OP7
OP8
24V / 155.1:B
24V / 147.10:B
GND / 153.1:B
GND / 149.10:B
-Y17/110
-Y18/110
-Y19/110
-Y20/110
-Y21/110
-Y22/110
-K42/151
-Y24/110
A1
A2
NO
NC
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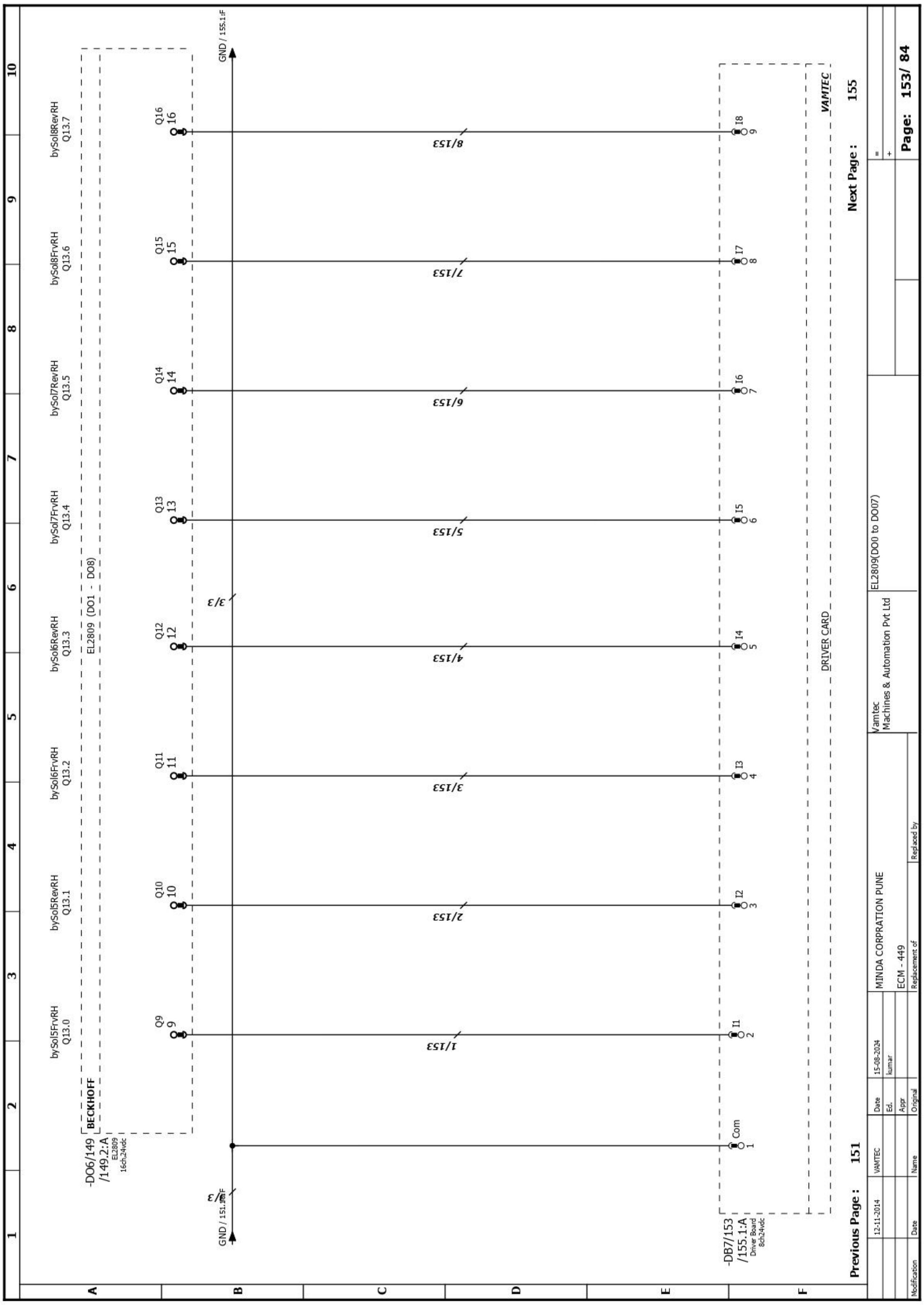

bySol5FrvRH Q13.0
bySol5RevRH Q13.1
bySol6FrvRH Q13.2
bySol6RevRH Q13.3
bySol7FrvRH Q13.4
bySol7RevRH Q13.5
bySol8FrvRH Q13.6
bySol8RevRH Q13.7
-DO6/149 /149.2:A
EL2809 16ch,24vdc
BECKHOFF
EL2809 (DO1 - DO8)
Q9 9
Q10 10
Q11 11
Q12 12
Q13 13
Q14 14
Q15 15
Q16 16
GND / 155.1:F
GND / 151.10:F
3/3
1/153
2/153
3/153
4/153
5/153
6/153
7/153
8/153
Com 1
I1 2
I2 3
I3 4
I4 5
I5 6
I6 7
I7 8
I8 9
-DB7/153 /155.1:A
Driver Board 8ch,24vdc
DRIVER CARD
VAMTEC
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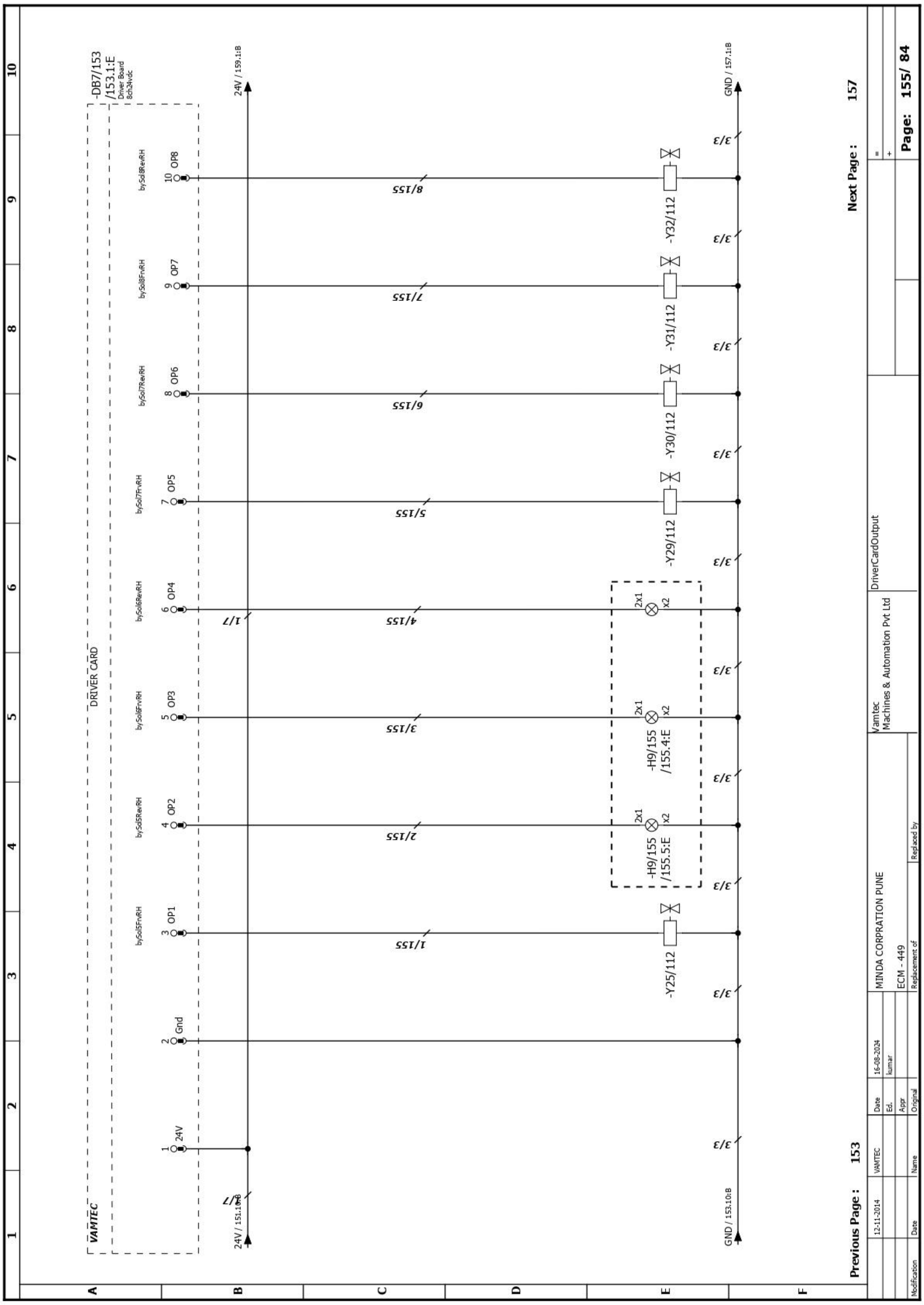
-DB7/153
/153.1:E
DRIVER CARD
24V
Gnd
OP1
OP2
OP3
OP4
OP5
OP6
OP7
OP8
-Y25/112
-H9/155
/155.5:E
-H9/155
/155.4:E
-Y29/112
-Y30/112
-Y31/112
-Y32/112
24V / 159.1:B
GND / 157.1:B
24V / 151.10:B
GND / 153.10:B
Previous Page : 153
Next Page : 157
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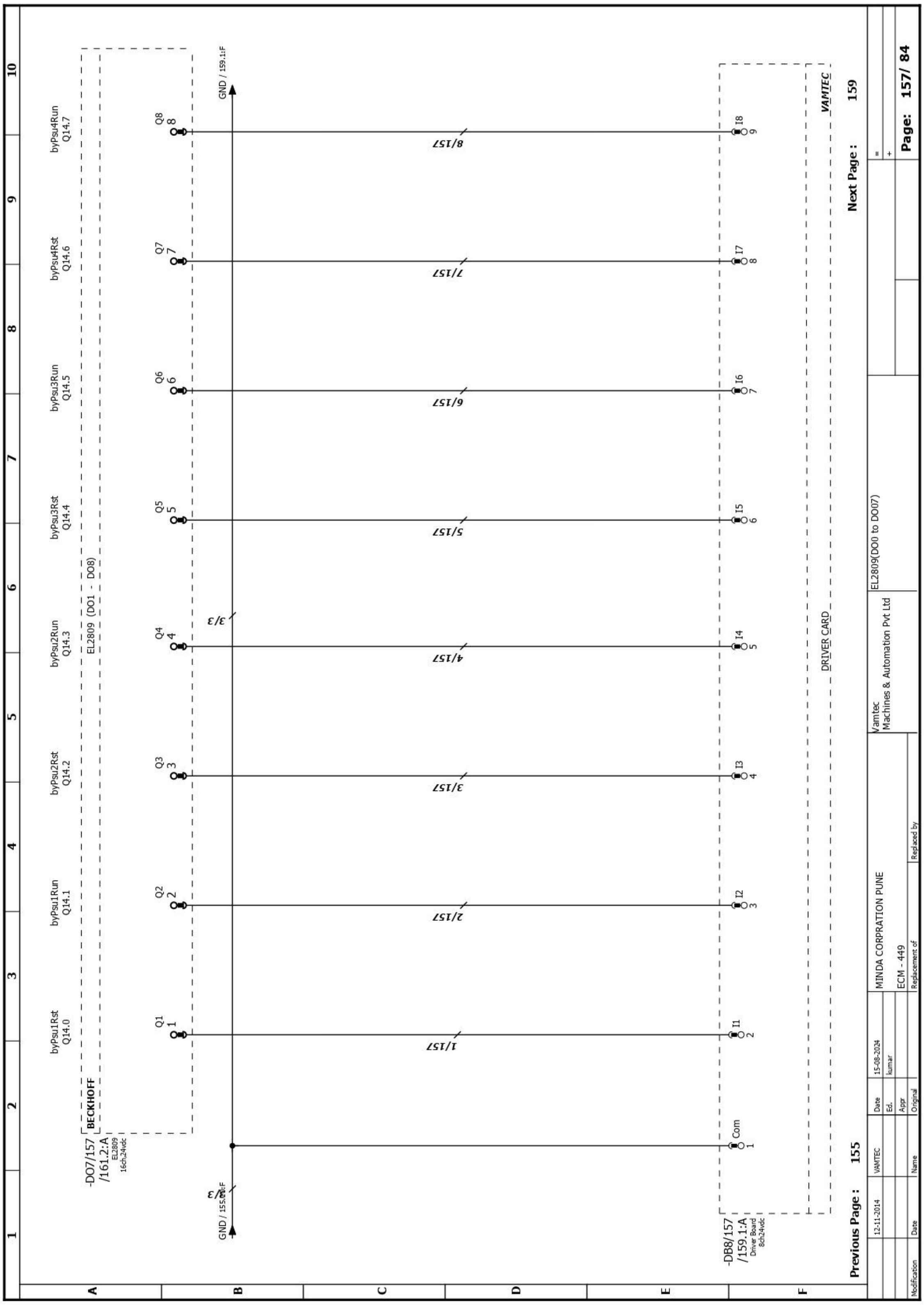

byPsu1Rst Q14.0
byPsu1Run Q14.1
byPsu2Rst Q14.2
byPsu2Run Q14.3
byPsu3Rst Q14.4
byPsu3Run Q14.5
byPsu4Rst Q14.6
byPsu4Run Q14.7
-DO7/157 /161.2:A
EL2809 16ch,24vdc
BECKHOFF
EL2809 (DO1 - DO8)
Q1 1
Q2 2
Q3 3
Q4 4
Q5 5
Q6 6
Q7 7
Q8 8
GND / 159.1:F
GND / 155.8:F
3/3
1/157
2/157
3/157
4/157
5/157
6/157
7/157
8/157
Com 1
I1 2
I2 3
I3 4
I4 5
I5 6
I6 7
I7 8
I8 9
-DB8/157 /159.1:A
Driver Board 8ch,24vdc
DRIVER CARD
VAMTEC
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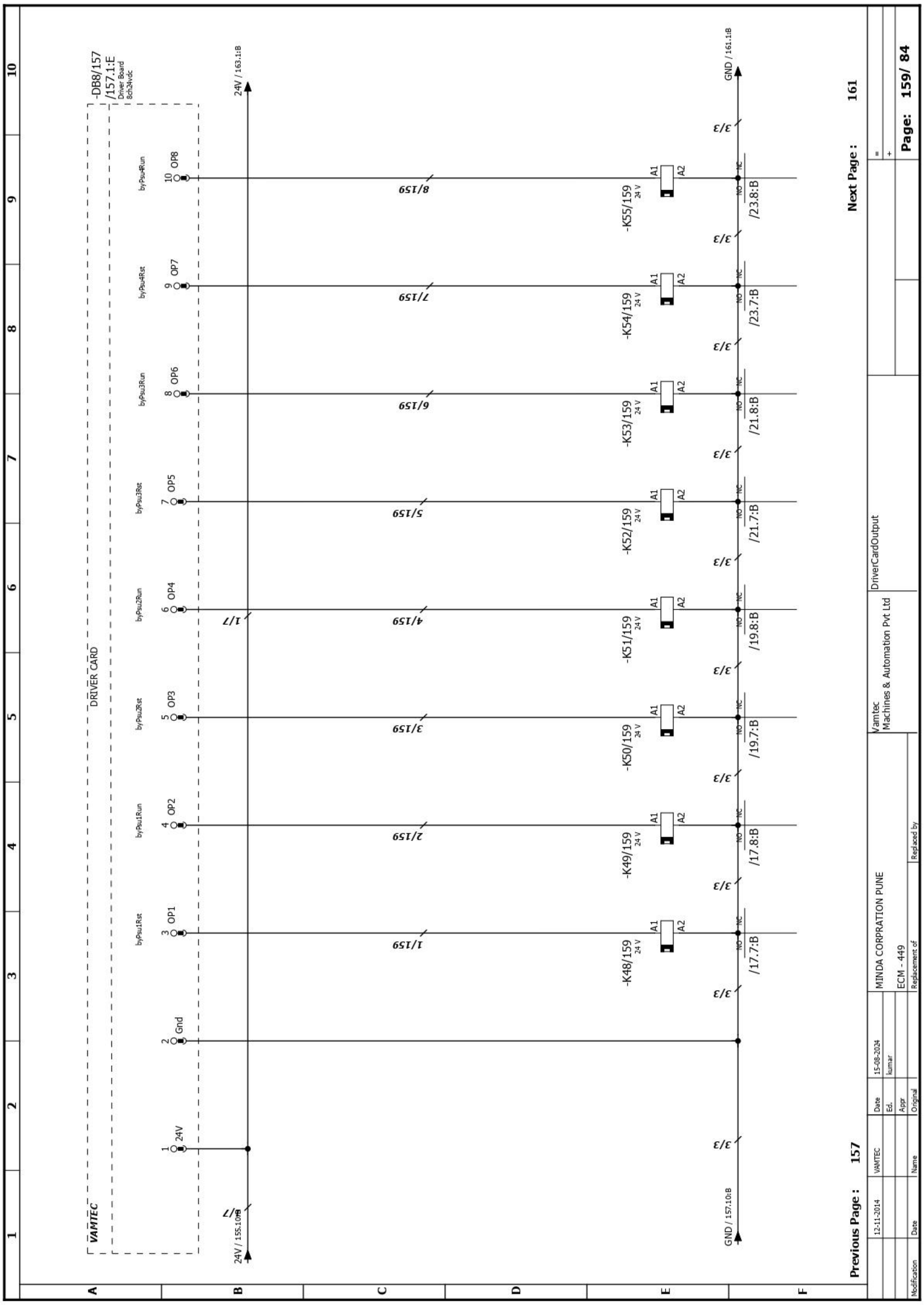
-DB8/157
/157.1:E
DRIVER CARD
VAMTEC
24V / 163.1:B
24V / 155.10:B
GND / 161.1:B
GND / 157.10:B
1 24V
2 Gnd
3 OP1
4 OP2
5 OP3
6 OP4
7 OP5
8 OP6
9 OP7
10 OP8
byPsu1Rst
byPsu1Run
byPsu2Rst
byPsu2Run
byPsu3Rst
byPsu3Run
byPsu4Rst
byPsu4Run
-K48/159
-K49/159
-K50/159
-K51/159
-K52/159
-K53/159
-K54/159
-K55/159
/17.7:B
/17.8:B
/19.7:B
/19.8:B
/21.7:B
/21.8:B
/23.7:B
/23.8:B
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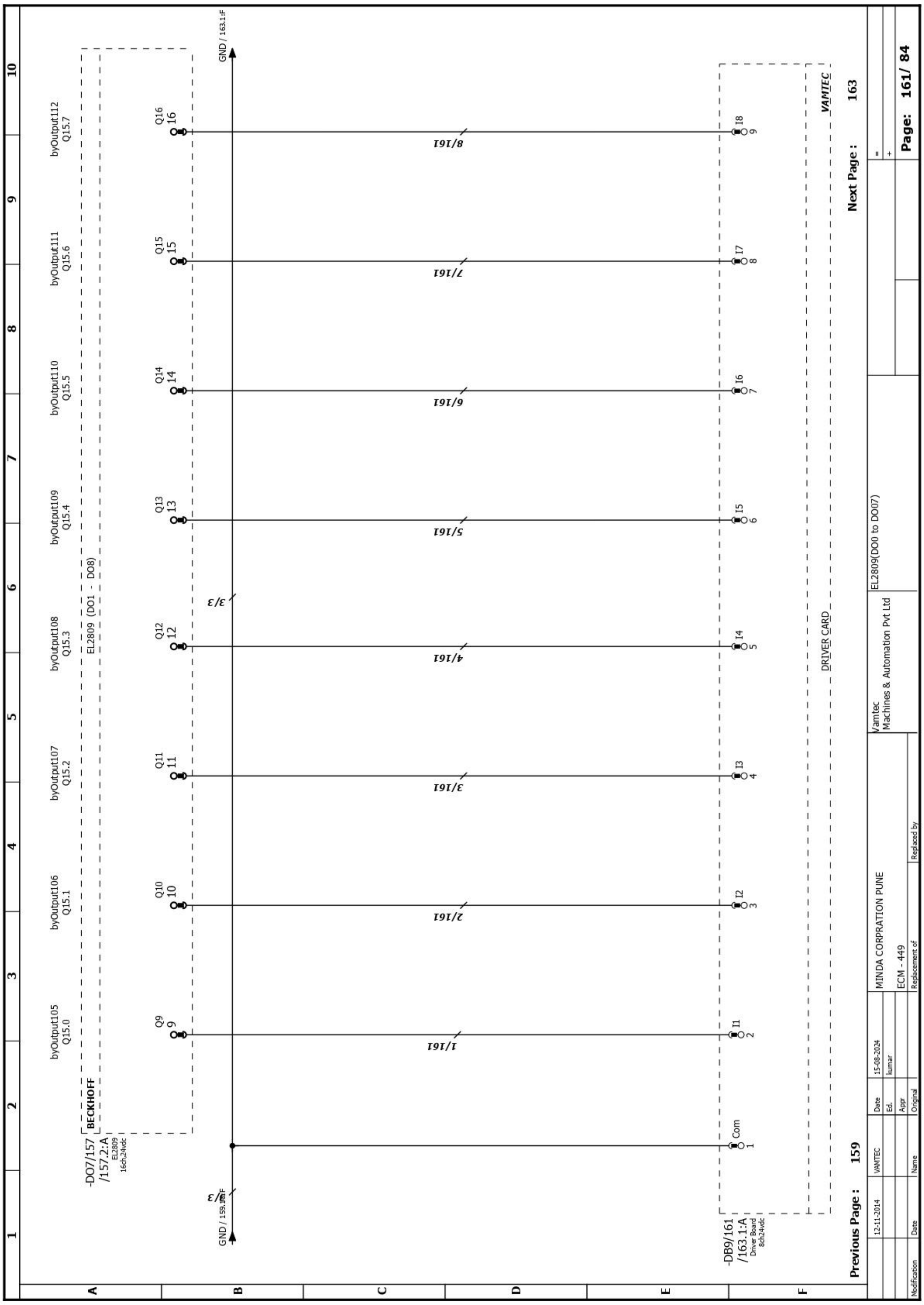

-DO7/157
/157.2:A
EL2809
16ch,24vdc
BECKHOFF
EL2809 (DO1 - DO8)
byOutput105
Q15.0
byOutput106
Q15.1
byOutput107
Q15.2
byOutput108
Q15.3
byOutput109
Q15.4
byOutput110
Q15.5
byOutput111
Q15.6
byOutput112
Q15.7
Q9
9
Q10
10
Q11
11
Q12
12
Q13
13
Q14
14
Q15
15
Q16
16
GND / 163.1:F
GND / 159.2:F
3/3
1/161
2/161
3/161
4/161
5/161
6/161
7/161
8/161
Com
1
I1
2
I2
3
I3
4
I4
5
I5
6
I6
7
I7
8
I8
9
-DB9/161
/163.1:A
Driver Board
8ch,24vdc
DRIVER CARD
VAMTEC
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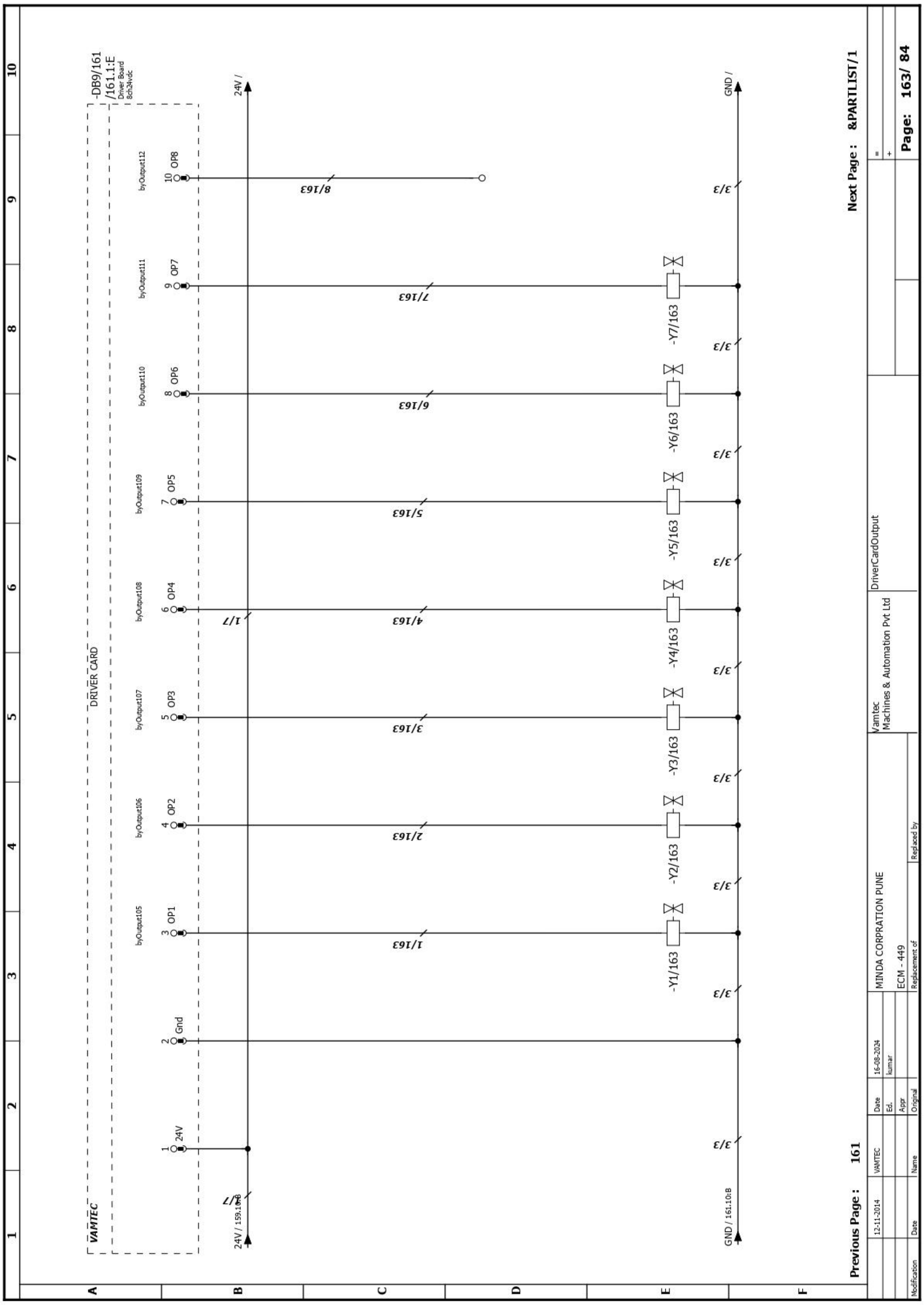

-DB9/161
/161.1:E
DRIVER CARD
VAMTEC
24V
Gnd
OP1
OP2
OP3
OP4
OP5
OP6
OP7
OP8
-Y1/163
-Y2/163
-Y3/163
-Y4/163
-Y5/163
-Y6/163
-Y7/163
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Summarized parts list

Carry over: VamtecSummaryPartList

ERP Number	Quantity	Part number	Designation	Manufacturer Supplier	Order Number	Serial no..
	1	TELTO.TSW010	5 Port Industrial Ethernet Switch	TELTONIKA	TSW010	1
E13629	4	BEC.C6015-0010	"Economy" control cabinet Industrial PC C6015-0010	Beckhoff Automation	C6015-0010	2
	5	BEC.EL3104	4-channel analog input terminal -10 V...+10 V, differential input, 16 bit	BECKHOFF	EL3104	3
L03081	4	BEC.EL3124	4-channel analog input terminal 4...20 mA, differential input, 16 bit	BECKHOFF	EL3124	4
	1	BEC.EL4012	2-channel analog output terminal 0...20 mA, 12 bit	BECKHOFF	EL4012	5
	1	RTD (Item to be created)				6
EB21202	9	KEY.FD-Q10C	Clamp-on Flow Sensor	KEYENCE	FD-Q10C	7
EB18165	1	KEY.FD-Q32C	Clamp-on Flow Sensor	KEYENCE	FD-Q32C	8
MB22009	3	FLPRO.LIM18PR	Proximity For FloatSensor HIGH		LIM18 PR	9
MB22008	3	FLPRO.LIM18PF	Proximity For FloatSensor LOW		LIM18 PF	10
L08749	34	SMC.D-G5BAL	Auto switch solid state, 2-wire 10-28V DC,	SMC Corporation	D-G5BAL	11
EB18070	2	VAM.M8PEEKPNP	Proximity ConnectorType(M18, Capacitive,PNPNO)	Vamtec	VAM.M8PEEKPNP	12
	3	BEC.EK1100	EtherCAT Coupler for E-bus terminals (ELxxxx)	BECKHOFF	EK1100	13
EB013	2	IFM.PT5403	Electronic pressure sensor(0-25 Bar,4-20mA)	ifm electronic	PT5403	14
	2	VAM.CSPCB8CH	Current Sensor PCb 8Ch	Vamtec		15
	1	VAM.CSPCB4CH	Current Sensor PCb 4Ch	Vamtec		16
L06219	9	DriverCard8Ch	Driver Card	Vamtec		17
L02975	7	BEC.EL1809	16-channel digital input terminal 24 V DC, filter 3.0 ms, type 3	BECKHOFF	EL1809	18
L02977	7	BEC.EL2809	16-channel digital output terminal 24 V DC, 0.5 A, 1-wire system	BECKHOFF	EL2809	19

Grand total: 0.00

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Summarized parts list

VamtecSummaryPartList
Carry over: 0.00

ERP Number	Quantity	Part number	Designation	Manufacturer Supplier	Order Number	Serial no..
L11084	1	TEL.MCB A9N2P04C	MCB 2Pole,4A	Télémecanique	A9N2P04C	20
L00088	3	TEL.MCB A9N2P06C	MCB,2Pole,6A	Télémecanique	A9N2P06C	21
L00089	1	TEL.MCB A9N3P06C	MCB,3Pole,6A	Télémecanique	A9N3P06C	22
L00889	1	TEL.MCB A9N3P63C	MCB,3Pole,63A	Télémecanique	A9N3P63C	23
EB18800	1	TEL.MCB A9N4P16C	MCB 4Pole,16A	Télémecanique	A9N4P16C	24
L11825	2	TEL.MCB A9N3P10C	MCB,3Pole,10A	Télémecanique	A9N3P10C	25
L00087	1	TEL.MCB A9N3P02C	MCB,3Pole,2A	Télémecanique	A9N3P02C	26
L01372	4	TEL.MCB A9N3P20C	MCB,3Pole,20A	Télémecanique	A9N3P20C	27
EB24003	1	MW.DRS-480-24	UPS 24V/20A, 1Φ-,100~277VAC	Meanwell	DRS-480-24	28
	0					29
EB18224	1	MW.TDR-240-24	3Φ,24V/10A	Meanwell	TDR-240-24	30
EB18216	2	CRT.CR-RMLL-24-D-18W	24Vdc/18W,LED Lamp,IP65		CR-RMLL-24-D-18W	31
EB21204	4	VSE.9W	24Vdc/9W,LED Lamp,	VSELECT	9W	32
EB23671	2	GRN.T63-HGV0T2-RAG	Tower Lamp Red,Yellow,Green,Buzzer	GREEN ELECTRONICS	T63-HGV0T2-RAG	33
L09901	28	ERI.020KDD-10100-00	100Vdc/100Adc(24VdcCoil)		012JDA482500	34
EB14150	20	FIN.34.51.7.024.0010	Electromechanical Relay, 1NO/NC	Finder CZ, s.r.o.	34.51.7.024.0010	35
L11085	2	SE.LC1D09BD	Contactor LC1-D-3P-AC3/440V,9A,Coil 24 V DC	Schneider Electric	LC1D09BD	36
	1	SMC.ISE20-P-01-LD	ISE20, Digital Pressure Switch, 3-Screen/3-Color Display	SMC Corporation	ISE20-P-01-LD	37
L13532	1	GRNF.CM1-2A-R-A-V-AVBV	MonoBlockPump 0.46kw/0.616HP	GRUNDFOS	A-97516508-P1-2113	38

Grand total: 0.00

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Date	17-08-2024	MINDA CORPRATION PUNE	Vamtec Machines & Automation Pvt Ltd	Summarised Part List
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Summarized parts list

VamtecSummaryPartList
Carry over: 0.00

ERP Number	Quantity	Part number	Designation	Manufacturer Supplier	Order Number	Serial no..
MB21550	1	FX 4002	MistCollector, SS, 3Φ1.1Kw/1.5HP		FX 4002	39
MB21505	2	BABJ.CM/L7800028518	Squirrel-cage motor 3HP/202kW	Bharath Bijlee	CM/L7800028518	40
MB21504	1	PRC.PED-01	Acid Dosing Pump 1Φ, 230V/ 60Watts	Precision	A-97516508-P1-2113	41
	1	LBL32409B33RDYR(1)(1)	4Pole, RotarySwitch,400A	SALZER	LBL32409B33RDYR	42
L00486	2	TEL.MPCB GV2ME08	2.5-4.0A(1.1KW/2HP)	Télémecanique	GV2-ME08AE11TQ	43
L09897	2	TEL.MPCB GVAE11	Add ON Block	Télémecanique	GVAE11	44
EB13107	1	ASHE.TX-2-2-W	RTD Signal Conditioner,Head Mountable,0-250 DegC	ASHE Controls	TX-2-2-W	45
EB21225	2	Push to off	Door Limit Switch NC	SHREE SURAJ	Push to Off	46
L00186	1	XB5AW74B2N	Emergency,Illuminated,Turn to Release	Schneider Electric	XB5AW74B2N	47
L00184	4	XB5AW33B1N	PushBotton,Green,Illuminated,	Schneider Electric	XB5AW33B1N	48
L00185	1	XB5AA51N	PushBotton,Flush,Yellow,Non Illuminated,	Schneider Electric	XB5AA51N	49
L00183	2	XB5AC31N	Mushroom Push Button,Green Non Illuminated	Schneider Electric	XB5AA31N	50
L06651	1	XB5AD53N	SelctorSwitch,Three Position Center Stay Put	Schneider Electric	XB5AD53N	51
L01915	1	XB5AG21N	KeySelctorSwitch,TwoStayPut	Schneider Electric	XB5AG21N	52
	1	BEC.EL9550	Surge filter system and field supply	BECKHOFF	EL9550	53
L00080	1	400/200-500VA	1Φ,400/200-500VA	JaynthiTransformer	400/200-500VA	54
	1	3Φ-Δ/Δ,400/80-16KVA	3Φ-Δ/Δ,400/80-16KVA	JaynthiTransformer	3Φ-Δ/Δ,400/80-16KVA	55
	1	Parrt To be Created				56
L09326	1	TURBO1000M2360Hz	Panel AC		TURBO1000M2360Hz	57

Grand total: 0.00

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VamtecSummaryPartList

Summarized parts list

Carry over: 0.00

ERP Number	Quantity	Part number	Designation	Manufacturer Supplier	Order Number	Serial no..
L12489	2	DAN.FC-051P2K2T4E20H3BXCXXXSXXX	Static Frequency Converter = < 1 kV	DANFOSS	132F0022	58
EB11543	4	VAM.ECMPSU	ECM generator 100V/90A	Vamtec		59
	1	VAM.SC CHECKV1.0		Vamtec		60
EB18206	1	KEY.SL-V24H	Safety Light Curtain, 450mm Hand,Type-4, Protection, 25-dia. mm	KEYENCE	SL-V24H	61
	1	SENSEL.PH-600-1 (230V)	pH Instrument	Sensel Instruments	PH-600-1 (230V)	62
	1	SENSEL.CC603-1(230V)	Conductivity Instrument	Sensel Instruments	CC603-1(230V)	63

Grand total: 0.00

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Modification	Date	Name		
Date	17-08-2024	MINDA CORPRATION PUNE	Vamtec Machines & Automation Pvt Ltd	Summarised Part List
Ed.	kumar	ECM - 449		
Appr		Replacement of	Replaced by	
Original				

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Device tag list

F03_003

DT	Quantity	Designation	Type number	Manufacturer	Part number	Page / path	Function text
						&/13.2:C	
-A1/53	1	5 Port Industrial Ethernet Switch	TSW010	TELTO	TELTO.TSW010	&/53.1:C	
-A3/67	1	"Economy" control cabinet Industrial PC C6015-0010	C6015-0010	Beckhoff Automation	BEC.C6015-0010	&/67.2:D	
-A4/67	1	"Economy" control cabinet Industrial PC C6015-0010	C6015-0010	Beckhoff Automation	BEC.C6015-0010	&/67.4:D	
-A5/69	1	"Economy" control cabinet Industrial PC C6015-0010	C6015-0010	Beckhoff Automation	BEC.C6015-0010	&/69.2:D	
-A6/69	1	"Economy" control cabinet Industrial PC C6015-0010	C6015-0010	Beckhoff Automation	BEC.C6015-0010	&/69.4:D	
-AI1/75	1	4-channel analog input terminal -10 V...+10 V, differential input, 16 bit	EL3104	BECKHOFF	BEC.EL3104	&/75.2:E	
-AI2/77	1	4-channel analog input terminal -10 V...+10 V, differential input, 16 bit	EL3104	BECKHOFF	BEC.EL3104	&/77.2:E	
-AI3/79	1	4-channel analog input terminal -10 V...+10 V, differential input, 16 bit	EL3104	BECKHOFF	BEC.EL3104	&/79.2:E	
-AI4/81	1	4-channel analog input terminal -10 V...+10 V, differential input, 16 bit	EL3104	BECKHOFF	BEC.EL3104	&/81.2:E	
-AI5/83	1	4-channel analog input terminal -10 V...+10 V, differential input, 16 bit	EL3104	BECKHOFF	BEC.EL3104	&/83.2:E	
-AI6/85	1	4-channel analog input terminal 4...20 mA, differential input, 16 bit	EL3124	BECKHOFF	BEC.EL3124	&/85.1:E	
-AI7/87	1	4-channel analog input terminal 4...20 mA, differential input, 16 bit	EL3124	BECKHOFF	BEC.EL3124	&/87.1:E	
-AI8/89	1	4-channel analog input terminal 4...20 mA, differential input, 16 bit	EL3124	BECKHOFF	BEC.EL3124	&/89.1:E	
-AI9/91	1	4-channel analog input terminal 4...20 mA, differential input, 16 bit	EL3124	BECKHOFF	BEC.EL3124	&/91.1:E	
-AI10						&/57.2:B	
-AI11						&/55.1:A	
-AO1/93	1	2-channel analog output terminal 0...20 mA, 12 bit	EL4012	BECKHOFF	BEC.EL4012	&/93.2:A	
-B1/71	1				RTD (Item to be created)	&/71.4:D	Electrolyte Temperature
-B2/87	1	Clamp-on Flow Sensor	FD-Q10C	KEYENCE	KEY.FD-Q10C	&/87.2:B	
-B3/87	1	Clamp-on Flow Sensor	FD-Q10C	KEYENCE	KEY.FD-Q10C	&/87.4:B	
-B4/87	1	Clamp-on Flow Sensor	FD-Q10C	KEYENCE	KEY.FD-Q10C	&/87.6:B	
-B5/87	1	Clamp-on Flow Sensor	FD-Q10C	KEYENCE	KEY.FD-Q10C	&/87.8:B	
-B6/89	1	Clamp-on Flow Sensor	FD-Q10C	KEYENCE	KEY.FD-Q10C	&/89.2:B	
-B7/89	1	Clamp-on Flow Sensor	FD-Q10C	KEYENCE	KEY.FD-Q10C	&/89.4:B	
-B8/89	1	Clamp-on Flow Sensor	FD-Q10C	KEYENCE	KEY.FD-Q10C	&/89.6:B	
-B9/89	1	Clamp-on Flow Sensor	FD-Q10C	KEYENCE	KEY.FD-Q10C	&/89.8:B	
-B10/91	1	Clamp-on Flow Sensor	FD-Q10C	KEYENCE	KEY.FD-Q10C	&/91.2:B	
-B11/91	1	Clamp-on Flow Sensor	FD-Q32C	KEYENCE	KEY.FD-Q32C	&/91.4:B	Dirty Water Flow
-B12/97	1	Proximity For FloatSensor HIGH	LIM18 PR	FLPRO	FLPRO.LIM18PR	&/97.5:B	
-B13/97	1	Proximity For FloatSensor LOW	LIM18 PF	FLPRO	FLPRO.LIM18PF	&/97.6:B	
-B14/97	1	Proximity For FloatSensor HIGH	LIM18 PR	FLPRO	FLPRO.LIM18PR	&/97.7:B	
-B15/97	1	Proximity For FloatSensor LOW	LIM18 PF	FLPRO	FLPRO.LIM18PF	&/97.8:B	
-B16/97	1	Proximity For FloatSensor HIGH	LIM18 PR	FLPRO	FLPRO.LIM18PR	&/97.9:B	
-B17/99	1	Proximity For FloatSensor LOW	LIM18 PF	FLPRO	FLPRO.LIM18PF	&/99.2:B	
-B18	1	Auto switch solid state, 2-wire 10-28V DC,	D-G5BAL	SMC	SMC.D-G5BAL	&/105.2:B	
-B19	1	Auto switch solid state, 2-wire 10-28V DC,	D-G5BAL	SMC	SMC.D-G5BAL	&/105.3:B	
-B20	1	Auto switch solid state, 2-wire 10-28V DC,	D-G5BAL	SMC	SMC.D-G5BAL	&/105.4:B	
-B21	1	Auto switch solid state, 2-wire 10-28V DC,	D-G5BAL	SMC	SMC.D-G5BAL	&/105.5:B	
-B22	1	Auto switch solid state, 2-wire 10-28V DC,	D-G5BAL	SMC	SMC.D-G5BAL	&/105.6:B	
-B23	1	Auto switch solid state, 2-wire 10-28V DC,	D-G5BAL	SMC	SMC.D-G5BAL	&/105.7:B	
-B24	1	Auto switch solid state, 2-wire 10-28V DC,	D-G5BAL	SMC	SMC.D-G5BAL	&/105.8:B	
-B25	1	Auto switch solid state, 2-wire 10-28V DC,	D-G5BAL	SMC	SMC.D-G5BAL	&/105.9:B	
-B26	1	Proximity ConnectorType(M18, Capacitive,PNPNO)	VAM.M8PEEKPNP	VAMTEC	VAM.M8PEEKPNP	&/107.2:B	
-B27	1	Auto switch solid state, 2-wire 10-28V DC,	D-G5BAL	SMC	SMC.D-G5BAL	&/109.5:C	
-B28	1	Auto switch solid state, 2-wire 10-28V DC,	D-G5BAL	SMC	SMC.D-G5BAL	&/109.6:C	
-B29	1	Auto switch solid state, 2-wire 10-28V DC,	D-G5BAL	SMC	SMC.D-G5BAL	&/109.7:C	
-B30	1	Auto switch solid state, 2-wire 10-28V DC,	D-G5BAL	SMC	SMC.D-G5BAL	&/109.8:C	
-B31	1	Auto switch solid state, 2-wire 10-28V DC,	D-G5BAL	SMC	SMC.D-G5BAL	&/109.9:C	
-B32	1	Auto switch solid state, 2-wire 10-28V DC,	D-G5BAL	SMC	SMC.D-G5BAL	&/111.2:B	
-B33	1	Auto switch solid state, 2-wire 10-28V DC,	D-G5BAL	SMC	SMC.D-G5BAL	&/111.3:B	
-B34	1	Auto switch solid state, 2-wire 10-28V DC,	D-G5BAL	SMC	SMC.D-G5BAL	&/111.4:B	
-B35	1	Auto switch solid state, 2-wire 10-28V DC,	D-G5BAL	SMC	SMC.D-G5BAL	&/111.5:B	
-B36	1	Auto switch solid state, 2-wire 10-28V DC,	D-G5BAL	SMC	SMC.D-G5BAL	&/111.6:B	
-B37	1	Auto switch solid state, 2-wire 10-28V DC,	D-G5BAL	SMC	SMC.D-G5BAL	&/111.7:B	
-B38	1	Auto switch solid state, 2-wire 10-28V DC,	D-G5BAL	SMC	SMC.D-G5BAL	&/111.8:B	
-B39	1	Auto switch solid state, 2-wire 10-28V DC,	D-G5BAL	SMC	SMC.D-G5BAL	&/111.9:B	
-B40	1	Auto switch solid state, 2-wire 10-28V DC,	D-G5BAL	SMC	SMC.D-G5BAL	&/101.2:B	

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Modification	Date	Name				
			Date	17-08-2024	MINDA CORPRATION PUNE	Vamtec Machines & Automation Pvt Ltd
			Ed.	kumar	ECM - 449	Device Tag List
			Appr		Replacement of	Replaced by
			Original			

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Device tag list

F03_003

DT	Quantity	Designation	Type number	Manufacturer	Part number	Page / path	Function text
-B41	1	Auto switch solid state, 2-wire 10-28V DC,	D-G5BAL	SMC	SMC.D-G5BAL	8/101.3:B	
-B42	1	Auto switch solid state, 2-wire 10-28V DC,	D-G5BAL	SMC	SMC.D-G5BAL	8/103.2:B	
-B43	1	Auto switch solid state, 2-wire 10-28V DC,	D-G5BAL	SMC	SMC.D-G5BAL	8/103.3:B	
-B44	1	Auto switch solid state, 2-wire 10-28V DC,	D-G5BAL	SMC	SMC.D-G5BAL	8/103.4:B	
-B45	1	Auto switch solid state, 2-wire 10-28V DC,	D-G5BAL	SMC	SMC.D-G5BAL	8/103.5:B	
-B46	1	Auto switch solid state, 2-wire 10-28V DC,	D-G5BAL	SMC	SMC.D-G5BAL	8/103.6:B	
-B47	1	Auto switch solid state, 2-wire 10-28V DC,	D-G5BAL	SMC	SMC.D-G5BAL	8/103.7:B	
-B48	1	Auto switch solid state, 2-wire 10-28V DC,	D-G5BAL	SMC	SMC.D-G5BAL	8/103.8:B	
-B49	1	Auto switch solid state, 2-wire 10-28V DC,	D-G5BAL	SMC	SMC.D-G5BAL	8/103.9:B	
-B50	1	Auto switch solid state, 2-wire 10-28V DC,	D-G5BAL	SMC	SMC.D-G5BAL	8/113.2:B	
-B51	1	Auto switch solid state, 2-wire 10-28V DC,	D-G5BAL	SMC	SMC.D-G5BAL	8/113.3:B	
-B52	1	Auto switch solid state, 2-wire 10-28V DC,	D-G5BAL	SMC	SMC.D-G5BAL	8/113.4:B	
-B53	1	Proximity ConnectorType(M18, Capacitive,PNPNO)	VAM.M8PEEKPNP	VAMTEC	VAM.M8PEEKPNP	8/113.5:B	
-BC1/59	1	EtherCAT Coupler for E-bus terminals (ELxxxx)	EK1100	BECKHOFF	BEC.EK1100		
-BC2/63	1	EtherCAT Coupler for E-bus terminals (ELxxxx)	EK1100	BECKHOFF	BEC.EK1100		
-BC3	1	EtherCAT Coupler for E-bus terminals (ELxxxx)	EK1100	BECKHOFF	BEC.EK1100	8/57.1:B	
-BP1/85	1	Electronic pressure sensor(0-25 Bar,4-20mA)	PT-025-SEG14-A-ZVG/US/ /W	IFM	IFM.PT5403	8/85.1:A	Clear Water Pressure
-BP2/85	1	Electronic pressure sensor(0-25 Bar,4-20mA)	PT-025-SEG14-A-ZVG/US/ /W	IFM	IFM.PT5403	8/85.3:A	Dirty Water Pressure
-CS1/45	1	Current Sensor PCb 8Ch		VAMTEC	VAM.CSPCB8CH	8/45.1:E	
-CS2/47	1	Current Sensor PCb 8Ch		VAMTEC	VAM.CSPCB8CH	8/47.1:E	
-CS3/49	1	Current Sensor PCb 4Ch		VAMTEC	VAM.CSPCB4CH	8/49.1:E	
-DB1/129	1	Driver Card		VAMTEC	DriverCard8Ch	8/129.1:E	
-DB2/133	1	Driver Card		VAMTEC	DriverCard8Ch	8/133.1:E	
-DB3/137	1	Driver Card		VAMTEC	DriverCard8Ch	8/137.1:E	
-DB4/141	1	Driver Card		VAMTEC	DriverCard8Ch	8/141.1:E	
-DB5/145	1	Driver Card		VAMTEC	DriverCard8Ch	8/145.1:E	
-DB6/149	1	Driver Card		VAMTEC	DriverCard8Ch	8/149.1:E	
-DB7/153	1	Driver Card		VAMTEC	DriverCard8Ch	8/153.1:E	
-DB8/157	1	Driver Card		VAMTEC	DriverCard8Ch	8/157.1:E	
-DB9/161	1	Driver Card		VAMTEC	DriverCard8Ch	8/161.1:E	
-DI1/95	1	16-channel digital input terminal 24 V DC, filter 3.0 ms, type 3	EL1809	BECKHOFF	BEC.EL1809	8/95.1:E	
-DI2/99	1	16-channel digital input terminal 24 V DC, filter 3.0 ms, type 3	EL1809	BECKHOFF	BEC.EL1809	8/99.1:E	
-DI3/103	1	16-channel digital input terminal 24 V DC, filter 3.0 ms, type 3	EL1809	BECKHOFF	BEC.EL1809	8/103.1:E	
-DI4/107	1	16-channel digital input terminal 24 V DC, filter 3.0 ms, type 3	EL1809	BECKHOFF	BEC.EL1809	8/107.1:E	
-DI5/111	1	16-channel digital input terminal 24 V DC, filter 3.0 ms, type 3	EL1809	BECKHOFF	BEC.EL1809	8/111.1:E	
-DI6/115	1	16-channel digital input terminal 24 V DC, filter 3.0 ms, type 3	EL1809	BECKHOFF	BEC.EL1809	8/115.1:E	
-DI7/118	1	16-channel digital input terminal 24 V DC, filter 3.0 ms, type 3	EL1809	BECKHOFF	BEC.EL1809	8/118.1:E	
-DO1/119	1	16-channel digital output terminal 24 V DC, 0.5 A, 1-wire system	EL2809	BECKHOFF	BEC.EL2809	8/119.1:A	
-DO2/123	1	16-channel digital output terminal 24 V DC, 0.5 A, 1-wire system	EL2809	BECKHOFF	BEC.EL2809	8/123.1:A	
-DO3/127	1	16-channel digital output terminal 24 V DC, 0.5 A, 1-wire system	EL2809	BECKHOFF	BEC.EL2809	8/127.1:A	
-DO4/133	1	16-channel digital output terminal 24 V DC, 0.5 A, 1-wire system	EL2809	BECKHOFF	BEC.EL2809	8/133.2:A	
-DO5/141	1	16-channel digital output terminal 24 V DC, 0.5 A, 1-wire system	EL2809	BECKHOFF	BEC.EL2809	8/141.2:A	
-DO6/149	1	16-channel digital output terminal 24 V DC, 0.5 A, 1-wire system	EL2809	BECKHOFF	BEC.EL2809	8/149.2:A	
-DO7/157	1	16-channel digital output terminal 24 V DC, 0.5 A, 1-wire system	EL2809	BECKHOFF	BEC.EL2809	8/157.2:A	
-F1/1	1	MCB 2Pole,4A	A9N2P04C	TELEMECANIQUE	TEL.MCB A9N2P04C	8/1.3:B	
-F3/1	1	MCB,2Pole,6A	A9N2P06C	TELEMECANIQUE	TEL.MCB A9N2P06C	8/1.5:B	
-F4/1	1	MCB,2Pole,6A	A9N2P06C	TELEMECANIQUE	TEL.MCB A9N2P06C	8/1.8:B	
-F6/5	1	MCB,3Pole,6A	A9N3P06C	TELEMECANIQUE	TEL.MCB A9N3P06C	8/5.2:B	
-F7/7	1	MCB,3Pole,63A	A9N3P63C	TELEMECANIQUE	TEL.MCB A9N3P63C	8/7.2:B	
-F8/7	1	MCB 4Pole,16A	A9N4P16C	TELEMECANIQUE	TEL.MCB A9N4P16C	8/7.4:B	
-F9/11	1	MCB,3Pole,10A	A9N3P10C	TELEMECANIQUE	TEL.MCB A9N3P10C	8/11.2:B	
-F10/13	1	MCB,3Pole,10A	A9N3P10C	TELEMECANIQUE	TEL.MCB A9N3P10C	8/13.2:B	
-F11/15	1	MCB,3Pole,2A	A9N3P02C	TELEMECANIQUE	TEL.MCB A9N3P02C	8/15.2:B	
-F12/1	1	MCB,2Pole,6A	A9N2P06C	TELEMECANIQUE	TEL.MCB A9N2P06C	8/1.3:D	
-F12/17	1	MCB,3Pole,20A	A9N3P20C	TELEMECANIQUE	TEL.MCB A9N3P20C	8/17.2:B	
-F13/19	1	MCB,3Pole,20A	A9N3P20C	TELEMECANIQUE	TEL.MCB A9N3P20C	8/19.2:B	
-F14/21	1	MCB,3Pole,20A	A9N3P20C	TELEMECANIQUE	TEL.MCB A9N3P20C	8/21.2:B	
-F15/23	1	MCB,3Pole,20A	A9N3P20C	TELEMECANIQUE	TEL.MCB A9N3P20C	8/23.2:B	

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Date	17-08-2024	MINDA CORPRATION PUNE	Vamtec Machines & Automation Pvt Ltd	Device Tag List	=
Ed.	kumar	ECM - 449			+
Appr					
Original		Replacement of	Replaced by		Page: 2 / 5

Modification | Date | Name

Device tag list

F03_003

DT	Quantity	Designation	Type number	Manufacturer	Part number	Page / path	Function text
-G1/3	1	UPS 24V/20A, 1Φ-,100~277VAC	DRS-480-24	MEANWELL	MW.DRS-480-24	8/3.2:C	
-G2/3						8/3.3:E	
-G3/3						8/3.3:E	
-G4/5	1	3Φ,24V/10A	TDR-240-24	MEANWELL	MW.TDR-240-24	8/5.2:C	
-H2/9	1	24Vdc/18W,LED Lamp,IP65	CR-RMLL-24-D-18W	CRAFTECH	CRT.CR-RMLL-24-D-18W	8/9.3:E	
-H3/9	1	24Vdc/18W,LED Lamp,IP65	CR-RMLL-24-D-18W	CRAFTECH	CRT.CR-RMLL-24-D-18W	8/9.4:E	
-H4/9	1	24Vdc/9W,LED Lamp,	9W	VS ELECT	VSE.9W	8/9.6:E	
-H5/9	1	24Vdc/9W,LED Lamp,	9W	VS ELECT	VSE.9W	8/9.7:E	
-H6/9	1	24Vdc/9W,LED Lamp,	9W	VS ELECT	VSE.9W	8/9.8:E	
-H7/9	1	24Vdc/9W,LED Lamp,	9W	VS ELECT	VSE.9W	8/9.9:E	
-H8/143	1	Tower Lamp Red,Yellow,Green,Buzzer	T63-HGV0T2-RAG	GREEN	GRN.T63-HGV0T2-RAG	8/143.6:E	
-H9/155	1	Tower Lamp Red,Yellow,Green,Buzzer	T63-HGV0T2-RAG	GREEN	GRN.T63-HGV0T2-RAG	8/155.4:E	
-K1/119	1	100Vdc/100Adc(24VdcCoil)	012JDA482500	ERI	ERI.020KDD-10100-00	8/119.2:E	ST1 SCTestSSR5
-K2/119	1	100Vdc/100Adc(24VdcCoil)	012JDA482500	ERI	ERI.020KDD-10100-00	8/119.3:E	ST2 SCTestSSR5
-K3/119	1	100Vdc/100Adc(24VdcCoil)	012JDA482500	ERI	ERI.020KDD-10100-00	8/119.4:E	ST1 SCTestSSR6
-K4/119	1	100Vdc/100Adc(24VdcCoil)	012JDA482500	ERI	ERI.020KDD-10100-00	8/119.5:E	ST2 SCTestSSR6
-K5/119	1	100Vdc/100Adc(24VdcCoil)	012JDA482500	ERI	ERI.020KDD-10100-00	8/119.6:E	ST1 SCTestSSR7
-K6/119	1	100Vdc/100Adc(24VdcCoil)	012JDA482500	ERI	ERI.020KDD-10100-00	8/119.7:E	ST2 SCTestSSR7
-K7/119	1	100Vdc/100Adc(24VdcCoil)	012JDA482500	ERI	ERI.020KDD-10100-00	8/119.8:E	ST1 SCTestSSR8
-K8/119	1	100Vdc/100Adc(24VdcCoil)	012JDA482500	ERI	ERI.020KDD-10100-00	8/119.9:E	ST2 SCTestSSR8
-K9/121	1	100Vdc/100Adc(24VdcCoil)	012JDA482500	ERI	ERI.020KDD-10100-00	8/121.2:E	ST1 SCTestSSR9
-K10/121	1	100Vdc/100Adc(24VdcCoil)	012JDA482500	ERI	ERI.020KDD-10100-00	8/121.3:E	ST2 SCTestSSR9
-K11/121	1	100Vdc/100Adc(24VdcCoil)	012JDA482500	ERI	ERI.020KDD-10100-00	8/121.4:E	ST1 SCTestSSR10
-K12/121	1	100Vdc/100Adc(24VdcCoil)	012JDA482500	ERI	ERI.020KDD-10100-00	8/121.5:E	ST2 SCTestSSR10
-K13/121	1	100Vdc/100Adc(24VdcCoil)	012JDA482500	ERI	ERI.020KDD-10100-00	8/121.6:E	ST1 SCTestSSR11
-K14/121	1	100Vdc/100Adc(24VdcCoil)	012JDA482500	ERI	ERI.020KDD-10100-00	8/121.7:E	ST2 SCTestSSR11
-K15/121	1	100Vdc/100Adc(24VdcCoil)	012JDA482500	ERI	ERI.020KDD-10100-00	8/121.8:E	ST1 SCTestSSR12
-K16/121	1	100Vdc/100Adc(24VdcCoil)	012JDA482500	ERI	ERI.020KDD-10100-00	8/121.9:E	ST2 SCTestSSR12
-K17/123	1	100Vdc/100Adc(24VdcCoil)	012JDA482500	ERI	ERI.020KDD-10100-00	8/123.2:E	byST2SSR1On
-K18/123	1	100Vdc/100Adc(24VdcCoil)	012JDA482500	ERI	ERI.020KDD-10100-00	8/123.3:E	byST2SSR2On
-K19/123	1	100Vdc/100Adc(24VdcCoil)	012JDA482500	ERI	ERI.020KDD-10100-00	8/123.4:E	byST2SSR3On
-K20/123	1	100Vdc/100Adc(24VdcCoil)	012JDA482500	ERI	ERI.020KDD-10100-00	8/123.5:E	byST2SSR4On
-K21/123	1	Electromechanical Relay, 1NO/NC	34.51.7.024.0010	FIN	FIN.34.51.7.024.0010	8/123.6:E	byST2SSR5On
-K22/123	1	Electromechanical Relay, 1NO/NC	34.51.7.024.0010	FIN	FIN.34.51.7.024.0010	8/123.7:E	byST2SSR6On
-K23/123	1	Electromechanical Relay, 1NO/NC	34.51.7.024.0010	FIN	FIN.34.51.7.024.0010	8/123.8:E	byST2SSR7On
-K24/123	1	Electromechanical Relay, 1NO/NC	34.51.7.024.0010	FIN	FIN.34.51.7.024.0010	8/123.9:E	byST2SSR8On
-K25/125	1	Electromechanical Relay, 1NO/NC	34.51.7.024.0010	FIN	FIN.34.51.7.024.0010	8/125.2:E	byST2SSR9On
-K26/125	1	Electromechanical Relay, 1NO/NC	34.51.7.024.0010	FIN	FIN.34.51.7.024.0010	8/125.3:E	byST2SSR10On
-K27/125	1	Electromechanical Relay, 1NO/NC	34.51.7.024.0010	FIN	FIN.34.51.7.024.0010	8/125.4:E	byST2SSR11On
-K28/125	1	Electromechanical Relay, 1NO/NC	34.51.7.024.0010	FIN	FIN.34.51.7.024.0010	8/125.5:E	byST2SSR12On
-K29/125	1	Electromechanical Relay, 1NO/NC	34.51.7.024.0010	FIN	FIN.34.51.7.024.0010	8/125.6:E	byST2SSR13On
-K30/125	1	Electromechanical Relay, 1NO/NC	34.51.7.024.0010	FIN	FIN.34.51.7.024.0010	8/125.7:E	byST2SSR14On
-K31/125	1	Electromechanical Relay, 1NO/NC	34.51.7.024.0010	FIN	FIN.34.51.7.024.0010	8/125.8:E	byST2SSR15On
-K32/125	1	Electromechanical Relay, 1NO/NC	34.51.7.024.0010	FIN	FIN.34.51.7.024.0010	8/125.9:E	byST2SSR16On
-K33/127	1	Electromechanical Relay, 1NO/NC	34.51.7.024.0010	FIN	FIN.34.51.7.024.0010	8/127.2:E	byST1SCTestRly1
-K34/127	1	Electromechanical Relay, 1NO/NC	34.51.7.024.0010	FIN	FIN.34.51.7.024.0010	8/127.3:E	byST1SCTestRly2
-K35/127	1	Electromechanical Relay, 1NO/NC	34.51.7.024.0010	FIN	FIN.34.51.7.024.0010	8/127.4:E	byST1SCTestRly3
-K36/127	1	Electromechanical Relay, 1NO/NC	34.51.7.024.0010	FIN	FIN.34.51.7.024.0010	8/127.5:E	byST1SCTestRly4
-K37/127	1	Electromechanical Relay, 1NO/NC	34.51.7.024.0010	FIN	FIN.34.51.7.024.0010	8/127.6:E	byST1SCTestRly5
-K38/127	1	Electromechanical Relay, 1NO/NC	34.51.7.024.0010	FIN	FIN.34.51.7.024.0010	8/127.7:E	byST1SCTestRly6
-K39/127	1	Electromechanical Relay, 1NO/NC	34.51.7.024.0010	FIN	FIN.34.51.7.024.0010	8/127.8:E	byST1SCTestRly7
-K40/127	1	Electromechanical Relay, 1NO/NC	34.51.7.024.0010	FIN	FIN.34.51.7.024.0010	8/127.9:E	byST1SCTestRly8
-K41/143	1	Contactor LC1-D-3P-AC3/440V,9A,Coil 24 V DC	LC1-D 3P 9A	SE	SE.LC1D09BD	8/143.3:E	ST1 MIst Collector
-K42/151	1	Contactor LC1-D-3P-AC3/440V,9A,Coil 24 V DC	LC1-D 3P 9A	SE	SE.LC1D09BD	8/151.8:E	ST2 MIst Collector
-K43/167						8/7.6:C	
-K44/167						8/15.2:C	
-K45/167						8/7.4:C	
-K46/171						8/39.4:C	

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Modification	Date	Name	Original		Replacement of / Replaced by			

1	2	3	4	5	6	7	8	9	10

Device tag list

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DT	Quantity	Designation	Type number	Manufacturer	Part number	Page / path	Function text
-K47/171						&/39.7:B	
-K48/159	1	100Vdc/100Adc(24VdcCoil)	012JDA482500	ERI	ERI.020KDD-10100-00	&/159.3:E	
-K49/159	1	100Vdc/100Adc(24VdcCoil)	012JDA482500	ERI	ERI.020KDD-10100-00	&/159.4:E	
-K50/159	1	100Vdc/100Adc(24VdcCoil)	012JDA482500	ERI	ERI.020KDD-10100-00	&/159.5:E	
-K51/159	1	100Vdc/100Adc(24VdcCoil)	012JDA482500	ERI	ERI.020KDD-10100-00	&/159.6:E	
-K52/159	1	100Vdc/100Adc(24VdcCoil)	012JDA482500	ERI	ERI.020KDD-10100-00	&/159.7:E	
-K53/159	1	100Vdc/100Adc(24VdcCoil)	012JDA482500	ERI	ERI.020KDD-10100-00	&/159.8:E	
-K54/159	1	100Vdc/100Adc(24VdcCoil)	012JDA482500	ERI	ERI.020KDD-10100-00	&/159.8:E	
-K55/159	1	100Vdc/100Adc(24VdcCoil)	012JDA482500	ERI	ERI.020KDD-10100-00	&/159.9:E	
-KH1/99	1	ISE20, Digital Pressure Switch, 3-Screen/3-Color Display	ISE20-P-01-LD	SMC	SMC.ISE20-P-01-LD	&/99.3:A	
-M1/7	1	MonoBlockPump 0.46kw/0.616HP	A-97516508-P1-2113	GRNF	GRNF.CM1-2A-R-A-V-AVBV	&/7.6:E	WASH MOTOR
-M2/7	1	MistCollector, SS, 3Φ1.1Kw/1.5HP	FX 4002	FILTERMIST	FX 4002	&/7.7:E	MIst Collector
-M3/11	1	Squirrel-cage motor 3HP/202kW	CM/L7800028518	BABJ	BABJ.CM/L7800028518	&/11.2:F	CLEAN MOTOR
-M4/13	1	Squirrel-cage motor 3HP/202kW	CM/L7800028518	BABJ	BABJ.CM/L7800028518	&/13.2:F	DIRTY MOTOR
-M5/15	1	Acid Dosing Pump 1Φ, 230V/ 60Watts	A-97516508-P1-2113	PRC	PRC.PED-01	&/15.2:E	Acid Pump
-PE						&/1.9:C	
-PPC						&/61.2:E	
-Q1/1	1	4Pole, RotarySwitch,400A	LBL32409B33RDYR	SALZER	LBL32409B33RDYR(1)(1)	&/1.1:E	
-Q2/7	1	2.5-4.0A(1.1KW/2HP)	GV2-ME08AE11TQ	TELEMECANIQUE	TEL.MPCB GV2ME08	&/7.6:B	
-Q3/7	1	2.5-4.0A(1.1KW/2HP)	GV2-ME08AE11TQ	TELEMECANIQUE	TEL.MPCB GV2ME08	&/7.7:B	
-RTD1/71	1	RTD Signal Conditioner,Head Mountable,0-250 DegC	TX-2-2-W	ASHE	ASHE.TX-2-2-W	&/71.2:E	
-S1/9	1	Door Limit Switch NC		SURAJ	Push to off	&/9.6:B	
-S2/9	1	Door Limit Switch NC		SURAJ	Push to off	&/9.8:B	
-S3/95	1	Emergency,Illuminated,Turn to Release	XB5AW74B2N	SE	XB5AW74B2N	&/95.2:B	
-S4/95	1	PushBotton,Green,Illuminated,	XB5AW33B1N	SE	XB5AW33B1N	&/95.9:B	
-S5/97	1	PushBotton,Green,Illuminated,	XB5AW33B1N	SE	XB5AW33B1N	&/97.2:B	
-S6/97	1	PushBotton,Green,Illuminated,	XB5AW33B1N	SE	XB5AW33B1N	&/97.3:B	
-S7/97	1	PushBotton,Green,Illuminated,	XB5AW33B1N	SE	XB5AW33B1N	&/97.4:B	
-S8	1	PushBotton,Flush,Yellow,Non Illuminated,	XB5AA51N	SE	XB5AA51N	&/99.9:B	
-S9	1	Mushroom Push Button,Green Non Illuminated	XB5AC31N	SE	XB5AC31N	&/99.5:C	
-S10	1	Mushroom Push Button,Green Non Illuminated	XB5AC31N	SE	XB5AC31N	&/99.5:B	
-S11	1	SelctorSwitch,Three Position Center Stay Put	XB5AD53N	SE	XB5AD53N	&/99.6:B	
-S12	1	KeySelctorSwitch,TwoStayPut	XB5AG21N	SE	XB5AG21N	&/99.8:B	
-ST1/59	1	Surge filter system and field supply	EL9550	BECKHOFF	BEC.EL9550		
-ST7						&/63.3:D	
-T1/1	1	1Φ,400/200-500VA	400/200-500VA	JAYANTHI	400/200-500VA	&/1.3:C	
-T2/7	1	3Φ-Δ/Δ,400/80-16KVA	3Φ-Δ/Δ,400/80-16KVA	JAYANTHI	3Φ-Δ/Δ,400/80-16KVA	&/7.2:C	
-U1/1	1				Parrt To be Created	&/1.4:D	
-U2/1	1	Panel AC		Advance Cooling	TURBO1000M2360Hz	&/1.8:C	
-U3/7						&/7.3:D	CHILLER
-U3/11	1	Static Frequency Converter = < 1 kV	FC-051P2K2T4E20H3BXCXXXSXXX	DANFOSS	DAN.FC-051P2K2T4E20H3BXCXXXSXXX	&/11.1:C	
-U4/13	1	Static Frequency Converter = < 1 kV	FC-051P2K2T4E20H3BXCXXXSXXX	DANFOSS	DAN.FC-051P2K2T4E20H3BXCXXXSXXX	&/13.1:C	
-U5/17	1	ECM generator 100V/90A		VAMTEC	VAM.ECMPSU	&/17.2:D	
-U6/19	1	ECM generator 100V/90A		VAMTEC	VAM.ECMPSU	&/19.2:D	
-U6/39	1			VAMTEC	VAM.SC CHECKV1.0	&/39.2:D	
-U7/21	1	ECM generator 100V/90A		VAMTEC	VAM.ECMPSU	&/21.2:C	
-U8/23	1	ECM generator 100V/90A		VAMTEC	VAM.ECMPSU	&/23.2:D	
-U8/51	1	Safety Light Curtain, 450mm Hand,Type-4, Protection, 25-dia. mm	SL-V24H	KEYENCE	KEY.SL-V24H	&/51.2:D	
-U9/73	1	pH Instrument	PH-600-1 (230V)	SENSEL	SENSEL.PH-600-1 (230V)	&/73.2:D	
-U10/73	1	Conductivity Instrument	CC603-1(230V)	SENSEL	SENSEL.CC603-1(230V)	&/73.6:D	
-U15						&/35.1:E	
-U16						&/37.1:E	
-Y1/163						&/163.3:E	
-Y2/106						&/143.4:E	
-Y2/163						&/163.4:E	
-Y3/106						&/143.5:E	
-Y3/163						&/163.5:E	
-Y4/163						&/163.6:E	

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Device tag list

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DT	Quantity	Designation	Type number	Manufacturer	Part number	Page / path	Function text
-Y5/163						8/163.7:E	
-Y6/163						8/163.8:E	
-Y7/106						8/143.8:E	
-Y7/163						8/163.8:E	
-Y8/106						8/143.9:E	
-Y9/108						8/147.3:E	
-Y10/108						8/147.4:E	
-Y11/108						8/147.5:E	
-Y12/108						8/147.6:E	
-Y13/108						8/147.7:E	
-Y14/108						8/147.8:E	
-Y15/108						8/147.8:E	
-Y16/108						8/147.9:E	
-Y17/110						8/151.3:E	
-Y18/110						8/151.4:E	
-Y19/110						8/151.5:E	
-Y20/110						8/151.6:E	
-Y21/110						8/151.7:E	
-Y22/110						8/151.8:E	
-Y24/110						8/151.9:E	
-Y25/112						8/155.3:E	
-Y29/112						8/155.7:E	
-Y30/112						8/155.8:E	
-Y31/112						8/155.8:E	
-Y32/112						8/155.9:E	

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